

Publication Report for energycostoptimization Survey

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1 Introduction

This report is a companion document to the main report generated for the extracted information used in the survey of CP and Scheduling. This document is concerned with some of the summary statistics, and with data quality issues that are highlighted for correction by the authors.

2 Data Quality

This section gives an overall overview of the works covered by the survey. We first look at all works, and consider which entries cannot be fully analyzed. We consider the following status outcomes: no DOI, the bib entry does not give a DOI, this typically means that we cannot find the citation and reference counts for the work. A special case is the Thesis type, which typically do not have a DOI assigned by the university. Even entries with a DOI may not be covered, we distinguish entries that are covered by neither Crossref nor Scopus, or entries which are covered by one, but not the other. The OK status indicates that we can find the entry in all our sources.

Note that OpenCitations does not distinguish between a DOI that is not covered, and a DOI for which there are no references or citations. In both cases, an empty list is returned by the query.

We may be able to repair some of the entries by finding a DOI for entries which miss them, or by correcting a mistake in a DOI, where neither Crossref nor Scopus recognizes the entry. Note that the system responses are cached, and missing entries are not repeatedly queried by the system. This means that additions or corrections in the databases that occur after we first queried them for a specific entry are not automatically taken into account. It may be good practice to re-run all queries from time to time to reflect updates in the databases.

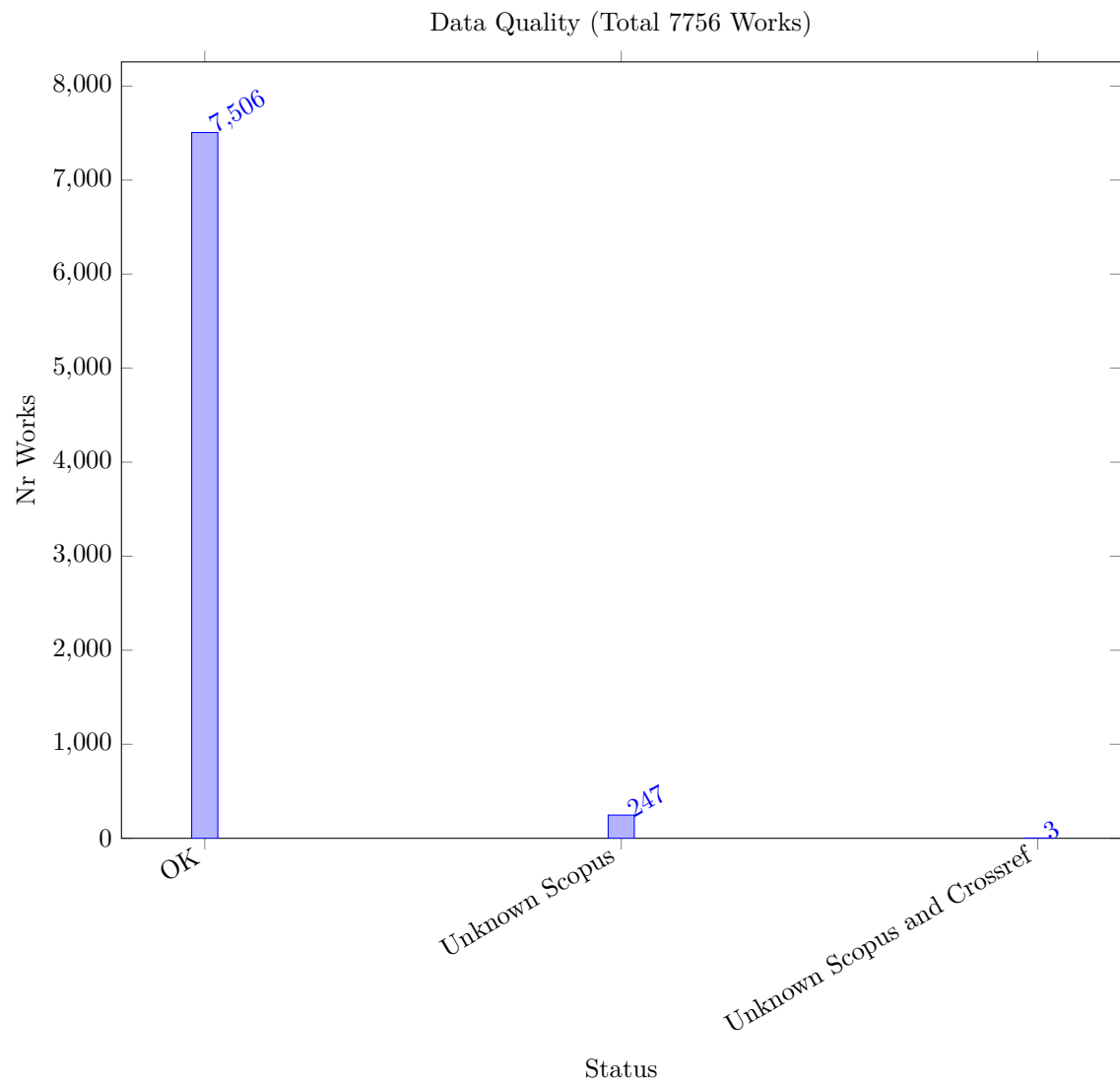


Table 1: Works Unknown to Crossref and Scopus

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
OConnor2025	10.48550/arxiv.2502.04935	OtherJournal	2025	0	0	0	null	0	NaN
OConnor2024a	10.48550/arxiv.2402.06714	OtherJournal	2024	0	0	0	null	0	NaN
OConnor2024b	10.48550/arxiv.2406.13851	OtherJournal	2024	0	0	0	null	0	NaN

Table 2: Works Unknown to Crossref

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
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Table 3: Works Unknown to Scopus

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Saib2025	10.2339/politeknik.1303193	OtherJournal	2025	0	0	0	null	0	NaN
Jin2024	10.1142/s2811034x24500060	OtherJournal	2024	0	24	0	null	24	100.00
Kosti2024	10.5937/tehnika2404453k	OtherJournal	2024	0	0	0	null	0	NaN
Ling2024a	10.1142/s2424786324500026	OtherJournal	2024	0	3	0	null	3	100.00
Pan2024	10.3389/ftther.2024.1391602	OtherJournal	2024	0	0	0	null	0	NaN
Omaji2024	10.53982/ajer.2024.0702.01-j	OtherJournal	2024	0	0	0	null	0	NaN
Ghazvini2024	10.1049/tje2.12372	OtherJournal	2024	0	1	0	null	1	100.00
Lytvyn2024	10.31548/energiya2(72).2024.168	OtherJournal	2024	0	0	0	null	0	NaN
Singha2024	10.24906/isc/2023/v37/i4/43717	OtherJournal	2024	0	0	0	null	0	NaN
Galeano2024	10.24857/rgsa.v18n10-307	OtherJournal	2024	0	0	0	null	0	NaN
Mubiru2024	10.4995/jarte.2024.20789	OtherJournal	2024	0	0	0	null	0	NaN
Jing2024	10.59717/j.xinn-energy.2024.100058	OtherJournal	2024	0	0	0	null	0	NaN
Sarwar2024	10.1049/enc2.12126	OtherJournal	2024	0	3	0	null	3	100.00
Usanova2024	10.1051/mateconf/202439201177	OtherJournal	2024	0	0	0	null	0	NaN
Rao2024a	10.1007/s41403-024-00498-z	OtherJournal	2024	0	0	0	null	0	NaN
Tunel2024	10.62802/b0ec2q73	OtherJournal	2024	0	0	0	null	0	NaN
Safari2024	10.1049/tje2.70012	OtherJournal	2024	0	0	0	null	0	NaN
Lv2024a	10.3389/frsgr.2024.1353732	OtherJournal	2024	0	0	0	null	0	NaN
Niu2024a	10.1007/s44295-024-00026-3	OtherJournal	2024	0	0	0	null	0	NaN
Adenekan2024	10.51594/ijmer.v6i5.1126	OtherJournal	2024	0	0	0	null	0	NaN
Katkar2024	10.3233/idt-240479	OtherJournal	2024	0	0	0	null	0	NaN
Xiao2024	10.59782/aai.v1i1.280	OtherJournal	2024	0	0	0	null	0	NaN
Kiianchuk2024	10.20998/2313-8890.2023.09.01	OtherJournal	2024	0	0	0	null	0	NaN
Sun2024b	10.1155/etep/5580709	OtherJournal	2024	0	0	0	null	0	NaN
Aljribi2024	10.2339/politeknik.886712	OtherJournal	2024	0	1	0	null	1	100.00
M. 2024	10.38124/ijisrt/ijisrt24jun019	OtherJournal	2024	0	1	0	null	1	100.00
Sobti2024	10.1051/mateconf/202439201182	OtherJournal	2024	0	2	0	null	2	100.00
Jalasri2024	10.4018/979-8-3693-5533-6.ch012	Inbook	2024	0	0	0	null	0	NaN
Qiu2024	10.1186/s40807-024-00103-5	OtherJournal	2024	0	0	0	null	0	NaN
Alotaibi2024	10.3390/fuels5020011	OtherJournal	2024	0	1	0	null	1	100.00
Mittal2024a	10.1051/mateconf/202439201187	OtherJournal	2024	0	0	0	null	0	NaN

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Shafiei2024	10.1049/tje2.12348	OtherJournal	2024	4	8	0	null	8	100.00
Nargeszar2024	10.1049/tje2.12387	OtherJournal	2024	0	1	0	null	1	100.00
Wang202426	10.1155/2024/3713139	OtherJournal	2024	0	0	0	null	0	NaN
Behera2024	10.12944/cwe.19.2.26	OtherJournal	2024	0	0	0	null	0	NaN
Lu2024e	10.37391/ijeer-120231	OtherJournal	2024	0	0	0	null	0	NaN
Petrovi2024	10.5937/imcsm24013p	OtherConf	2024	0	0	0	null	0	NaN
Zhang202423	10.1049/ein2.12018	OtherJournal	2024	0	0	0	null	0	NaN
Park2024a	10.1155/2024/1992135	OtherJournal	2024	0	0	0	null	0	NaN
Dekhici2024	10.4018/979-8-3693-7272-2.ch005	Inbook	2024	0	0	0	null	0	NaN
Vafaeva2024	10.1051/mateconf/202439201170	OtherJournal	2024	0	0	0	null	0	NaN
Gwaivangmin2024	10.20935/acadenergy7313	OtherJournal	2024	0	0	0	null	0	NaN
Ondachi2024	10.53982/ajer.2024.0701.30-j	OtherJournal	2024	0	0	0	null	0	NaN
Soni2024a	10.4018/979-8-3693-3884-1.ch008	Inbook	2024	0	0	0	null	0	NaN
elebi2024	10.47933/ijeir.1387314	OtherJournal	2024	1	2	0	null	2	100.00
Li202425	10.1016/j.heliyon.2024.e30018	OtherJournal	2024	0	0	0	null	0	NaN
Ustun2024	10.5772/intechopen.1005769	Inbook	2024	0	0	0	null	0	NaN
Pan2024c	10.59717/j.xinn-energy.2024.100040	OtherJournal	2024	0	7	0	null	7	100.00
Molakatala2024	10.4018/979-8-3693-3402-7.ch006	Inbook	2024	0	9	0	null	9	100.00
He2024d	10.1051/mateconf/202440401005	OtherJournal	2024	0	0	0	null	0	NaN
Abdelsattar2024b	10.21608/svusr.2023.240888.1153	OtherJournal	2024	4	6	0	null	6	100.00
Shaikh2023	10.48175/ijarsct-12060	OtherJournal	2023	1	0	0	null	1	100.00
Shu2023	10.54097/ije.v2i3.8807	OtherJournal	2023	0	0	0	null	0	NaN
Suneelgoutham2023	10.48175/ijarsct-9159	OtherJournal	2023	0	0	0	null	0	NaN
Liu2023d	10.32604/ee.2022.022610	OtherJournal	2023	2	2	0	null	2	100.00
Ahmadi2023a	10.1049/tje2.12237	OtherJournal	2023	11	13	0	null	13	100.00
Dursun2023	10.2339/politeknik.1074180	OtherJournal	2023	0	0	0	null	0	NaN
Bammeke2023	10.1007/978-3-031-30960-1_9	Inbook	2023	1	1	0	null	1	100.00
Kamoon2023	10.5772/intechopen.113001	Inbook	2023	2	2	0	null	2	100.00
2023a	10.12677/mos.2023.126504	OtherJournal	2023	0	2	0	null	2	100.00
kamel2023	10.21608/aej.2023.283030	OtherJournal	2023	1	1	0	null	1	100.00
Akka2023	10.36306/konjes.1377983	OtherJournal	2023	0	0	0	null	0	NaN
Bernadi2023	10.2478/bhee-2023-0004	OtherJournal	2023	0	0	0	null	0	NaN
Alizadeh2023	10.1186/s43067-023-00073-6	OtherJournal	2023	7	0	0	null	7	100.00
Eldessouky2023	10.38032/jea.2023.03.005	OtherJournal	2023	0	0	0	null	0	NaN
Gven2023	10.58559/ijes.1324236	OtherJournal	2023	4	13	0	null	13	100.00
2023b	10.21917/ijme.2023.0249	OtherJournal	2023	0	7	0	null	7	100.00
Suwi2023	10.52339/tjet.v42i1.889	OtherJournal	2023	0	2	0	null	2	100.00
Ma2023c	10.3724/j.issn.1674-4969.22092703	OtherJournal	2023	0	0	0	null	0	NaN
Yan2023e	10.1049/enc2.12093	OtherJournal	2023	2	3	0	null	3	100.00
Achiluzzi2023	10.1049/tje2.12295	OtherJournal	2023	1	1	0	null	1	100.00
Min2023	10.53941/ijamm.2023.100018	OtherJournal	2023	0	1	0	null	1	100.00
Joel2023	10.32628/ijrsrset2310229	OtherJournal	2023	0	0	0	null	0	NaN
Celtek2023	10.17798/bitlisfen.1317696	OtherJournal	2023	0	0	0	null	0	NaN
Bonkougou2023	10.1155/2023/8892122	OtherJournal	2023	0	0	0	null	0	NaN
Pavan2023	10.26634/jcir.11.1.19971	OtherJournal	2023	0	0	0	null	0	NaN
Akbari2022	10.1155/2022/7217478	OtherJournal	2022	3	3	0	null	3	100.00
Zhang2022a	10.1155/2022/7041778	OtherJournal	2022	6	9	0	null	9	100.00
Dassa2022	10.4018/978-1-7998-8561-0.ch005	Inbook	2022	8	8	0	null	8	100.00
Momayezi2022	10.1007/978-3-030-72322-4_67-1	Inbook	2022	0	0	0	null	0	NaN
Reciou2022	10.4018/978-1-6684-4829-8.ch005	Inbook	2022	0	0	0	null	0	NaN
Ouedraogo2022	10.2139/ssrn.4048104	OtherJournal	2022	0	0	0	null	0	NaN
Kumar2022a	10.1201/b22884-2	Inbook	2022	0	1	0	null	1	100.00
Xu2022d	10.54097/hset.v3i.705	OtherJournal	2022	1	1	0	null	1	100.00

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Mohan2022	10.1049/enc2.12059	OtherJournal	2022	2	3	0	null	3	100.00
2022	10.56082/annalsarsciinfo.2022.1-2.15	OtherJournal	2022	0	0	0	null	0	NaN
Ferro2022	10.4018/978-1-7998-6858-3.ch007	Inbook	2022	0	0	0	null	0	NaN
Alsharif2022	10.11113/elektrika.v21n3.353	OtherJournal	2022	6	5	0	null	6	100.00
Alfiah2022	10.34306/ijcitsm.v2i1.79	OtherJournal	2022	3	3	0	null	3	100.00
Akindeji2022	10.4028/p-13gc8e	OtherJournal	2022	4	4	0	null	4	100.00
Kama2022	10.1002/est2.332	OtherJournal	2022	20	20	0	null	20	100.00
Han2022a	10.1049/enc2.12058	OtherJournal	2022	5	12	0	null	12	100.00
Thogaru2022	10.1155/2022/3676438	OtherJournal	2022	2	2	0	null	2	100.00
Daragmeh2022	10.7148/2022-0254	OtherConf	2022	0	1	0	null	1	100.00
Reciou2022a	10.51485/ajss.v7i3.167	OtherJournal	2022	3	2	0	null	3	100.00
xi2022	10.2139/ssrn.4137325	OtherJournal	2022	0	0	0	null	0	NaN
Ma2022e	10.1049/enc2.12071	OtherJournal	2022	6	10	0	null	10	100.00
Rasheed2022	10.2139/ssrn.4118135	OtherJournal	2022	0	0	0	null	0	NaN
Souhe2022	10.1049/tje2.12146	OtherJournal	2022	16	16	0	null	16	100.00
Singh2022a	10.2139/ssrn.4145310	OtherJournal	2022	0	0	0	null	0	NaN
Hazra2022	10.4018/ijeoe.295983	OtherJournal	2022	1	1	0	null	1	100.00
Eidiani2022	10.1007/978-3-030-72322-4_61-1	Inbook	2022	0	0	0	null	0	NaN
Ding2022a	10.2139/ssrn.4165796	OtherJournal	2022	0	0	0	null	0	NaN
Braun2022	10.1007/978-3-030-72322-4_163-1	Inbook	2022	0	0	0	null	0	NaN
Gao2022c	10.2139/ssrn.4173878	OtherJournal	2022	0	0	0	null	0	NaN
Guneser2022	10.4018/978-1-7998-8561-0.ch008	Inbook	2022	5	6	0	null	6	100.00
Sharma2022f	10.1016/j.cep.2022.109088	OtherJournal	2022	5	4	0	null	5	100.00
Shezan2022	10.47981/j.mijst.10(01)2022.318(15-24)	OtherJournal	2022	16	18	0	null	18	100.00
Reciou2021	10.4018/978-1-7998-8048-6.ch051	Inbook	2021	6	6	0	null	6	100.00
Chaudhary2021	10.1063/9780735422827_005	Inbook	2021	2	2	0	null	2	100.00
Amoura2021	10.1007/978-981-16-3246-4_10	Inbook	2021	3	3	0	null	3	100.00
Pansota2021	10.51846/vol4iss3pp8-17	OtherJournal	2021	10	11	0	null	11	100.00
Hashmi2021	10.51846/vol4iss2pp21-27	OtherJournal	2021	3	3	0	null	3	100.00
Ali2021c	10.22581/muet1982.2104.09	OtherJournal	2021	1	1	0	null	1	100.00
Deng2021b	10.1049/tje2.12011	OtherJournal	2021	3	3	0	null	3	100.00
Ramalingam2021	10.1007/978-981-16-2712-5_43	Inbook	2021	5	4	0	null	5	100.00
Sadeeq2021	10.38094/jastt20285	OtherJournal	2021	78	85	0	null	85	100.00
Zhu2021d	10.19206/ce-131967	OtherJournal	2021	59	75	0	null	75	100.00
Deming2021	10.18034/apjee.v8i2.762	OtherJournal	2021	0	10	0	null	10	100.00
Kamal2021	10.1002/est2.314	OtherJournal	2021	12	12	0	null	12	100.00
Thomas2021	10.5772/intechopen.94989	Inbook	2021	1	1	0	null	1	100.00
Kannan2021	10.1016/j.matpr.2021.03.698	OtherJournal	2021	3	4	0	null	4	100.00
Alsharif2021	10.31580/sps.v3i1.1886	OtherJournal	2021	1	0	0	null	1	100.00
2021a	10.35940/ijeat.c2265.0210321	OtherJournal	2021	4	5	0	null	5	100.00
Fang2021b	10.1109/jsyst.2020.3008316	OtherJournal	2021	47	49	0	null	49	100.00
Pluihin2021	10.33042/2079-424x.2021.60.2.02	OtherJournal	2021	3	3	0	null	3	100.00
Das2021a	10.1002/est2.223	OtherJournal	2021	36	45	0	null	45	100.00
Ullah2021d	10.22581/muet1982.2103.06	OtherJournal	2021	0	0	0	null	0	NaN
Abdi2021	10.52460/issc.2021.026	OtherConf	2021	1	1	0	null	1	100.00
2021b	10.35940/ijitee.c8365.0110321	OtherJournal	2021	3	3	0	null	3	100.00
Vu2021	10.2139/ssrn.3943537	OtherJournal	2021	0	0	0	null	0	NaN
Ferencz2021	10.2478/auseme-2021-0007	OtherJournal	2021	2	2	0	null	2	100.00
Swami2021	10.1007/978-981-16-4103-9_23	Inbook	2021	0	0	0	null	0	NaN
Wang2021u	10.2139/ssrn.3954338	OtherJournal	2021	0	0	0	null	0	NaN
Hossain2021a	10.1007/978-3-030-68154-8_59	Inbook	2021	1	1	0	null	1	100.00
Suriadi2021	10.1088/1757-899x/1087/1/012080	OtherJournal	2021	1	0	0	null	1	100.00
Yuan2021b	10.1145/3508467.3508472	OtherJournal	2021	3	3	0	null	3	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Cusa2021	10.31686/ijier.vol9.iss3.3015	OtherJournal	2021	1	1	0	null	1	100.00
Chaudhary2021a	10.1007/978-981-16-1220-6_28	Inbook	2021	1	1	0	null	1	100.00
Valencia2020	10.22517/23447214.23541	OtherJournal	2020	6	6	0	null	6	100.00
Lasmari2020	10.6001/energetika.v66i1.4294	OtherJournal	2020	13	14	0	null	14	100.00
Gler2020	10.18038/estubtda.648767	OtherJournal	2020	0	1	0	null	1	100.00
Molla2020	10.4018/978-1-7998-1230-2.ch011	Inbook	2020	3	3	0	null	3	100.00
Aubeck2020	10.23919/ecc51009.2020.9143843	OtherConf	2020	5	0	0	null	5	100.00
Aktas2020	10.31593/ijeat.680639	OtherJournal	2020	0	0	0	null	0	NaN
Chen2020a	10.1049/enc2.12002	OtherJournal	2020	30	33	0	null	33	100.00
Sintianingrum2020	10.23960/jesr.v2i1.34	OtherJournal	2020	1	0	0	null	1	100.00
Liu2020e	10.1049/enc2.12010	OtherJournal	2020	12	15	0	null	15	100.00
Zhuang2020	10.1142/9789811223334_0161	OtherConf	2020	1	1	0	null	1	100.00
Dzobo2020	10.4018/978-1-5225-8551-0.ch001	Inbook	2020	1	1	0	null	1	100.00
Faria2020	10.18502/keg.v5i6.7099	OtherJournal	2020	3	3	0	null	3	100.00
Shuai2020	10.1109/pesgm41954.2020.9281545	OtherConf	2020	0	2	0	null	2	100.00
Arefin2020	10.5772/intechopen.91841	Inbook	2020	9	9	0	null	9	100.00
Mallikarjuna2020	10.4018/ijfc.2020010101	OtherJournal	2020	14	14	0	null	14	100.00
Hazra2020	10.4018/978-1-7998-3222-5.ch001	Inbook	2020	0	0	0	null	0	NaN
Oyekanlu2020	10.5772/intechopen.91837	Inbook	2020	3	3	0	null	3	100.00
Vorwerk2020	10.37394/23202.2020.19.28	OtherJournal	2020	0	0	0	null	0	NaN
Obukhov2020	10.17588/2072-2672.2020.6.025-038	OtherJournal	2020	1	1	0	null	1	100.00
Dey2020b	10.1186/s43067-020-00023-6	OtherJournal	2020	17	0	0	null	17	100.00
Kalananda2020	10.1007/978-981-15-9968-2_6	Inbook	2020	3	4	0	null	4	100.00
Khalaf2020	10.26480/aem.01.2020.11.23	OtherJournal	2020	13	12	0	null	13	100.00
kugor2020	10.1504/ijpt.2020.10031902	OtherJournal	2020	0	1	0	null	1	100.00
Reciouli2019	10.4018/978-1-5225-8030-0.ch005	Inbook	2019	1	2	0	null	2	100.00
Mushtaq2019	10.22581/muet1982.1903.06	OtherJournal	2019	1	1	0	null	1	100.00
Bayat2019	10.3233/jifs-190666	OtherJournal	2019	9	8	0	null	9	100.00
2019	10.17775/cseejpes.2019.00340	OtherJournal	2019	38	34	0	null	38	100.00
Shan2019	10.3390/sul1072124	OtherJournal	2019	11	10	0	null	11	100.00
Sujil2019	10.1049/joe.2018.9323	OtherJournal	2019	34	29	0	null	34	100.00
Doroudchi2019	10.1049/joe.2018.8153	OtherJournal	2019	4	2	0	null	4	100.00
Gundogdu2019a	10.1049/joe.2018.8226	OtherJournal	2019	0	12	0	null	12	100.00
Elkazaz2019a	10.3390/en12142712	OtherJournal	2019	11	12	0	null	12	100.00
Dey2019	10.4018/ijeoe.2019010104	OtherJournal	2019	5	5	0	null	5	100.00
Adjei2019	10.9734/jamcs/2019/v33i230175	OtherJournal	2019	1	1	0	null	1	100.00
Parashar2019	10.1049/joe.2018.9237	OtherJournal	2019	9	7	0	null	9	100.00
Ye2019	10.1049/joe.2018.5043	OtherJournal	2019	12	10	0	null	12	100.00
Srivastava2019	10.1109/locs.2019.2931967	OtherJournal	2019	9	11	0	null	11	100.00
Khan2019a	10.4018/978-1-5225-8030-0.ch014	Inbook	2019	22	23	0	null	23	100.00
Shetty2019	10.36548/jeea.2019.2.004	OtherJournal	2019	1	1	0	null	1	100.00
Chedid2019	10.1002/est2.46	OtherJournal	2019	49	59	0	null	59	100.00
Khan2019c	10.3390/app9235197	OtherJournal	2019	17	19	0	null	19	100.00
Lei2019	10.1109/tsg.2019.2892201	OtherJournal	2019	35	37	0	null	37	100.00
Diab2019	10.14569/ijacsa.2019.0100817	OtherJournal	2019	10	8	0	null	10	100.00
Du2019	10.1109/tii.2019.2899885	OtherJournal	2019	57	65	0	null	65	100.00
Zhang2019f	10.1109/tii.2019.2915051	OtherJournal	2019	19	21	0	null	21	100.00
Rivera-Rojo2019	10.4018/978-1-5225-8131-4.ch023	Inbook	2019	1	1	0	null	1	100.00
Singh2019e	10.4018/978-1-5225-5709-8.ch013	Inbook	2019	2	1	0	null	2	100.00
Zhu2019d	10.1049/joe.2018.9231	OtherJournal	2019	10	9	0	null	10	100.00
Elgammal2018	10.1049/joe.2018.5036	OtherJournal	2018	44	43	0	null	44	100.00
Chamandoust2018	10.2478/ece-2018-0012	OtherJournal	2018	9	9	0	null	9	100.00
N.2018	10.21817/ijet/2018/v10i5/181005013	OtherJournal	2018	0	11	0	null	11	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Fina2018	10.1155/2018/9759680	OtherJournal	2018	41	36	0	null	41	100.00
Gaonkar2018	10.1007/s40012-018-0201-4	OtherJournal	2018	2	2	0	null	2	100.00
Wang2018j	10.1049/joe.2018.8276	OtherJournal	2018	2	2	0	null	2	100.00
An2018	10.3390/en11113039	OtherJournal	2018	22	19	0	null	22	100.00
Shan2018	10.1049/joe.2018.8849	OtherJournal	2018	12	12	0	null	12	100.00
Fan2018b	10.1049/joe.2018.8684	OtherJournal	2018	3	2	0	null	3	100.00
Khalid2018	10.1109/icpesg.2018.8384505	OtherConf	2018	4	4	0	null	4	100.00
Behrens2018	10.48084/etasr.1819	OtherJournal	2018	7	7	0	null	7	100.00
2018	10.17775/cseepjes.2017.01250	OtherJournal	2018	65	68	0	null	68	100.00
P.2018	10.1016/j.jesit.2017.06.001	OtherJournal	2018	90	98	0	null	98	100.00
Kumar2017	10.1049/joe.2017.0447	OtherJournal	2017	22	20	0	null	22	100.00
Ying2017	10.1049/joe.2017.0766	OtherJournal	2017	0	12	0	null	12	100.00
Li2017b	10.1049/iet-cps.2017.0050	OtherJournal	2017	0	10	0	null	10	100.00
Sheng2017	10.1049/joe.2017.0696	OtherJournal	2017	17	11	0	null	17	100.00
Parashar2017	10.1049/joe.2017.0673	OtherJournal	2017	13	12	0	null	13	100.00
Liu2017c	10.24251/hicss.2017.369	OtherConf	2017	6	6	0	null	6	100.00
Khemakhem2017	10.1109/sta.2017.8314936	OtherConf	2017	8	8	0	null	8	100.00
Meena2017	10.1049/joe.2017.0659	OtherJournal	2017	17	19	0	null	19	100.00
Reciou2017	10.51485/ajss.v2i2.34	OtherJournal	2017	9	11	0	null	11	100.00
Scalfati2017	10.1109/icrera.2017.8191125	OtherConf	2017	16	16	0	null	16	100.00
Shahryari2017	10.24017/science.2017.3.35	OtherJournal	2017	2	2	0	null	2	100.00
Cheng2017c	10.1186/s41601-017-0056-y	OtherJournal	2017	39	43	0	null	43	100.00
Kinsella2016	10.1609/aaai.v30i2.19079	OtherConf	2016	2	2	0	null	2	100.00
Ahmad2016	10.4018/978-1-5225-0072-8.ch001	Inbook	2016	3	2	0	null	3	100.00
Salehin2016	10.4028/www.scientific.net/amm.819.518	OtherJournal	2016	7	7	0	null	7	100.00
Iyer2016	10.4018/978-1-4666-9911-3.ch019	Inbook	2016	0	0	0	null	0	NaN
Rajarajeswari2016	10.17485/ijst/2016/v9i43/101858	OtherJournal	2016	13	13	0	null	13	100.00
Tesfaye2016	10.7753/ijsea0503.1005	OtherJournal	2016	24	20	0	null	24	100.00
Kumar2016a	10.1109/ictfecn.2016.8052753	OtherConf	2016	7	7	0	null	7	100.00
He2016	10.1049/iet-cps.2016.0019	OtherJournal	2016	339	368	0	null	368	100.00
Yan2015	10.1155/2015/384528	OtherJournal	2015	8	8	0	null	8	100.00
Kawashima2015	10.9746/jcmsi.8.52	OtherJournal	2015	14	14	0	null	14	100.00
Ahmed2015	10.4028/www.scientific.net/amm.785.526	OtherJournal	2015	16	16	0	null	16	100.00
Kumar2015	10.1155/2015/856075	OtherJournal	2015	24	26	0	null	26	100.00
Kumar2015a	10.1049/cp.2015.0332	OtherConf	2015	8	0	0	null	8	100.00
Kan2015	10.1007/978-3-319-13356-0_40	Inbook	2015	4	4	0	null	4	100.00
Manasseh2014	10.1049/joe.2014.0199	OtherJournal	2014	3	2	0	null	3	100.00
Ye2014	10.1177/0954407013512289	OtherJournal	2014	10	11	0	null	11	100.00
Cirocco2014	10.3390/challe5020473	OtherJournal	2014	9	9	0	null	9	100.00
Silva2014	10.14807/ijmp.v5i3.122	OtherJournal	2014	0	1	0	null	1	100.00
.2014	10.15623/ijret.2014.0319051	OtherJournal	2014	12	13	0	null	13	100.00
Cervantes2014	10.1201/b17506-18	Inbook	2014	0	0	0	null	0	NaN
Gudelj2013	10.7225/toms.v02.n01.004	OtherJournal	2013	13	16	0	null	16	100.00
Tomoiaga2013	10.4028/www.scientific.net/aef.8-9.185	OtherJournal	2013	7	6	0	null	7	100.00
Sou2013	10.23919/ecc.2013.6669409	OtherConf	2013	9	9	0	null	9	100.00
Hassan2013	10.3390/en6126608	OtherJournal	2013	53	50	0	null	53	100.00
Tahmasebi2013	10.17485/ijst/2013/v6i11.2	OtherJournal	2013	16	16	0	null	16	100.00
Lu2013	10.1145/2465529.2465551	OtherConf	2013	62	60	0	null	62	100.00
Balan2013	10.12720/sgce.2.3.343-349	OtherJournal	2013	4	4	0	null	4	100.00
Chang2013	10.4236/epe.2013.54b003	OtherJournal	2013	15	16	0	null	16	100.00
Ahmad2012	10.12720/sgce.1.1.15-21	OtherJournal	2012	16	15	0	null	16	100.00
Kani2011	10.1109/pes.2011.6039025	OtherConf	2011	12	13	0	null	13	100.00
Caron2010	10.1109/smartgrid.2010.5622073	OtherConf	2010	210	207	0	null	210	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Livengood2009	10.1287/serv.1.1.1	OtherJournal	2009	111	112	0	null	112	100.00
Chasse2009	10.3182/20091130-3-fr-4008.00015	OtherJournal	2009	13	13	0	null	13	100.00
Gong2008	10.1115/dscc2008-2241	OtherConf	2008	7	7	0	null	7	100.00
Chedid1996	10.1002/(sici)1099-114x(199607)20:7<609::aid-er176>3.0.co;2-o	OtherJournal	1996	76	75	0	null	76	100.00

2.1 Range of Citation Counts

We get citation counts for the works included in the survey from different sources. OpenCitations provides the set of papers citing a reference, but only if both have DOIs. Crossref gives a count of how many papers cite a reference, they include some papers without DOI. Scopus gives a citation count, but does not give access to the actual citations. In this table we show the works with the largest range of citation count, excluding all background works. A typical issue is that one source does not cover the work, and has a zero count. An alternative is where papers with many citations give a slightly different count depending on which links are included in their database.

The results seem to indicate the using multiple sources is required, to avoid leaving out works that are not covered by one specific source. Note that the WoS numbers are only present for a few works, we show them, but do not include them in computing range.

Table 4: Works with largest Range of Citation Counts

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Gad2022	10.1007/s11831-021-09694-4	OtherJournal	2022	524	0	894	null	894	100.00
Zhao2013	10.1109/tsg.2013.2251018	OtherJournal	2013	553	0	653	null	653	100.00
He2016	10.1049/iet-cps.2016.0019	OtherJournal	2016	339	368	0	null	368	100.00
Murty2020	10.1186/s41601-019-0147-z	OtherJournal	2020	204	0	332	null	332	100.00
Byrne2018	10.1109/access.2017.2741578	OtherJournal	2018	0	301	323	null	323	100.00
Anoune2018	10.1016/j.rser.2018.05.032	OtherJournal	2018	280	304	0	null	304	100.00
Zhu2012	10.1109/isgt.2012.6175785	OtherConf	2012	34	0	275	null	275	100.00
Mariano-Hernandez2021	10.1016/j.jobe.2020.101692	OtherJournal	2021	249	300	465	null	216	46.45
Caron2010	10.1109/smartgrid.2010.5622073	OtherConf	2010	210	207	0	null	210	100.00
Wieczorek2017	10.1016/j.apenergy.2017.02.022	OtherJournal	2017	0	185	201	null	201	100.00
Un-Noor2017	10.3390/en10081217	OtherJournal	2017	478	423	621	null	198	31.88
Huang2020c	10.1186/s41601-020-0154-0	OtherJournal	2020	94	0	195	null	195	100.00
Mohsenian-Rad2010a	10.1109/tsg.2010.2089069	OtherJournal	2010	2185	2232	2375	null	190	8.00
2020a	10.17775/cseejpes.2019.02890	OtherJournal	2020	0	61	166	null	166	100.00
Abdolrasol2021a	10.3390/electronics10212689	OtherJournal	2021	254	373	419	null	165	39.38
Teng2020	10.1016/j.ijhydene.2019.12.202	OtherJournal	2020	0	156	164	null	164	100.00
Ahmed2020a	10.1016/j.rser.2020.109792	OtherJournal	2020	678	808	838	null	160	19.09
Amin2020	10.3390/su122310160	OtherJournal	2020	0	137	145	null	145	100.00
Motlagh2020	10.3390/en13020494	OtherJournal	2020	431	489	569	null	138	24.25
Logenthiran2012	10.1109/tsg.2012.2195686	OtherJournal	2012	955	990	1091	null	136	12.47
Yu2015b	10.1109/tsg.2015.2501316	OtherJournal	2015	137	136	267	null	131	49.06
Rao2010	10.1109/infcom.2010.5461933	OtherConf	2010	339	331	460	null	129	28.04
Xiong2011	10.1109/isgt.2011.5759154	OtherConf	2011	73	0	129	null	129	100.00
Bouktif2018	10.3390/en11071636	OtherJournal	2018	578	649	706	null	128	18.13
Luo2014	10.1109/tpds.2013.69	OtherJournal	2014	92	0	126	null	126	100.00
Song2015a	10.1016/j.apenergy.2015.08.120	OtherJournal	2015	0	117	126	null	126	100.00
Fu2020b	10.1016/j.ijhydene.2020.01.017	OtherJournal	2020	0	118	124	null	124	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Hannan2021	10.1016/j.est.2021.103023	OtherJournal	2021	311	410	434	null	123	28.34
Bakare2023	10.1186/s42162-023-00262-7	OtherJournal	2023	63	0	116	null	116	100.00
Haghnegahdar2019	10.1007/s00521-019-04453-w	OtherJournal	2019	91	0	114	null	114	100.00
Livengood2009	10.1287/serv.1.1.1	OtherJournal	2009	111	112	0	null	112	100.00
Ma2015a	10.1109/tsg.2015.2419818	OtherJournal	2015	70	55	166	null	111	66.87
Cai2018	10.1186/s41601-018-0083-3	OtherJournal	2018	53	0	110	null	110	100.00
Liu2021d	10.1186/s41601-021-00184-0	OtherJournal	2021	72	0	109	null	109	100.00
Thirunavukkarasu2023	10.1016/j.rser.2023.113192	OtherJournal	2023	180	283	289	null	109	37.72
Melhem2017	10.1109/cjece.2017.2716780	OtherJournal	2017	0	90	107	null	107	100.00
Mesbahi2017	10.1109/tiv.2017.2720464	OtherJournal	2017	0	44	107	null	107	100.00
Zhang2019c	10.1016/j.apenergy.2019.113891	OtherJournal	2019	0	90	102	null	102	100.00
Li2020c	10.1109/tsg.2020.2968747	OtherJournal	2020	0	89	100	null	100	100.00
Yuan2018	10.1016/j.ijhydene.2018.03.033	OtherJournal	2018	0	91	98	null	98	100.00
Duan2023a	10.1016/j.segan.2023.101004	OtherJournal	2023	198	242	296	null	98	33.11
P.2018	10.1016/j.jesit.2017.06.001	OtherJournal	2018	90	98	0	null	98	100.00
Amir2023a	10.1016/j.est.2023.108694	OtherJournal	2023	121	209	216	null	95	43.98
Li2021h	10.1109/tste.2021.3090463	OtherJournal	2021	340	425	434	null	94	21.66
Khan2019	10.1007/s12652-018-01169-y	OtherJournal	2019	85	0	92	null	92	100.00
Zhang2012	10.1109/isgt.2012.6175655	OtherConf	2012	56	0	91	null	91	100.00
L2020	10.1016/j.enconman.2020.112474	OtherJournal	2020	412	466	503	null	91	18.09
Cardoso2013	10.1016/j.epsr.2013.05.005	OtherJournal	2013	76	0	90	null	90	100.00
Ajoulabadi2020	10.1016/j.energy.2020.117024	OtherJournal	2020	76	0	90	null	90	100.00
Adefarati2016	10.1049/iet-rpg.2015.0378	OtherJournal	2016	430	459	520	null	90	17.31

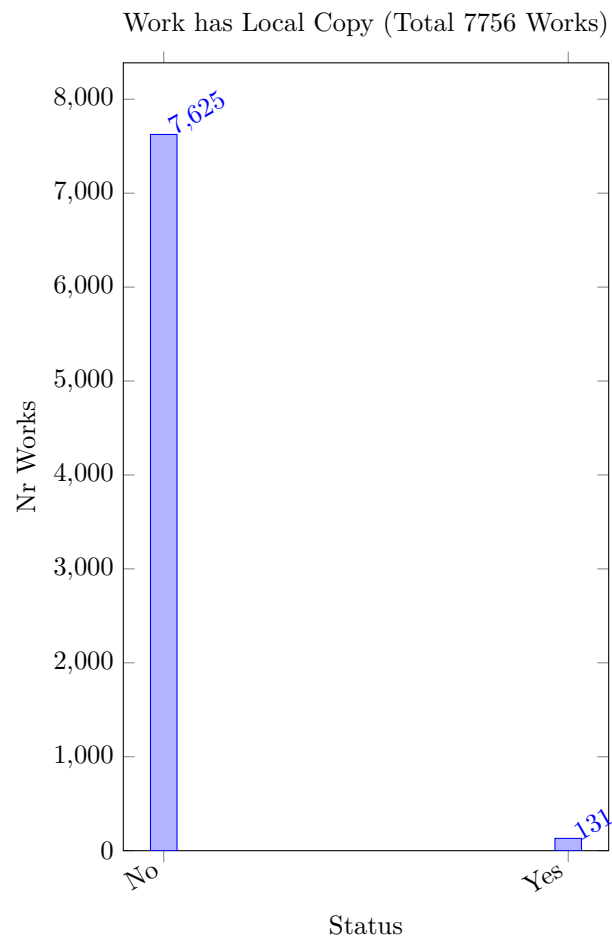
We only have Web of Science data in a few bibtex entries, we here try to evaluate their citation numbers on those bib entries which are from WoS.

Table 5: Works with WoS Citation Counts

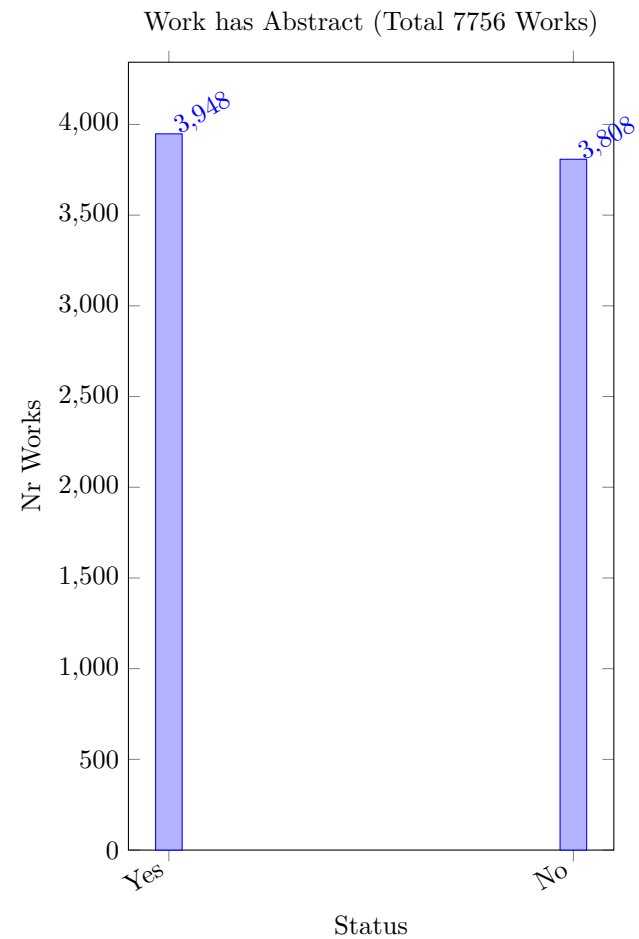
Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
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2.2 Local Copies

The tool relies on local pdf copies of works to perform a detailed analysis of the content of the work. We have collected our own private copies of works for that purpose. The following plot shows how many entries do not have a local copy, or which do not extract any concepts from the local copy. A detailed list of all missing entries is given in the main report. Note that in some cases we use an open access version of the work, which might differ slightly from the published version.



2.3 Presence of Abstracts



2.4 Orphan Files

The following list shows entries for which we have a pdf file in the works directory, but the name of the file does not match any key in the bibliography. These orphans should be resolved, either by correcting the name, or adding a bib entry for the work, or by removing the file, if it is not required.

If there are no files listed, then all pdf files in the works directory correspond to a bib entry, and no clean-up is required.

Table 6: Orphan Files

Key	File
Rahmani-andebili2017	Rahmani-andebili2017.xml

2.5 Missing Publisher

Table 7: Missing Publisher

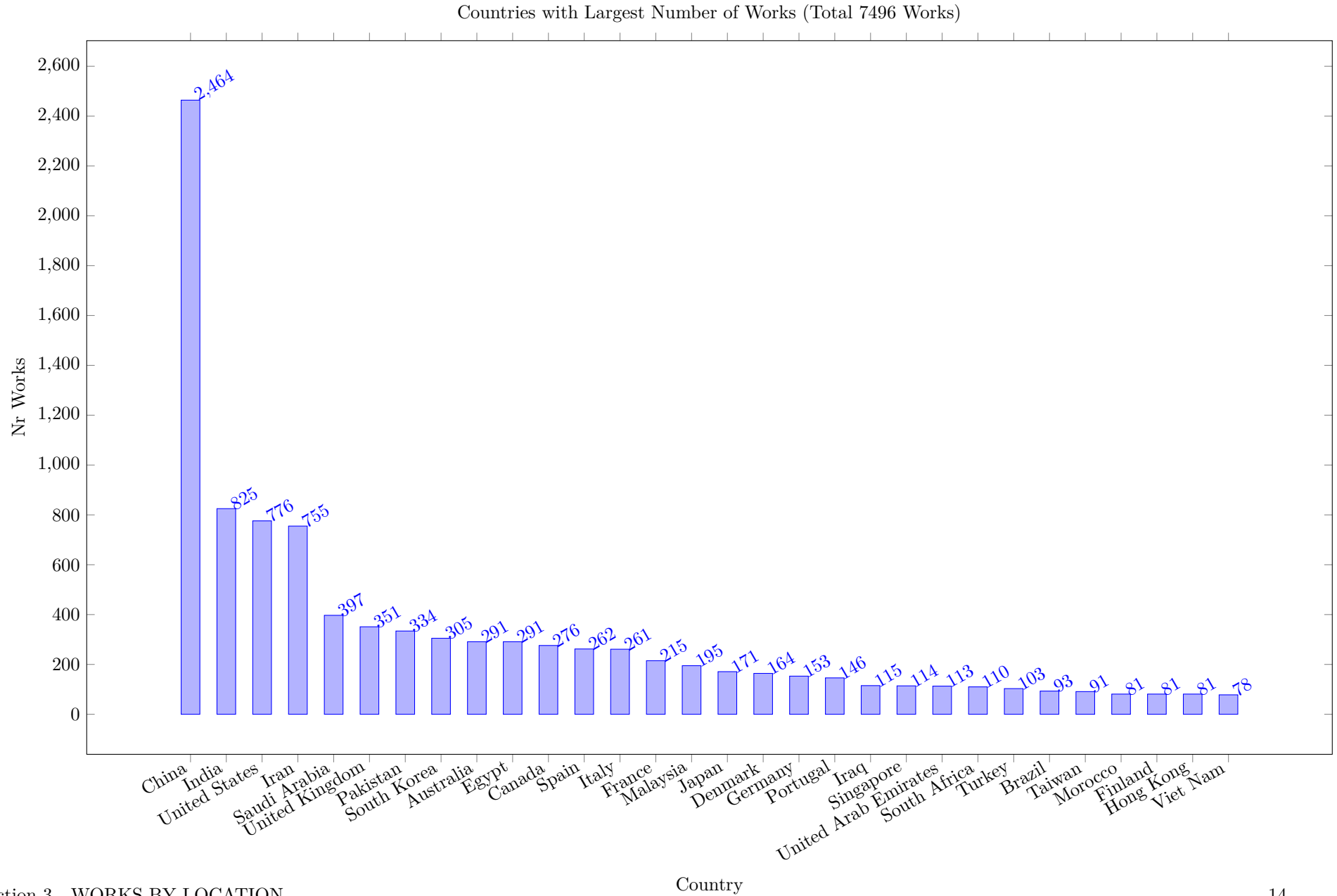
Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
OConnor2024a	10.48550/arxiv.2402.06714	OtherJournal	2024	0	0	0	null	0	NaN
OConnor2024b	10.48550/arxiv.2406.13851	OtherJournal	2024	0	0	0	null	0	NaN
OConnor2025	10.48550/arxiv.2502.04935	OtherJournal	2025	0	0	0	null	0	NaN

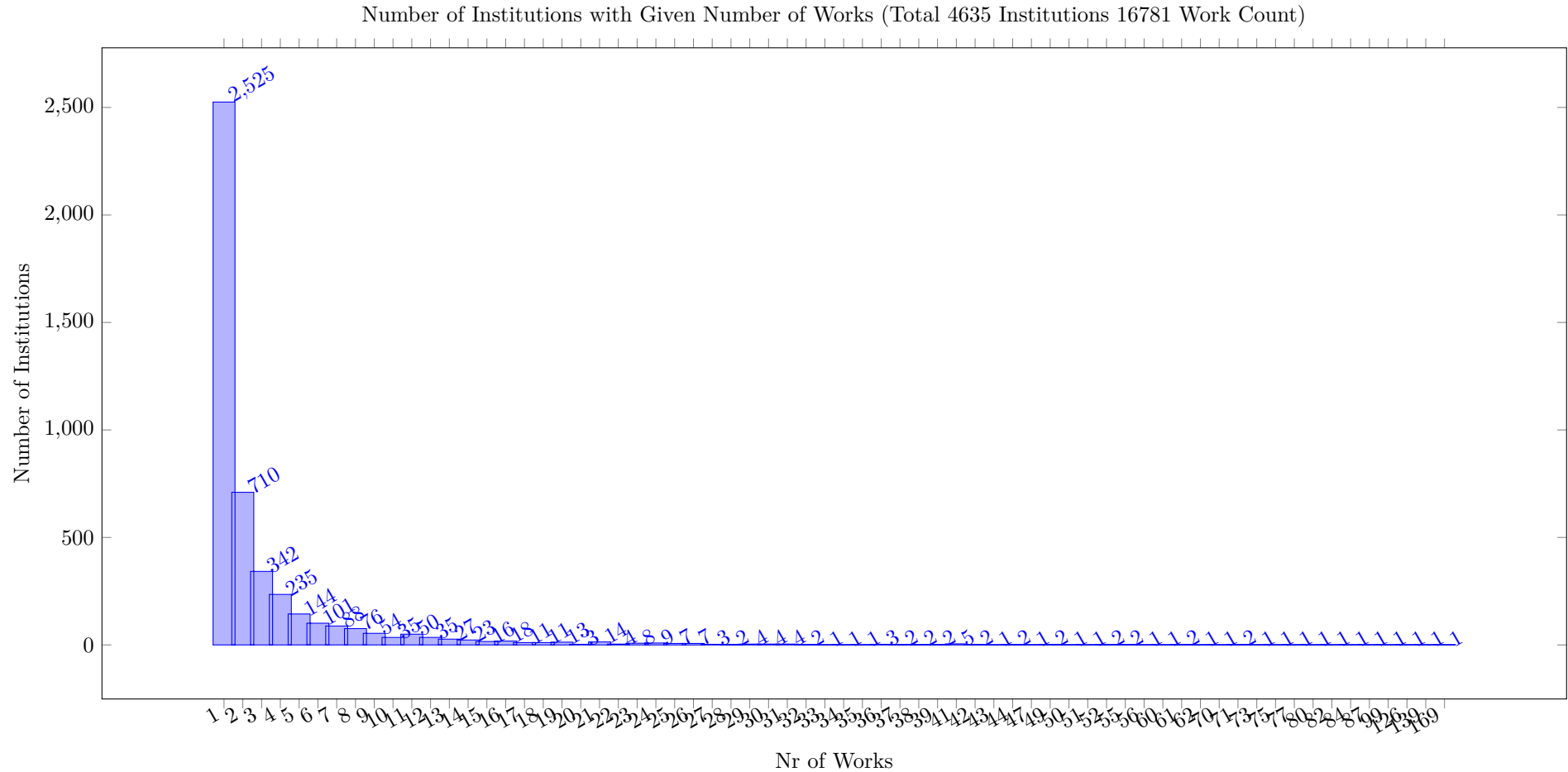
3 Works by Location



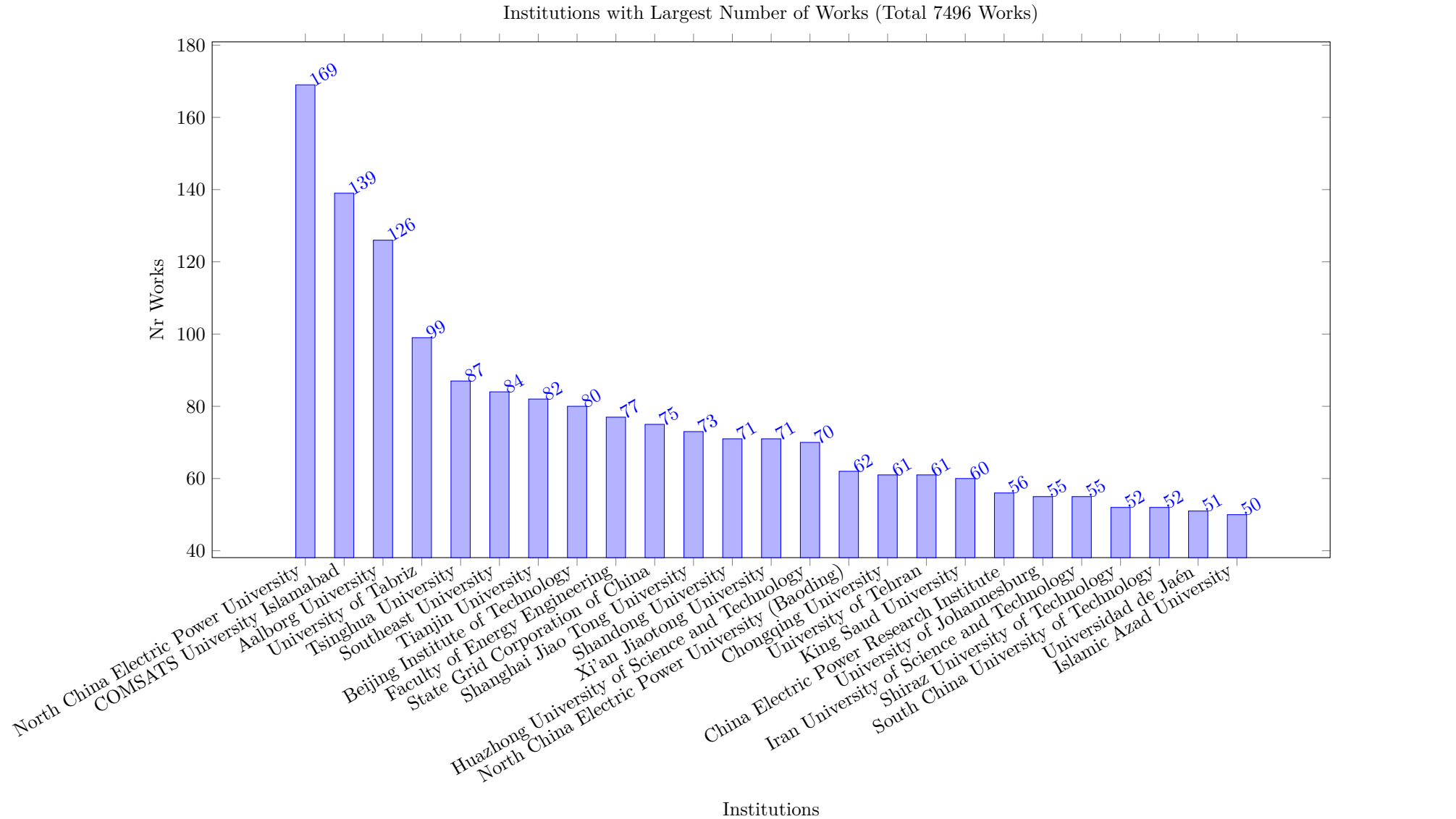
This section analyzes papers by affiliation, which is given by the Scopus data only. Only works which are covered by Scopus are included. We first present the number of papers by country. A paper is counted in this analysis (once), if at least one of the affiliations is from the country. Multiple affiliations from the same country only count once. The 30 countries with the largest counts are shown.

Note that one work will be counted for multiple countries, if the affiliations are from different countries. So the sum of the bar heights typically exceeds the total number of works considered.





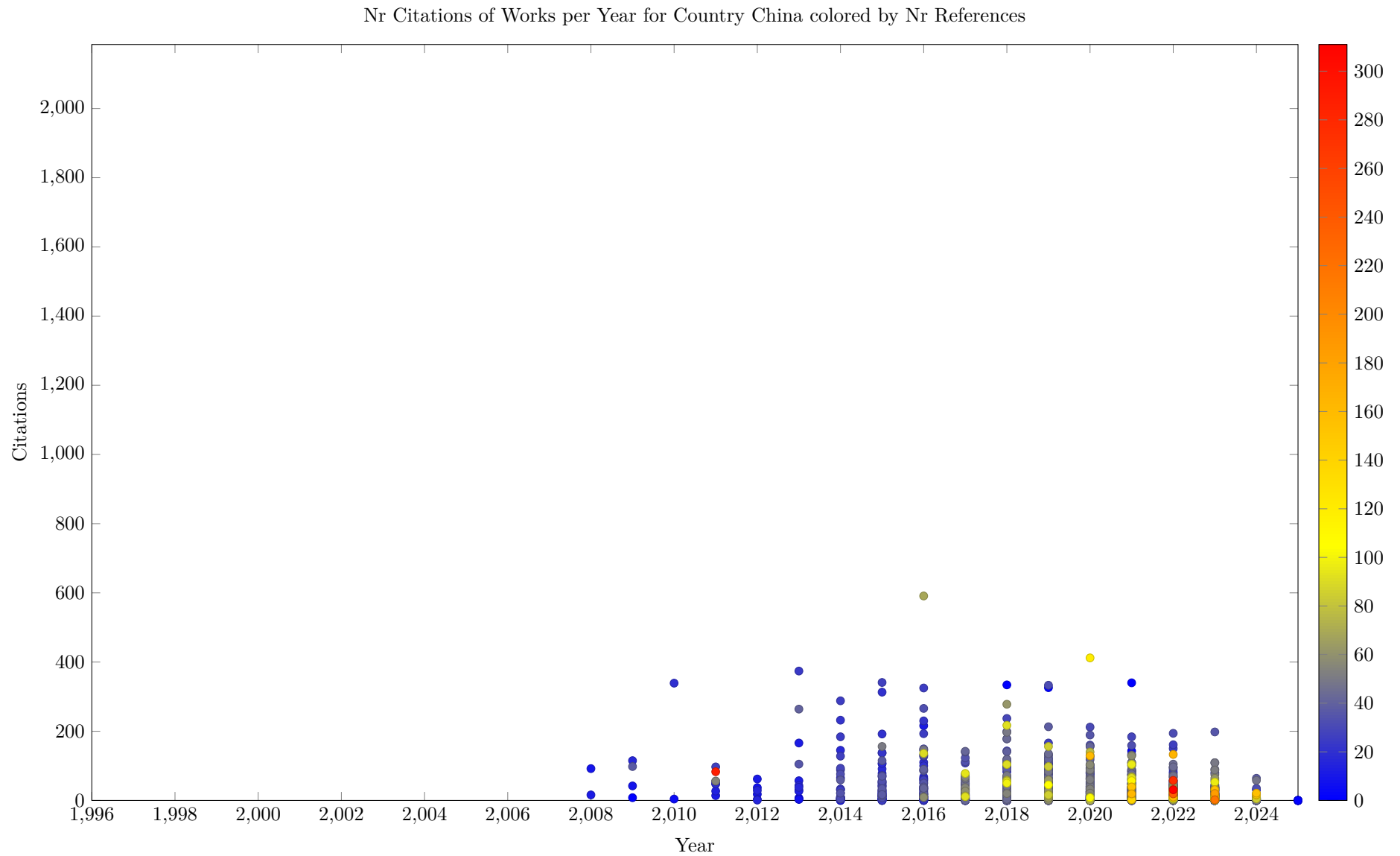
The next plot shows the number of papers associated to institutions, as stated in the Scopus affiliation. A work is counted, if at least one of the affiliations is from a given institution. Due to the format of the Scopus data, we cannot fractionally assign a paper based on the author affiliations, each paper is counted one for every institution for which an affiliation is given. If some author has multiple affiliations listed, we (mis)count the work for each of them.

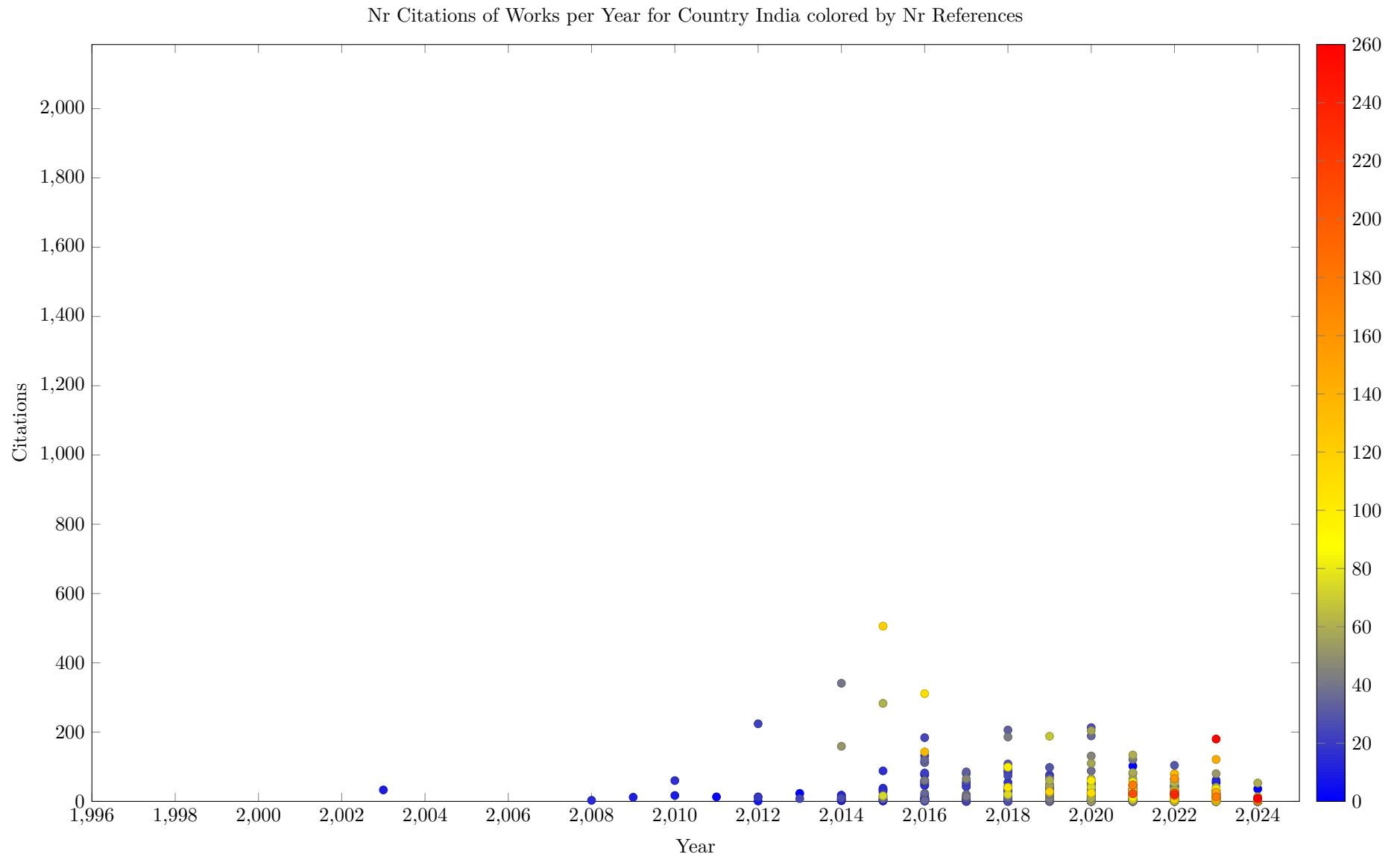


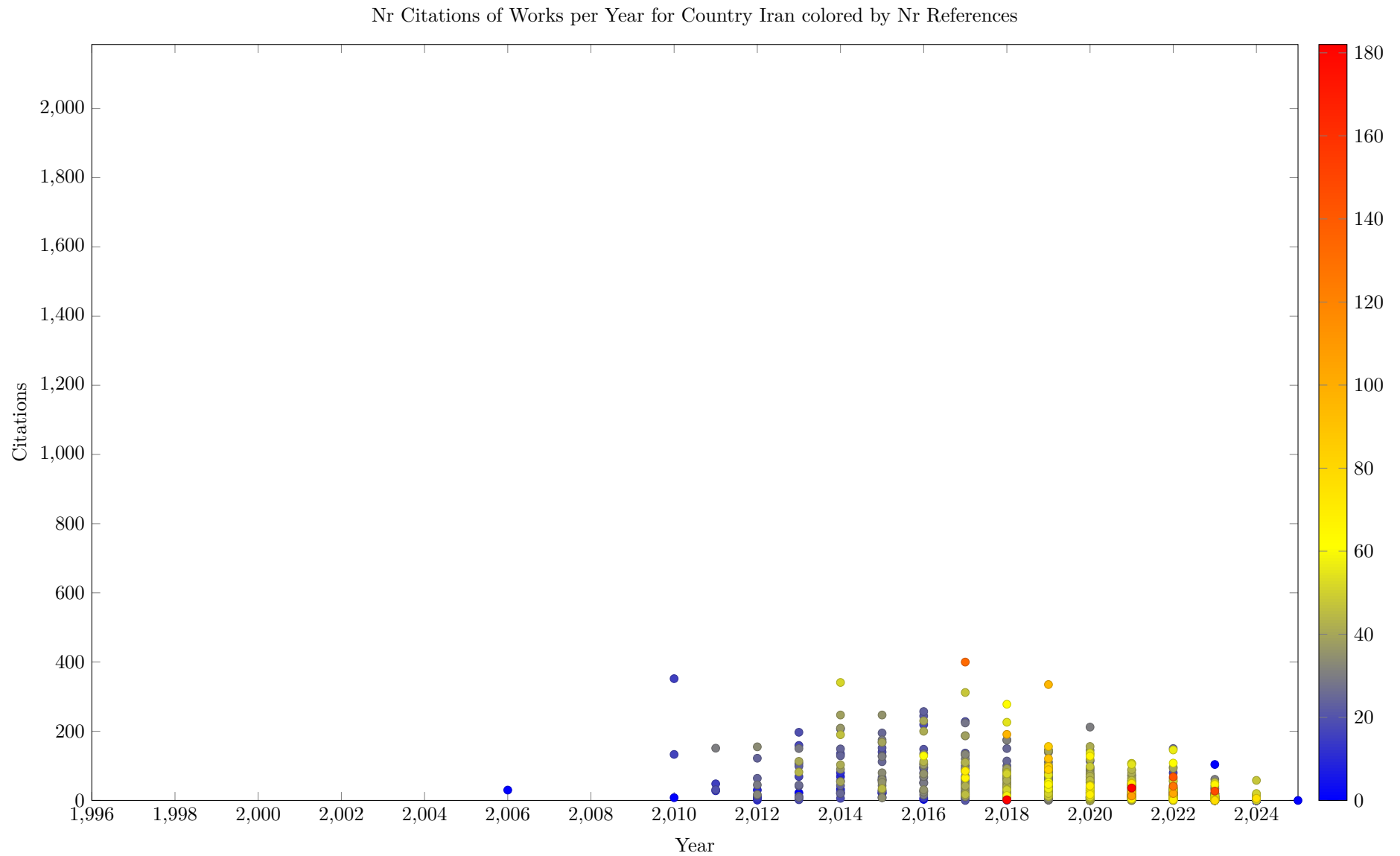
The following plots show for the top 30 countries when the works included were published, and how many citations (OpenCitation count) each paper had. The scatter plots are colored by the number of references (OpenCitation count), this help to identify surveys more easily. The plot gives an indication in which

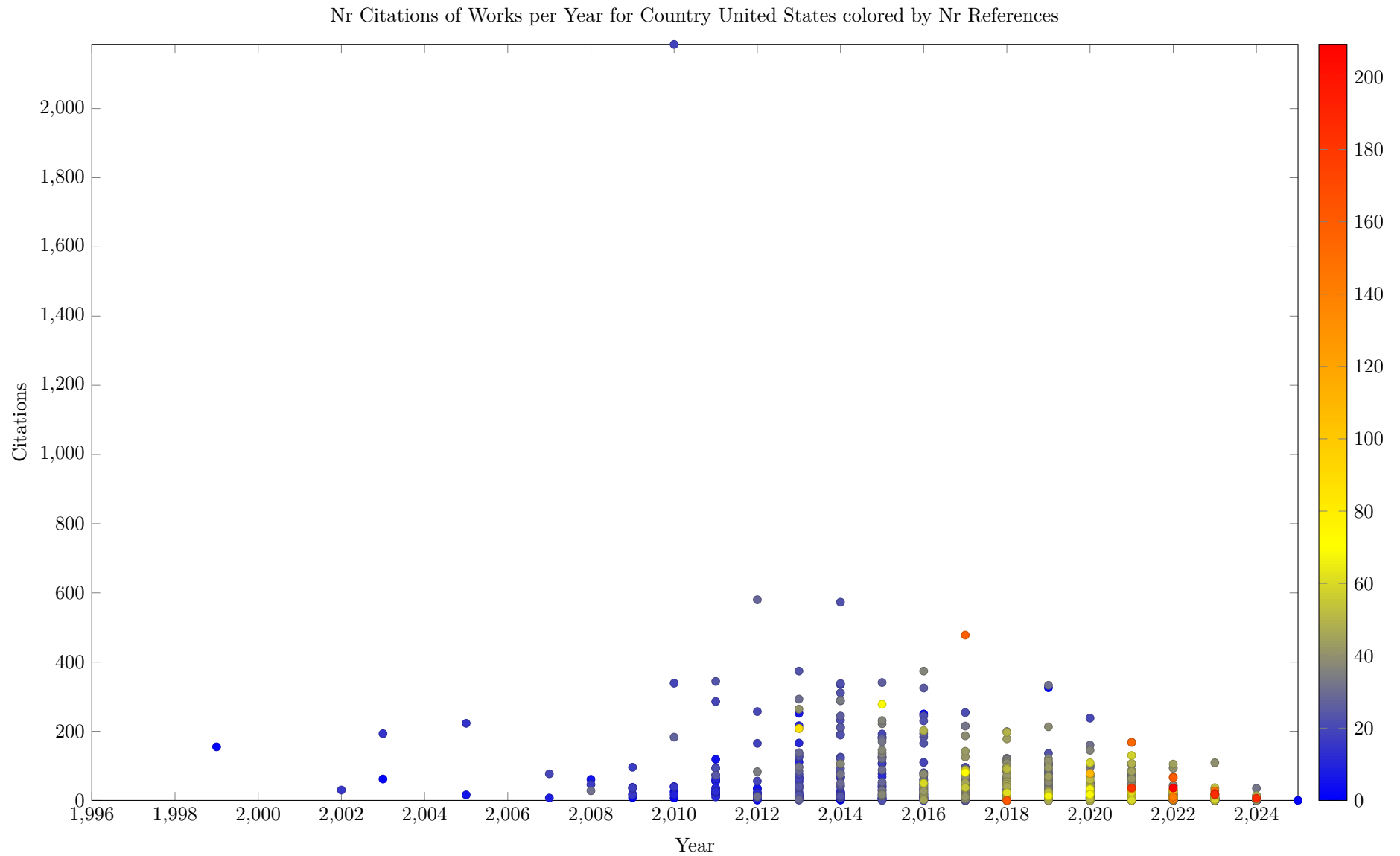
period the work from the country falls, and how influential the published works are. The x and y ranges of all plots are uniform to allow comparison between plots.

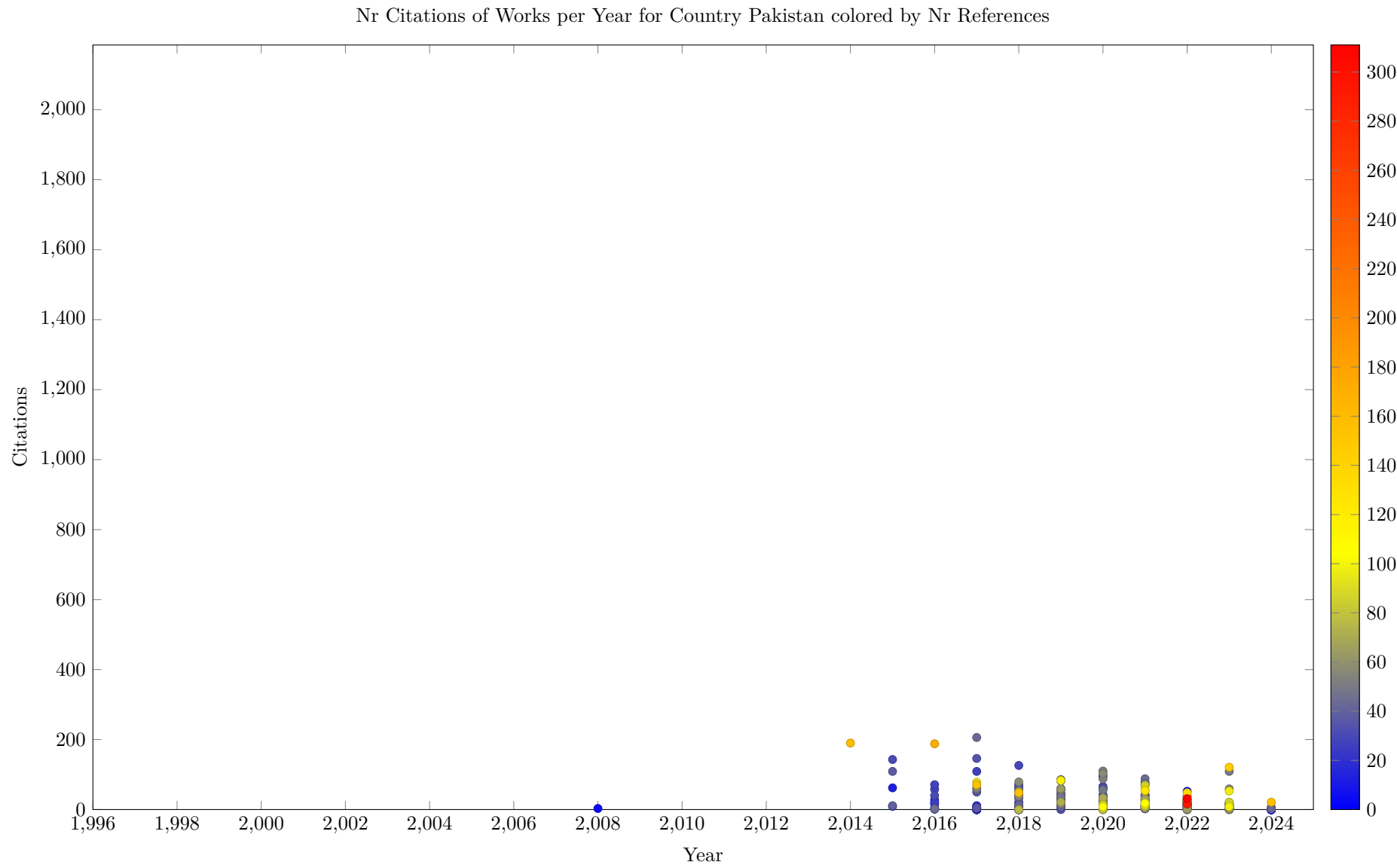
It would be nice to have tooltips on the plots, so identify specific works in the plots. This is currently not supported by the framework library used.

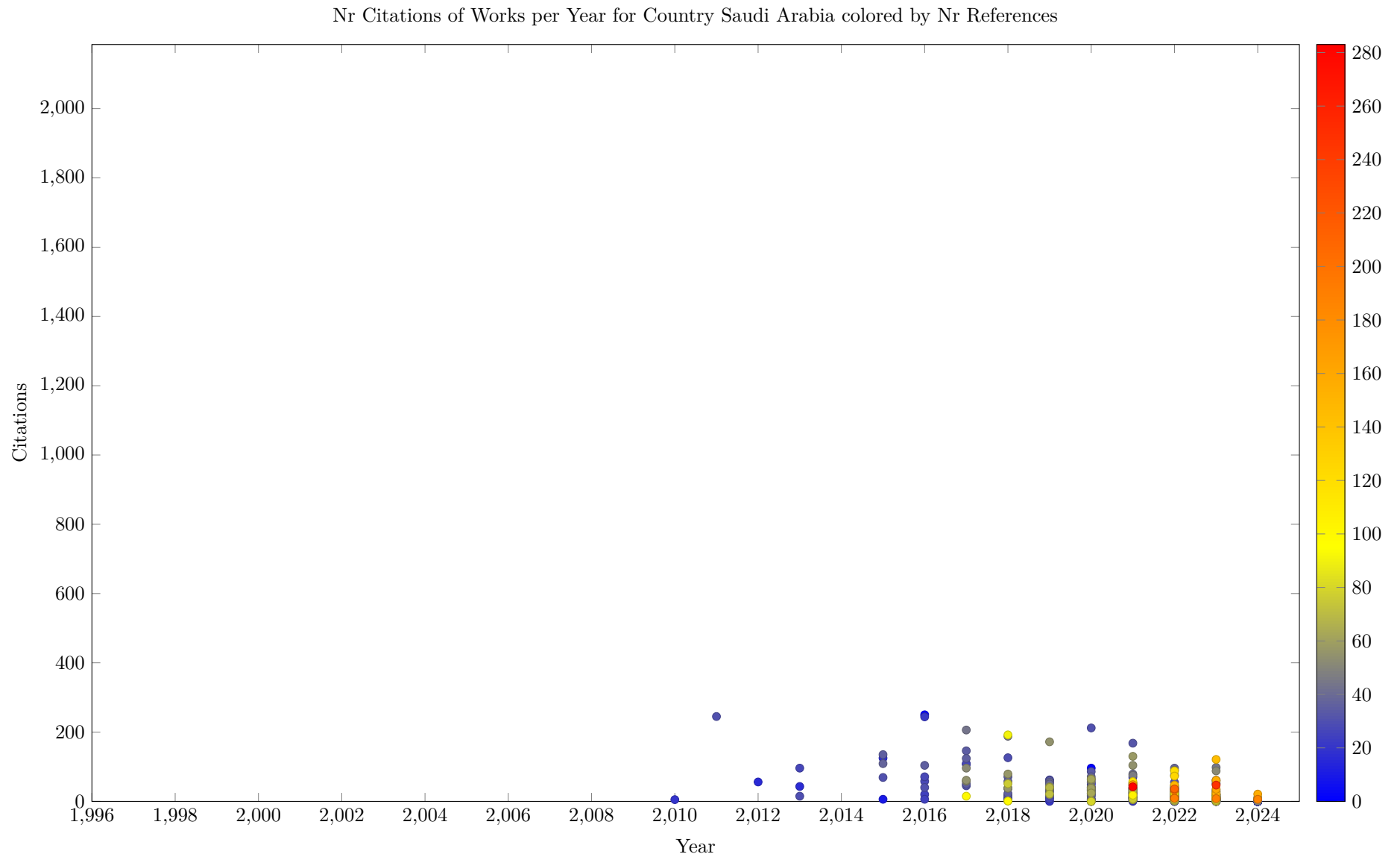


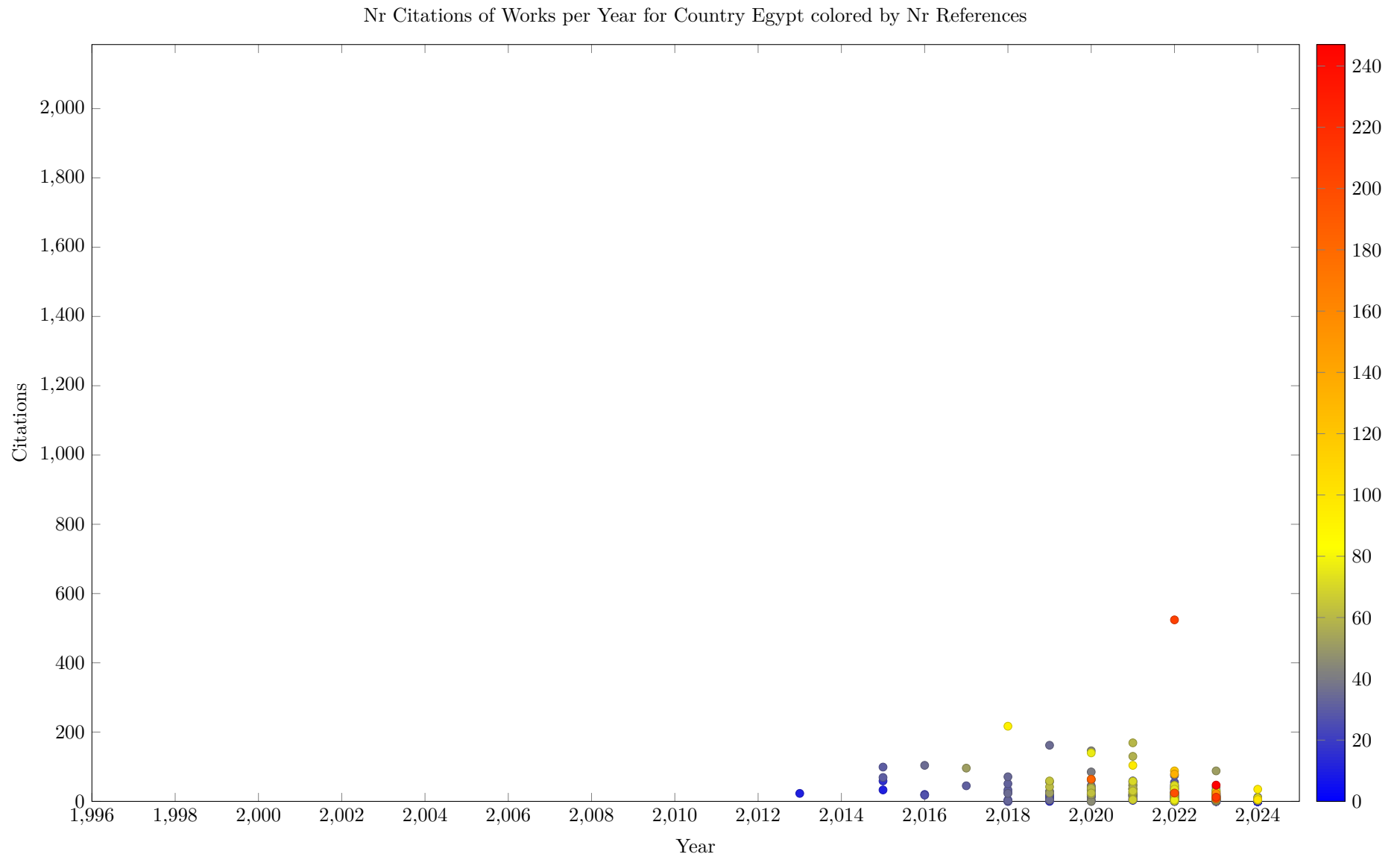


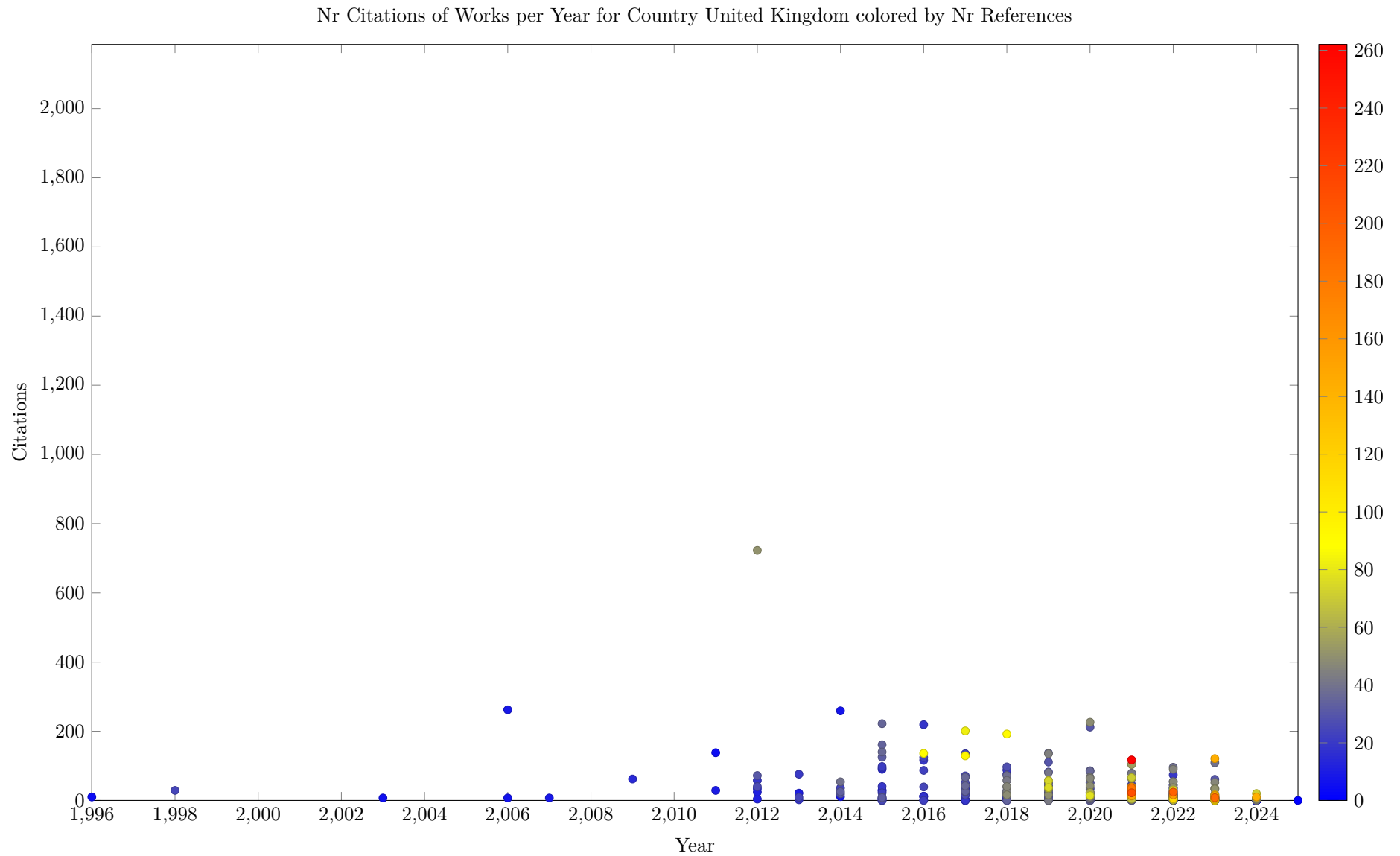


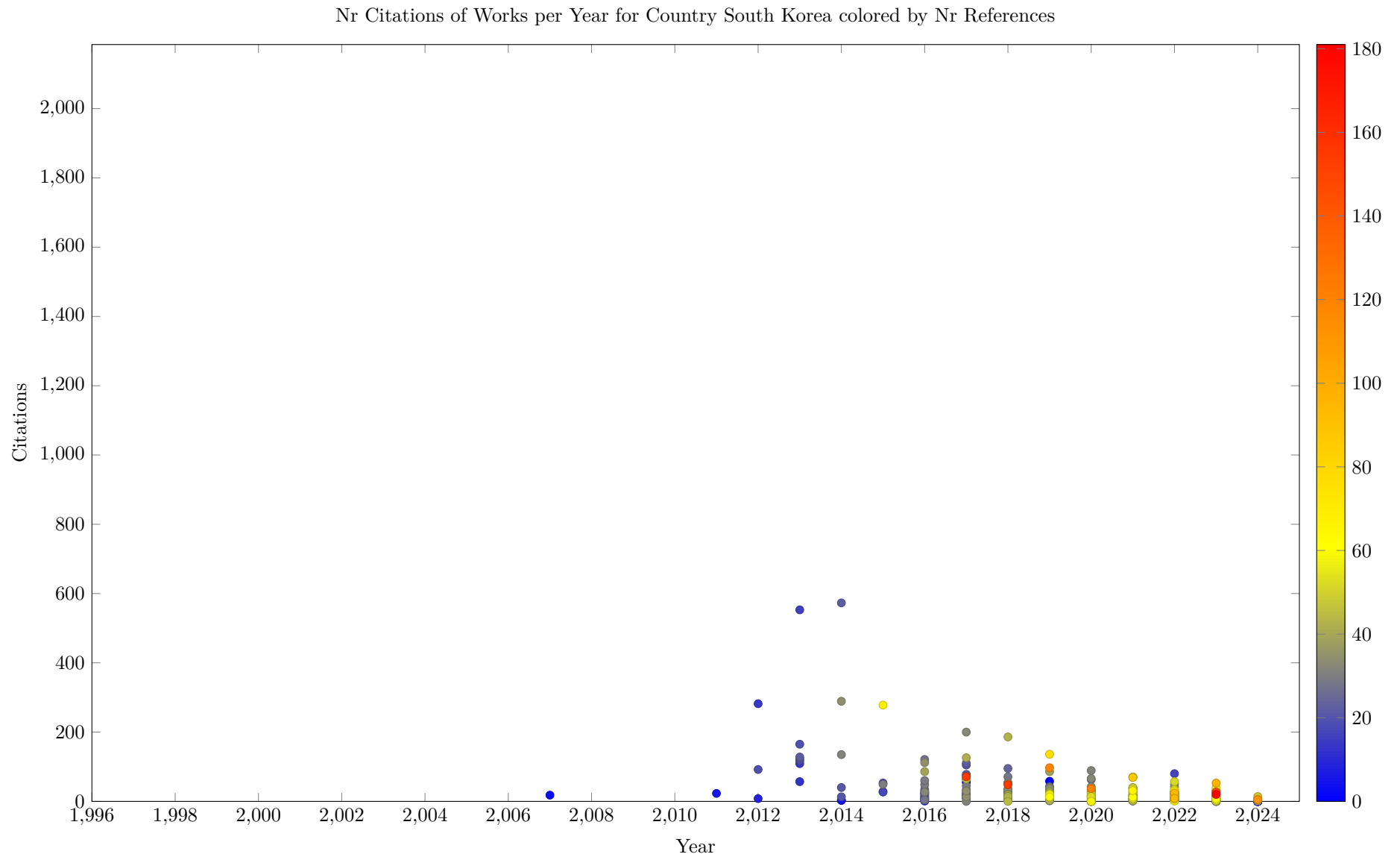


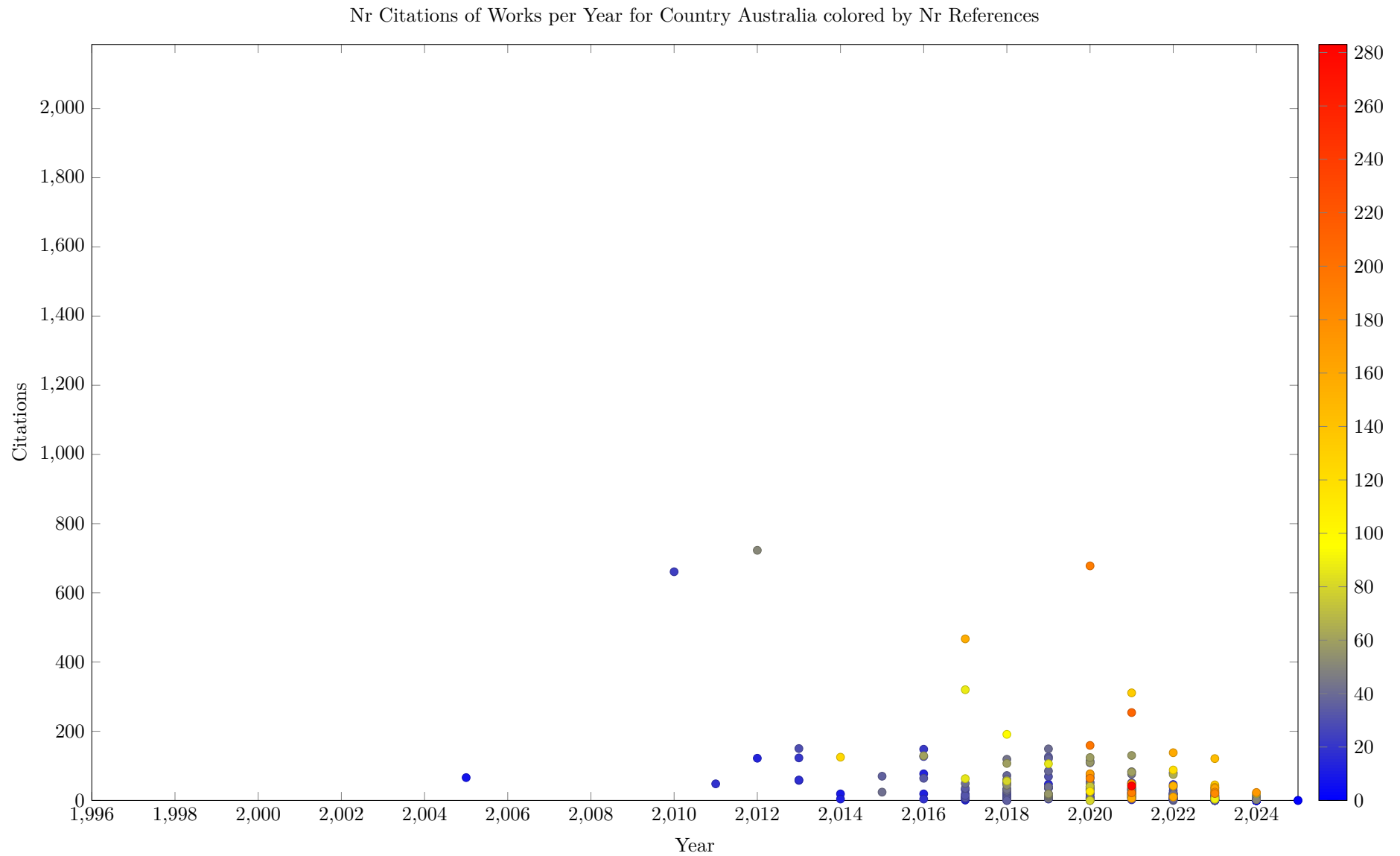


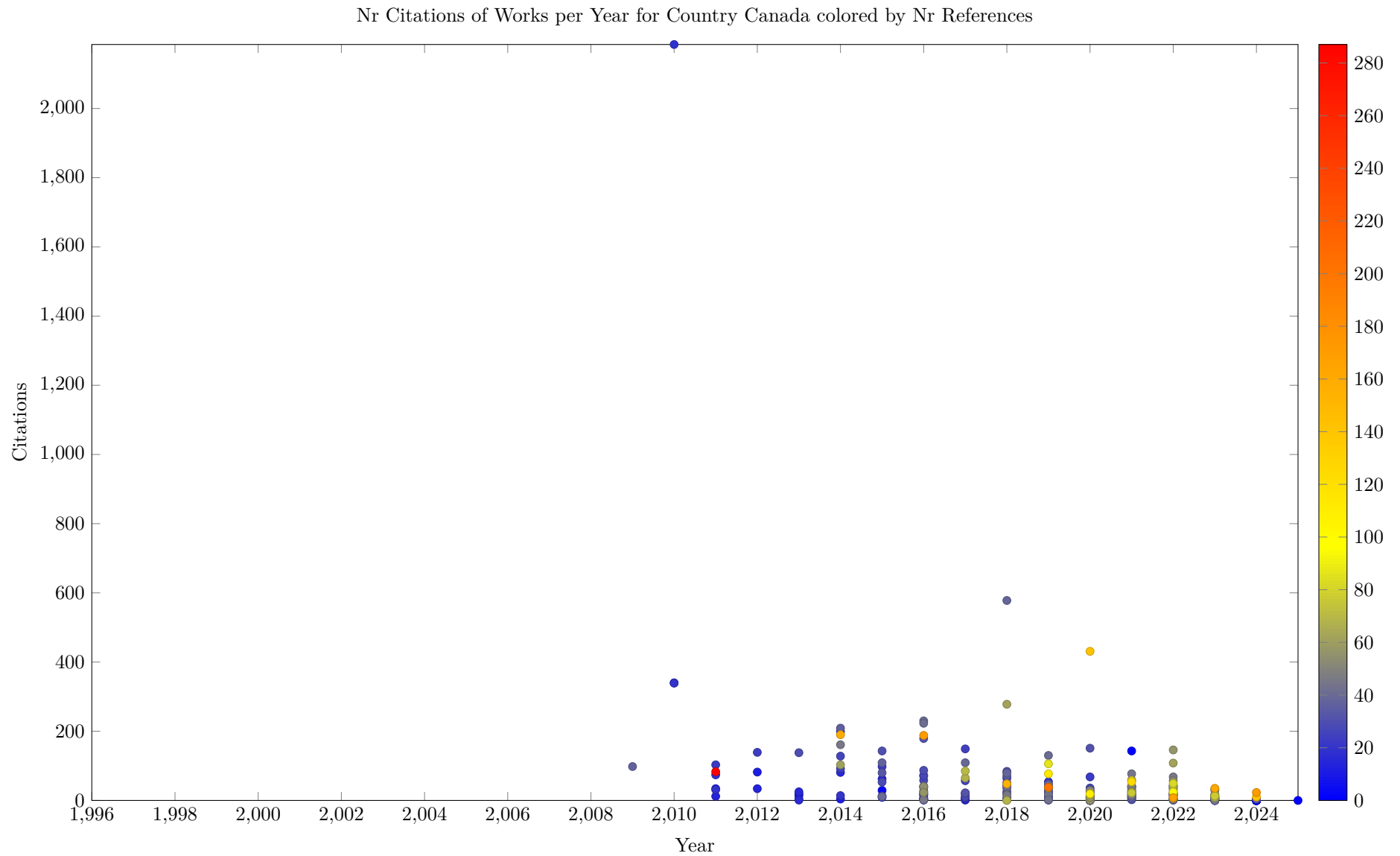


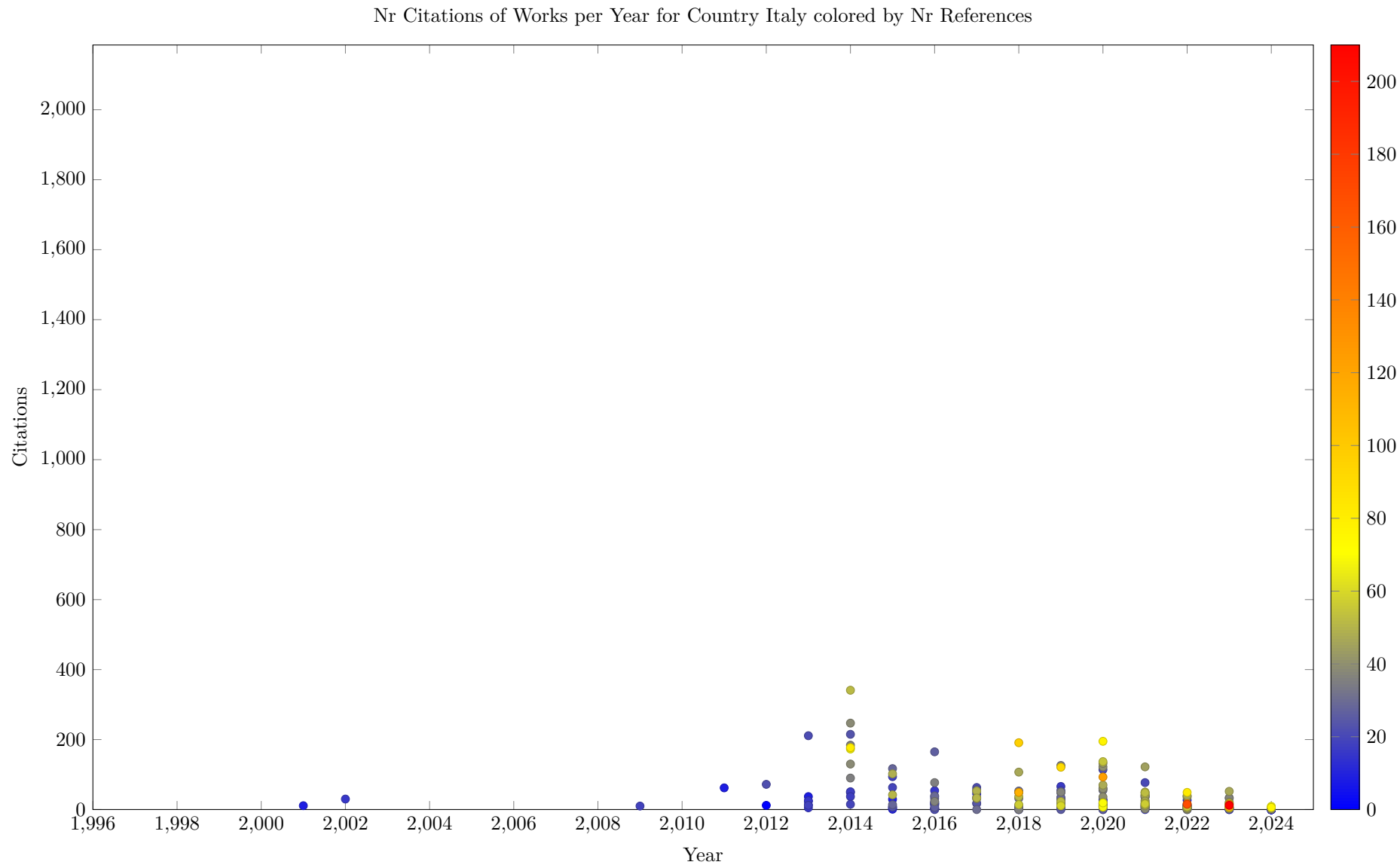


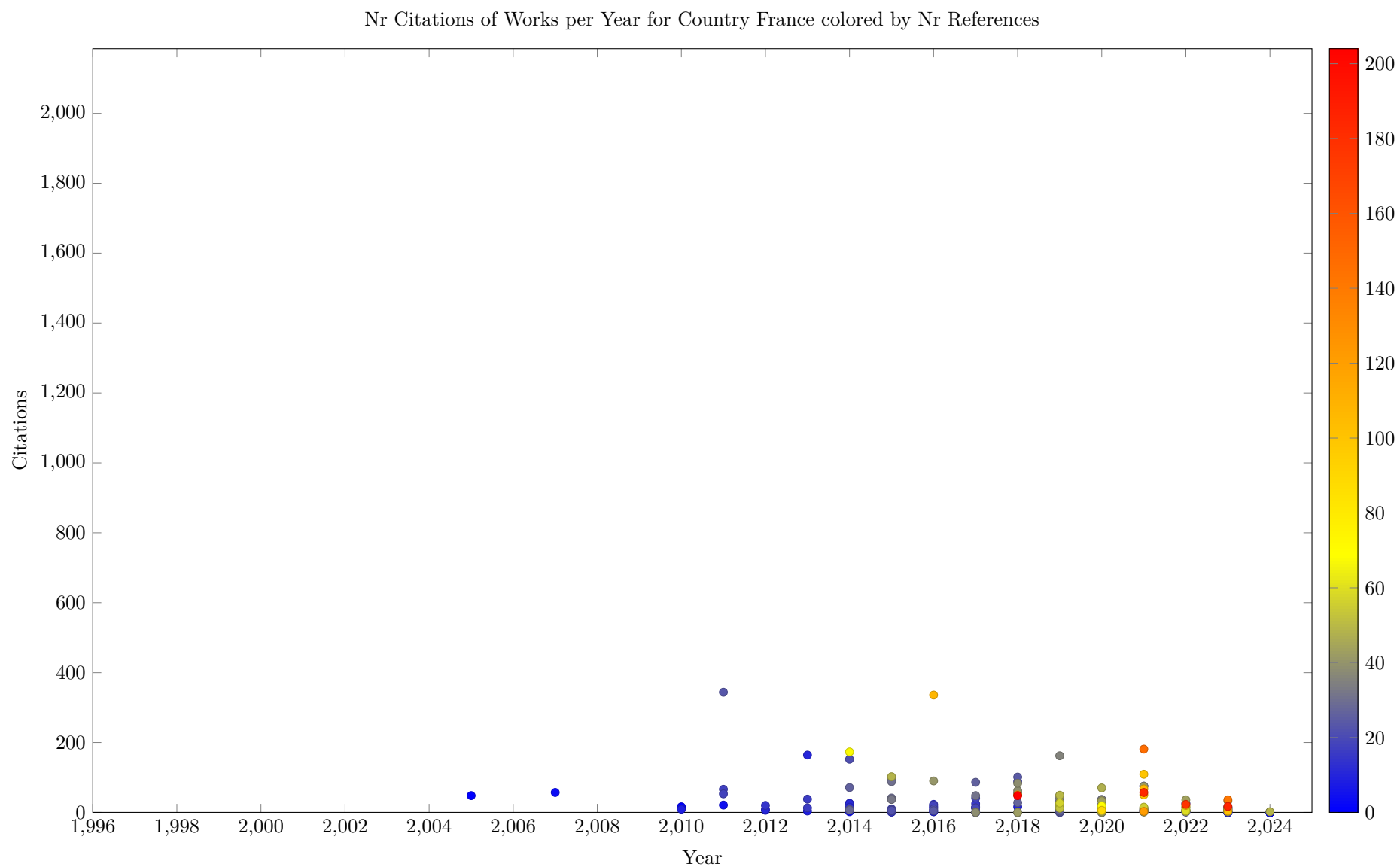


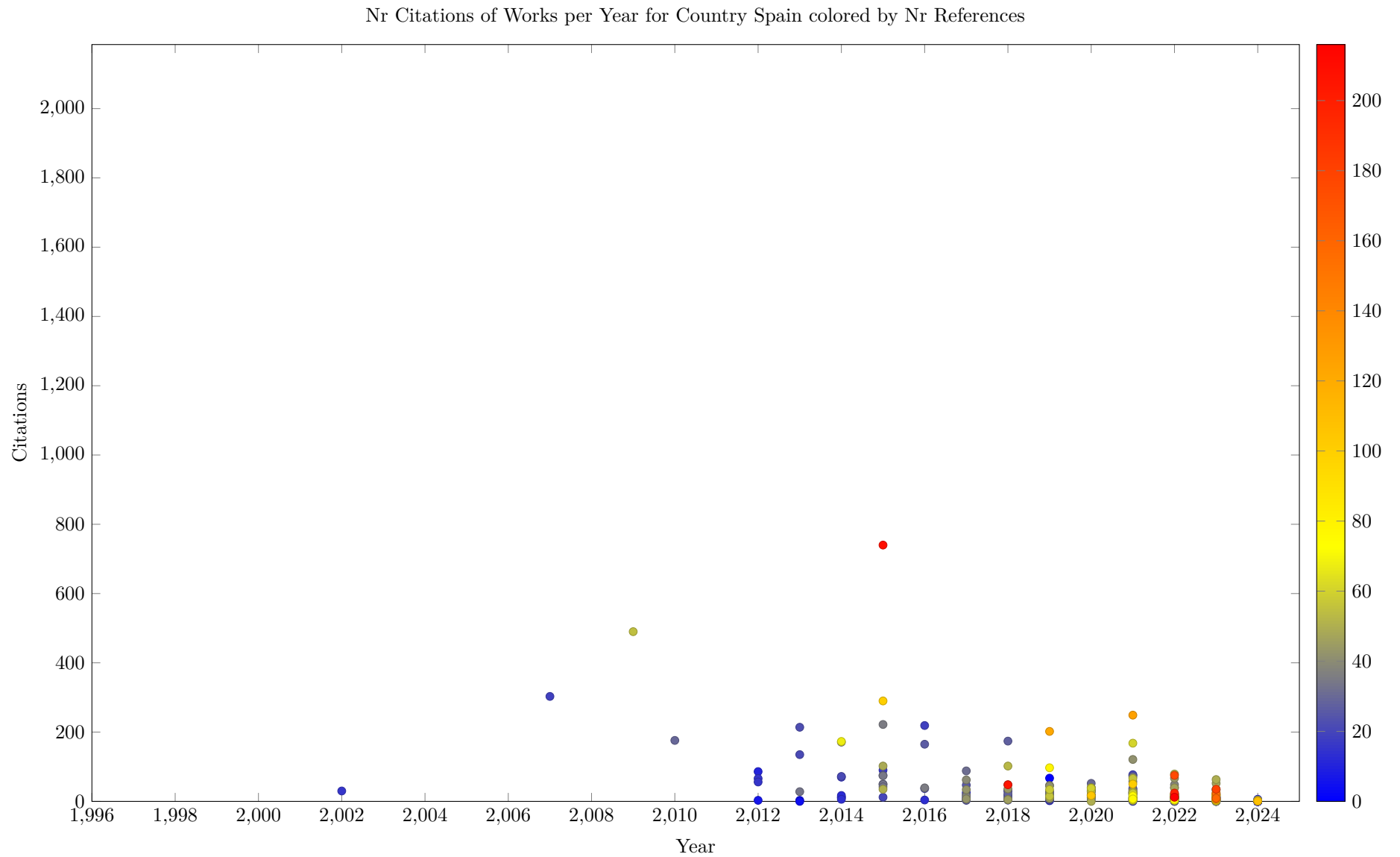


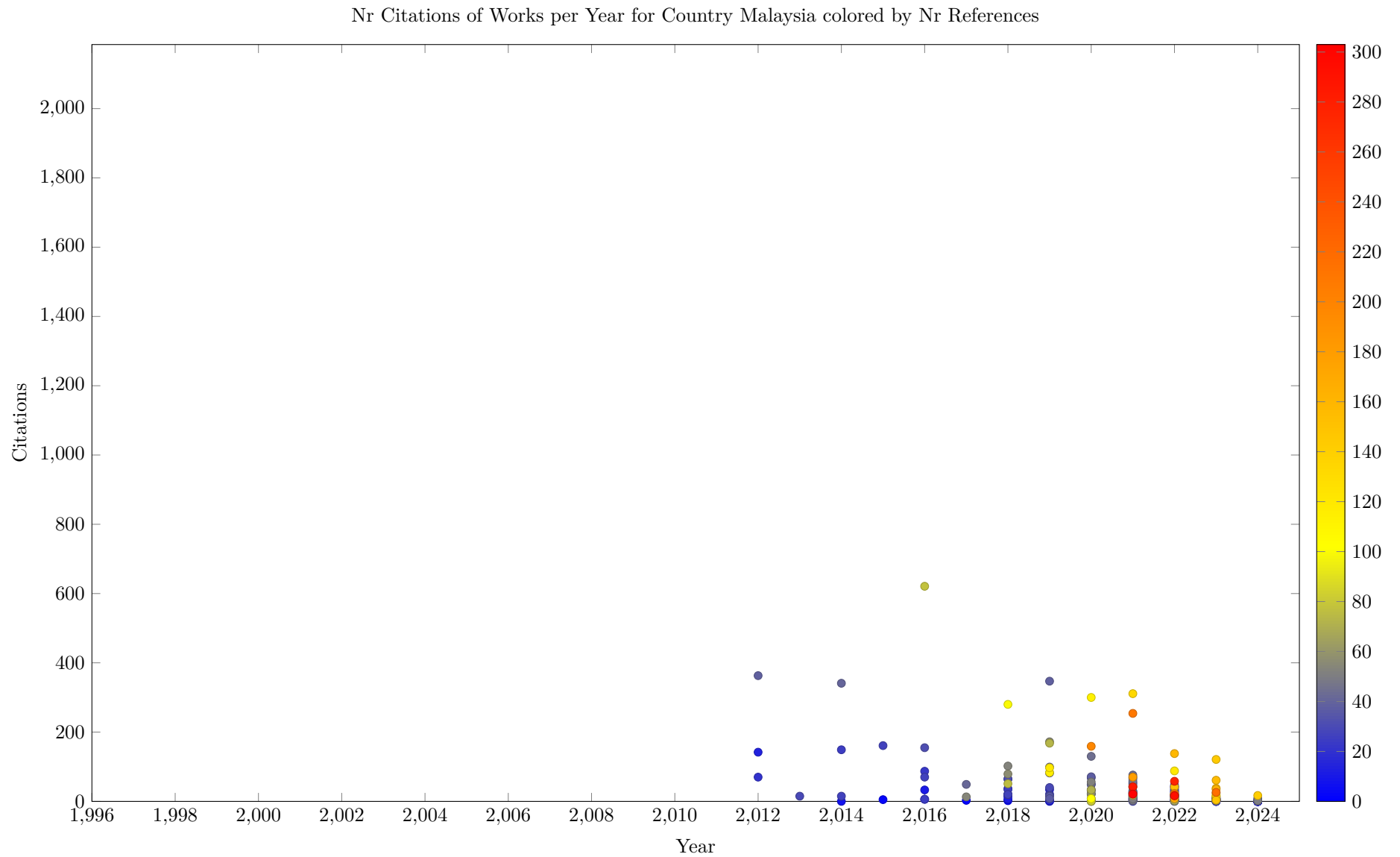


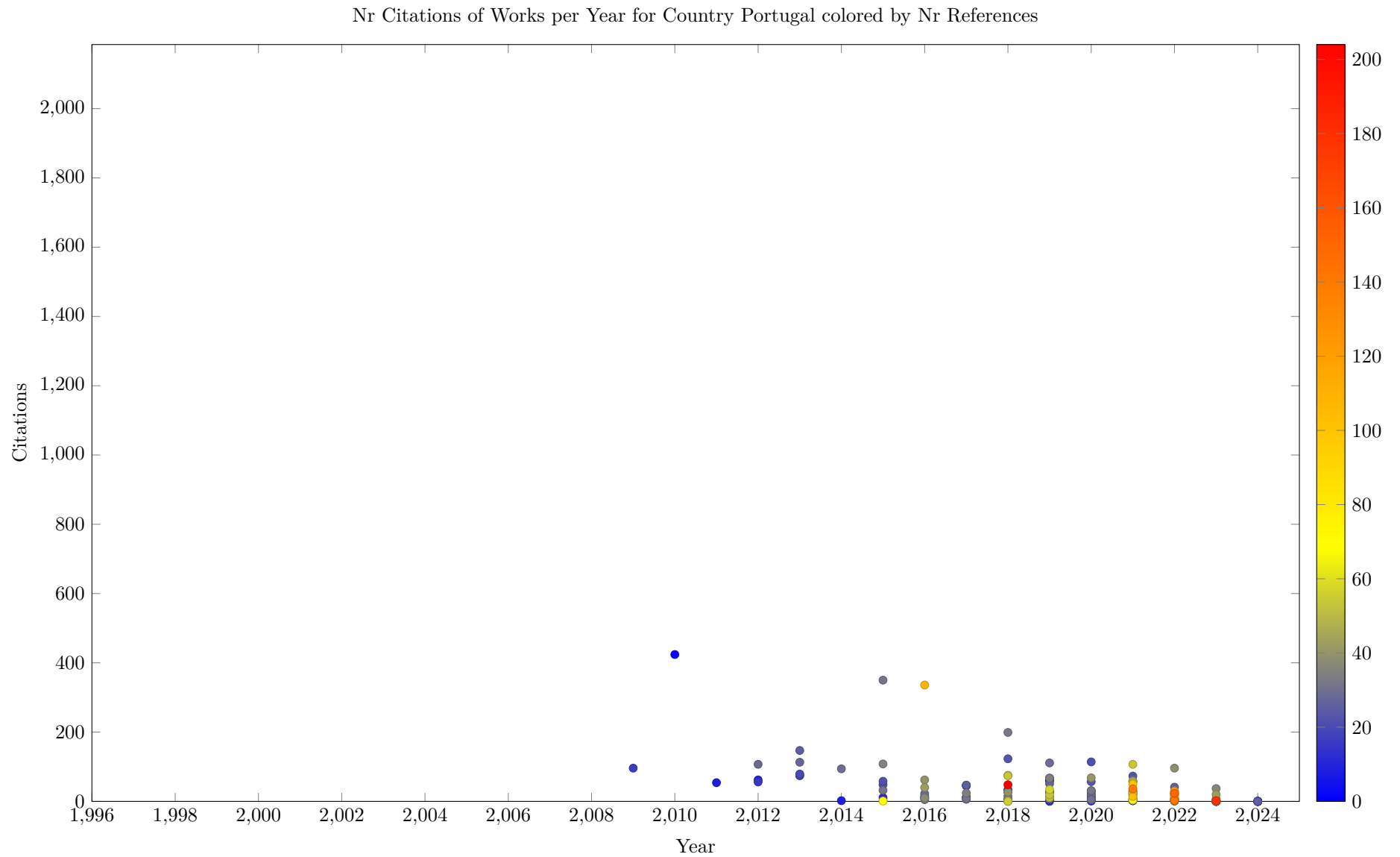


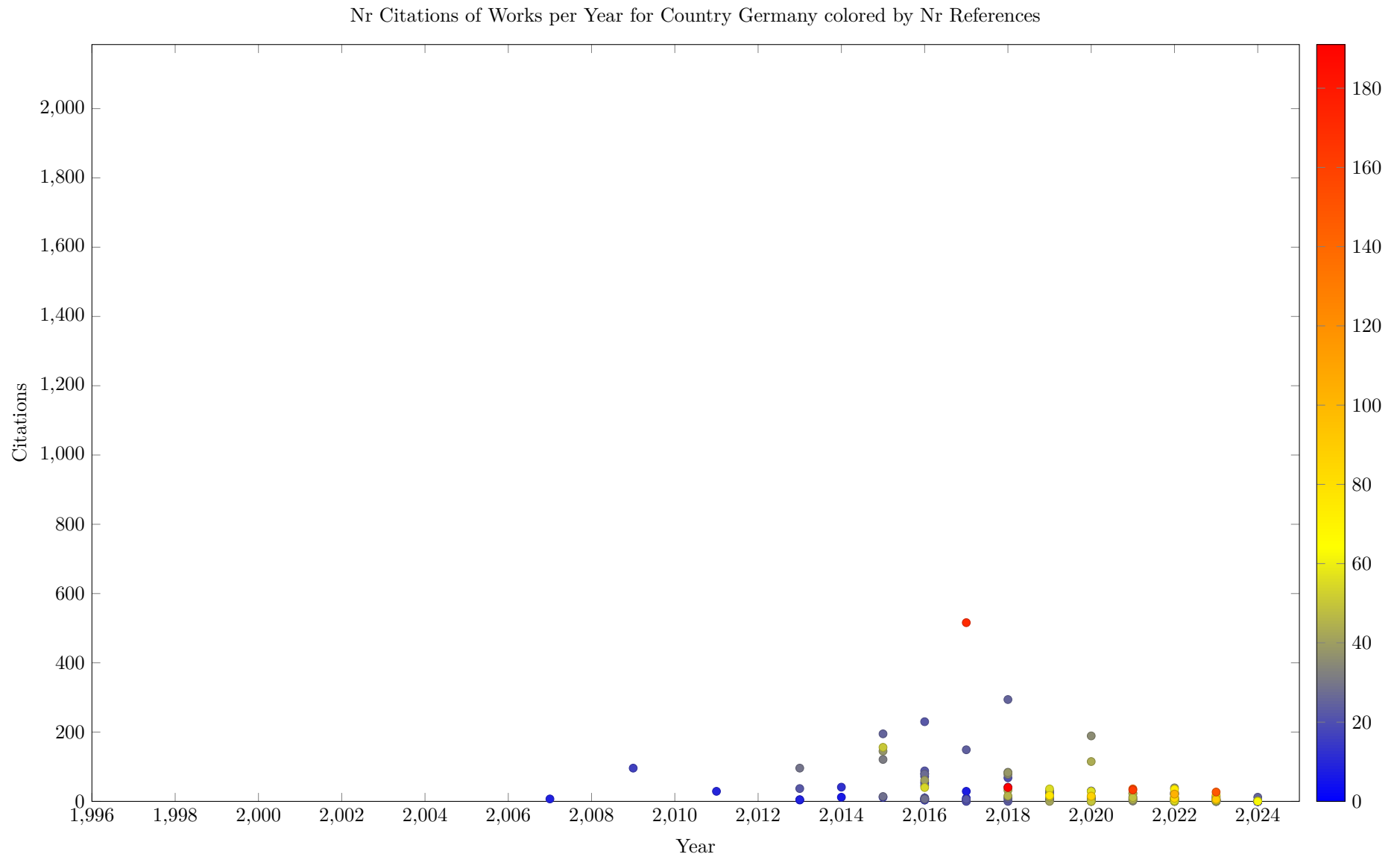


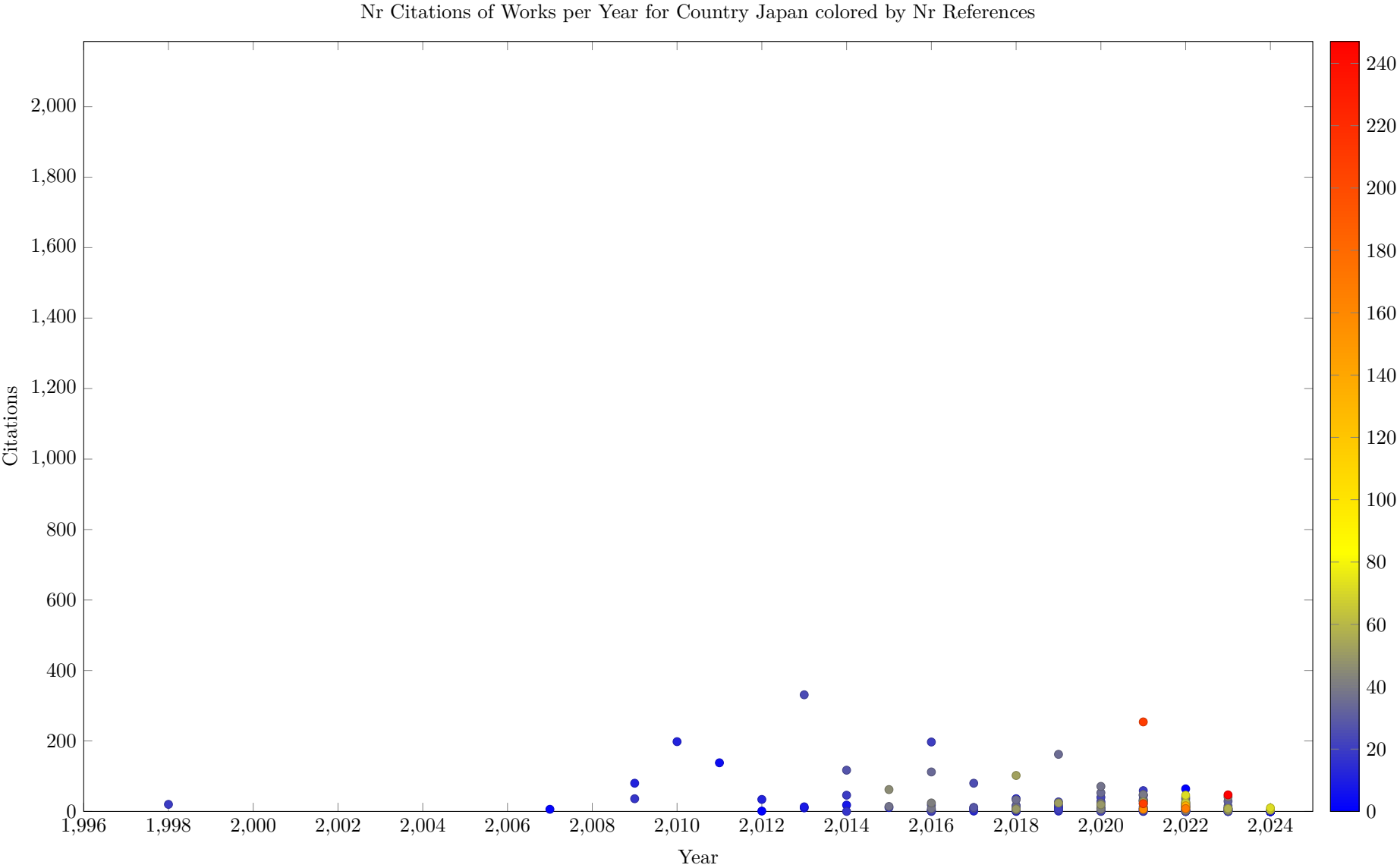


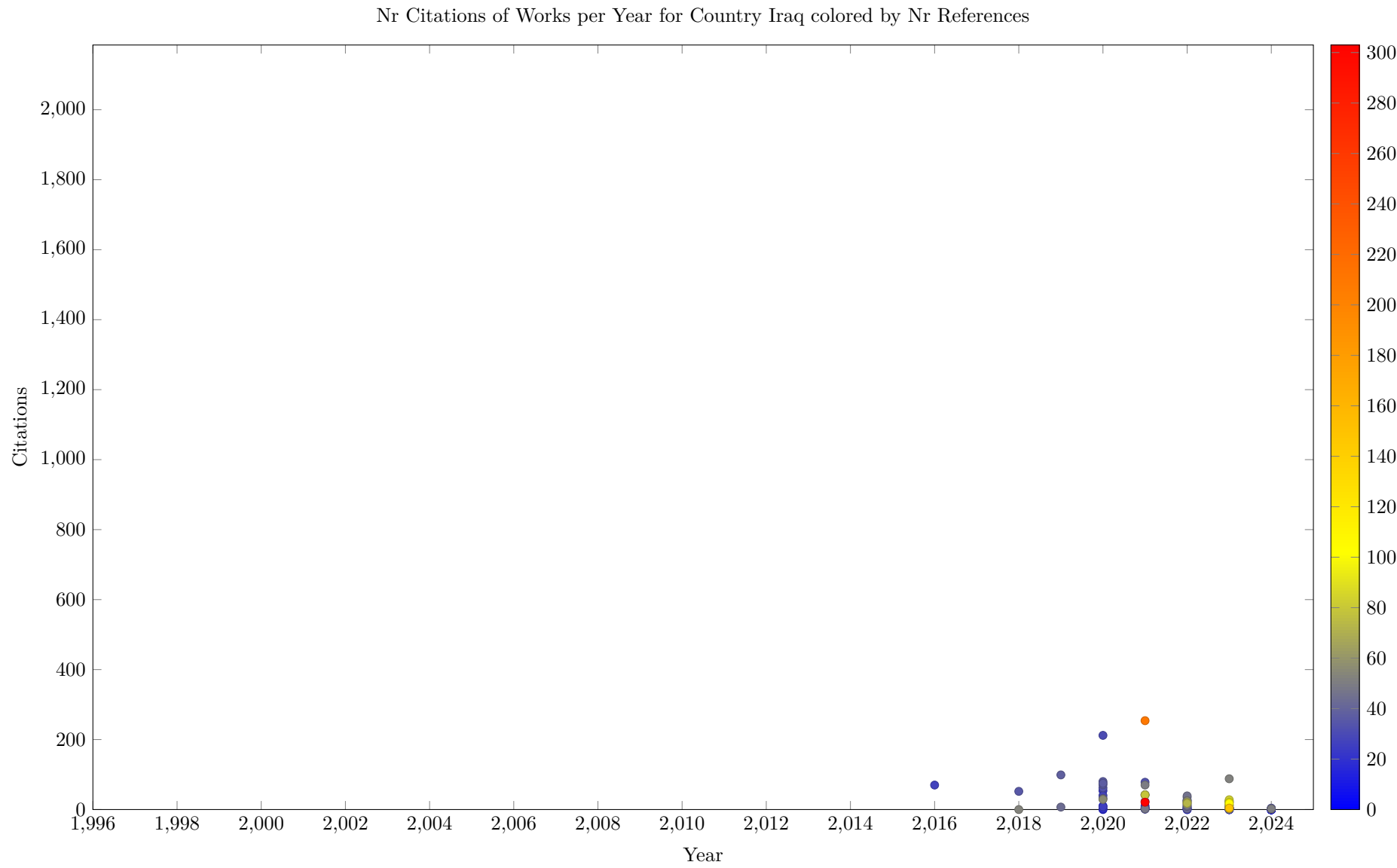


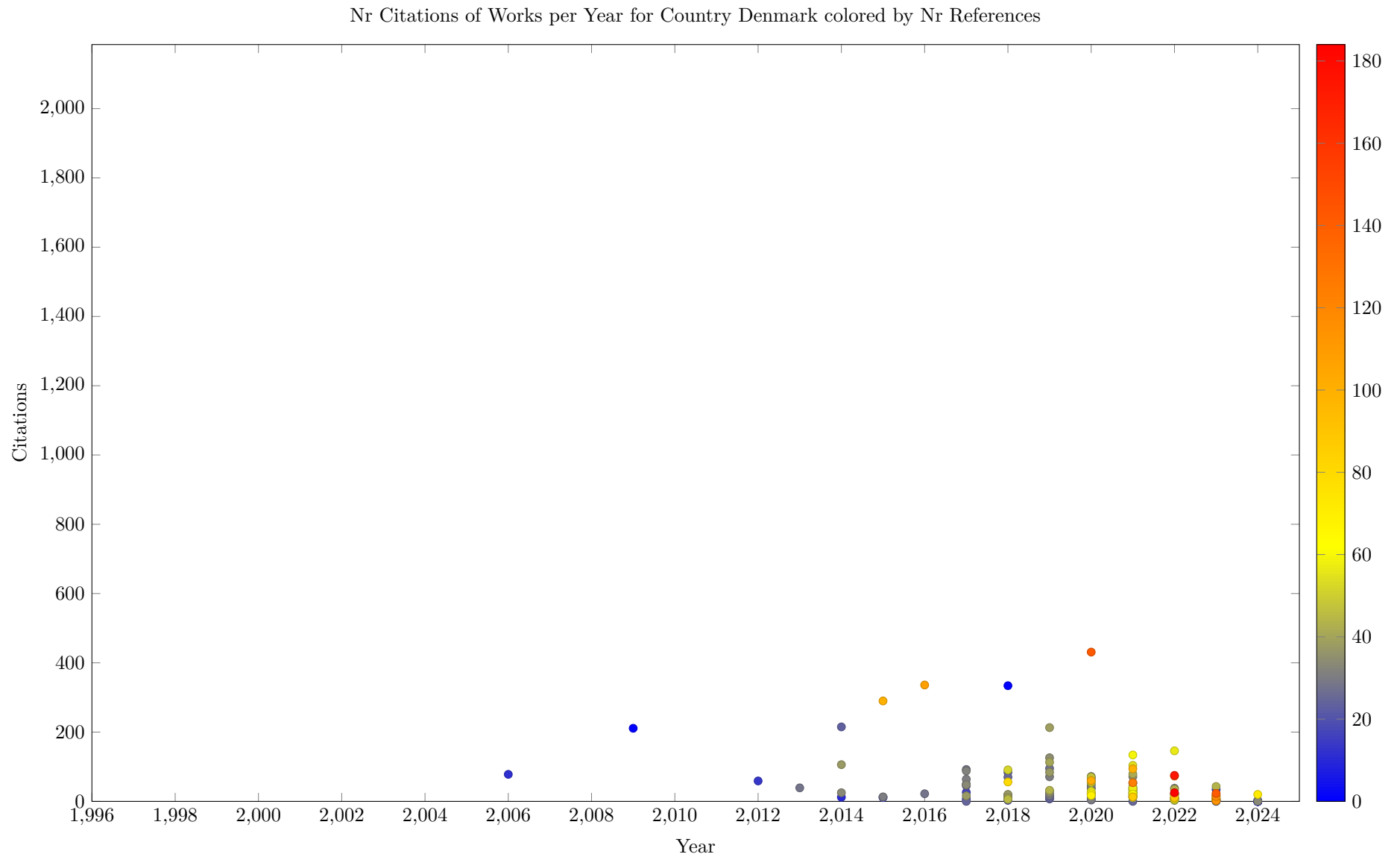


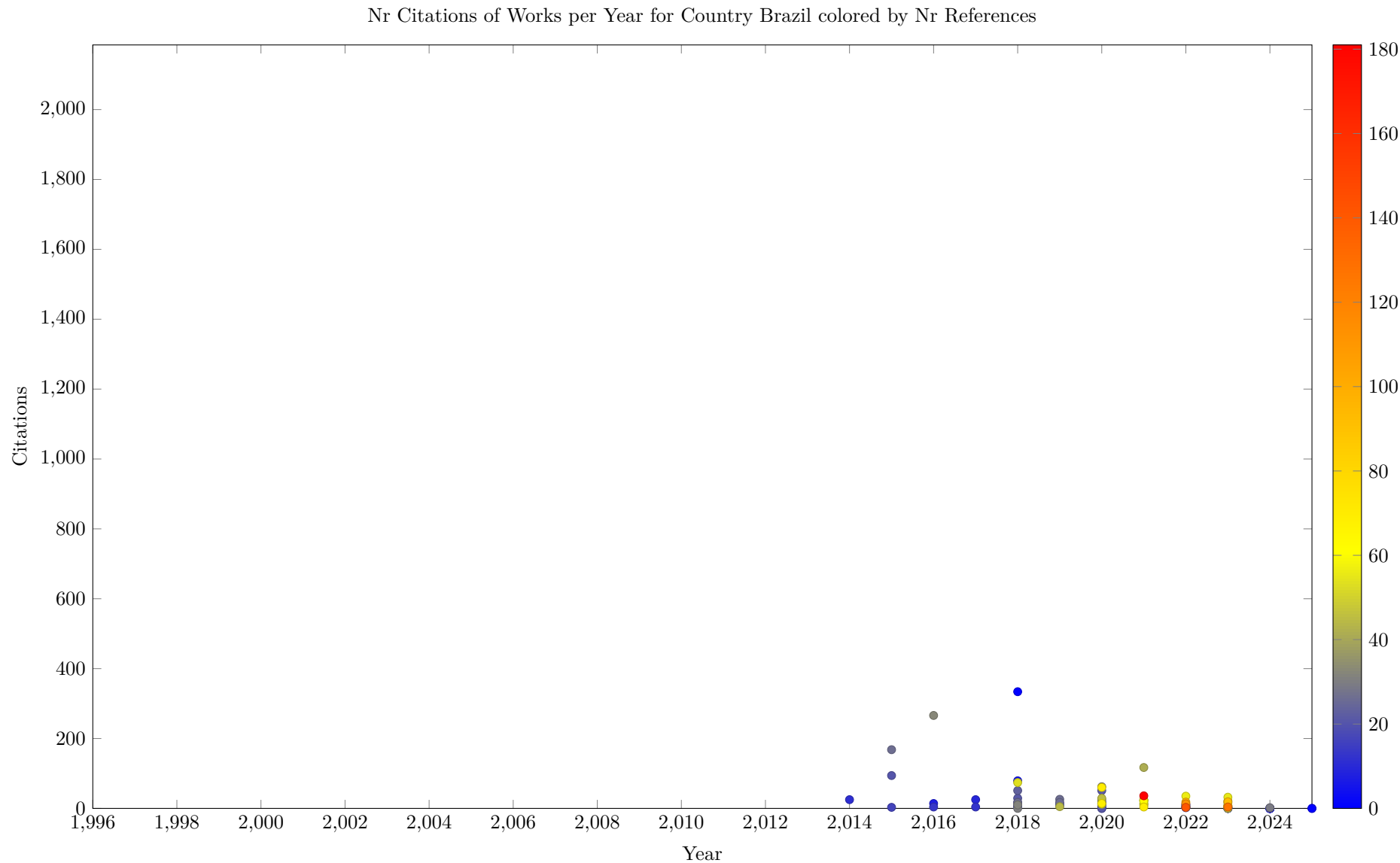


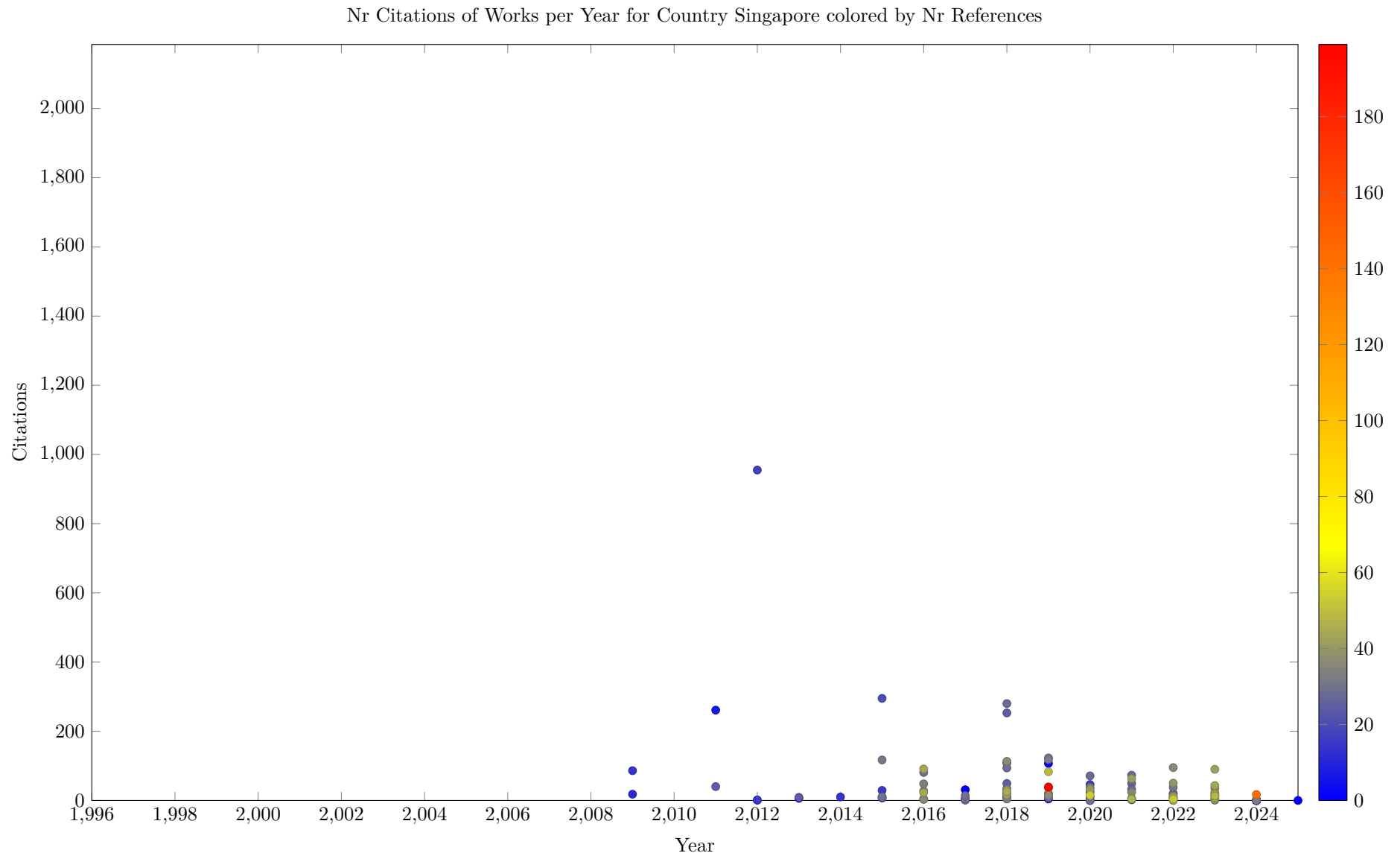


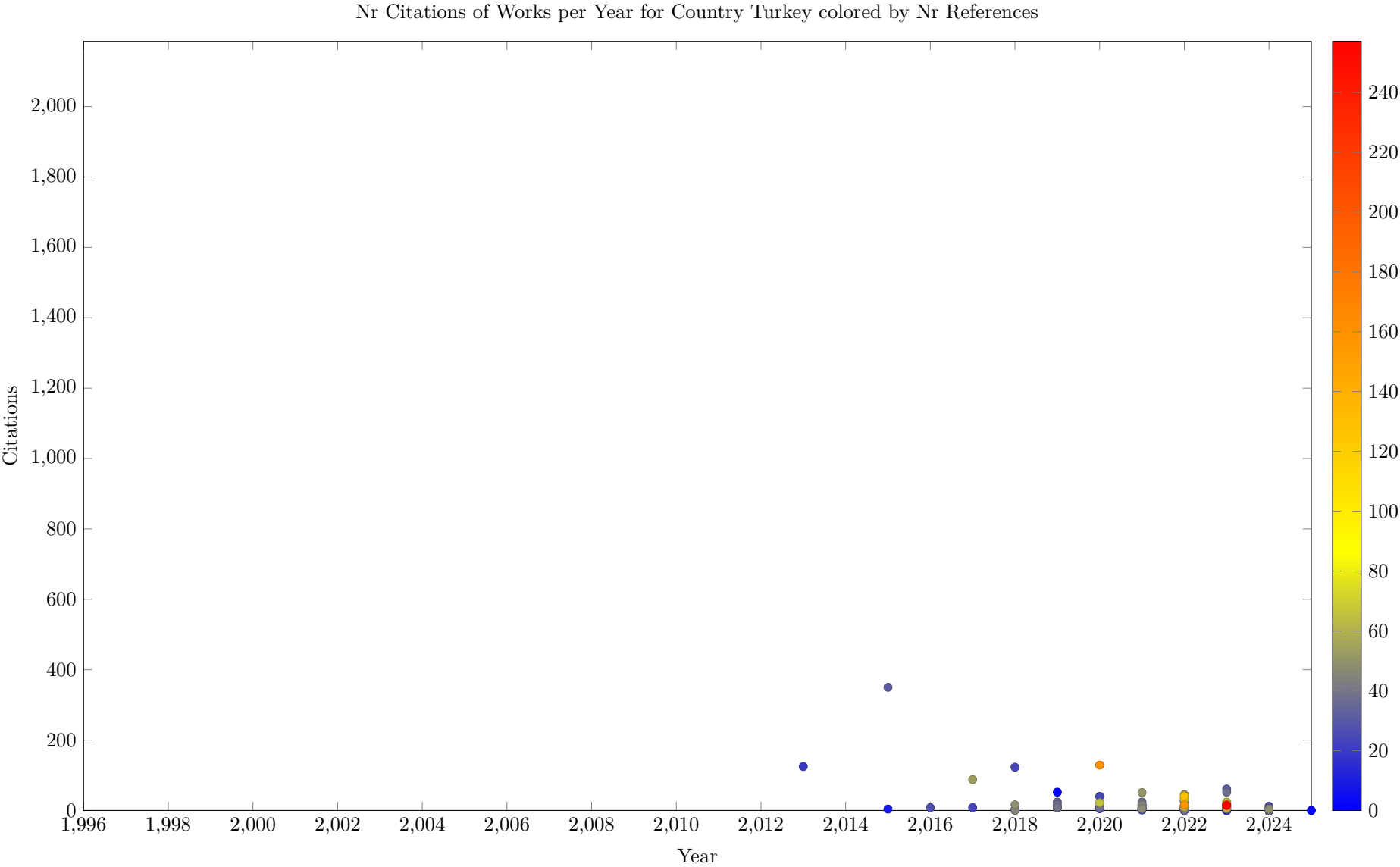


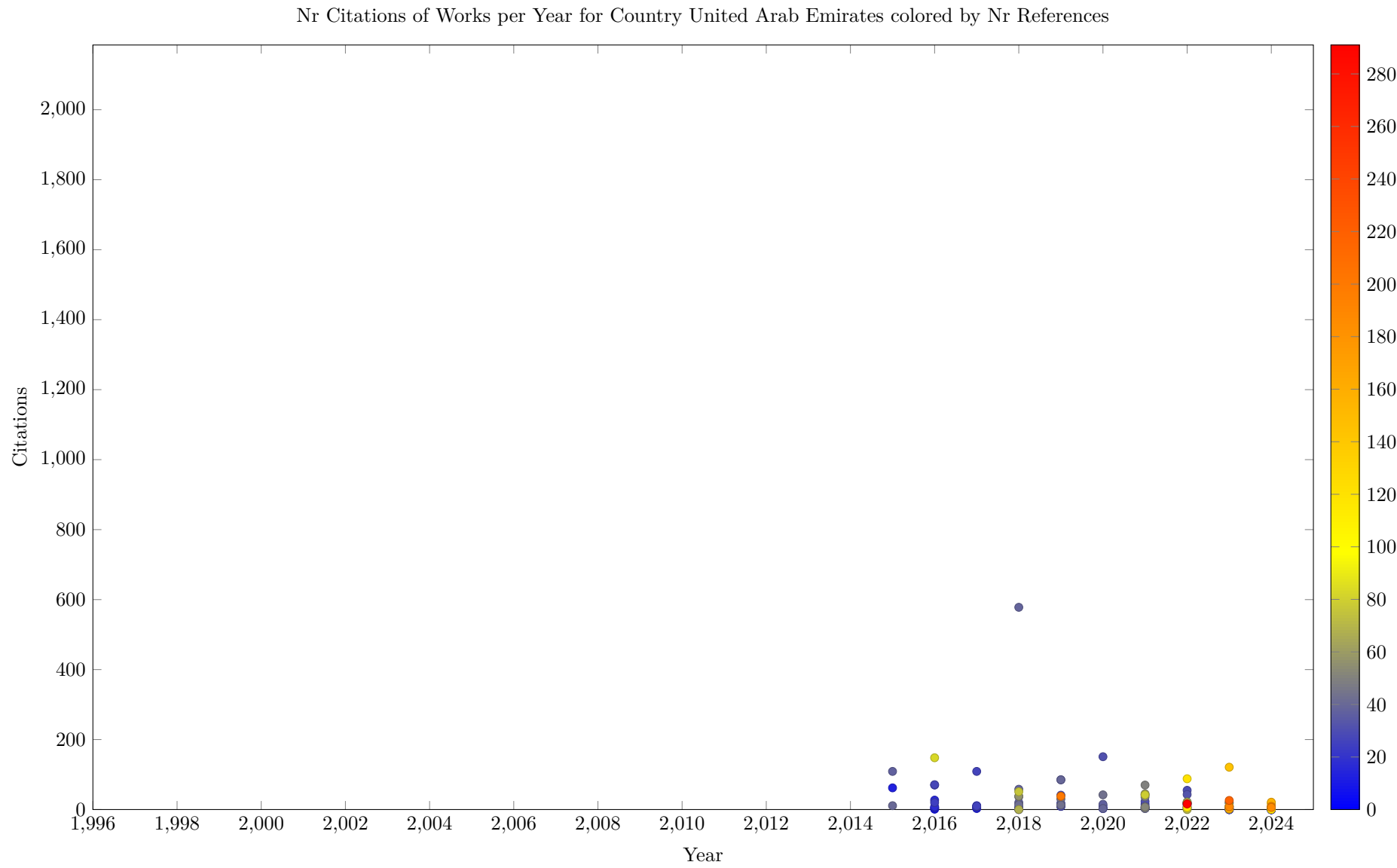


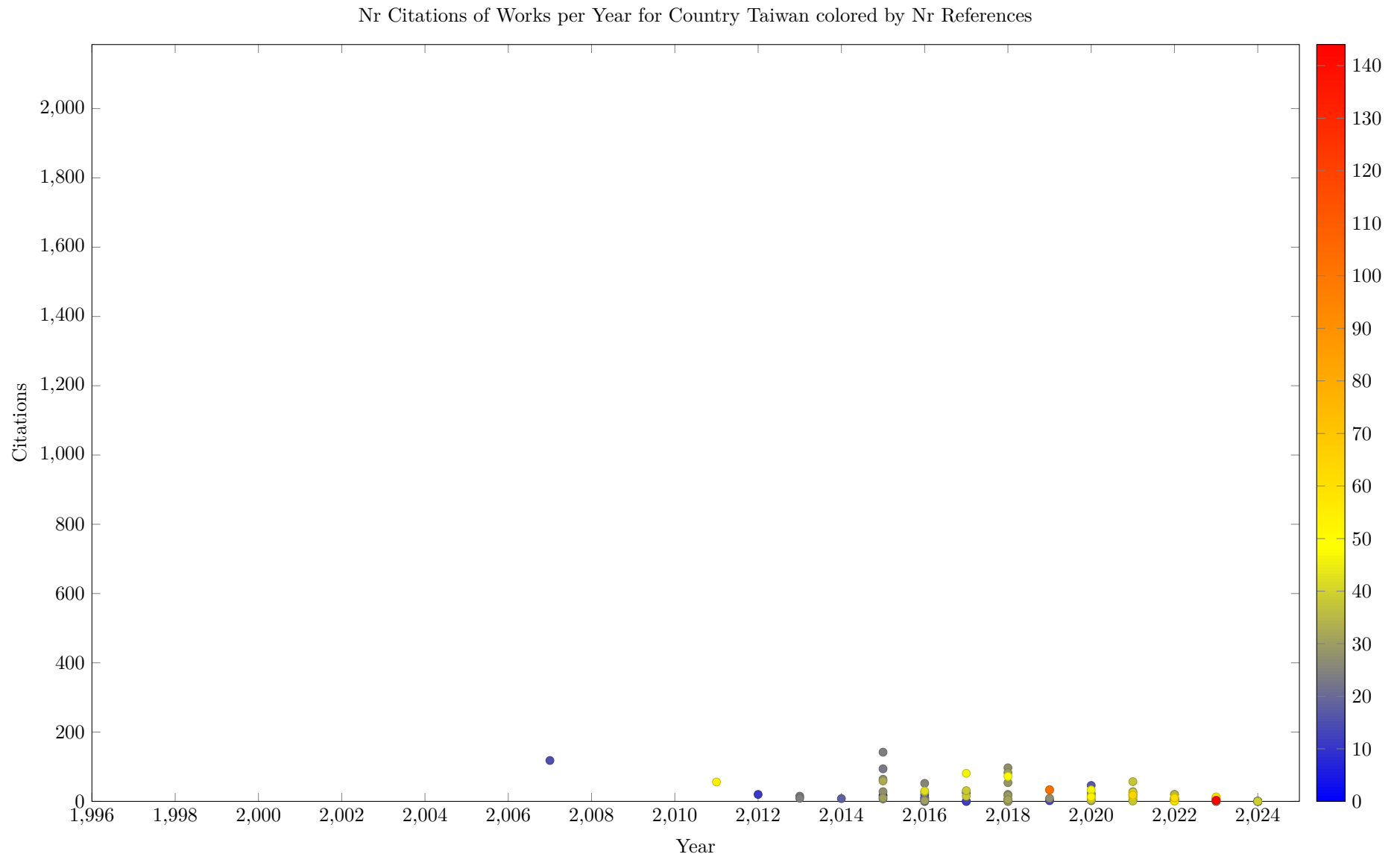


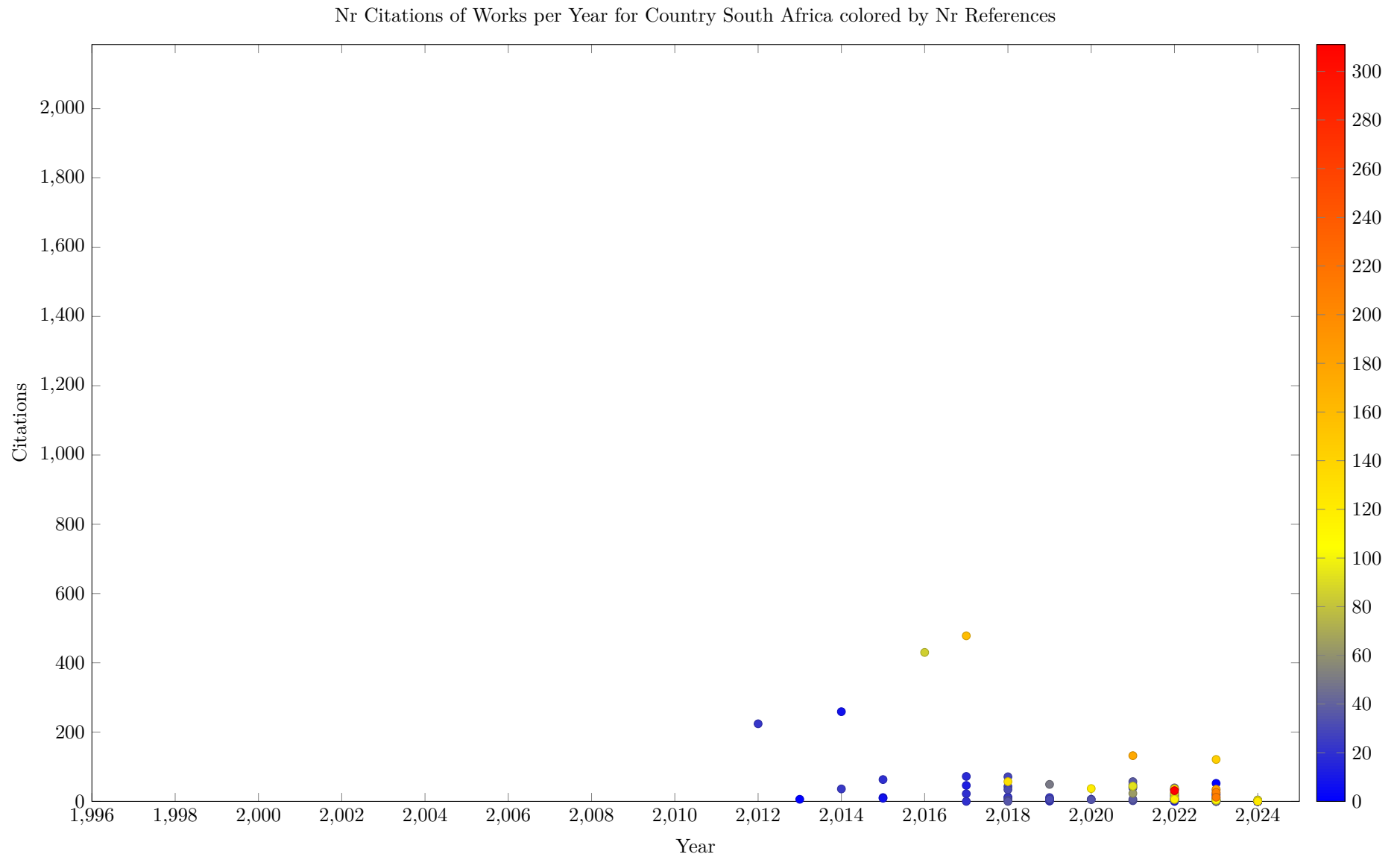


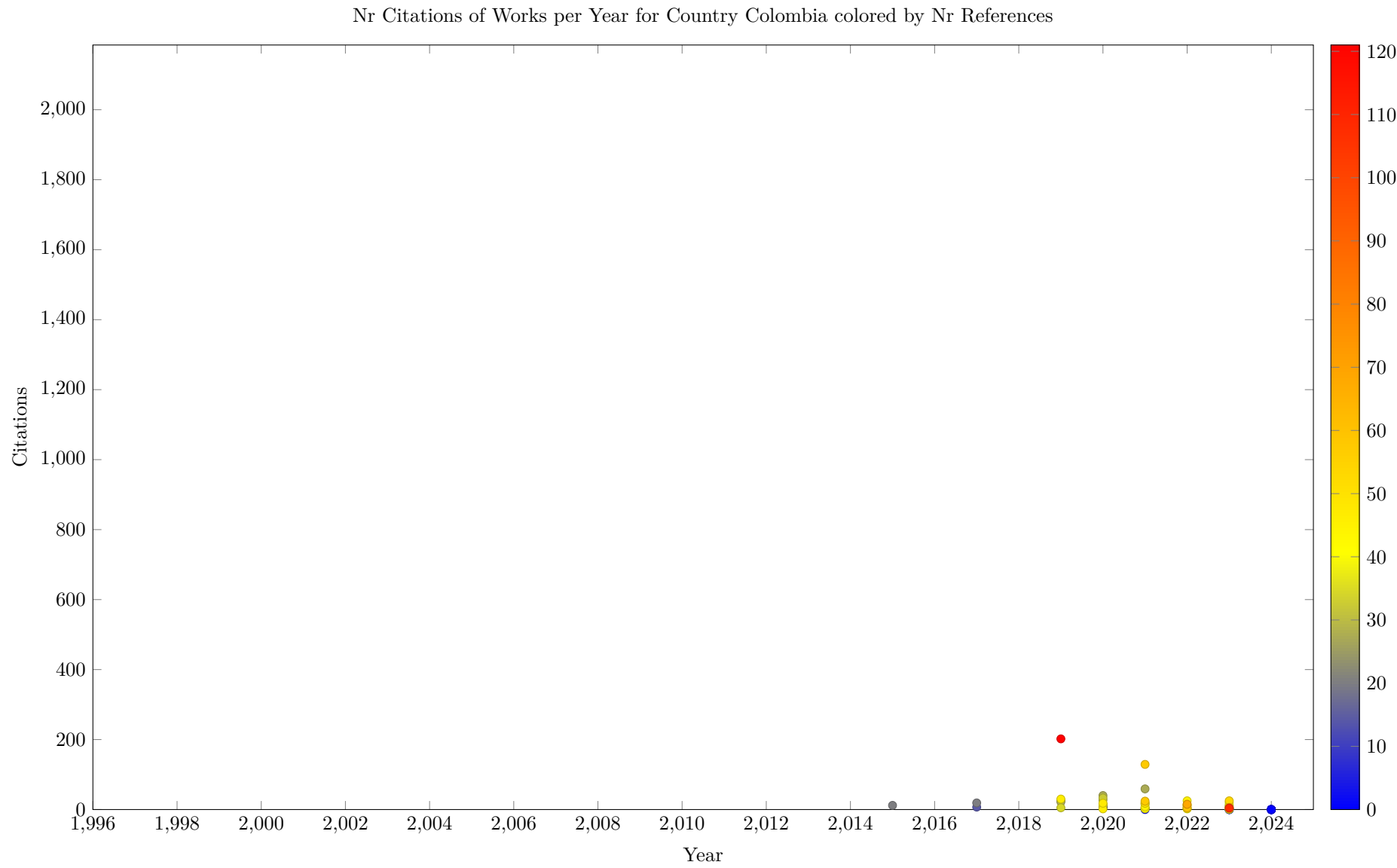


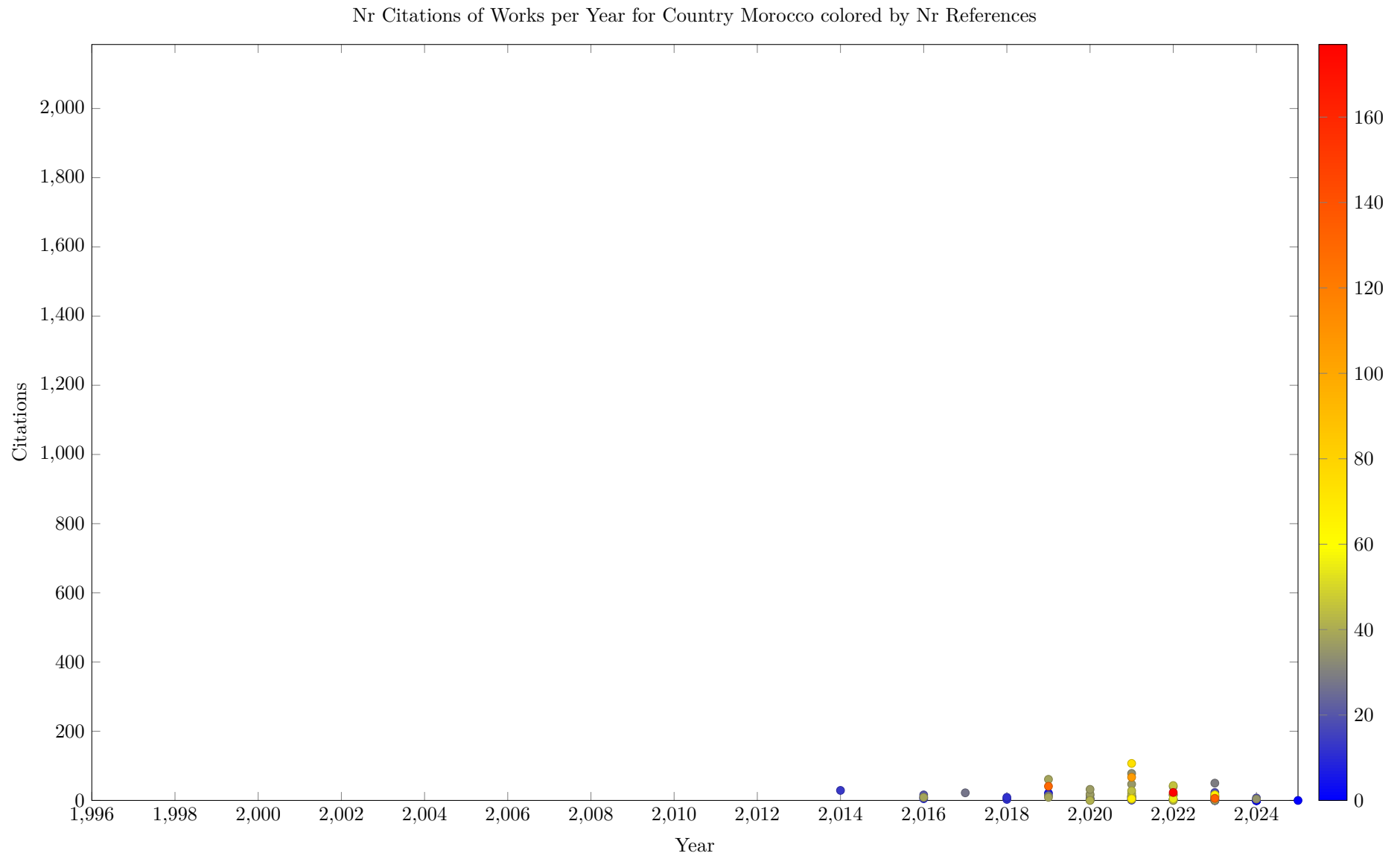


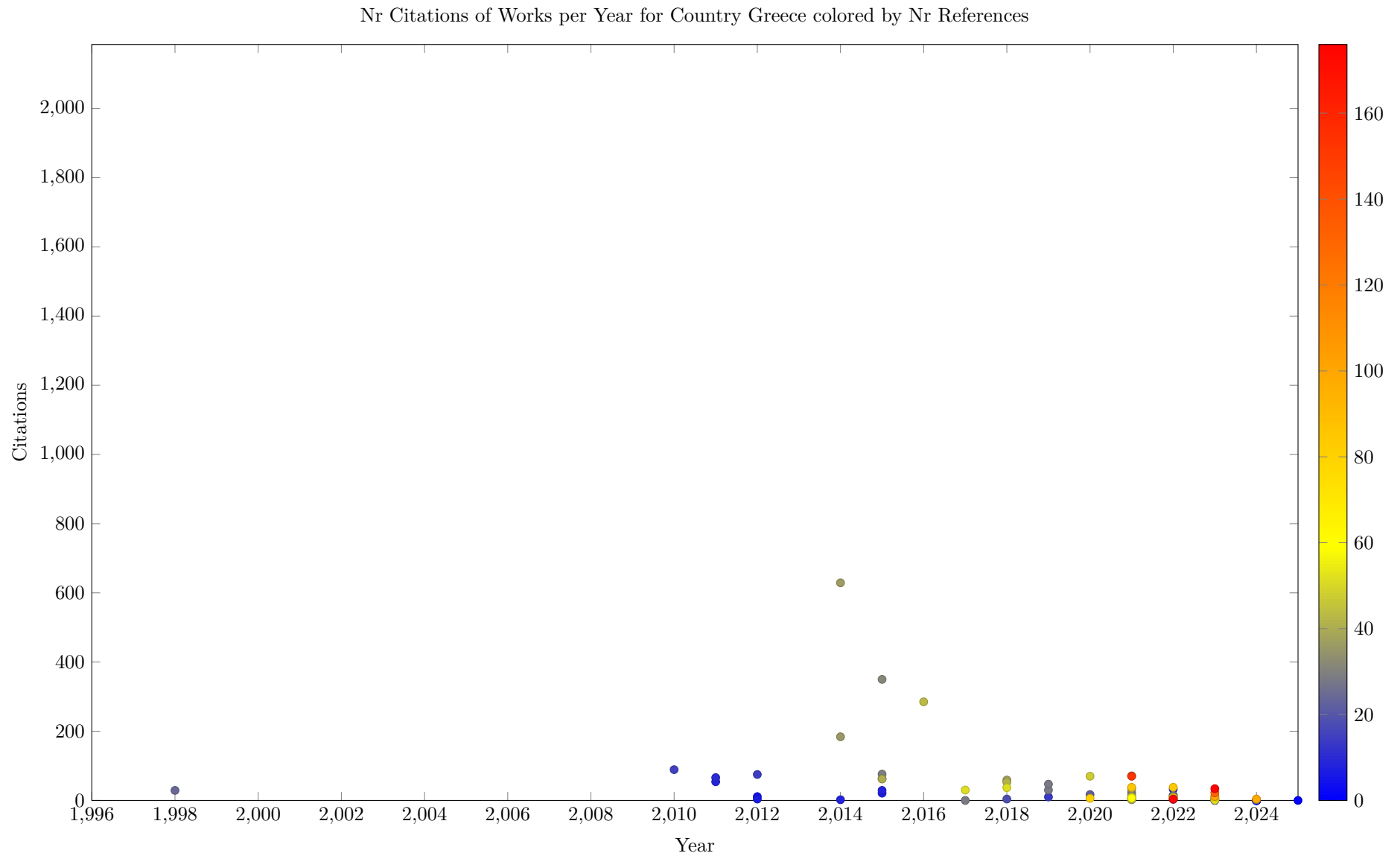


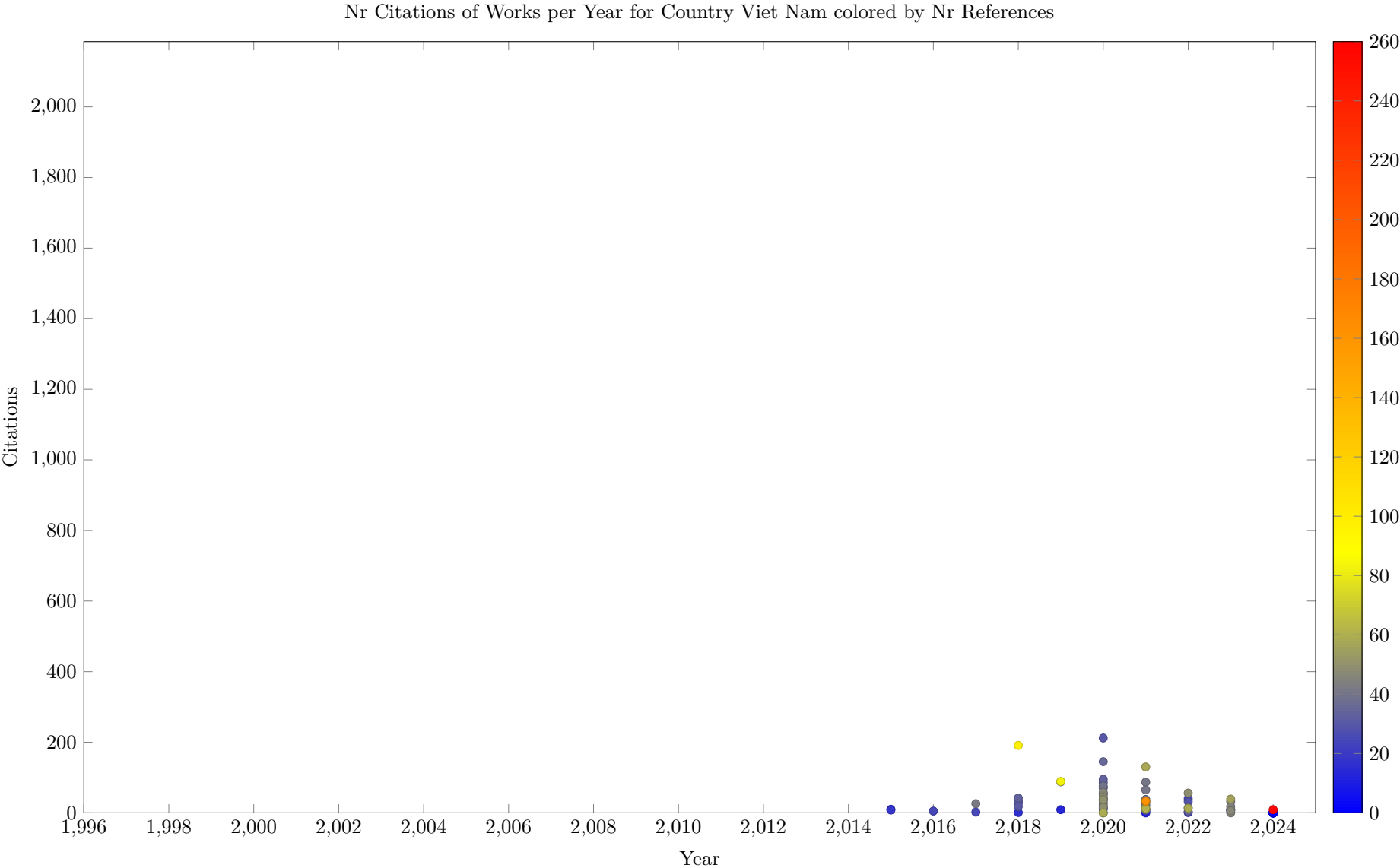


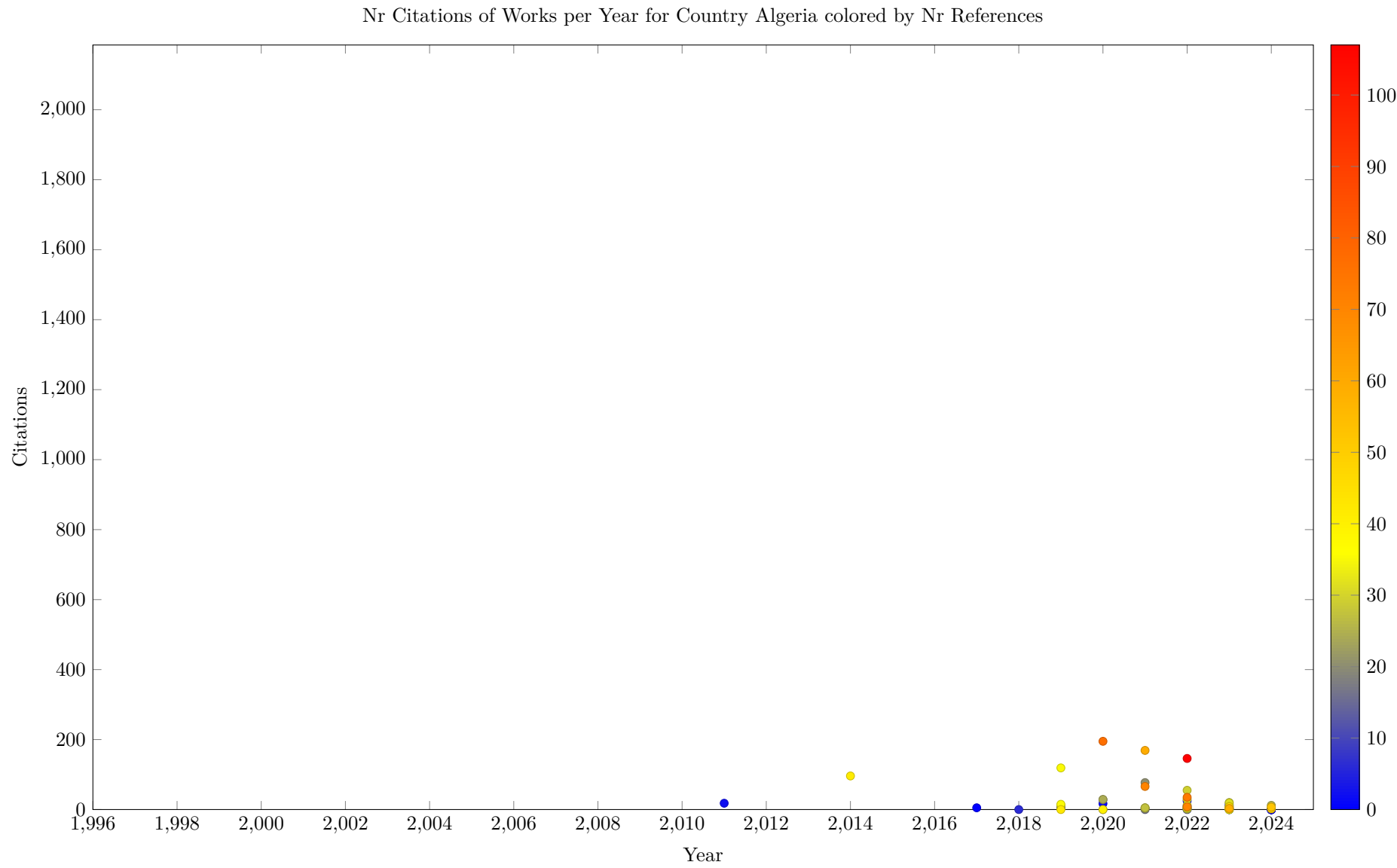


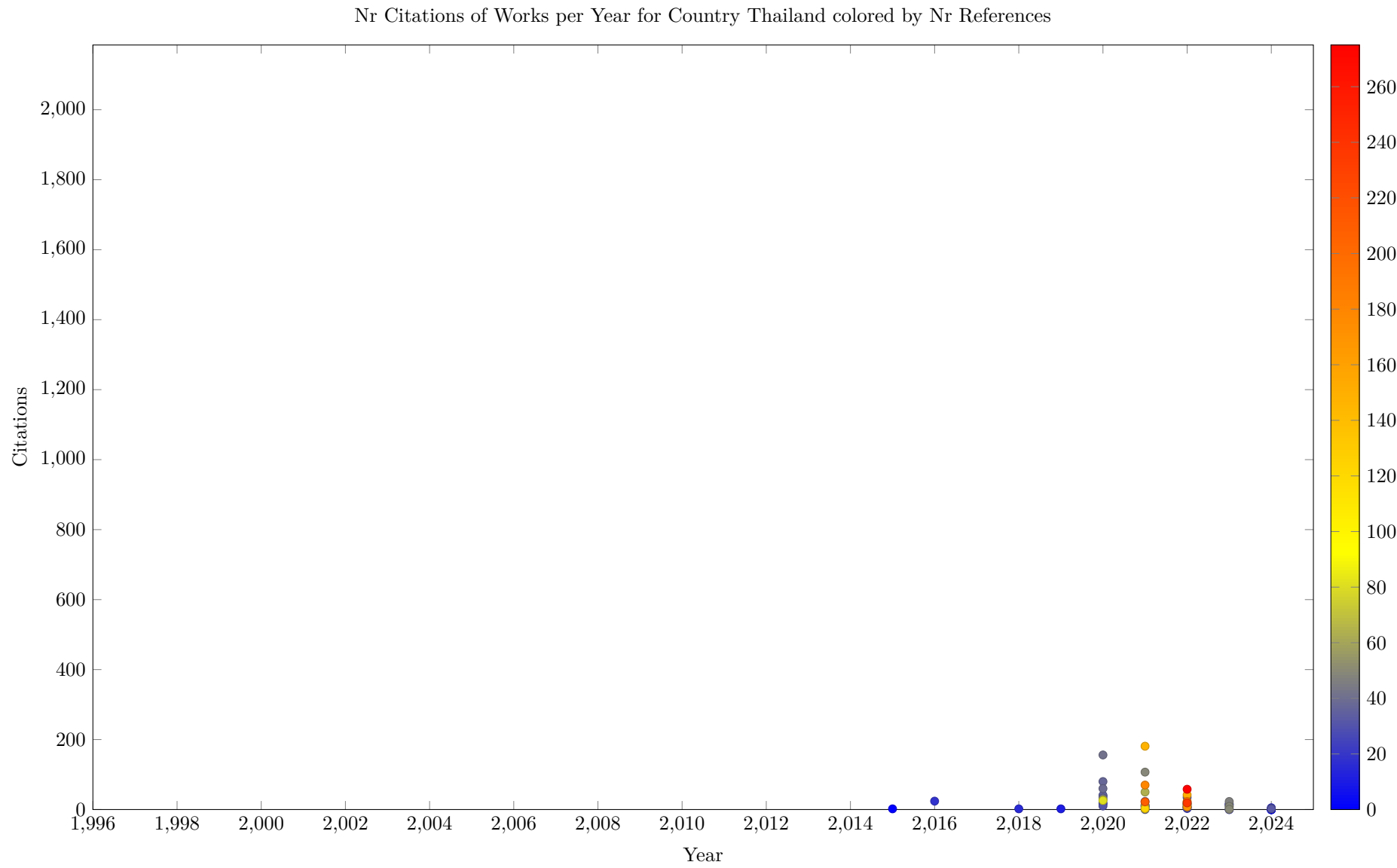


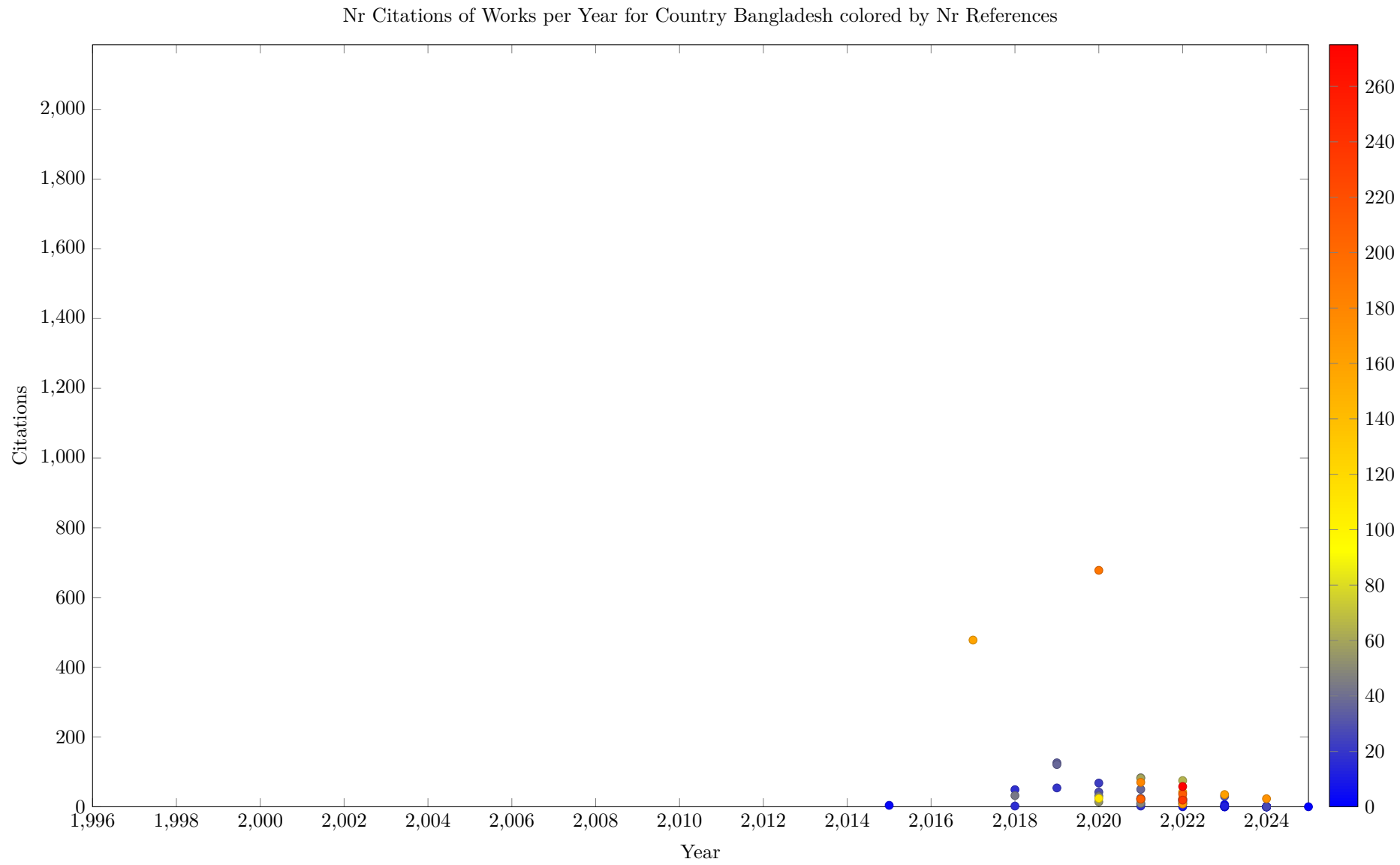


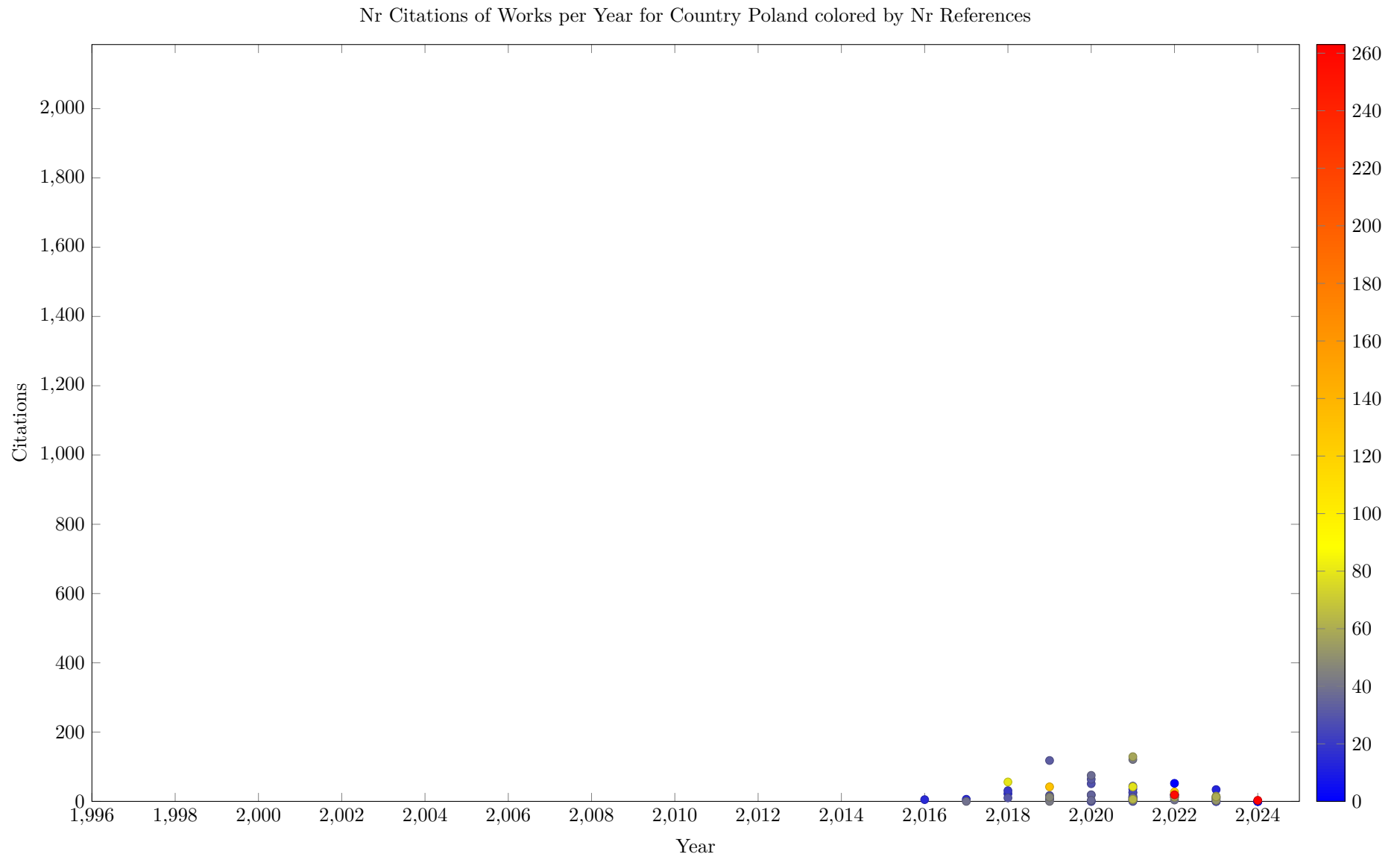


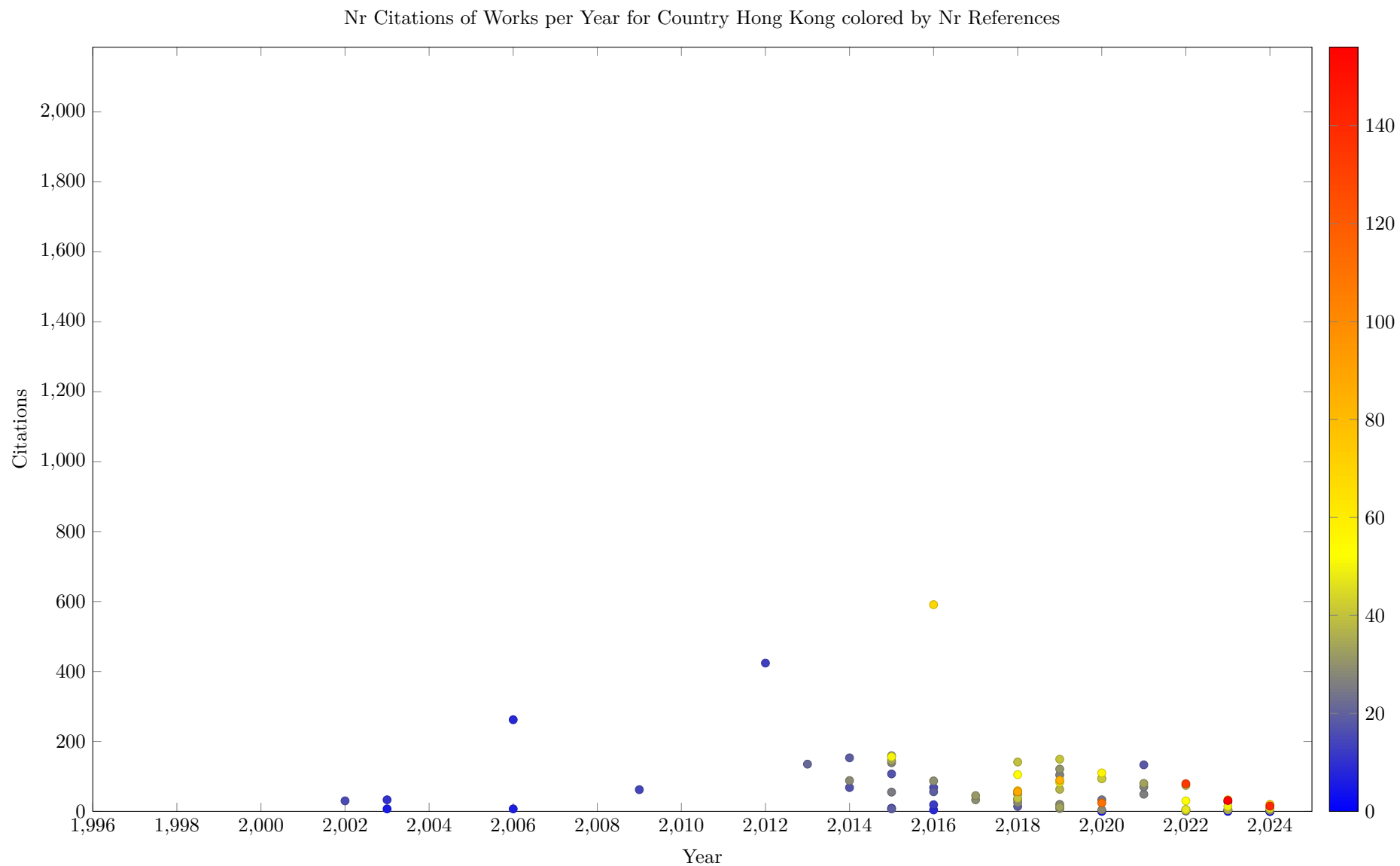


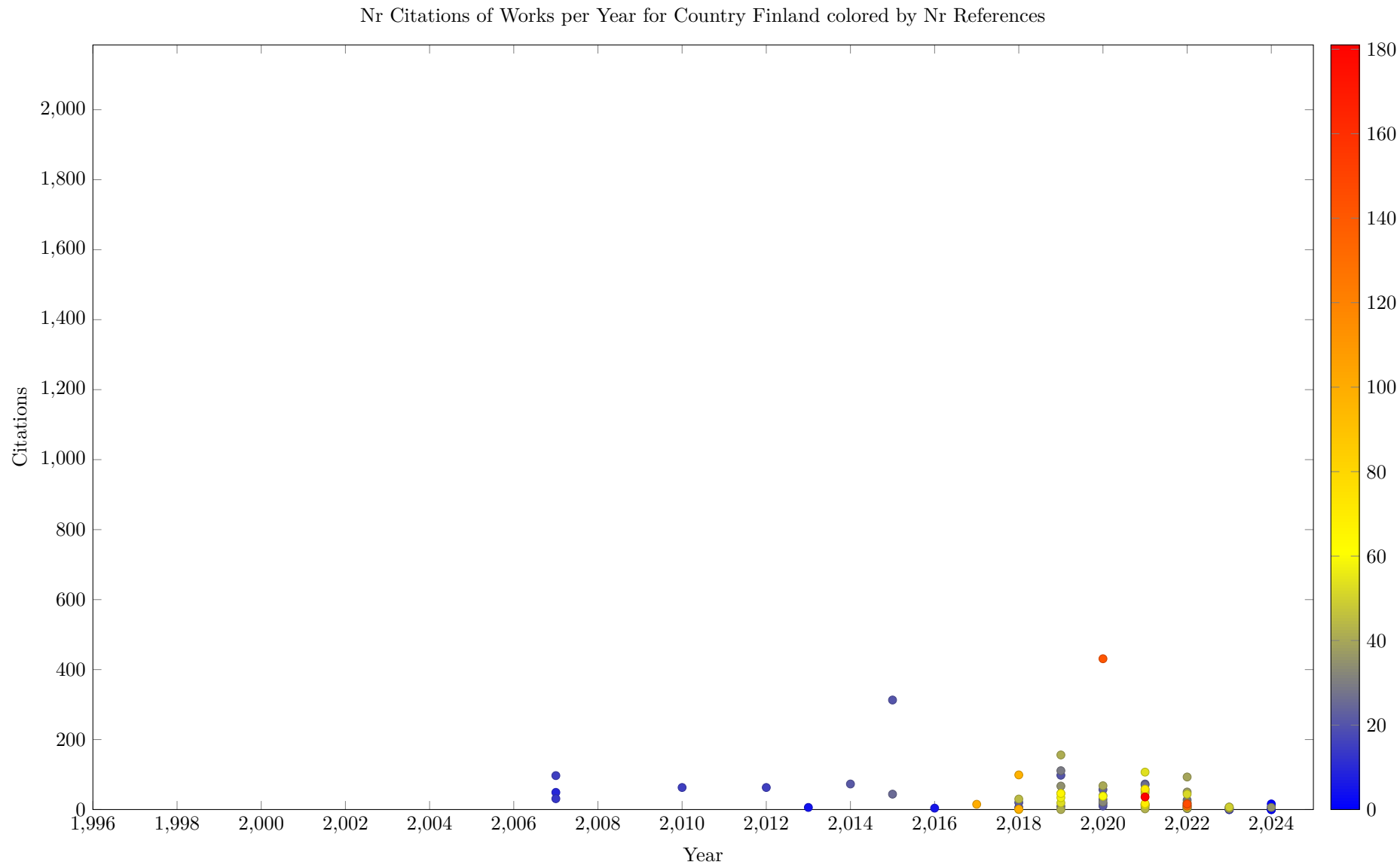


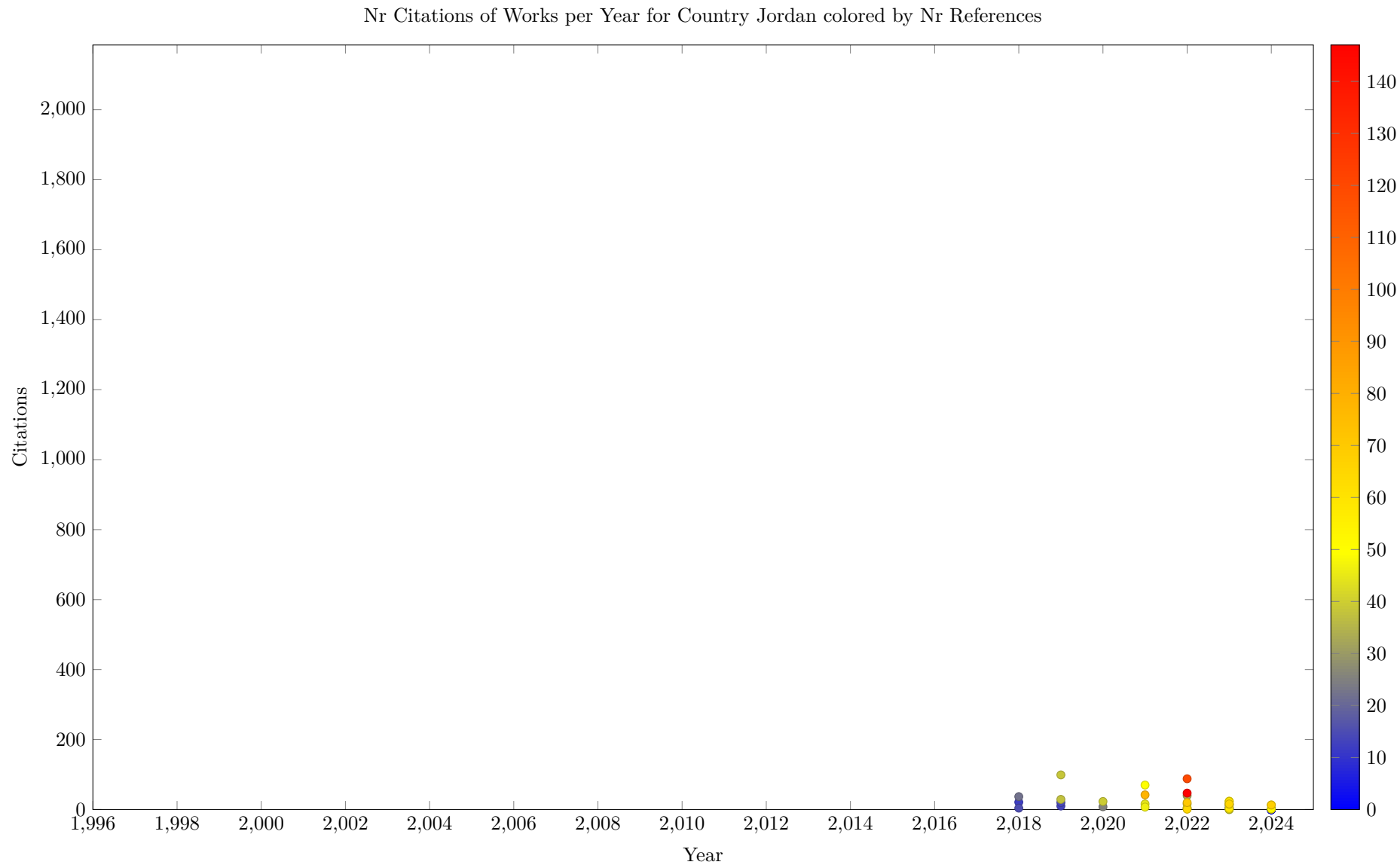


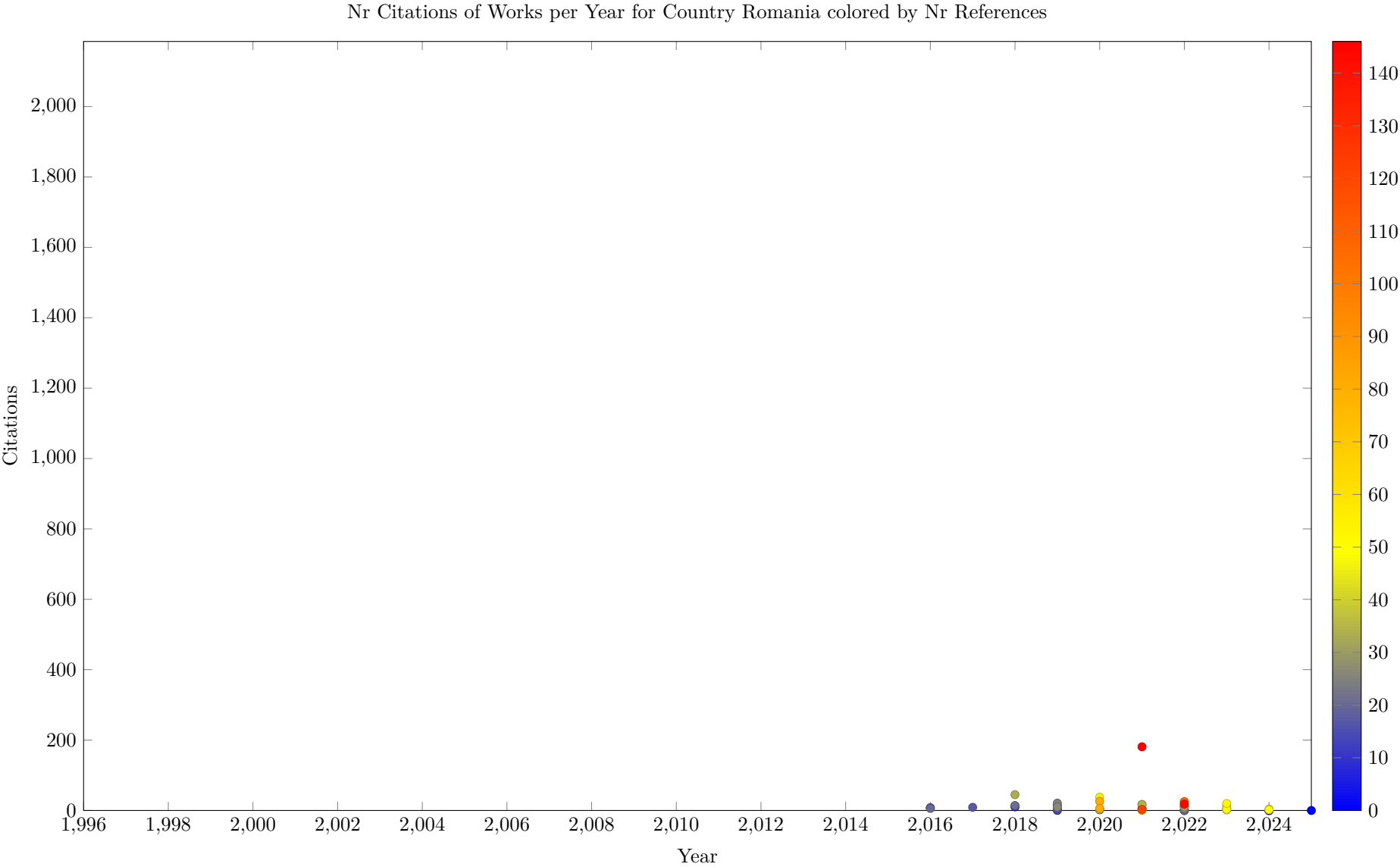


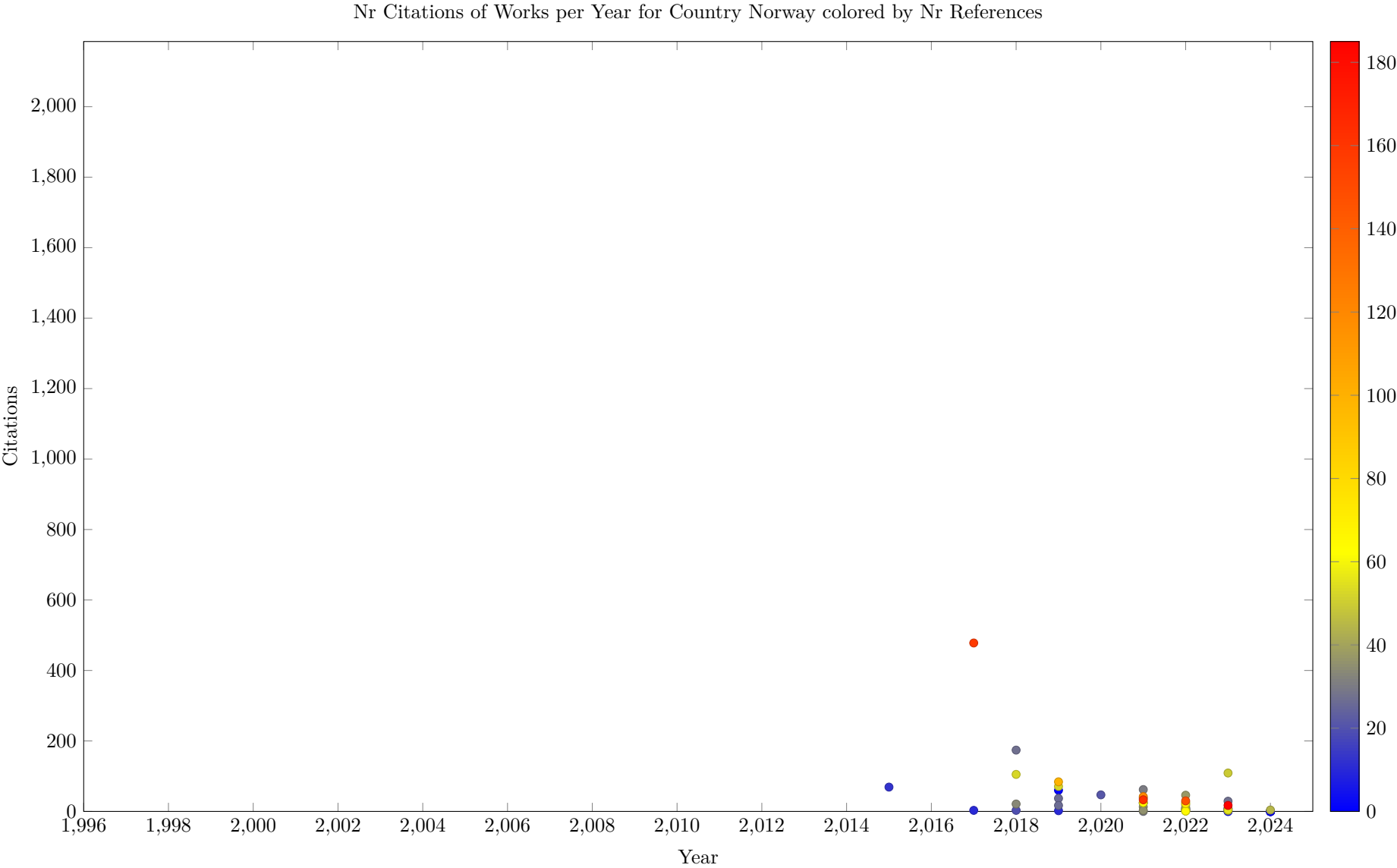


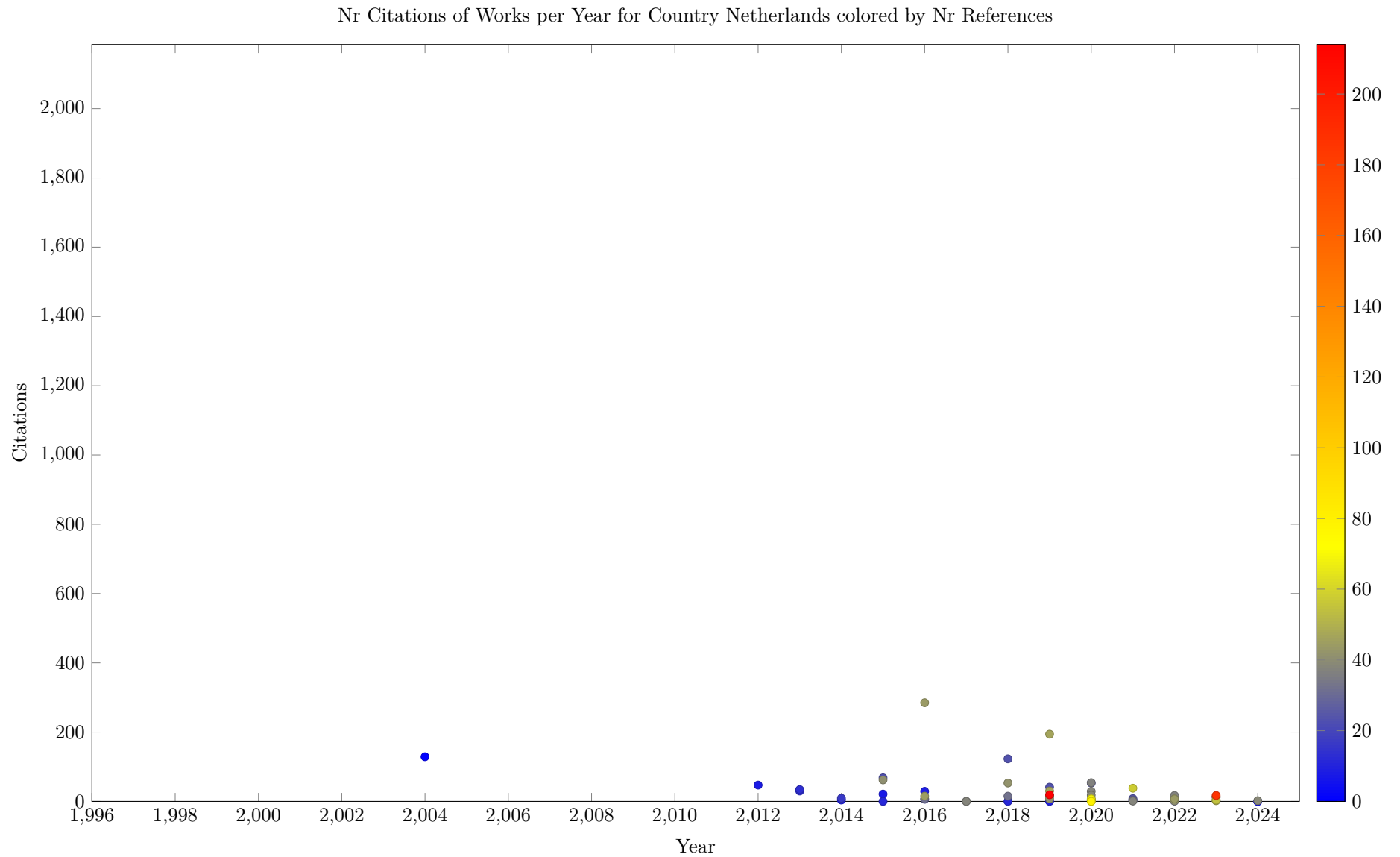


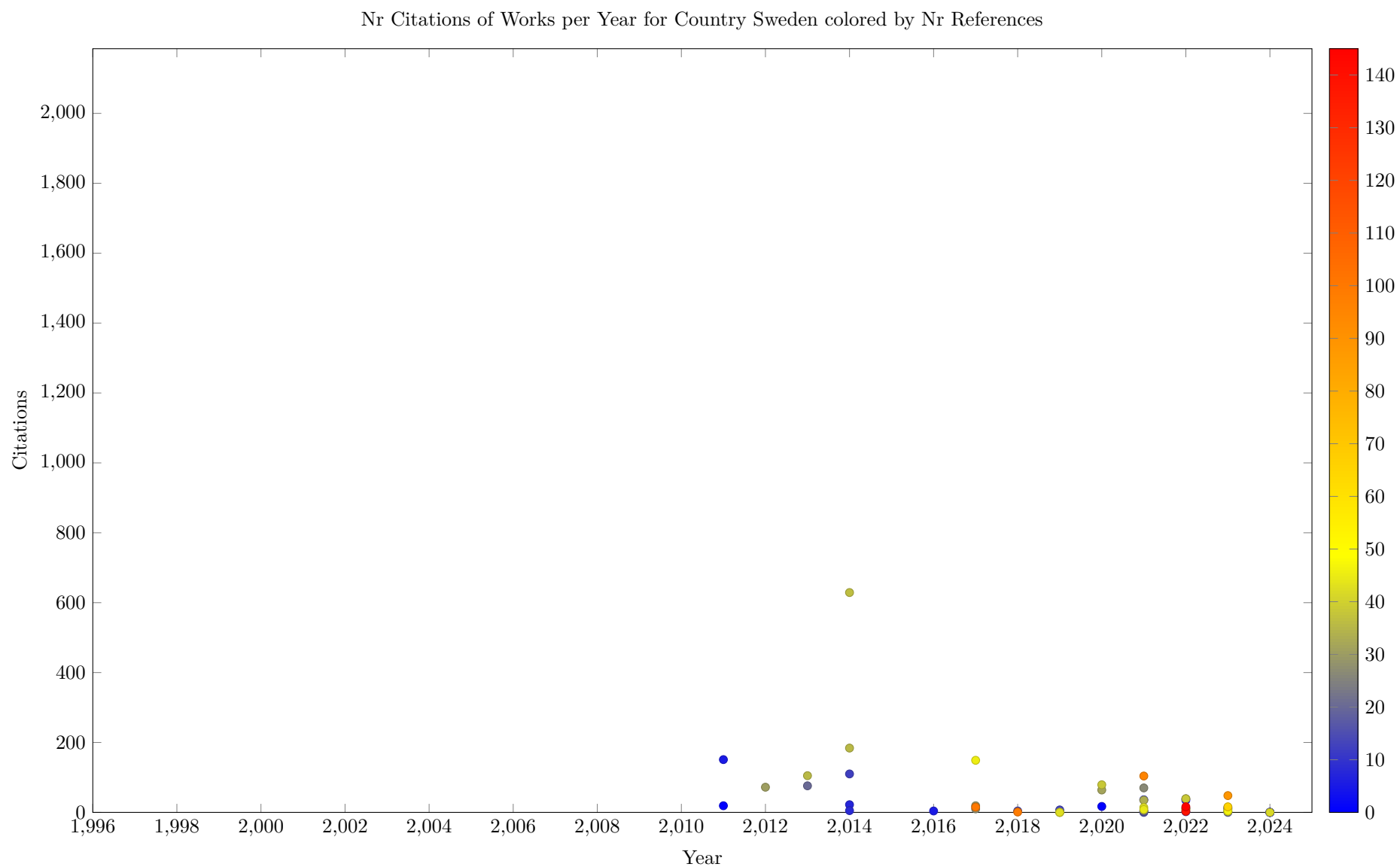


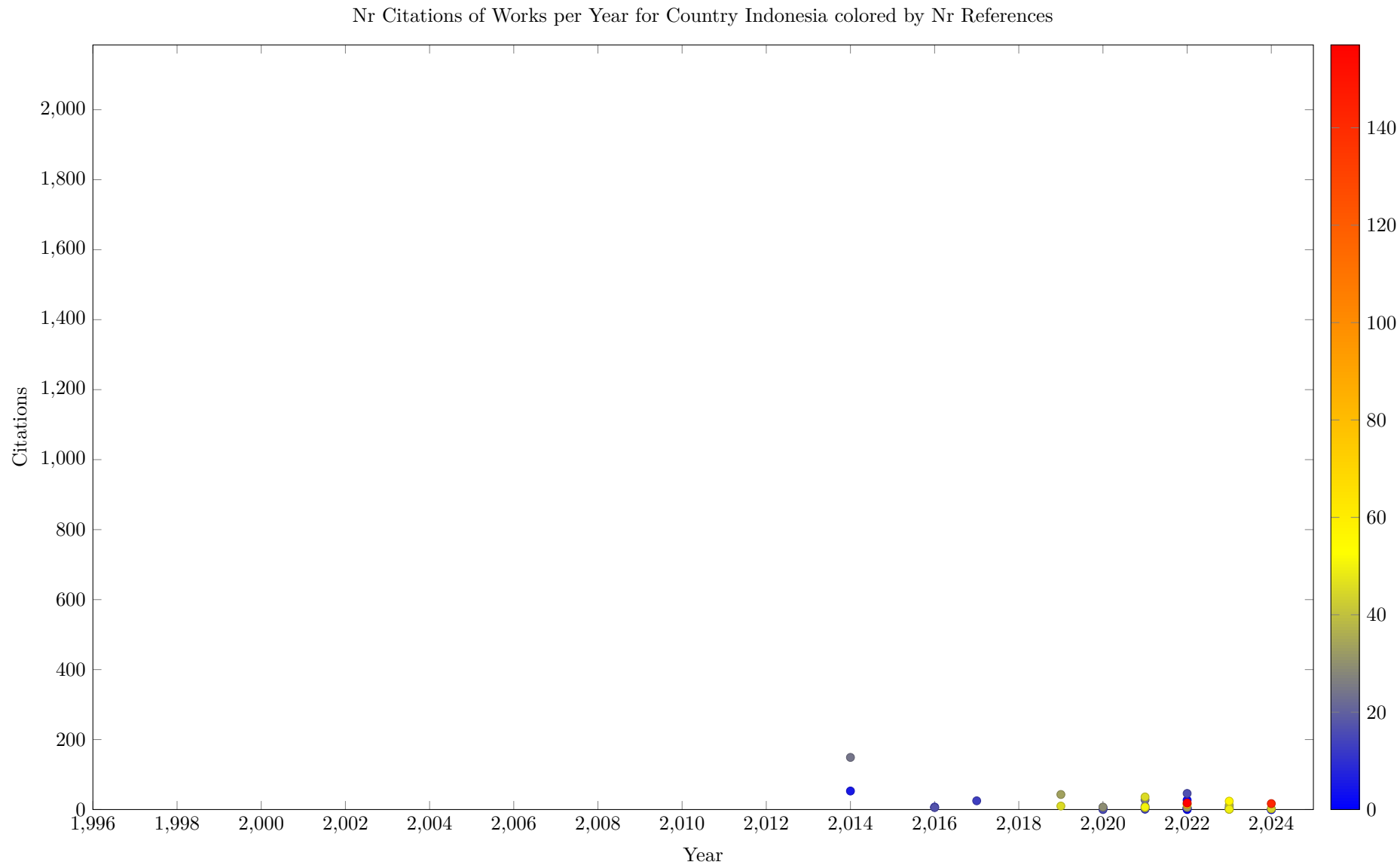


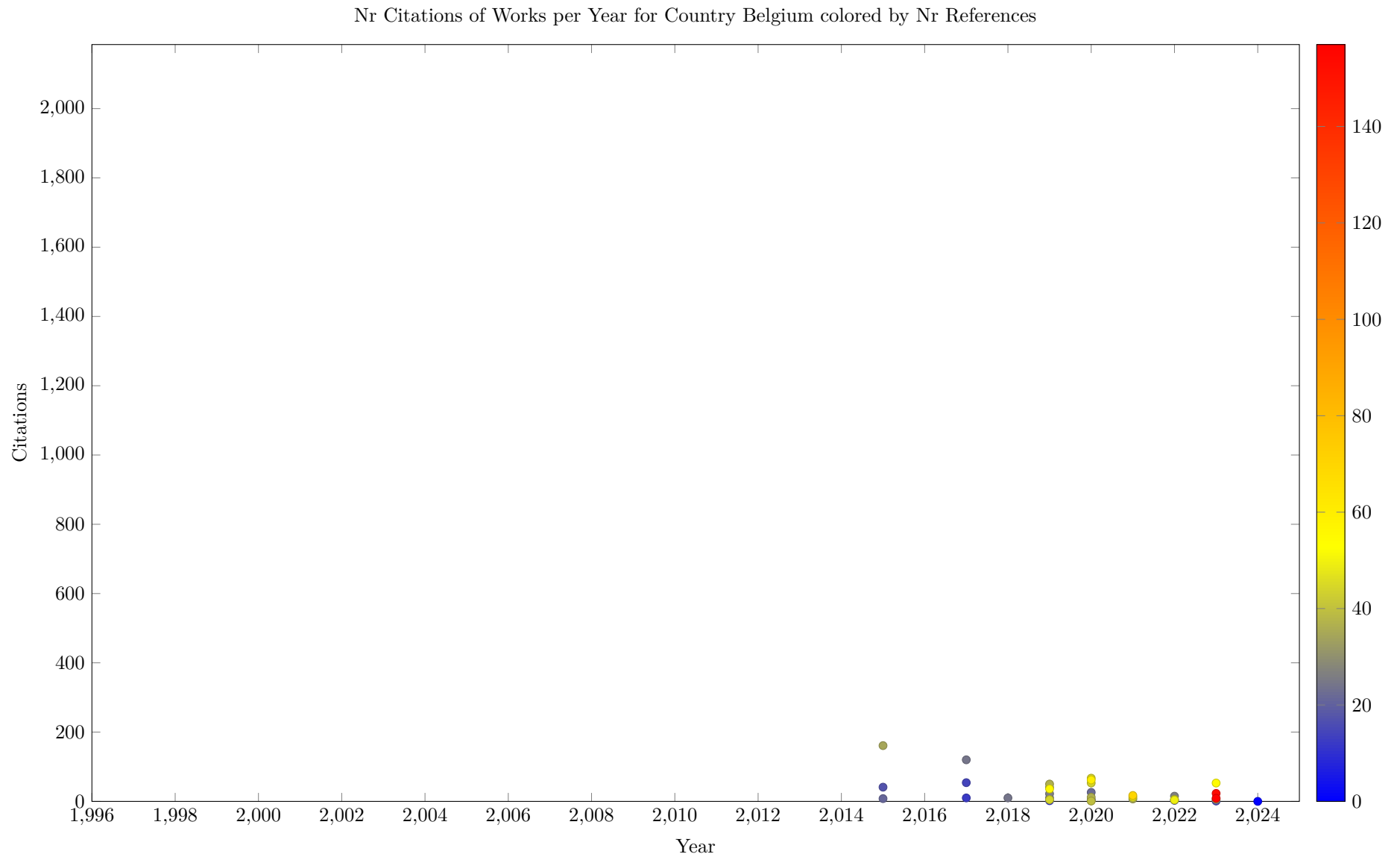


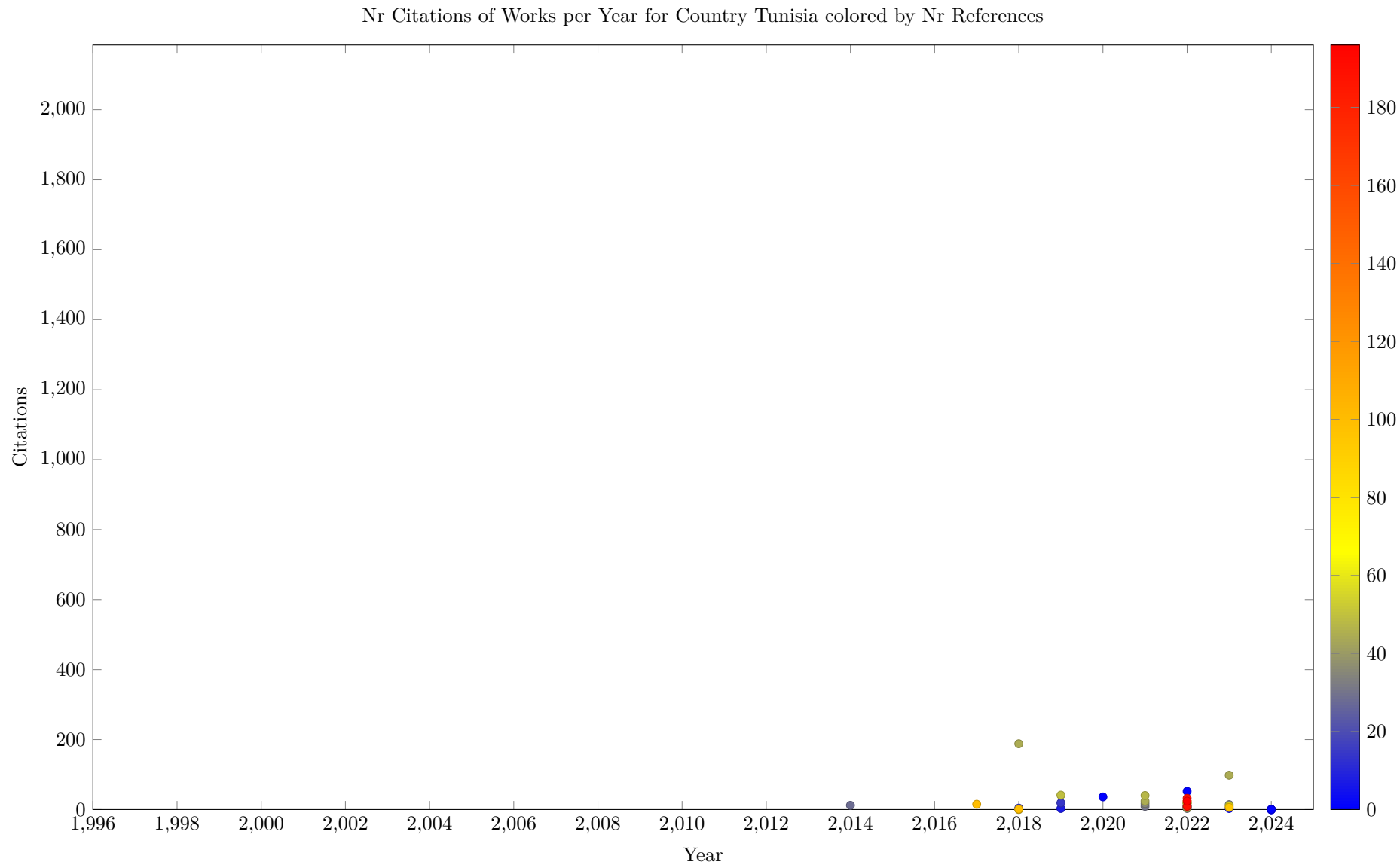


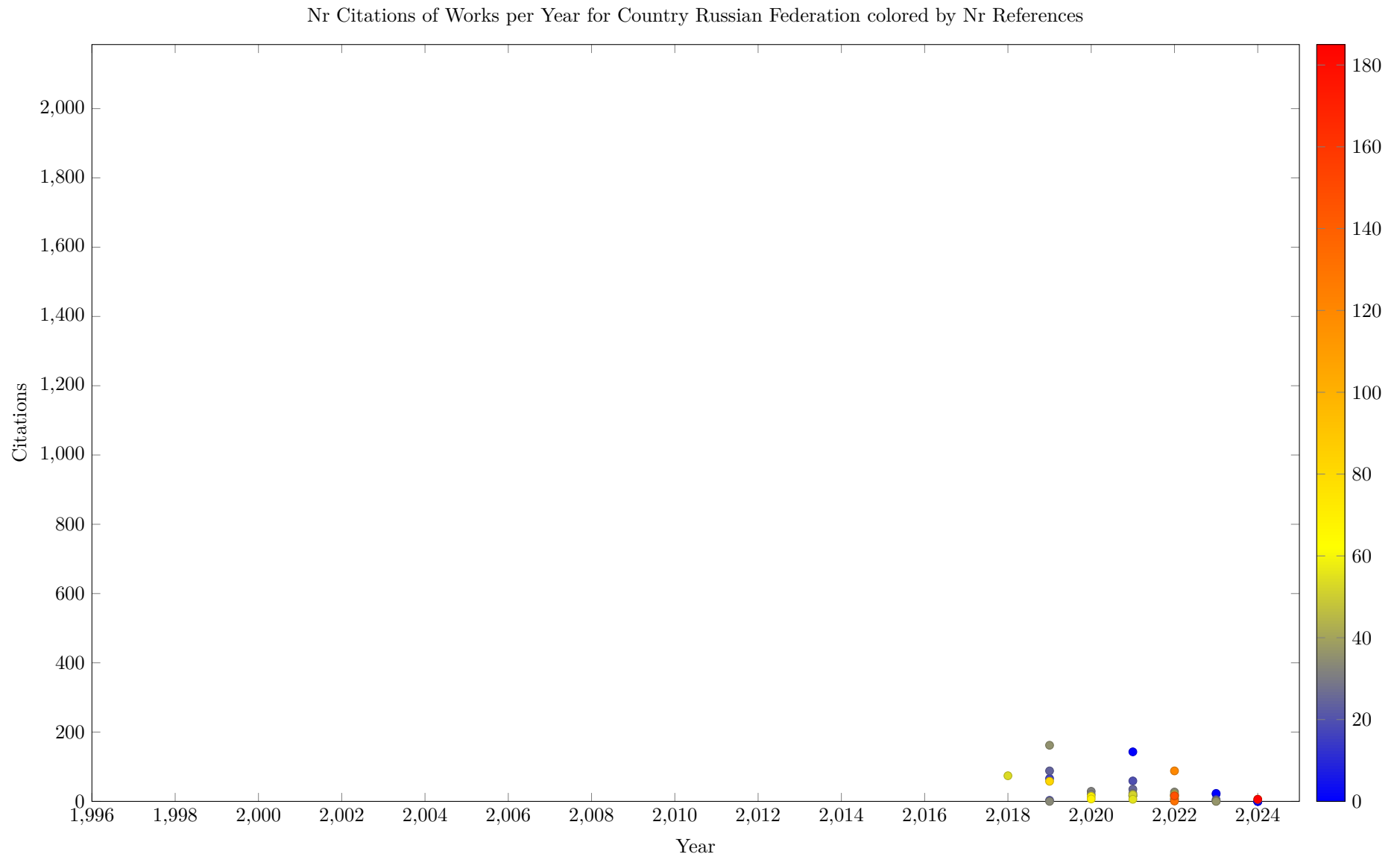


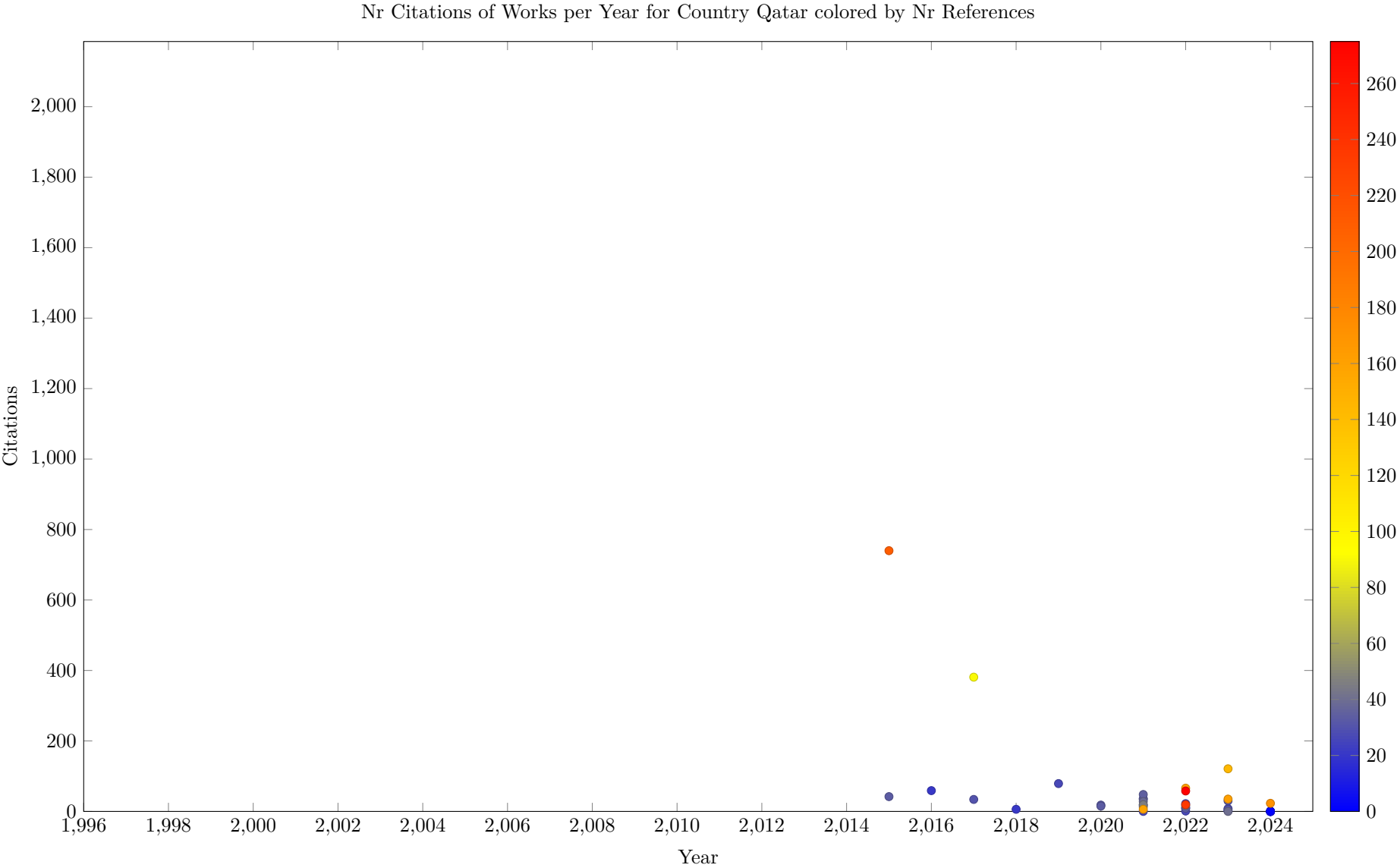


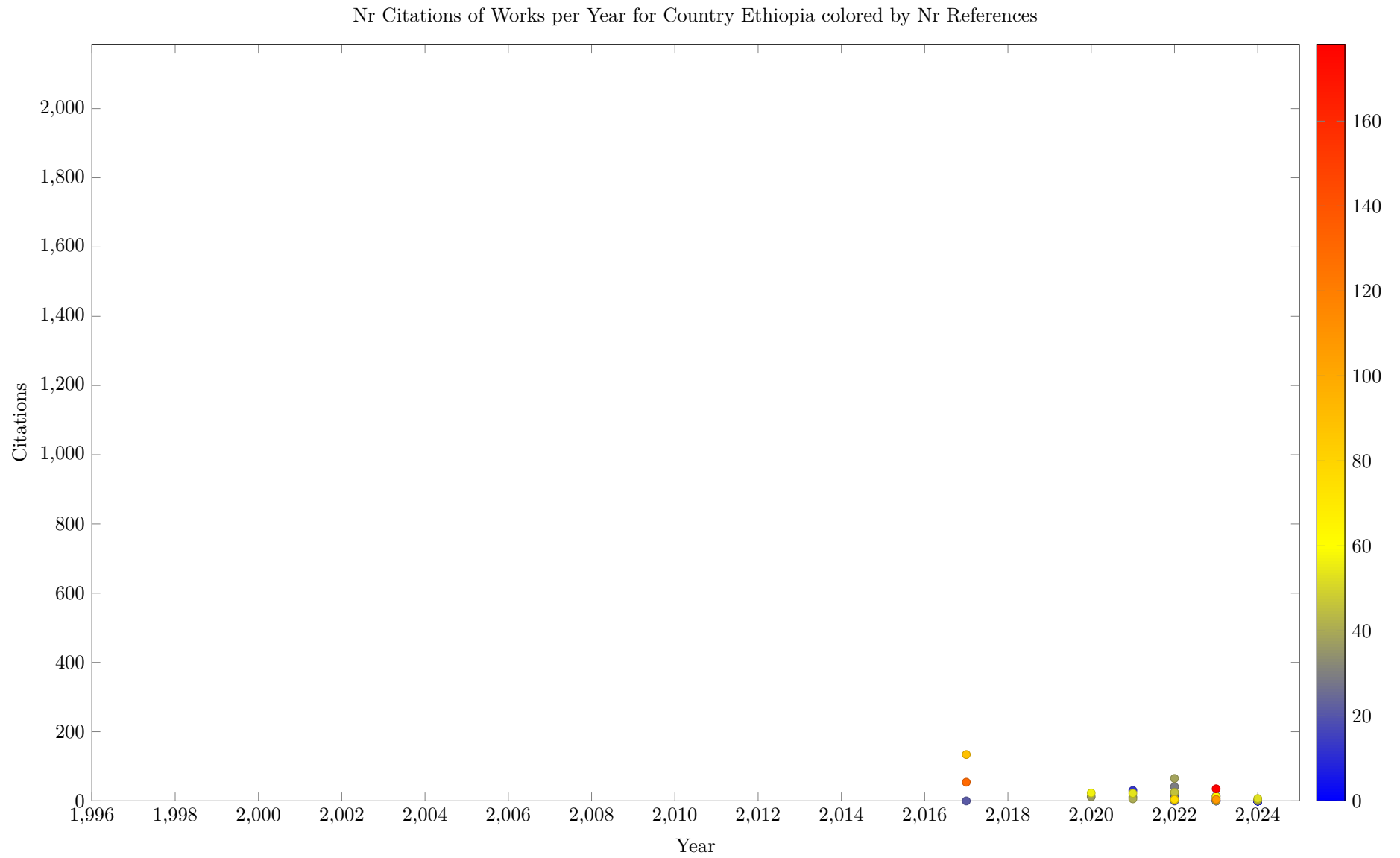


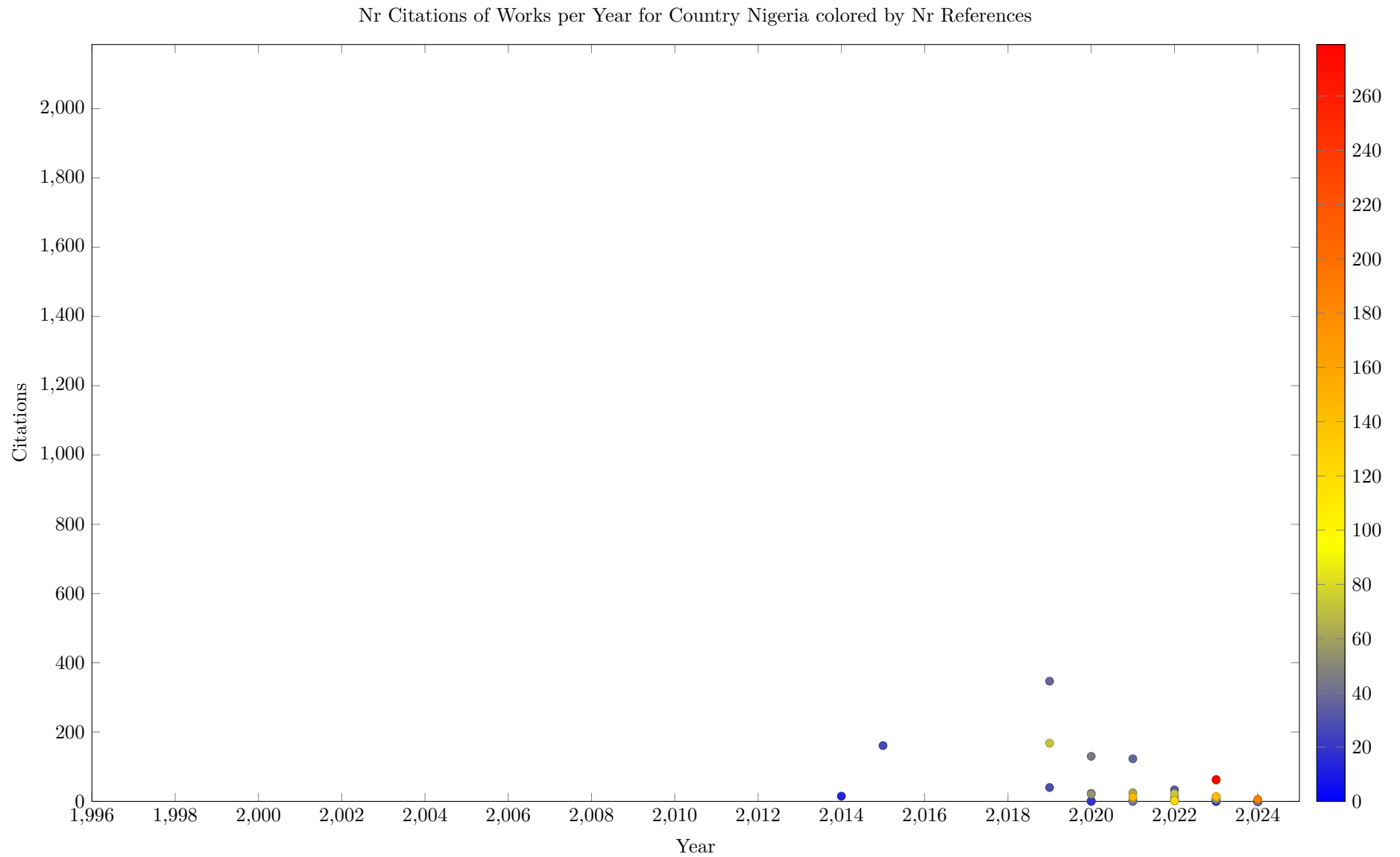


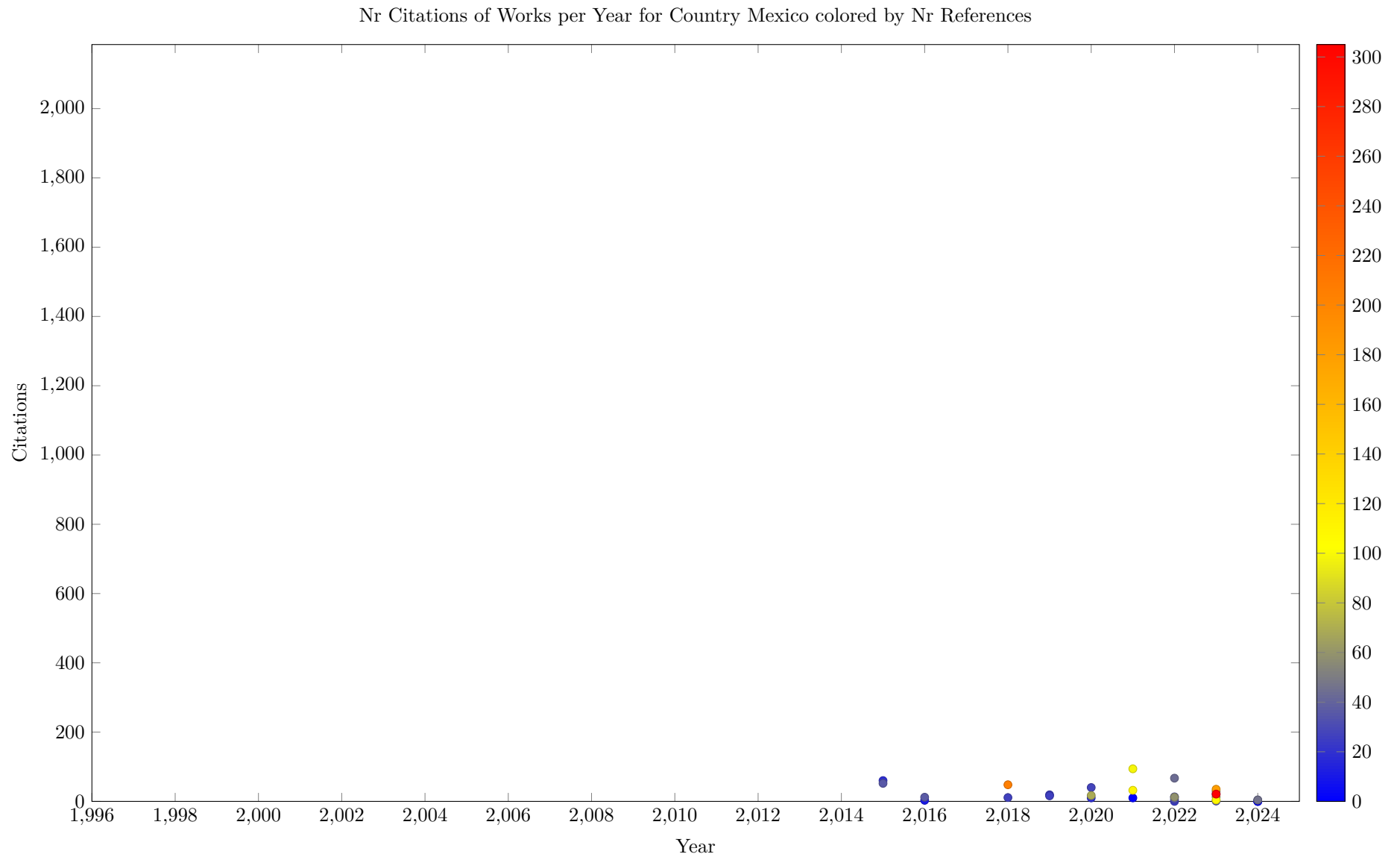


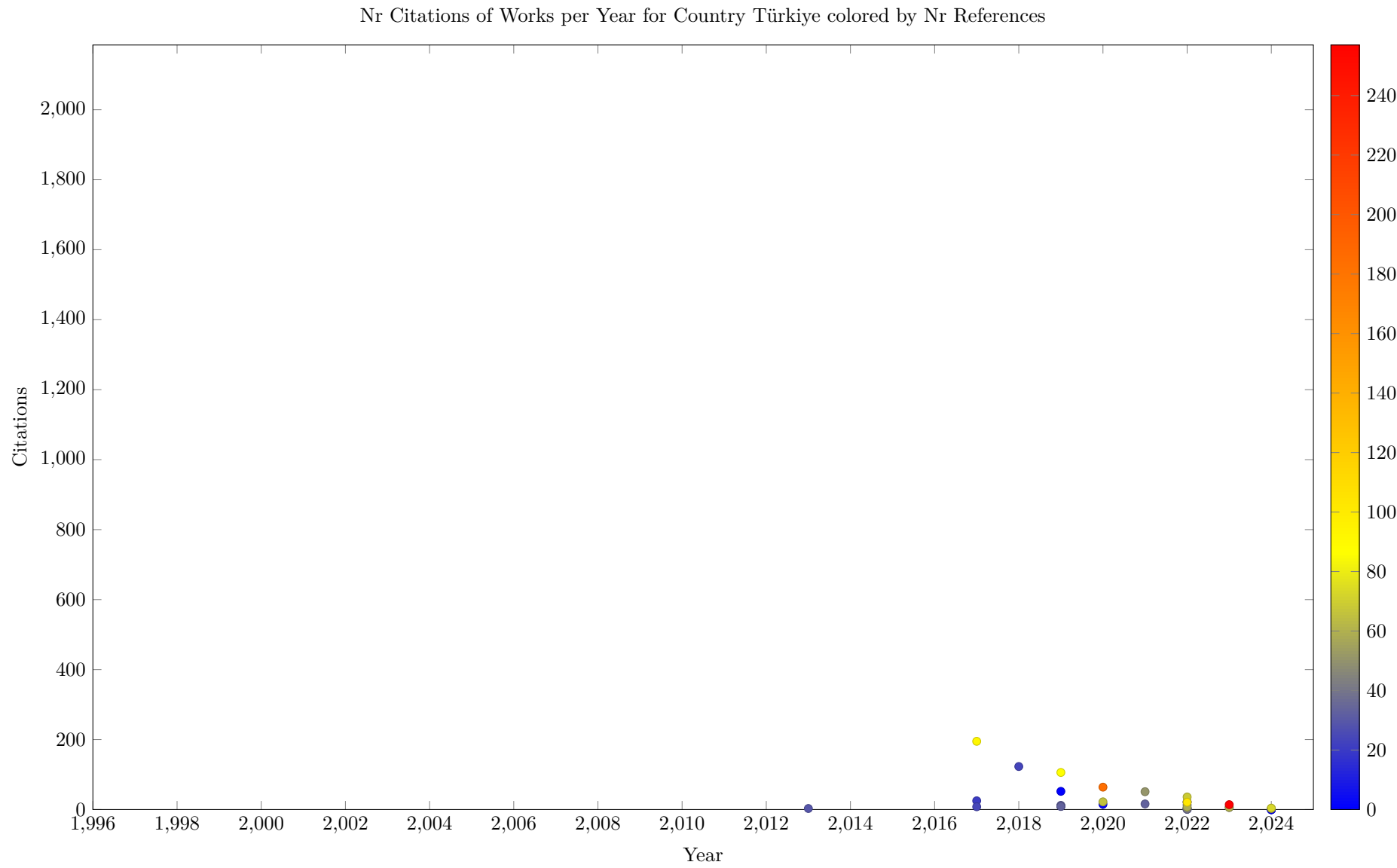


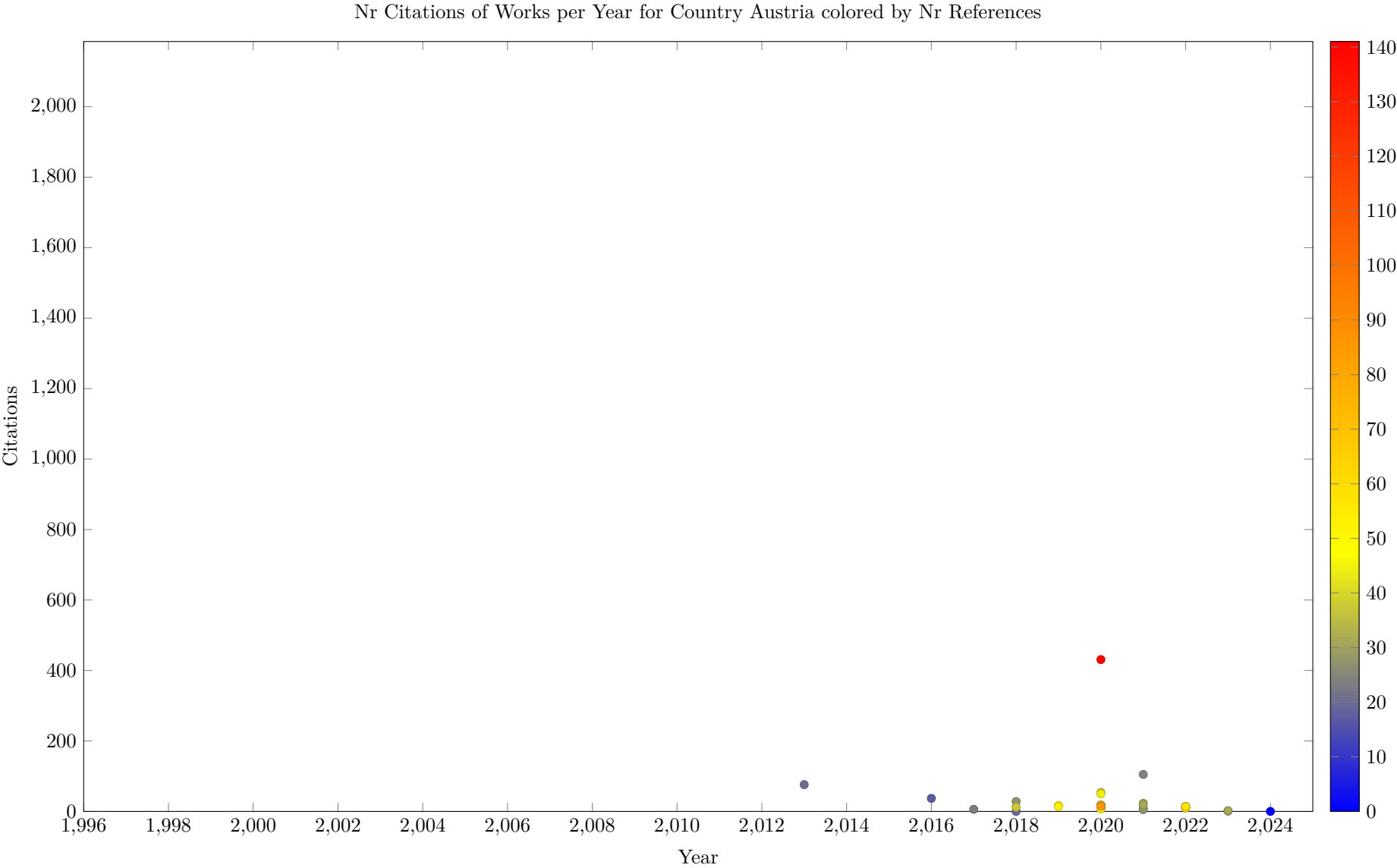


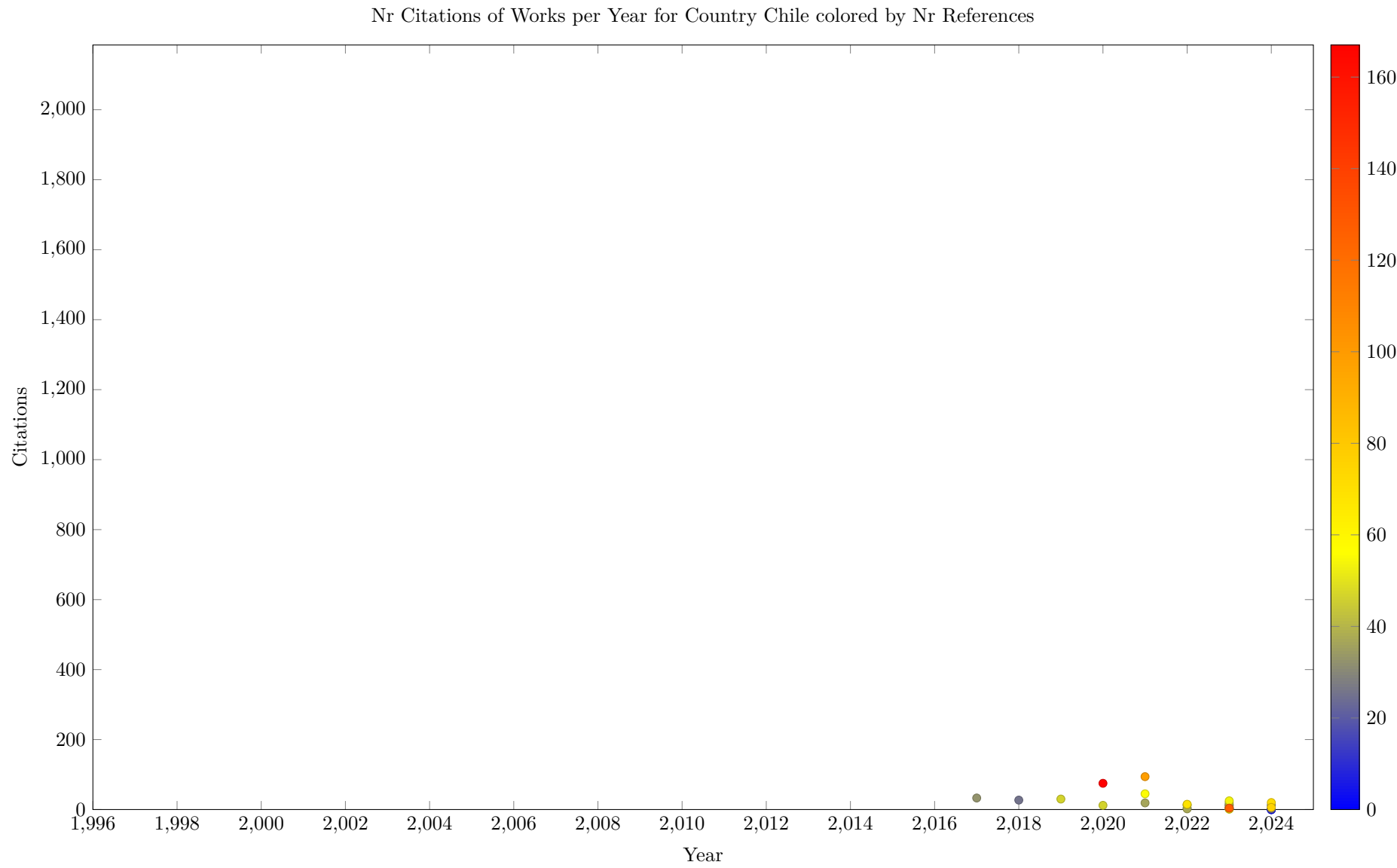


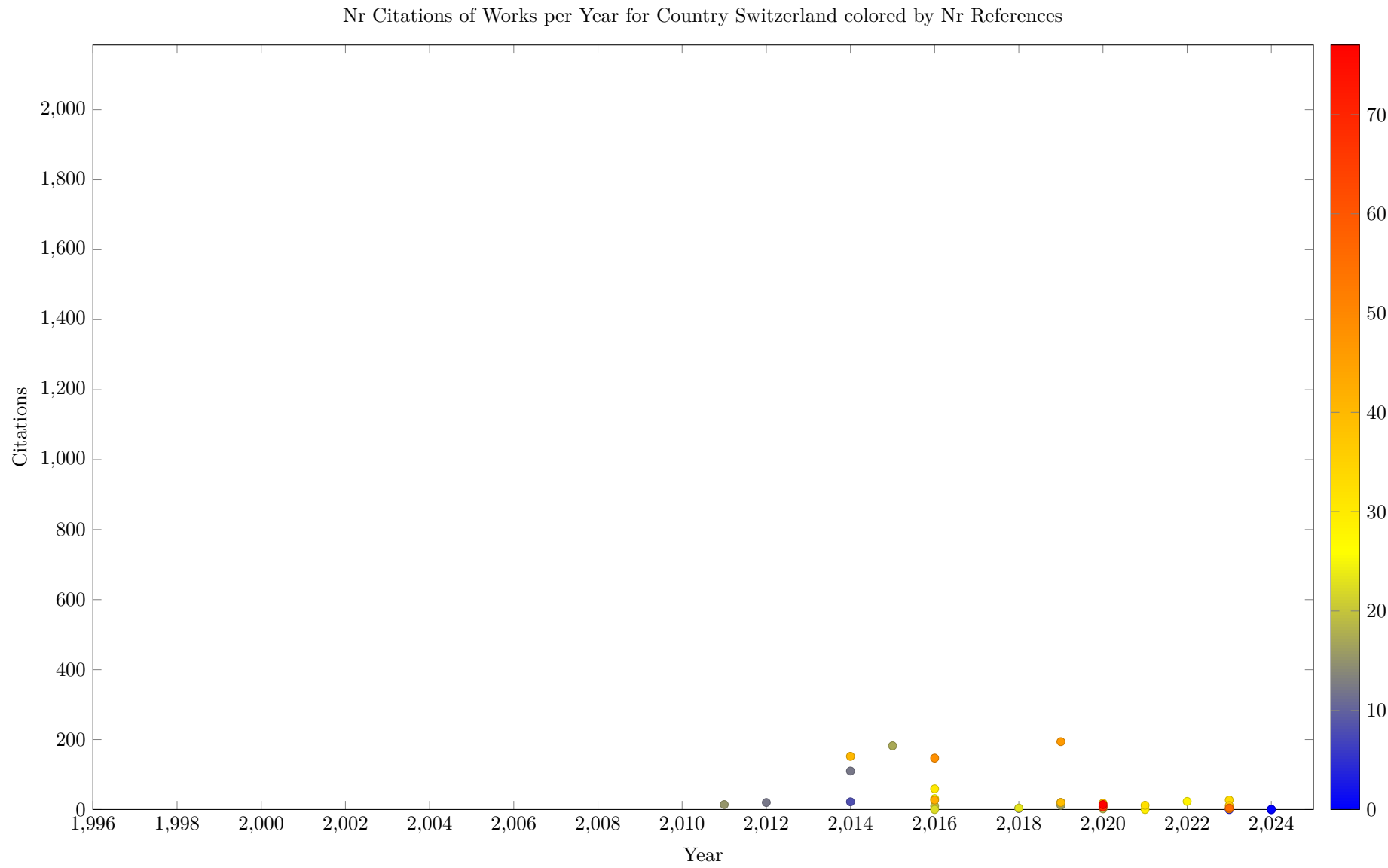


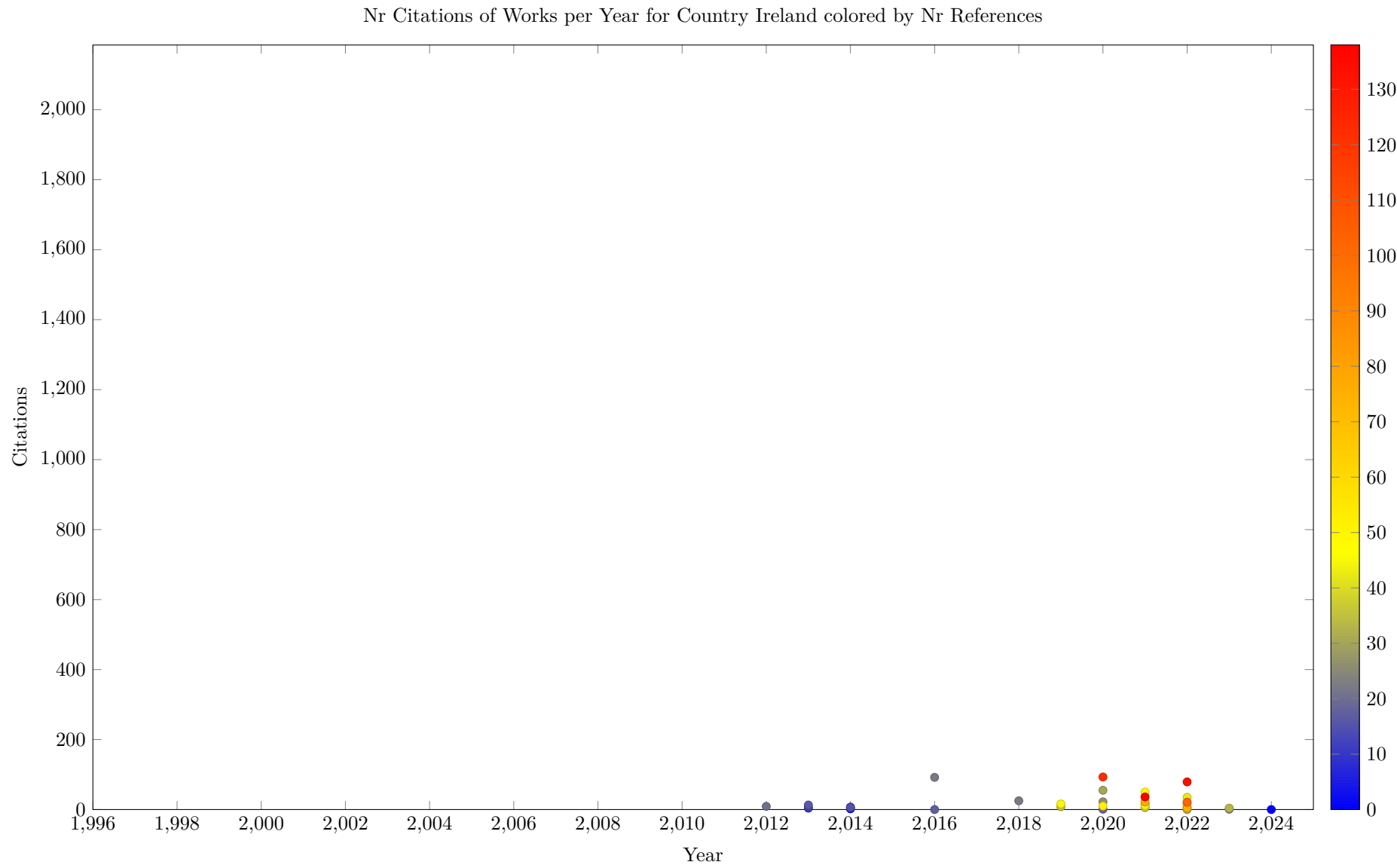


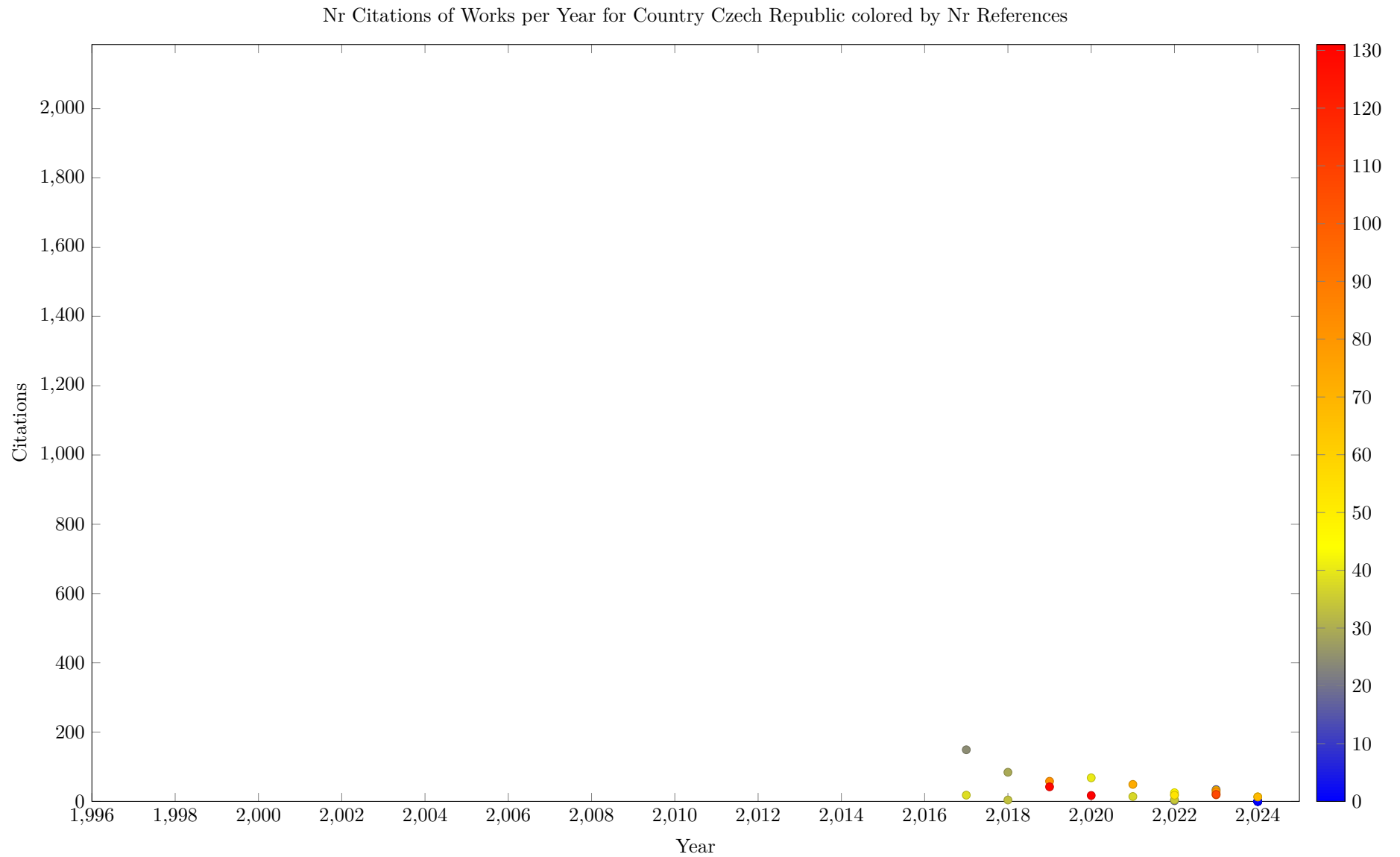


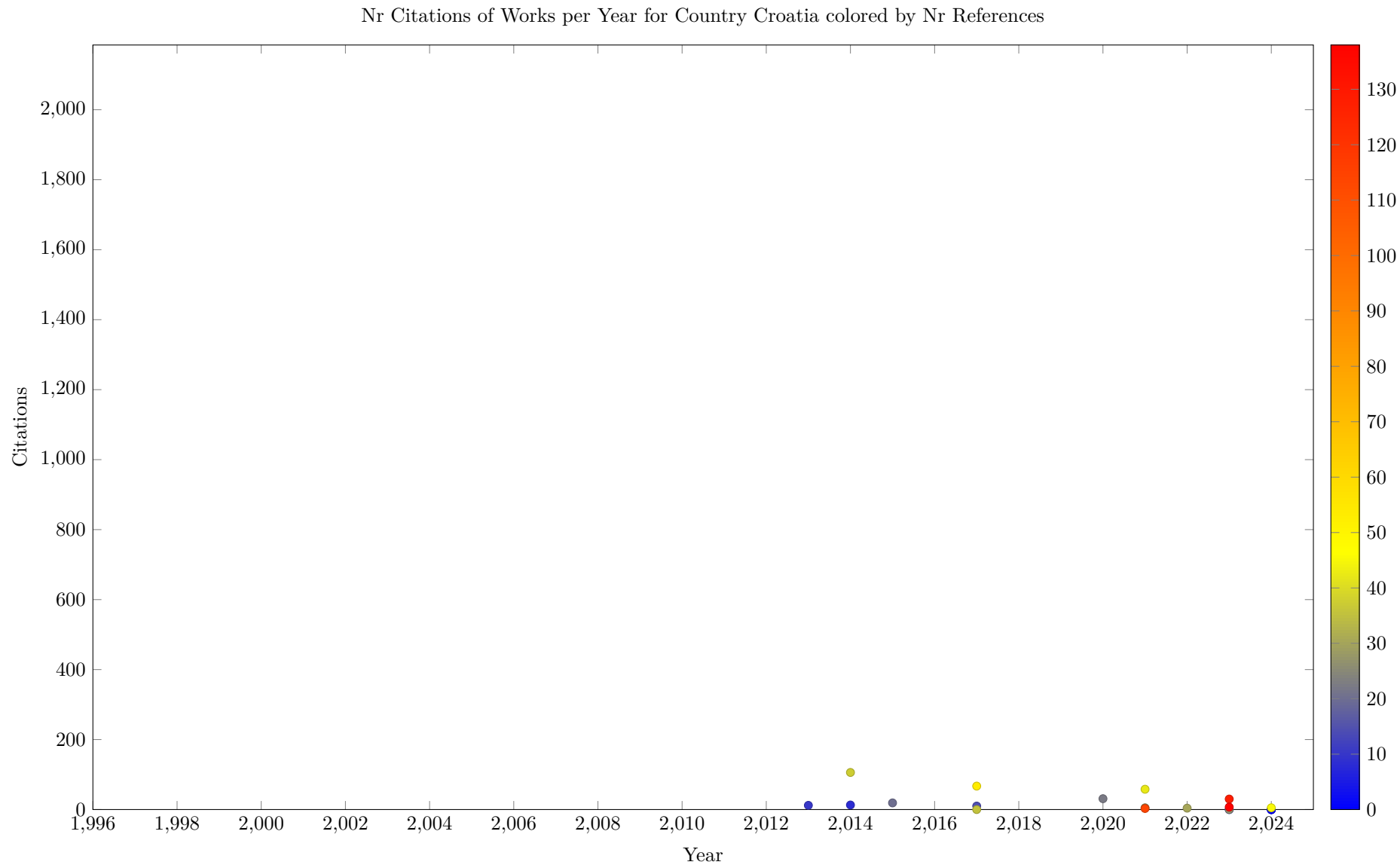


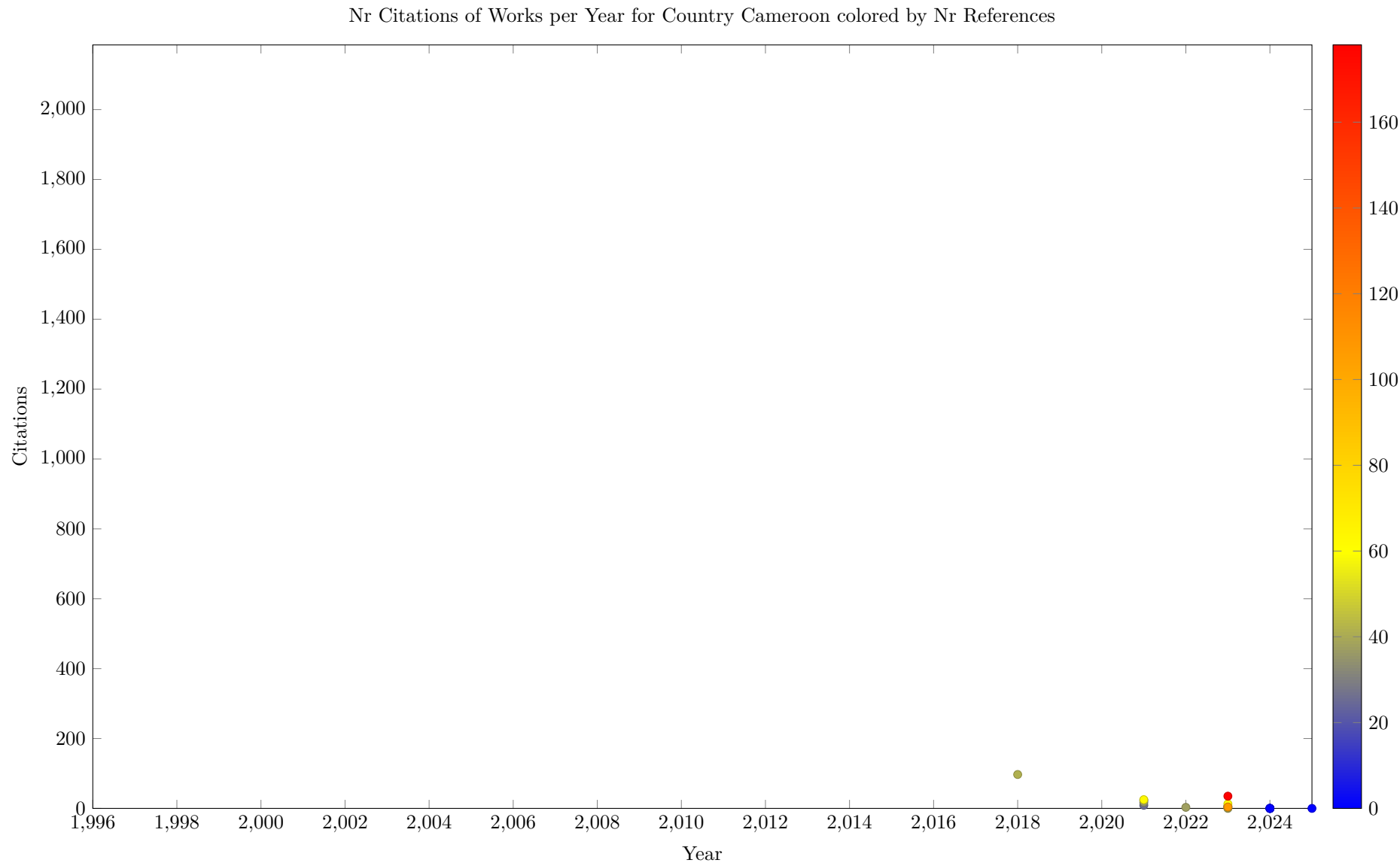


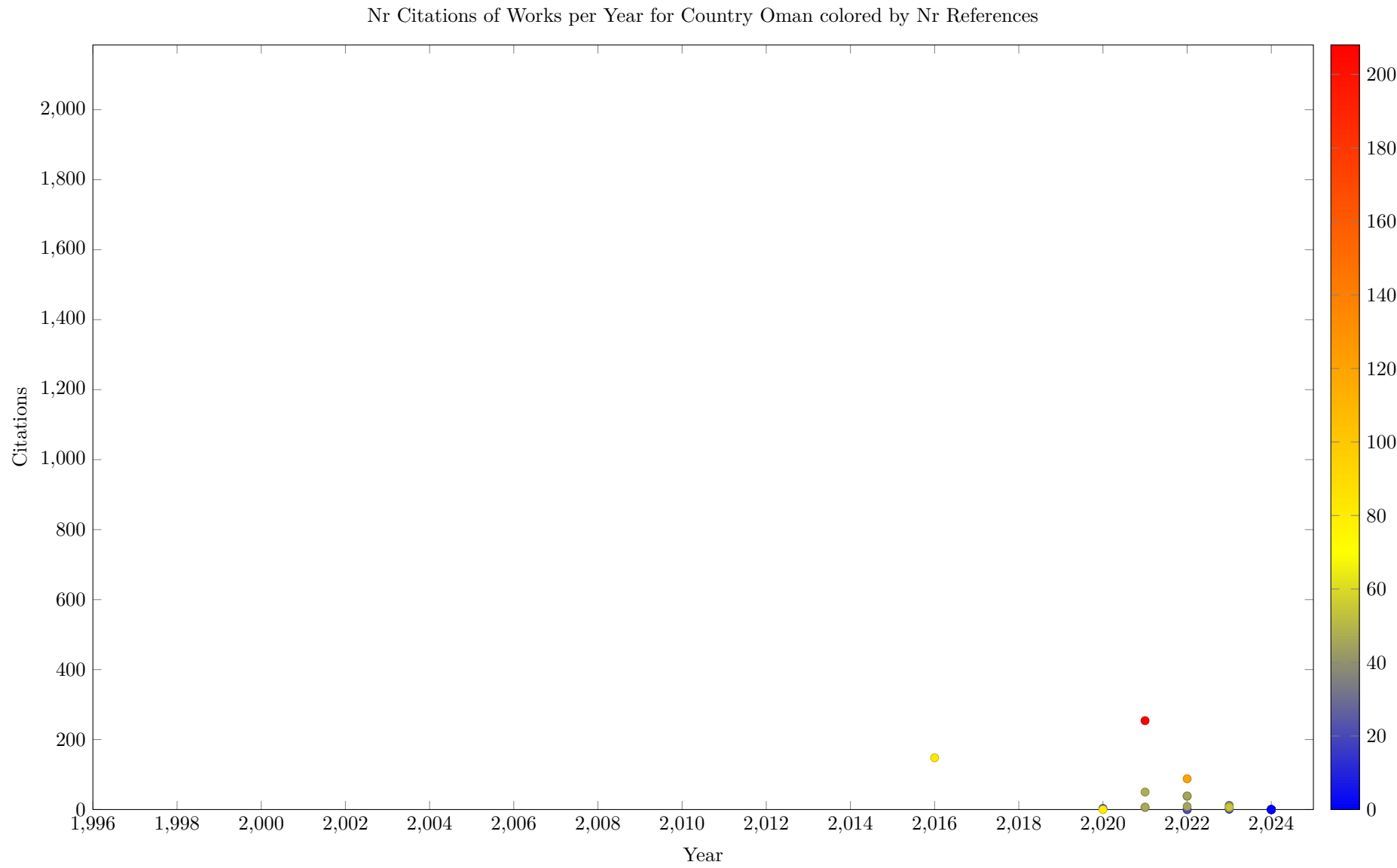


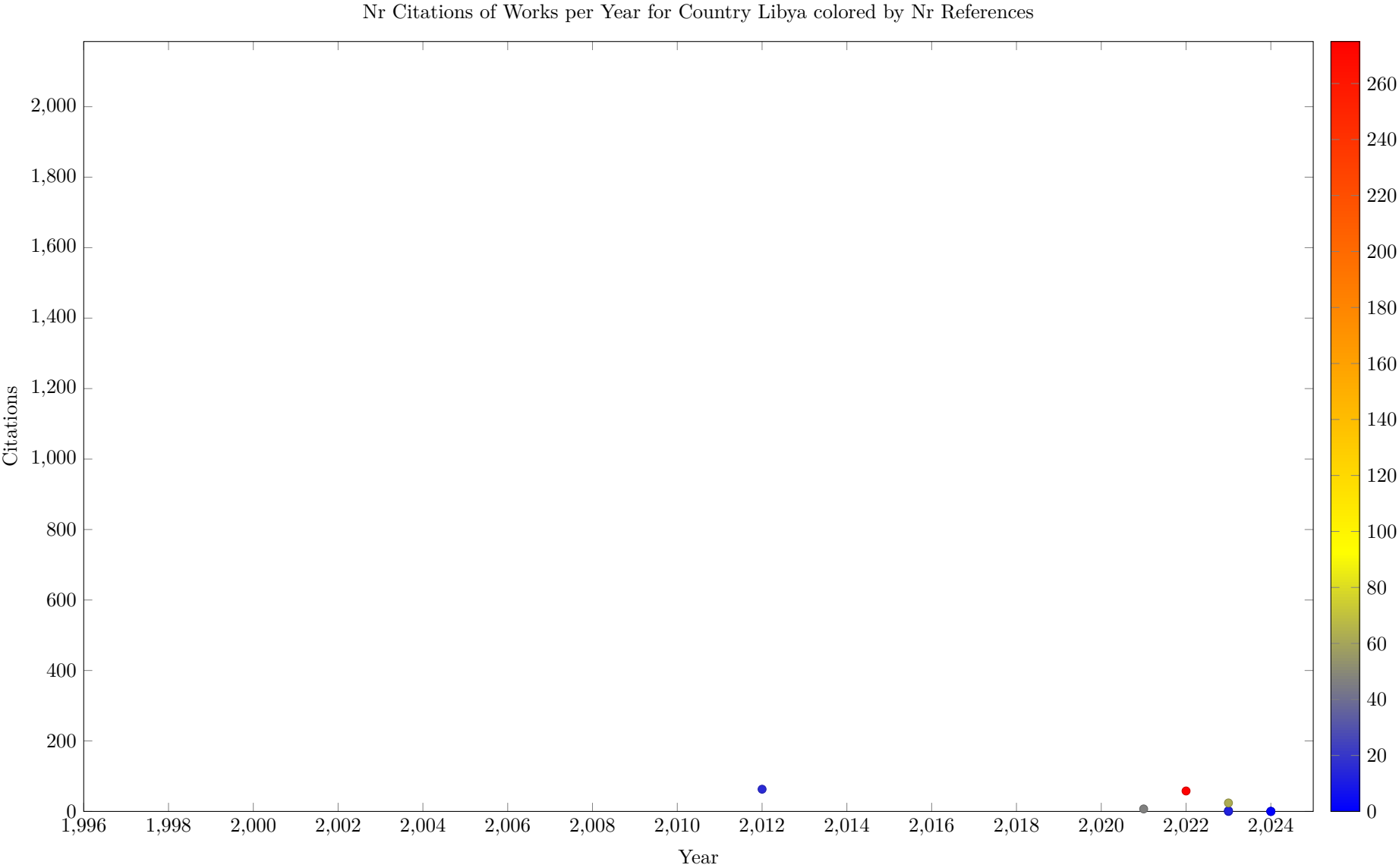


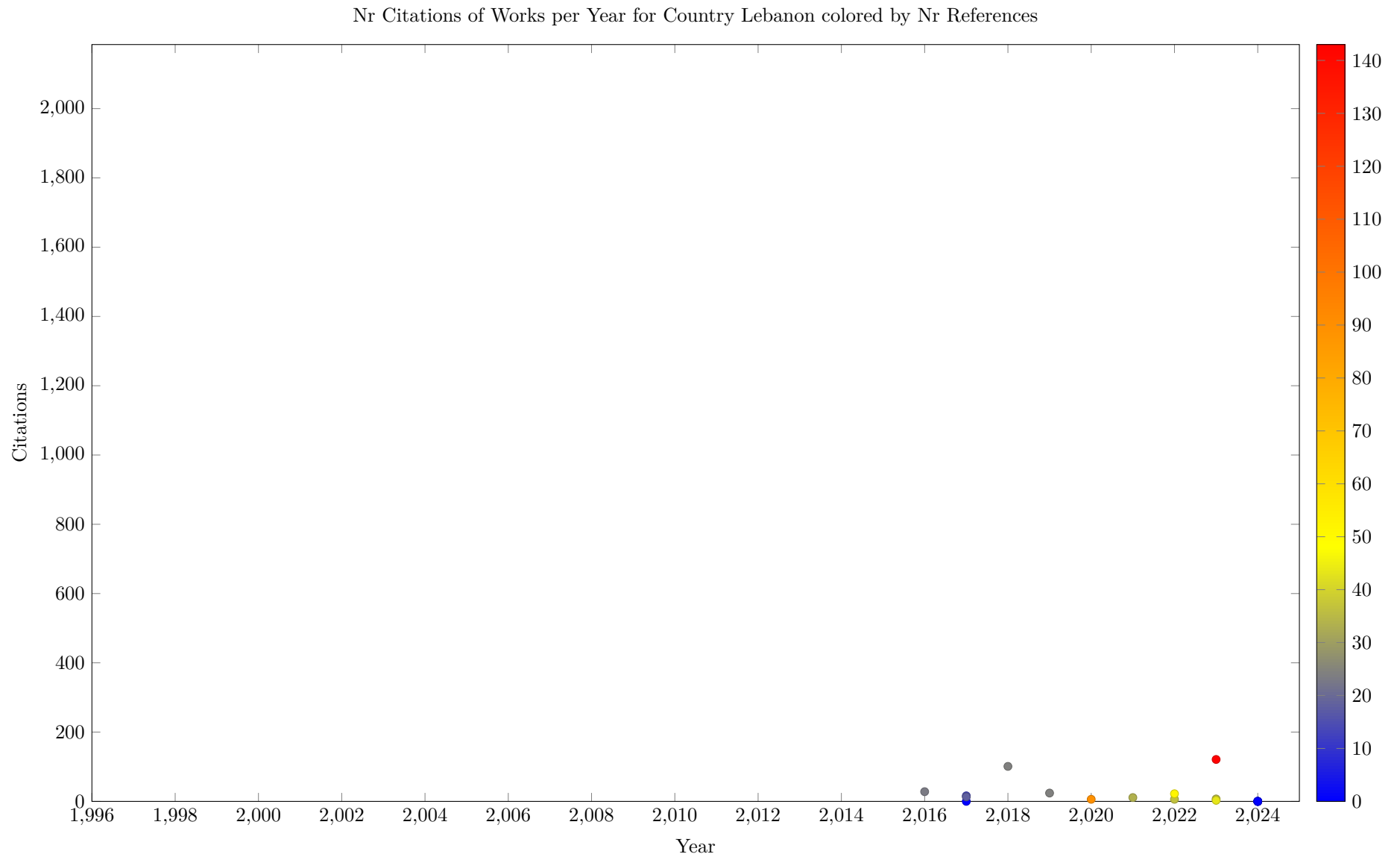


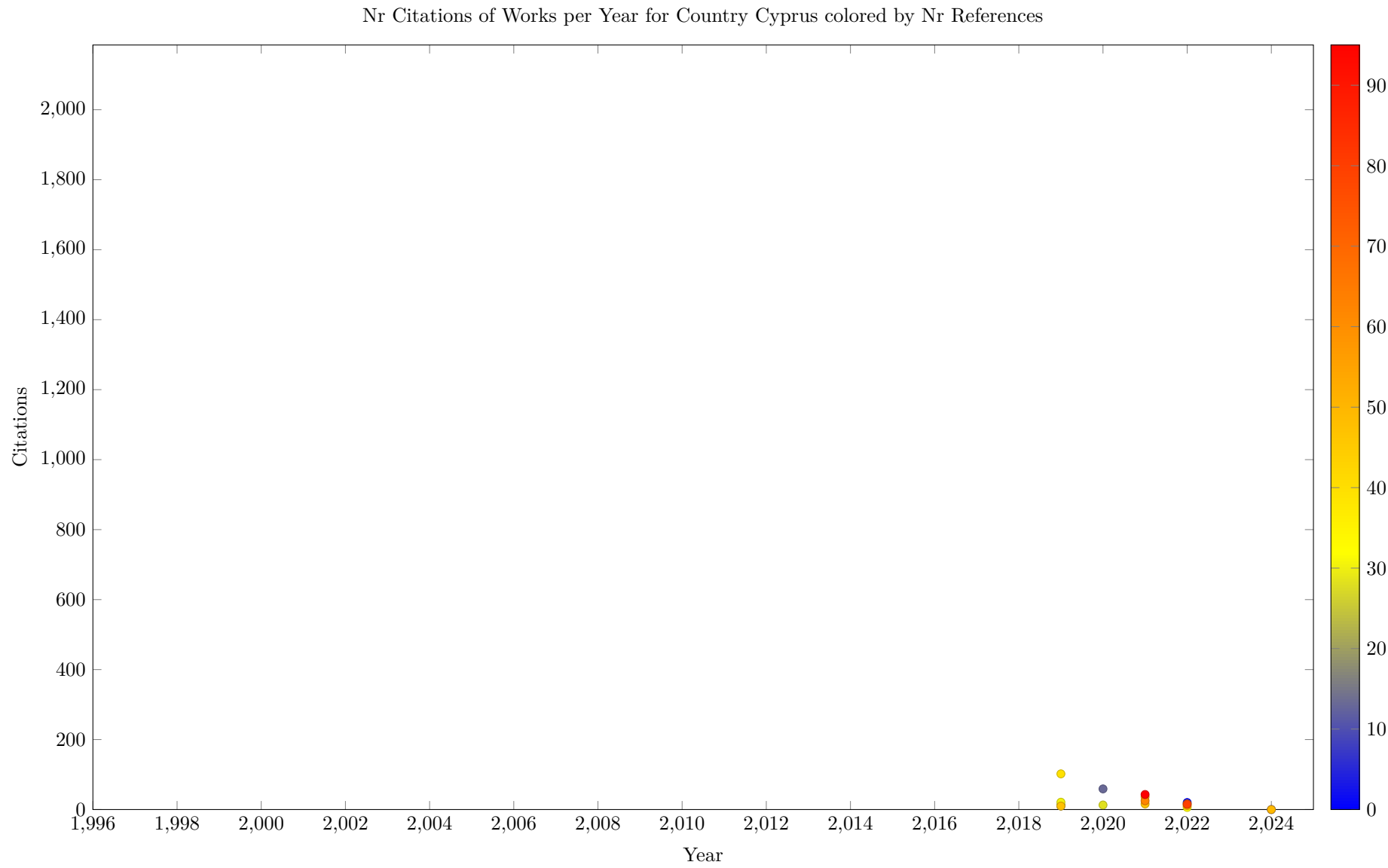


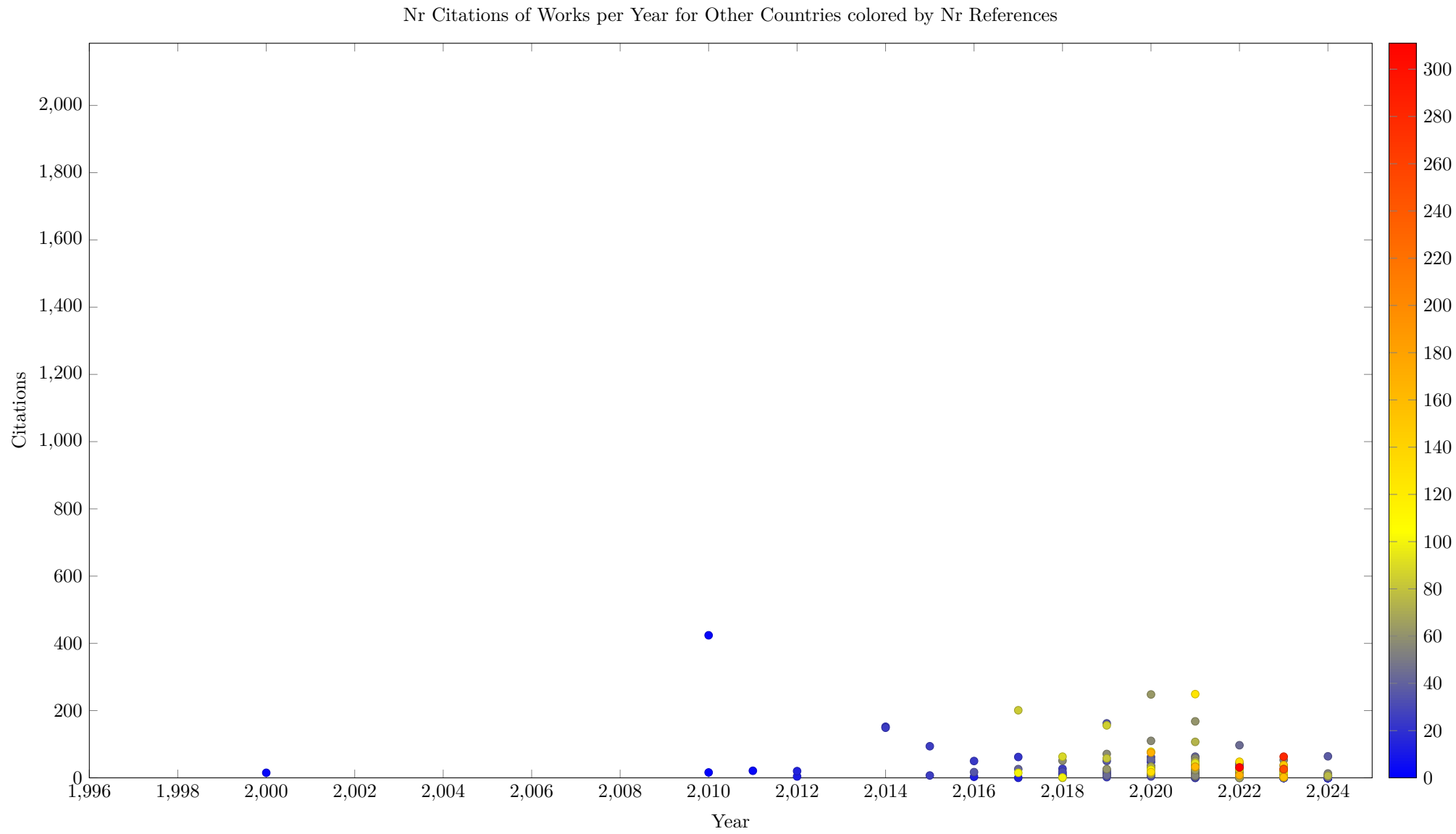












4 Collaborations

This section shows data about collaborations between multiple affiliations for the same work. This is based on Scopus data, which associates the affiliation with the work, not with each author of the work. The analysis excludes background work.

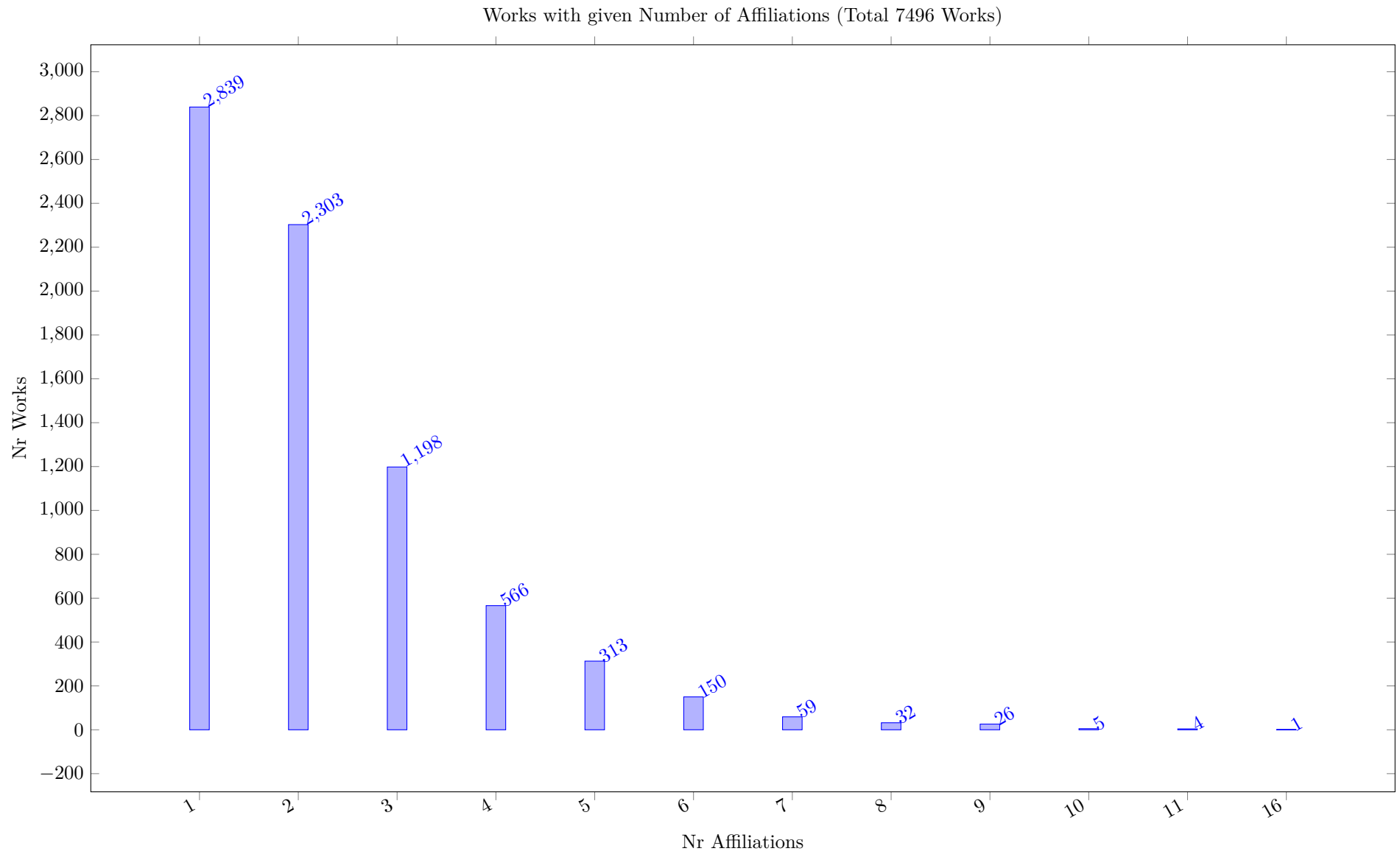


Table 8: Collaboration Data (Top 45 Inst by Decreasing Collab Fraction)

Inst	Nr Works	Collab Count	Domestic Collab	International Collab	Collab Fraction	Domestic Fraction	International Fraction	Collab Percentage	International Percentage
Aalborg University, Aalborg, Denmark	126	273	4	269	119.00	1.33	117.67	94.44	93.39
North China Electric Power University, Beijing, China	169	216	159	57	112.00	93.32	18.68	66.27	11.06
COMSATS University Islamabad, Islamabad, Pakistan	139	272	124	148	100.00	48.11	51.89	71.94	37.33
Faculty of Energy Engineering, Aswan, Egypt	77	281	94	187	77.00	26.65	50.35	100.00	65.39
Southeast University, Nanjing, China	84	131	98	33	72.00	56.25	15.75	85.71	18.75
University of Tabriz, Tabriz, Iran	99	157	50	107	71.00	22.15	48.85	71.72	49.34
State Grid Corporation of China, Beijing, China	75	128	110	18	69.00	62.80	6.20	92.00	8.27
Tsinghua University, Beijing, China	87	150	88	62	69.00	42.33	26.67	79.31	30.65
Tianjin University, Tianjin, China	82	124	79	45	66.00	46.33	19.67	80.49	23.99
King Saud University, Riyadh, Saudi Arabia	60	184	15	169	56.00	3.90	52.10	93.33	86.84
Huazhong University of Science and Technology, Wuhan, China	70	130	60	70	56.00	29.52	26.47	80.00	37.82
Beijing Institute of Technology, Beijing, China	80	110	88	22	53.00	42.93	10.07	66.25	12.58
Shanghai Jiao Tong University, Shanghai, China	73	86	55	31	53.00	33.25	19.75	72.60	27.05
China Electric Power Research Institute, Beijing, China	56	95	83	12	52.00	48.17	3.83	92.86	6.85
Chongqing University, Chongqing, China	61	108	64	44	52.00	35.08	16.92	85.25	27.73
Universidad de Jaén, Jaen, Spain	51	148	8	140	49.00	4.92	44.08	96.08	86.44
Shandong University, Jinan, China	71	102	70	32	49.00	33.68	15.32	69.01	21.57
Xi'an Jiaotong University, Xi'an, China	71	83	57	26	48.00	36.75	11.25	67.61	15.85
Prince Sattam Bin Abdulaziz University, Al Kharj, Saudi Arabia	49	205	42	163	47.00	8.68	38.32	95.92	78.20
Shiraz University of Technology, Shiraz, Iran	52	103	37	66	46.00	18.57	27.43	88.46	52.76
North China Electric Power University (Baoding), Baoding, China	62	92	57	35	44.00	33.78	10.22	70.97	16.48
University of Tehran, Tehran, Iran	61	93	36	57	44.00	22.03	21.97	72.13	36.01
Universiti Tenaga Nasional, Kajang, Malaysia	42	110	32	78	40.00	13.98	26.02	95.24	61.94
Islamic Azad University, Tehran, Iran	50	116	52	64	39.00	19.63	19.37	78.00	38.73
University of Engineering and Technology, Peshawar, Peshawar, Pakistan	38	125	58	67	37.00	16.72	20.28	97.37	53.38
University of Johannesburg, Johannesburg, South Africa	55	149	3	146	37.00	2.50	34.50	67.27	62.73
University of the Ryukyus, Nishihara, Japan	41	98	12	86	37.00	5.67	31.33	90.24	76.42
Hunan University, Changsha, China	38	71	55	16	37.00	27.33	9.67	97.37	25.44
Duy Tan University, Da Nang, Viet Nam	34	117	1	116	34.00	0.25	33.75	100.00	99.26
Northeast Electric Power University, Jilin, China	42	61	35	26	33.00	21.30	11.70	78.57	27.86
Institute for Systems and Computer Engineering, Technology and Science, Porto, Portugal	35	103	40	63	33.00	14.03	18.97	94.29	54.19
Shanghai University of Electric Power, Shanghai, China	47	41	33	8	32.00	27.50	4.50	68.09	9.57
Nanyang Technological University, Singapore City, Singapore	39	67	13	54	32.00	7.45	24.55	82.05	62.95
Politecnico di Milano, Milan, Italy	43	68	8	60	32.00	5.83	26.17	74.42	60.85
King Abdulaziz University, Jeddah, Saudi Arabia	32	83	23	60	32.00	9.23	22.77	100.00	71.15
University of Engineering and Technology Mardan, Mardan, Pakistan	32	120	67	53	32.00	19.51	12.49	100.00	39.05
Università degli Studi di Salerno, Salerno, Italy	36	86	4	82	32.00	3.25	28.75	88.89	79.86
Aalto University, Espoo, Finland	41	74	2	72	32.00	1.50	30.50	78.05	74.39
South China University of Technology, Guangzhou, China	52	54	42	12	31.00	25.50	5.50	59.62	10.58
Sharif University of Technology, Tehran, Iran	37	78	38	40	31.00	16.12	14.88	83.78	40.22
Wuhan University, Wuhan, China	44	66	46	20	30.00	21.21	8.79	68.18	19.98
Faculty of Engineering Ain Shams University, Cairo, Egypt	30	93	34	59	30.00	13.33	16.67	100.00	55.58
Graphic Era Deemed to be University, Dehradun, India	30	171	68	103	30.00	12.17	17.83	100.00	59.42
National University of Sciences and Technology, Islamabad, Pakistan	31	93	34	59	30.00	12.02	17.98	96.77	58.01
Tickle College of Engineering, Knoxville, United States	31	43	24	19	29.00	17.83	11.17	93.55	36.02

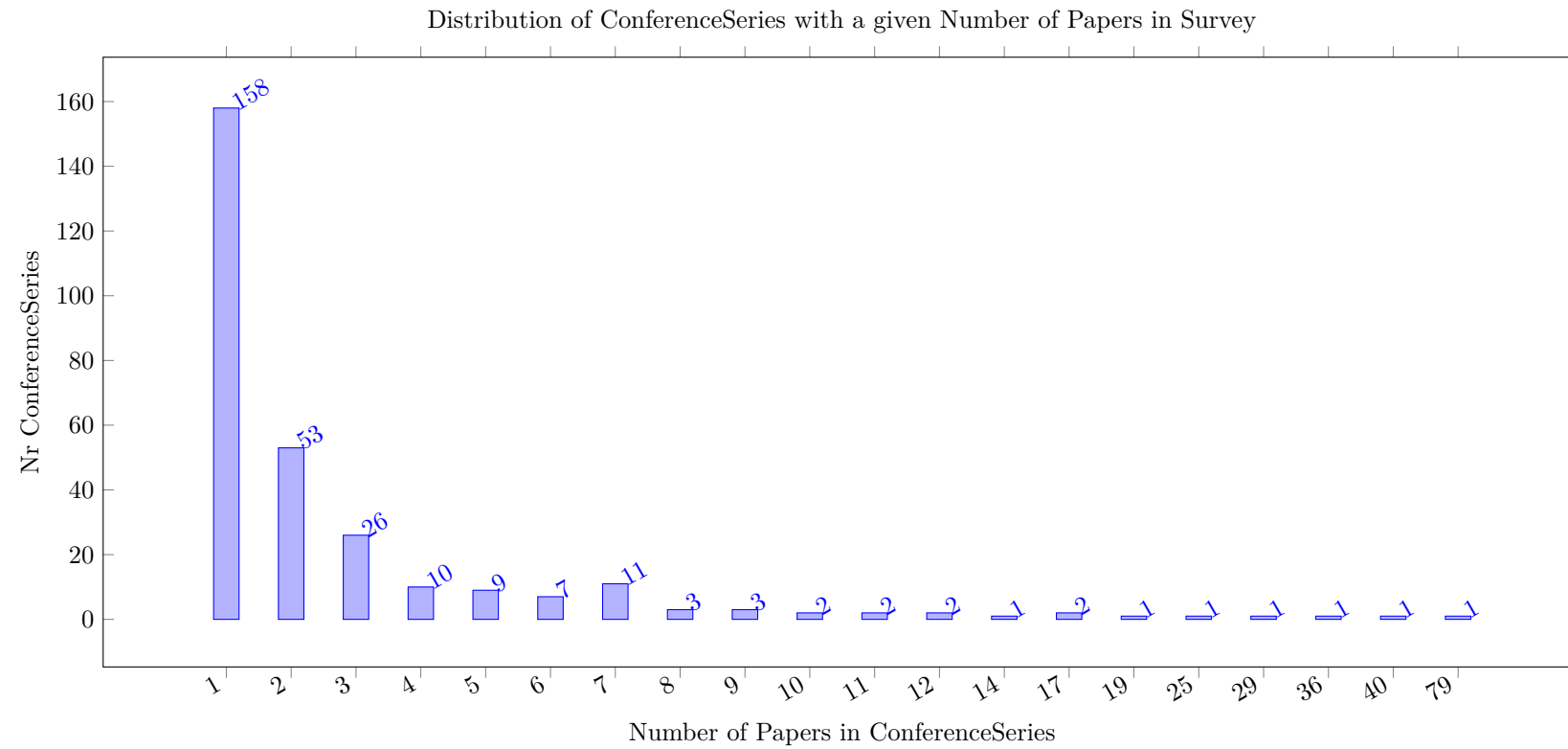
Table 9: Heat Map based on Collaboration between Countries (Fractional Count)

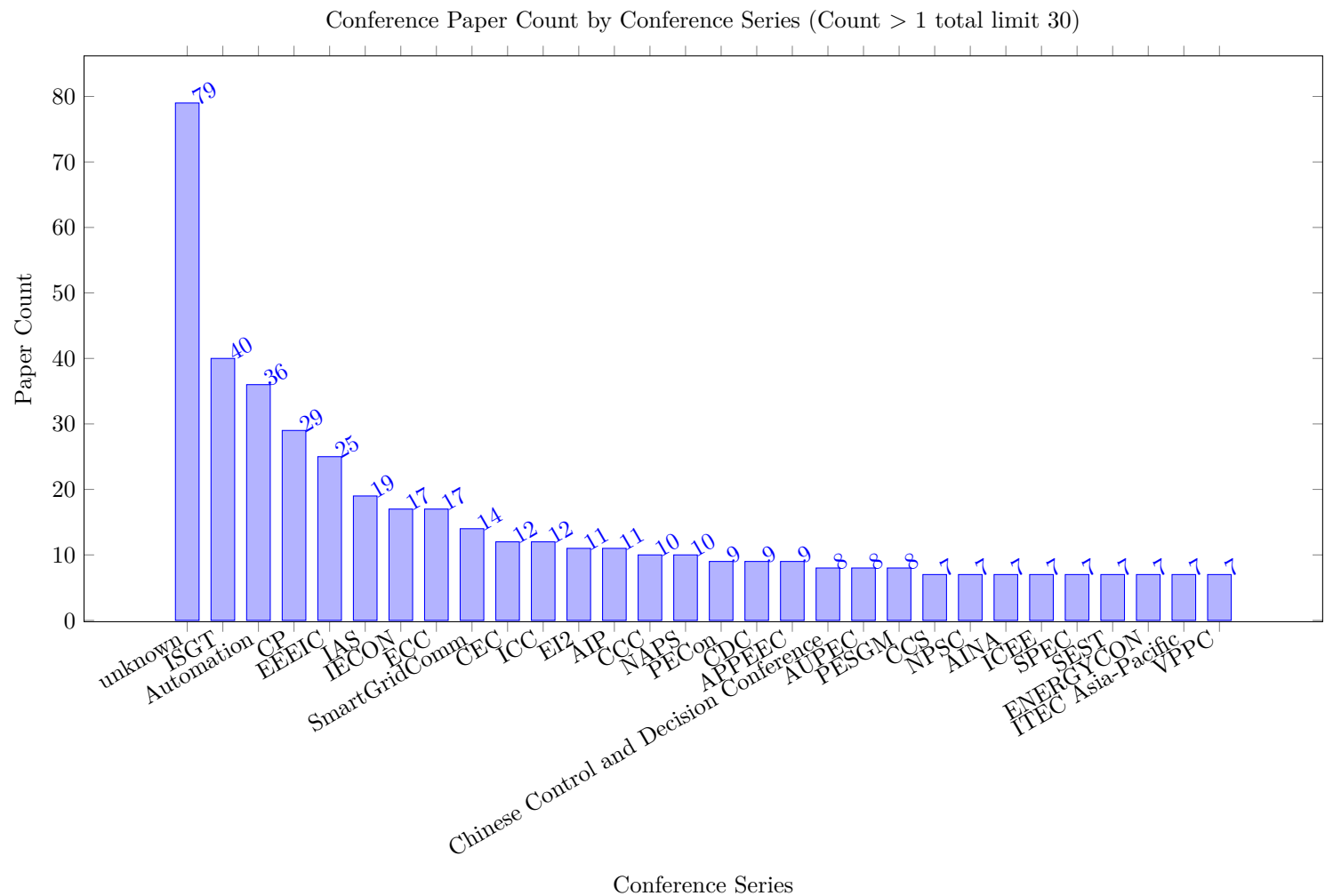
From/To Total	Total																								
		China	India	United States	Iran	Pakistan	Saudi Arabia	Egypt	United Kingdom	Australia	South Korea	France	Canada	Spain	Malaysia	Portugal	Italy	Iraq	Japan	Germany	Denmark	Brazil	Singapore	United Arab Emirates	
		3578.75	1145.00	852.00	821.00	570.00	487.00	467.00	361.00	310.00	296.00	278.00	265.00	258.00	248.00	243.00	242.00	194.00	171.00	160.00	157.00	129.00	126.00	114.00	108.00
China	3578.75	2783.88	10.09	152.02	44.86	41.07	16.14	18.74	80.07	55.03	8.60	18.48	33.17	3.45	16.87	6.57	7.08	12.51	18.05	7.89	27.70	3.50	51.08	2.93	5.31
India	1145.00	10.09	820.64	21.68	4.17	3.21	26.36	11.97	15.52	9.29	16.08	2.32	5.14	8.27	12.04	2.73	0.98	15.50	23.43	0.53	14.15	0.90	4.50	3.57	0.90
United States	852.00	152.02	21.68	406.37	45.11	9.60	17.80	6.50	7.80	7.74	19.08	9.58	18.35	4.42	1.37	6.13	8.45	1.75	13.03	12.67	7.16	1.83	4.07	3.74	5.31
Iran	821.00	44.86	4.17	45.11	423.79	0.00	6.76	1.95	18.81	23.93	4.44	4.42	23.48	11.77	6.42	17.08	17.05	9.19	5.20	1.53	40.08	4.25	0.00	6.27	6.31
Pakistan	570.00	41.07	3.21	9.60	0.00	259.90	87.51	6.08	21.28	5.10	28.29	0.25	20.91	2.20	17.67	0.75	1.57	3.46	0.00	0.77	3.02	0.00	0.60	22.12	0.63
Saudi Arabia	487.00	16.14	26.36	17.80	6.76	87.51	103.86	80.77	13.21	8.42	5.48	0.81	5.18	5.53	13.82	0.53	1.50	8.93	5.13	0.40	2.83	0.50	0.00	11.30	2.38
Egypt	467.00	18.74	11.97	6.50	1.95	6.08	80.77	206.39	4.61	2.09	3.14	5.68	5.82	16.93	1.84	0.60	3.40	4.79	15.07	0.56	0.25	0.25	0.00	8.35	1.48
United Kingdom	361.00	80.07	15.52	7.80	18.81	21.28	13.21	4.61	89.00	4.79	1.78	3.25	4.00	8.37	3.39	1.45	1.60	3.14	2.48	3.67	4.03	0.75	5.83	2.38	8.42
Australia	310.00	55.03	9.29	7.74	23.93	5.10	8.42	2.09	4.79	92.36	2.53	0.64	2.46	1.12	16.46	1.53	3.67	0.20	4.07	5.03	3.08	0.00	6.08	1.26	0.63
South Korea	296.00	8.60	16.08	19.08	4.44	28.29	5.48	3.14	1.78	2.53	160.73	0.00	3.19	1.00	4.11	0.00	0.50	0.62	0.50	0.20	1.33	0.00	1.67	0.95	1.37
France	278.00	18.48	2.32	9.58	4.42	0.25	0.81	5.68	3.25	0.64	0.00	149.32	3.50	4.42	0.00	2.33	7.75	0.67	0.25	2.62	0.83	1.00	0.00	1.00	0.63
Canada	265.00	33.17	5.14	18.35	23.48	20.91	5.18	5.82	4.00	2.46	3.19	3.50	74.03	0.33	3.55	1.60	8.70	0.00	0.14	3.17	2.02	1.00	0.50	13.05	0.63
Spain	258.00	3.45	8.27	4.42	11.77	2.20	5.53	16.93	8.37	1.12	1.00	4.42	0.33	85.65	1.07	13.00	9.12	0.75	2.92	2.14	8.42	5.67	0.50	0.33	0.63
Malaysia	248.00	16.87	12.04	1.37	6.42	17.67	13.82	1.84	3.39	16.46	4.11	0.00	3.55	1.07	63.85	0.00	0.00	14.36	4.49	0.17	4.25	0.00	0.45	4.93	2.09
Portugal	243.00	6.57	2.73	6.13	17.08	0.75	0.53	0.60	1.45	1.53	0.00	2.33	1.60	13.00	0.00	145.73	3.87	0.00	1.75	3.92	2.90	8.62	0.33	0.00	2.09
Italy	242.00	7.08	0.98	8.45	17.05	1.57	1.50	3.40	1.60	3.67	0.50	7.75	8.70	9.12	0.00	3.87	106.40	2.57	0.00	1.00	3.48	2.60	0.50	0.33	1.37
Iraq	194.00	12.51	15.50	1.75	9.19	3.46	8.93	4.79	3.14	0.20	0.62	0.67	0.00	0.75	14.36	0.00	2.57	67.37	1.47	1.00	1.57	0.00	0.00	2.14	2.38
Japan	171.00	18.05	23.43	13.03	5.20	0.00	5.13	15.07	2.48	4.07	0.50	0.25	0.14	2.92	4.49	1.75	0.00	1.47	51.00	0.00	0.25	0.00	0.00	0.33	0.90
Germany	160.00	7.89	0.53	12.67	1.53	0.77	0.40	0.56	3.67	5.03	0.20	2.62	3.17	2.14	0.17	3.92	1.00	1.00	0.00	81.20	0.00	2.33	0.50	0.00	0.31
Denmark	157.00	27.70	14.15	7.16	40.08	3.02	2.83	0.25	4.03	3.08	1.33	0.83	2.02	8.42	4.25	2.90	3.48	1.57	0.25	0.00	2.67	1.70	0.67	0.33	1.37
Brazil	129.00	3.50	0.90	1.83	4.25	0.00	0.50	0.25	0.75	0.00	0.00	1.00	1.00	5.67	0.00	8.62	2.60	0.00	0.00	2.33	1.70	75.07	0.00	0.00	0.63
Singapore	126.00	51.08	4.50	4.07	0.00	0.60	0.00	0.00	5.83	6.08	1.67	0.00	0.50	0.50	0.45	0.33	0.50	0.00	0.00	0.50	0.67	0.00	40.80	1.08	0.31
United Arab Emirates	114.00	2.93	3.57	3.74	6.27	22.12	11.30	8.35	2.38	1.26	0.95	1.00	13.05	0.33	4.93	0.00	0.33	2.14	0.33	0.00	0.33	0.00	1.08	7.41	0.31
Turkey	108.00	5.31	0.50	5.25	6.58	0.13	2.50	1.48	8.03	0.67	1.20	0.54	0.00	0.00	2.03	2.67	1.50	2.83	0.92	0.56	1.53	0.00	0.25	0.20	35.83
Colombia	106.00	1.00	2.42	1.75	3.00	0.00	0.00	0.00	0.40	1.00	0.00	1.00	1.00	12.09	0.00	0.67	2.50	0.17	0.00	0.00	2.00	3.92	0.00	1.00	0.63
Taiwan	96.00	15.65	3.67	1.50	1.17	0.33	0.89	0.13	3.52	0.00	0.33	0.00	0.14	0.13	5.97	0.00	0.00	0.67	0.50	0.00	0.95	0.00	3.17	0.25	0.50
Viet Nam	93.00	9.71	1.62	2.33	15.38	0.00	3.73	1.40	1.64	1.33	10.50	2.50	0.50	0.70	0.79	0.00	2.33	1.53	7.75	0.25	2.92	0.00	0.00	0.25	0.31
Thailand	91.00	4.37	1.56	1.70	2.77	2.17	0.17	0.00	1.27	4.96	0.33	1.25	1.14	0.50	3.11	0.00	0.33	1.11	1.62	0.00	1.00	0.00	0.00	0.00	0.63
Morocco	90.00	0.20	0.33	3.63	0.00	0.40	5.34	5.12	1.17	0.53	1.00	10.58	1.50	0.70	0.20	0.00	0.50	1.08	0.00	1.00	0.13	0.00	0.00	0.00	0.63
Bangladesh	87.00	3.50	2.64	2.70	0.00	0.00	5.88	0.00	0.48	18.02	1.83	0.00	4.15	0.60	6.55	0.00	1.67	0.00	2.22	0.50	0.33	0.00	0.00	0.00	0.63
South Africa	85.00	1.73	6.20	2.67	2.92	2.29	1.46	0.95	2.27	2.13	3.69	1.00	0.36	1.12	0.18	0.00	6.16	2.80	0.50	1.50	1.23	1.00	0.00	2.43	0.90
Jordan	80.00	0.34	13.54	1.50	1.17	2.05	5.35	8.68	1.12	0.75	1.05	0.29	0.00	0.80	4.53	0.00	0.00	3.75	0.00	0.14	0.00	0.00	0.00	4.12	0.90
Algeria	78.00	0.54	1.59	0.20	0.65	0.40	4.60	10.39	1.92	0.00	0.00	5.11	1.00	5.06	3.00	1.42	2.00	0.75	0.00	1.43	0.00	0.00	0.00	0.58	1.37
Hong Kong	76.00	36.33	3.15	4.03	0.67	1.25	1.20	1.61	6.65	5.79	0.00	0.00	0.50	0.83	0.00	0.00	0.33	0.00	0.50	0.33	1.13	0.00	0.83	0.00	0.63
Finland	76.00	5.33	1.00	2.92	21.68	1.58	0.83	4.08	2.00	0.25	0.00	0.33	1.33	3.00	0.20	6.33	0.83	0.00	0.00	2.67	0.67	1.25	1.00	0.25	0.63
Greece	73.00	0.67	0.00	4.00	0.33	0.00	0.00	0.00	1.75	0.00	0.00	1.33	0.00	1.23	0.00	2.67	5.83	0.00	0.00	0.14	0.67	0.00	0.00	0.00	0.31
Norway	70.00	3.48	8.40	3.23	3.28	2.63	1.11	0.35	1.03	1.00	0.55	0.75	3.14	2.00	0.72	0.60	0.75	0.14	0.33	0.75	0.42	1.00	0.00	0.00	0.40
Poland	62.00	3.67	3.17	2.33	1.25	0.87	0.20	0.25	1.00	1.83	0.50	0.00	0.00	1.08	2.12	1.00	1.67	1.38	0.00	0.33	1.33	0.00	0.50	0.13	1.48
Netherlands	62.00	1.83	0.00	0.45	1.17	0.00	0.00	0.00	3.37	0.00	0.00	0.25	1.00	0.33	0.00	1.10	2.62	0.00	0.00	3.00	0.50	1.00	0.00	0.00	2.43
Romania	60.00	0.00	1.50	0.00	0.00	0.00	0.00	0.25	0.00	0.00	0.00	4.35	0.00	0.00	0.00	0.00	1.25	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.63
Other		65.32	60.63	25.57	38.43	20.53	30.95	26.53	22.19	14.85	10.78	29.60	15.92	30.47	27.52	5.12	19.62	22.82	7.11	16.10	9.50	10.87	1.08	11.00	20.93

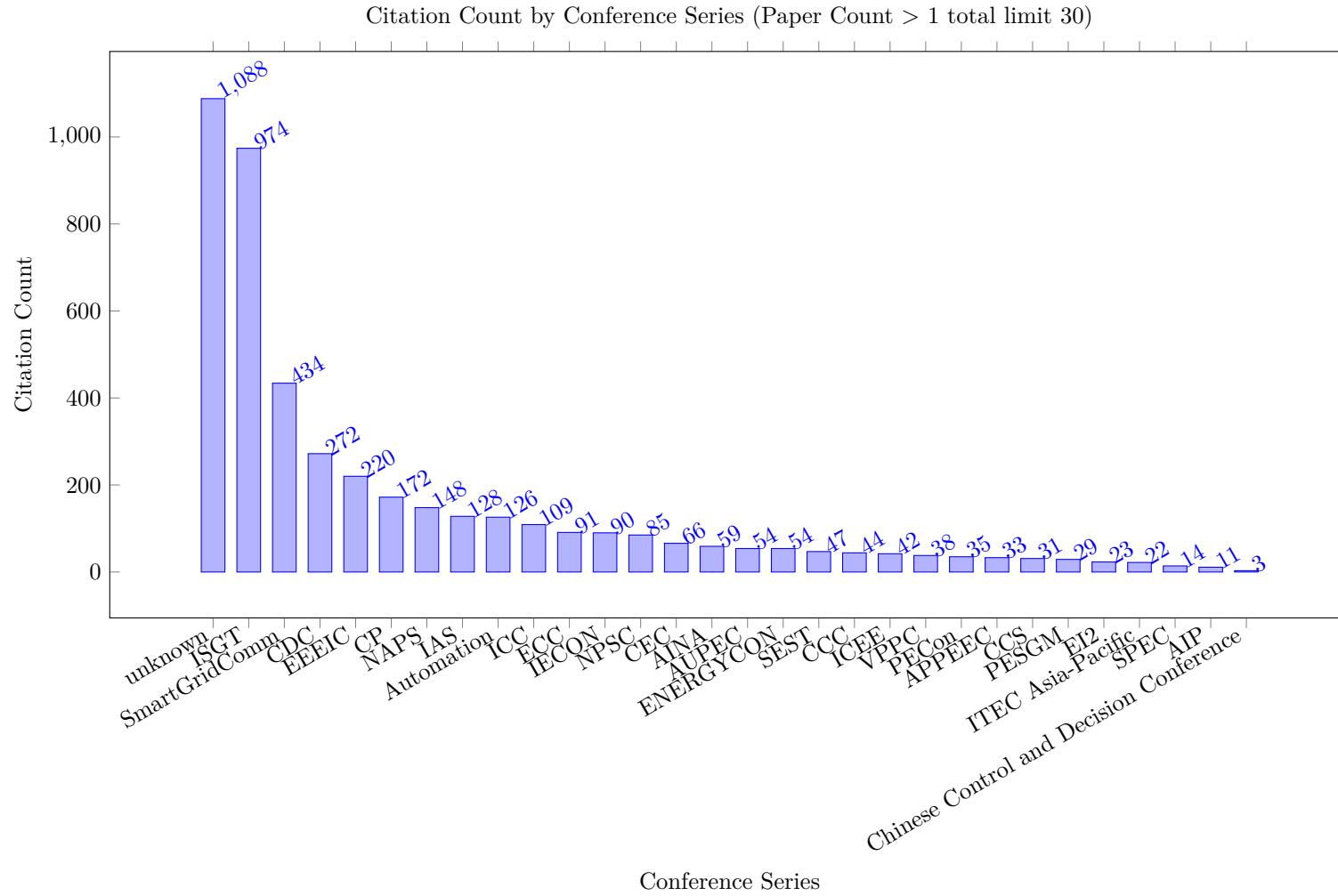
Table 10: Heat Map based on Collaboration between Countries (Integer Count)

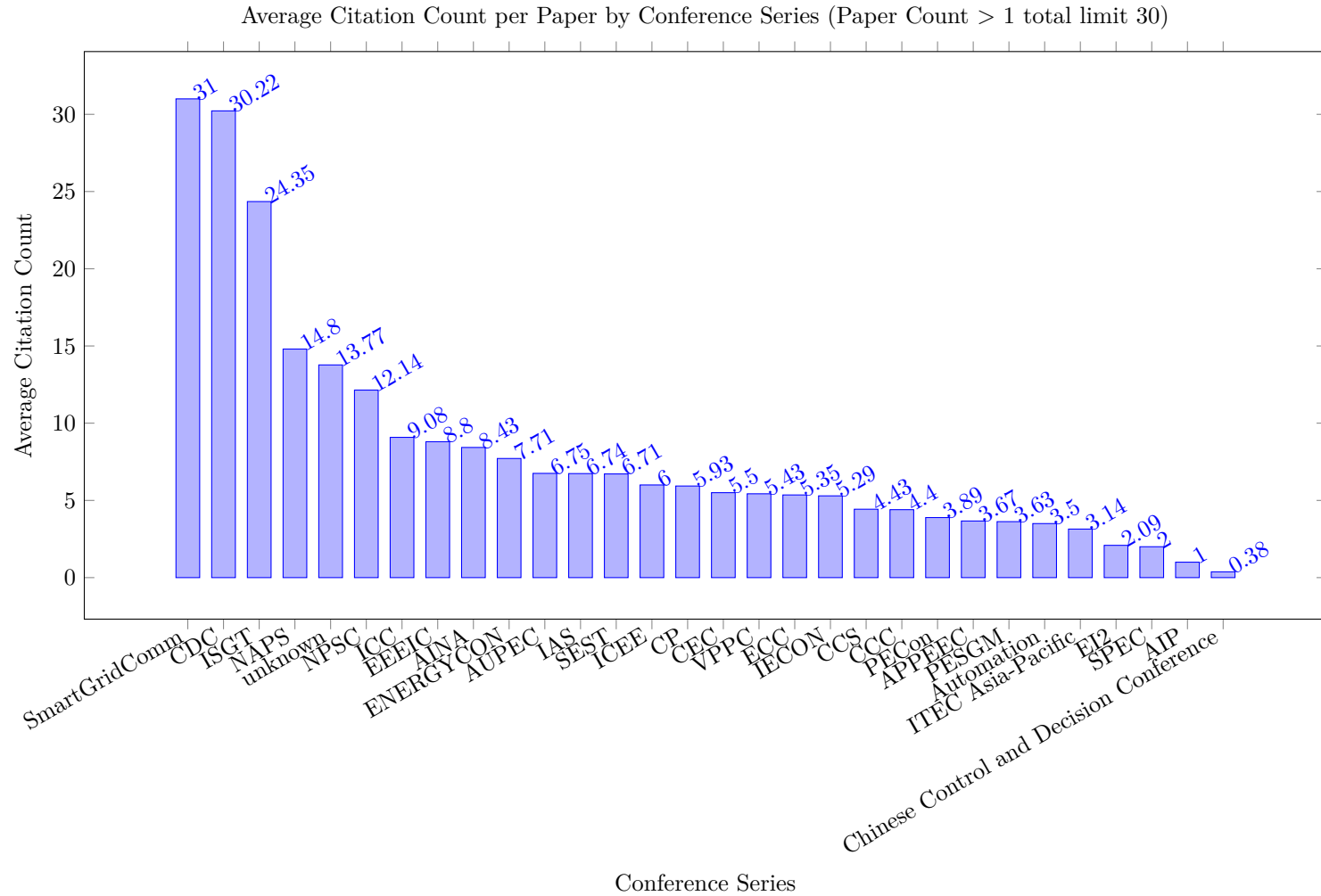
From/To Total	Total	China	Saudi Arabia	United States	Iran	India	Egypt	Pakistan	United Kingdom	Malaysia	Australia	Spain	Canada	Denmark	Iraq	South Korea	Italy	United Arab Emirates	Japan	France	Jordan	South Africa	Portugal	Viet Nam	Turkey
China	1064.00	0.00	37.00	167.00	64.00	14.00	34.00	60.00	89.00	30.00	63.00	6.00	37.00	36.00	27.00	15.00	12.00	9.00	25.00	21.00	2.00	4.00	8.00	19.00	8.00
Saudi Arabia	726.00	37.00	0.00	31.00	18.00	43.00	117.00	110.00	35.00	28.00	20.00	16.00	13.00	8.00	19.00	15.00	6.00	26.00	12.00	4.00	13.00	5.00	2.00	11.00	5.00
United States	635.00	167.00	31.00	0.00	59.00	28.00	15.00	17.00	17.00	3.00	20.00	10.00	24.00	12.00	3.00	23.00	13.00	7.00	18.00	12.00	4.00	5.00	9.00	6.00	9.00
Iran	617.00	64.00	18.00	59.00	0.00	10.00	6.00	0.00	31.00	12.00	33.00	19.00	29.00	52.00	21.00	9.00	26.00	8.00	9.00	5.00	3.00	7.00	21.00	27.00	10.00
India	530.00	14.00	43.00	28.00	10.00	0.00	23.00	5.00	24.00	23.00	16.00	12.00	8.00	20.00	20.00	20.00	3.00	7.00	29.00	4.00	23.00	13.00	4.00	4.00	1.00
Egypt	490.00	34.00	117.00	15.00	6.00	23.00	0.00	15.00	11.00	7.00	7.00	27.00	8.00	1.00	10.00	6.00	7.00	17.00	24.00	12.00	16.00	4.00	1.00	5.00	5.00
Pakistan	483.00	60.00	110.00	17.00	0.00	5.00	15.00	0.00	31.00	28.00	9.00	5.00	29.00	4.00	6.00	42.00	4.00	34.00	0.00	1.00	5.00	7.00	1.00	0.00	1.00
United Kingdom	455.00	89.00	35.00	17.00	31.00	24.00	11.00	31.00	0.00	9.00	10.00	13.00	6.00	9.00	4.00	4.00	4.00	7.00	4.00	5.00	4.00	6.00	5.00	11.00	11.00
Malaysia	370.00	30.00	28.00	3.00	12.00	23.00	7.00	28.00	9.00	0.00	24.00	3.00	8.00	9.00	25.00	10.00	0.00	14.00	13.00	0.00	11.00	2.00	0.00	3.00	7.00
Australia	368.00	63.00	20.00	20.00	33.00	16.00	7.00	9.00	10.00	24.00	0.00	3.00	6.00	9.00	1.00	5.00	8.00	3.00	9.00	3.00	3.00	5.00	3.00	4.00	1.00
Spain	305.00	6.00	16.00	10.00	19.00	12.00	27.00	5.00	13.00	3.00	3.00	0.00	1.00	12.00	2.00	1.00	15.00	2.00	6.00	7.00	4.00	5.00	2.00	0.00	0.00
Canada	299.00	37.00	13.00	24.00	29.00	8.00	8.00	29.00	6.00	8.00	6.00	1.00	0.00	6.00	0.00	9.00	13.00	23.00	1.00	4.00	0.00	3.00	2.00	1.00	0.00
Denmark	266.00	36.00	8.00	12.00	52.00	20.00	1.00	4.00	9.00	9.00	9.00	12.00	6.00	0.00	3.00	4.00	9.00	1.00	1.00	2.00	0.00	5.00	5.00	7.00	3.00
Iraq	249.00	27.00	19.00	3.00	21.00	20.00	10.00	6.00	4.00	25.00	1.00	2.00	0.00	3.00	0.00	3.00	5.00	6.00	6.00	1.00	8.00	5.00	0.00	6.00	4.00
South Korea	234.00	15.00	15.00	23.00	9.00	20.00	6.00	42.00	4.00	10.00	5.00	1.00	9.00	4.00	3.00	0.00	2.00	4.00	1.00	0.00	3.00	7.00	0.00	10.00	3.00
Italy	234.00	12.00	6.00	13.00	26.00	3.00	7.00	4.00	4.00	0.00	8.00	15.00	13.00	9.00	5.00	2.00	0.00	1.00	0.00	10.00	0.00	16.00	5.00	3.00	2.00
United Arab Emirates	228.00	9.00	26.00	7.00	8.00	7.00	17.00	34.00	7.00	14.00	3.00	2.00	23.00	1.00	6.00	4.00	1.00	0.00	1.00	1.00	8.00	5.00	0.00	1.00	1.00
Japan	203.00	25.00	12.00	18.00	9.00	29.00	24.00	0.00	4.00	13.00	9.00	6.00	1.00	1.00	6.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	3.00	10.00	3.00
France	187.00	21.00	4.00	12.00	5.00	4.00	12.00	1.00	5.00	0.00	3.00	7.00	4.00	2.00	1.00	0.00	10.00	1.00	1.00	0.00	2.00	1.00	5.00	3.00	2.00
Jordan	176.00	2.00	13.00	4.00	3.00	23.00	16.00	5.00	4.00	11.00	3.00	4.00	0.00	0.00	8.00	3.00	0.00	8.00	0.00	2.00	0.00	0.00	0.00	0.00	2.00
South Africa	158.00	4.00	5.00	5.00	7.00	13.00	4.00	7.00	4.00	2.00	5.00	5.00	3.00	5.00	5.00	7.00	16.00	5.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Portugal	149.00	8.00	2.00	9.00	21.00	4.00	1.00	1.00	6.00	0.00	3.00	16.00	2.00	5.00	0.00	0.00	5.00	0.00	3.00	5.00	0.00	0.00	0.00	0.00	3.00
Viet Nam	143.00	19.00	11.00	6.00	27.00	4.00	5.00	0.00	5.00	3.00	4.00	2.00	1.00	7.00	6.00	10.00	3.00	1.00	10.00	3.00	0.00	0.00	0.00	0.00	1.00
Turkey	133.00	8.00	5.00	9.00	10.00	1.00	5.00	1.00	11.00	7.00	1.00	0.00	0.00	3.00	4.00	3.00	2.00	1.00	3.00	2.00	2.00	0.00	3.00	1.00	0.00
Germany	132.00	10.00	1.00	16.00	3.00	3.00	3.00	2.00	6.00	1.00	8.00	3.00	4.00	0.00	1.00	1.00	1.00	0.00	0.00	5.00	1.00	2.00	5.00	1.00	3.00
Finland	127.00	6.00	3.00	7.00	31.00	1.00	5.00	3.00	6.00	1.00	1.00	6.00	2.00	2.00	0.00	0.00	2.00	1.00	0.00	1.00	0.00	1.00	15.00	0.00	0.00
Singapore	119.00	57.00	0.00	7.00	0.00	6.00	0.00	1.00	10.00	2.00	10.00	1.00	1.00	2.00	0.00	2.00	1.00	3.00	0.00	0.00	0.00	0.00	1.00	0.00	1.00
Ethiopia	111.00	4.00	5.00	3.00	4.00	26.00	6.00	2.00	0.00	1.00	1.00	4.00	0.00	4.00	1.00	1.00	0.00	1.00	0.00	2.00	7.00	8.00	1.00	0.00	0.00
Bangladesh	110.00	5.00	8.00	5.00	0.00	7.00	0.00	0.00	2.00	10.00	25.00	1.00	10.00	1.00	0.00	3.00	3.00	0.00	5.00	0.00	0.00	1.00	0.00	0.00	0.00
Hong Kong	109.00	44.00	3.00	8.00	1.00	5.00	4.00	2.00	10.00	0.00	10.00	2.00	1.00	2.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Norway	108.00	7.00	6.00	8.00	7.00	13.00	2.00	6.00	4.00	3.00	1.00	2.00	4.00	2.00	1.00	3.00	1.00	0.00	1.00	2.00	1.00	5.00	1.00	2.00	2.00
Algeria	102.00	2.00	10.00	1.00	2.00	4.00	16.00	1.00	3.00	3.00	0.00	9.00	1.00	0.00	1.00	0.00	3.00	2.00	0.00	8.00	5.00	0.00	2.00	0.00	3.00
Thailand	96.00	11.00	1.00	3.00	6.00	4.00	0.00	3.00	3.00	8.00	7.00	1.00	2.00	1.00	2.00	1.00	1.00	0.00	4.00	4.00	0.00	0.00	1.00	0.00	0.00
Taiwan	92.00	17.00	4.00	2.00	3.00	7.00	1.00	1.00	9.00	9.00	0.00	1.00	1.00	3.00	2.00	1.00	0.00	1.00	1.00	0.00	0.00	2.00	0.00	0.00	1.00
Colombia	86.00	2.00	0.00	2.00	3.00	4.00	0.00	0.00	1.00	0.00	1.00	16.00	1.00	3.00	1.00	0.00	3.00	1.00	0.00	1.00	0.00	1.00	2.00	0.00	0.00
Sweden	83.00	11.00	2.00	6.00	1.00	0.00	3.00	1.00	4.00	0.00	3.00	4.00	1.00	1.00	0.00	1.00	6.00	1.00	0.00	1.00	2.00	0.00	1.00	0.00	0.00
Russian Federation	82.00	7.00	4.00	2.00	7.00	6.00	10.00	2.00	0.00	1.00	1.00	0.00	1.00	0.00	4.00	1.00	2.00	2.00	3.00	1.00	2.00	0.00	1.00	0.00	0.00
Qatar	81.00	6.00	4.00	6.00	4.00	6.00	0.00	1.00	1.00	5.00	10.00	3.00	5.00	0.00	2.00	2.00	0.00	3.00	1.00	0.00	0.00	1.00	0.00	1.00	2.00
Tunisia	78.00	1.00	18.00	0.00	2.00	4.00	3.00	5.00	1.00	1.00	0.00	1.00	5.00	0.00	2.00	4.00	0.00	0.00	6.00	3.00	3.00	0.00	0.00	0.00	0.00
Brazil	77.00	4.00	1.00	4.00	6.00	2.00	1.00	0.00	2.00	0.00	0.00	8.00	1.00	3.00	0.00	0.00	2.00	0.00	1.00	0.00	1.00	11.00	0.00	0.00	0.00
Other		83.00	62.00	43.00	58.00	78.00	58.00	39.00	50.00	56.00	25.00	56.00	28.00	24.00	47.00	18.00	44.00	27.00	10.00	49.00	44.00	29.00	15.00	8.00	39.00

5 Conference Papers by Most Common Conference Series



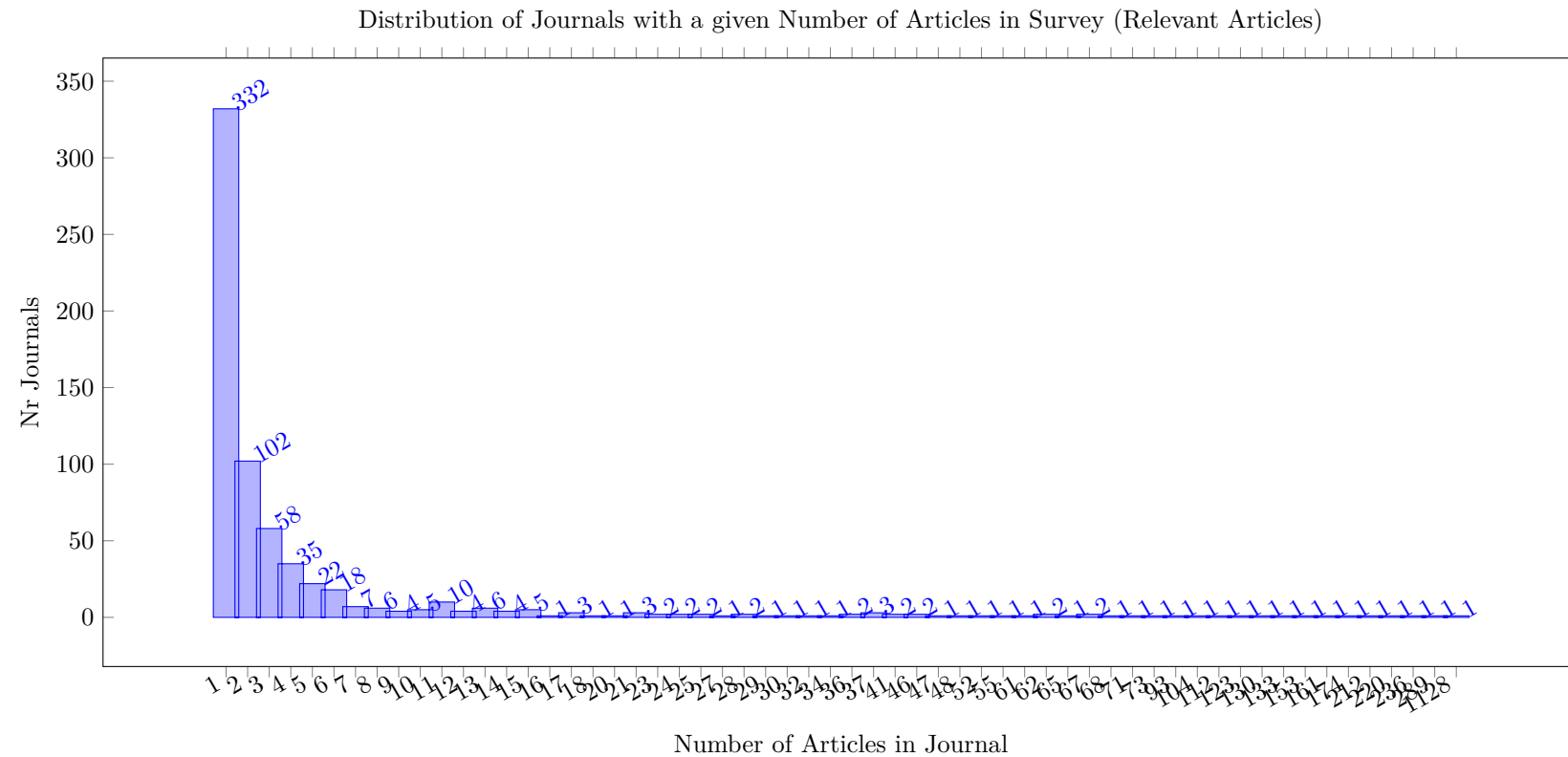


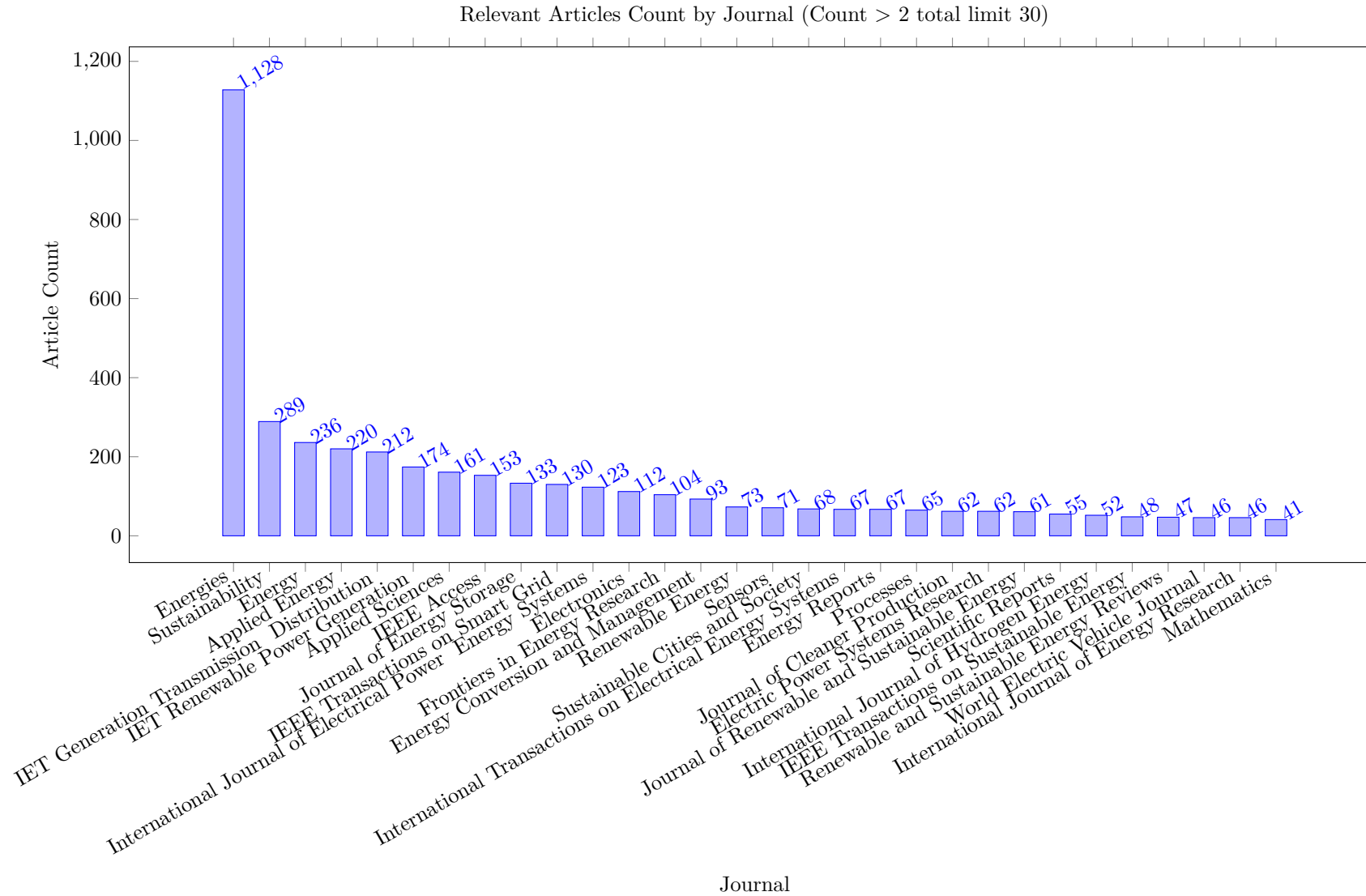


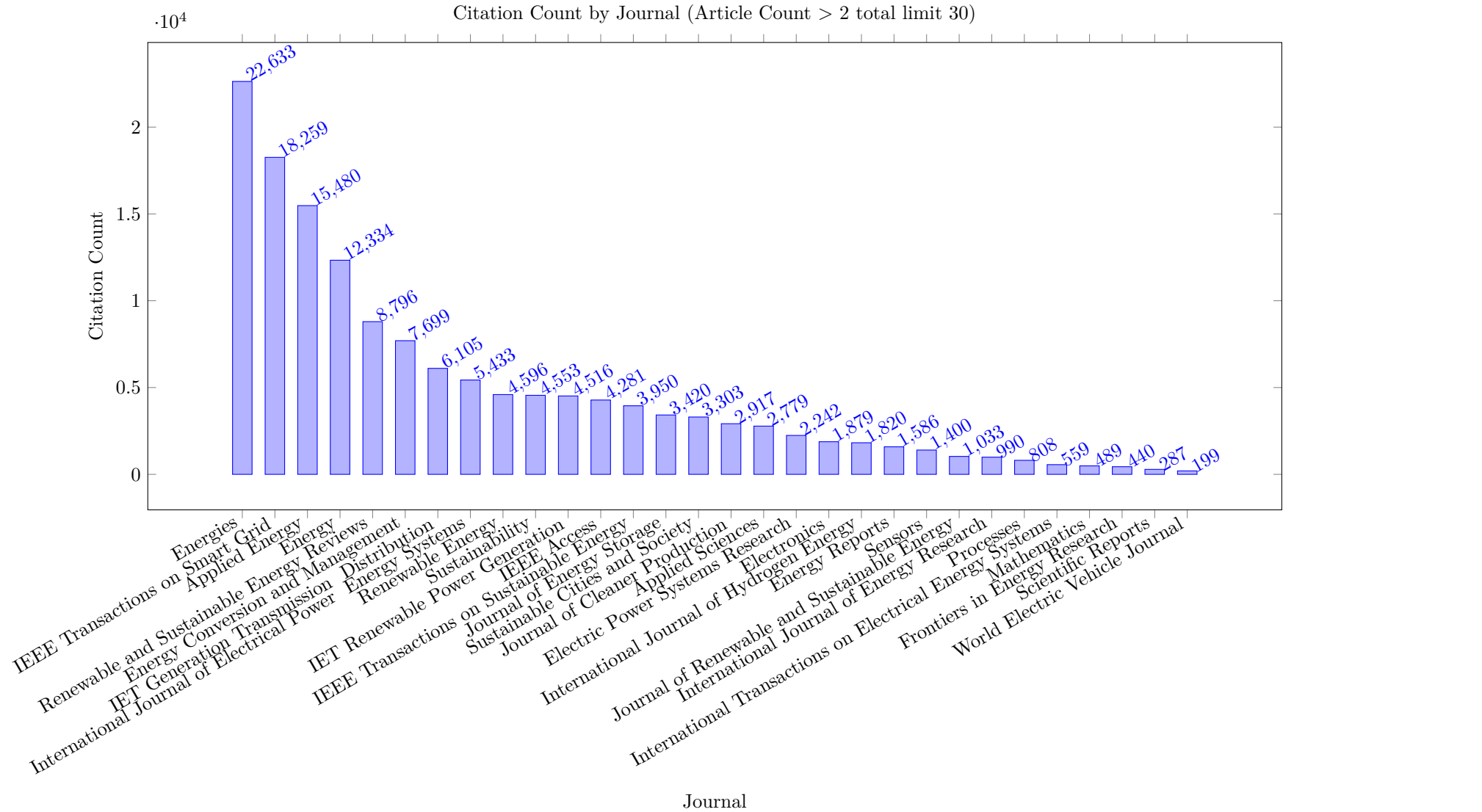


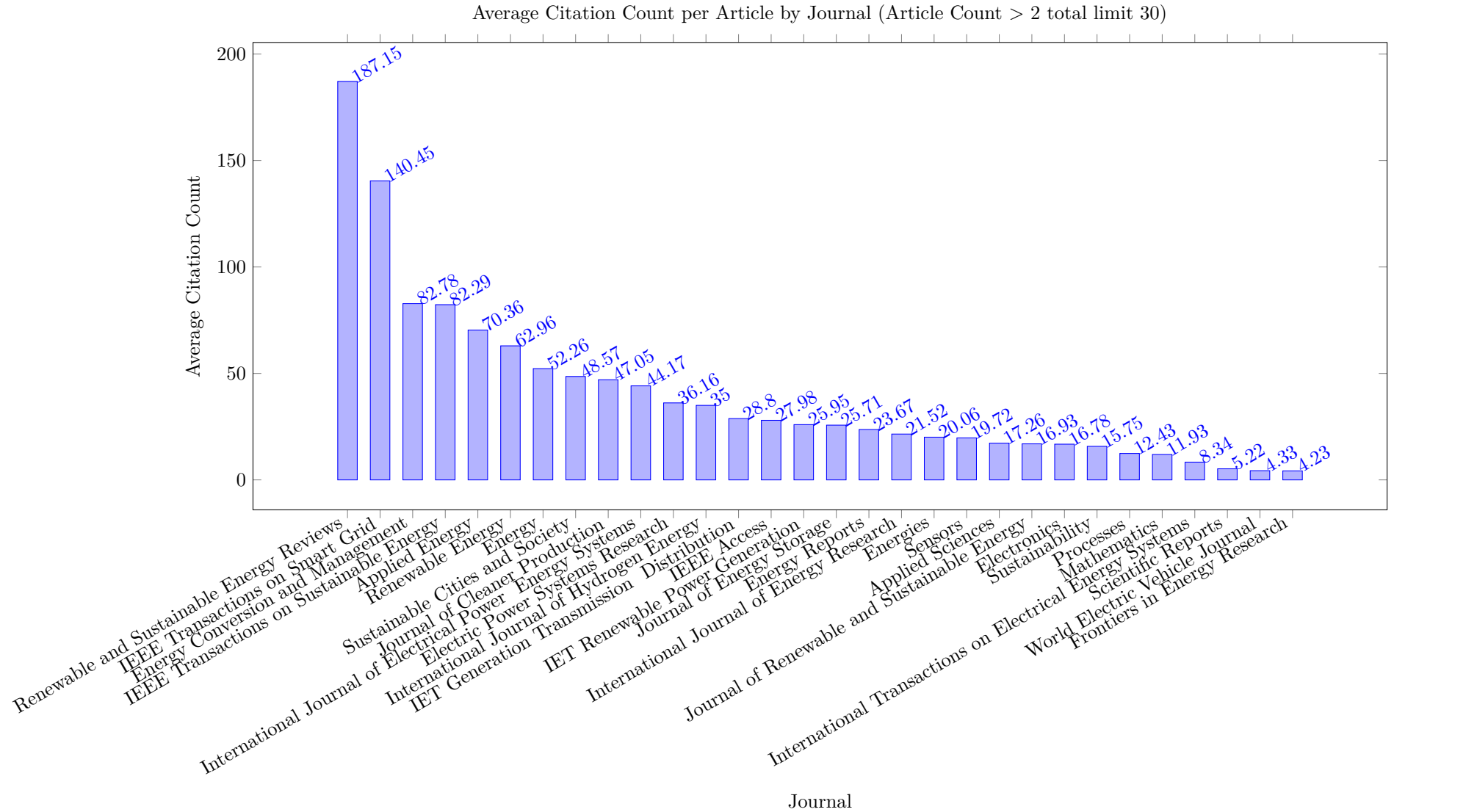
6 Journal Articles by Most Common Journals

6.1 Relevant Articles

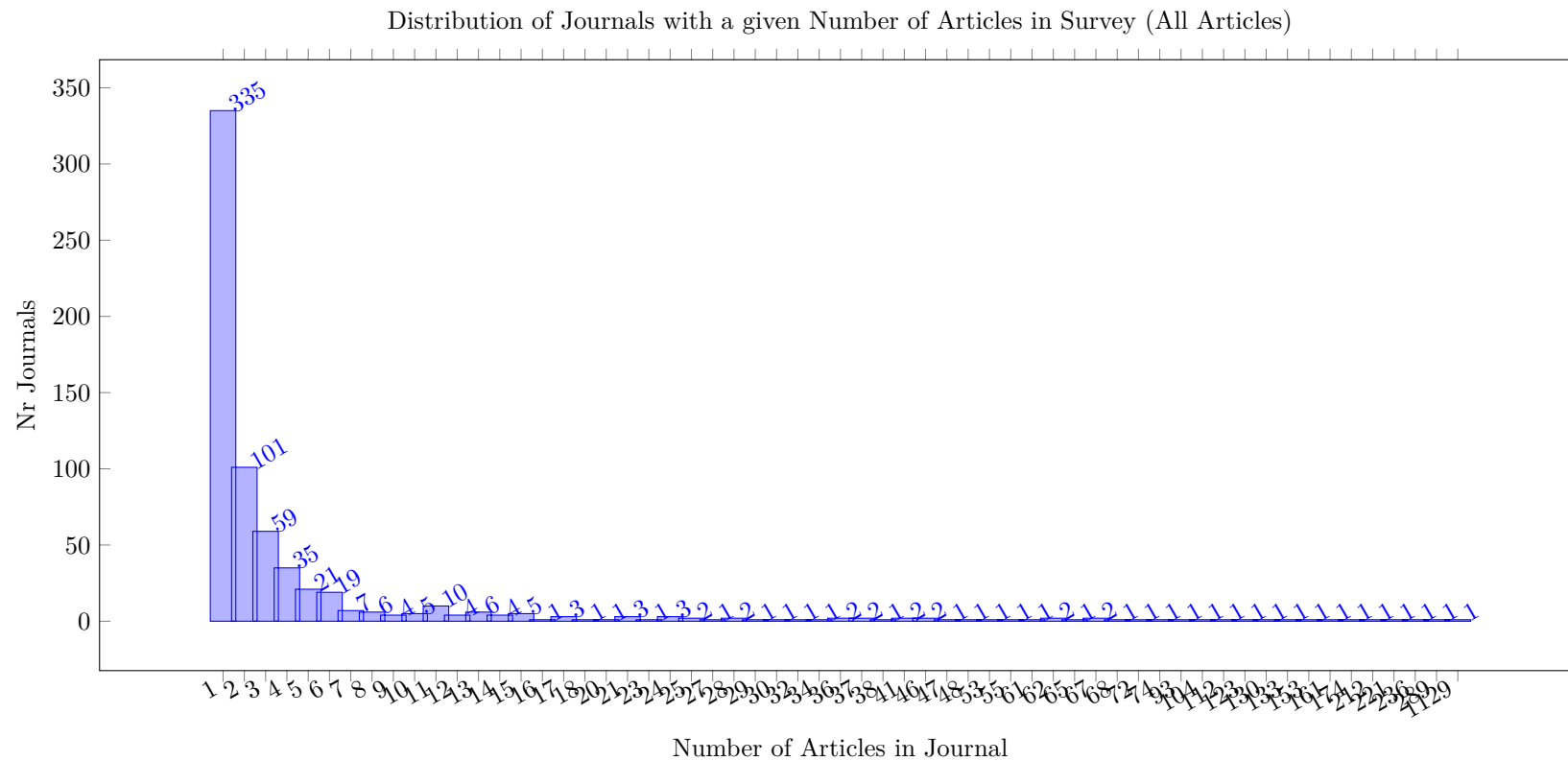


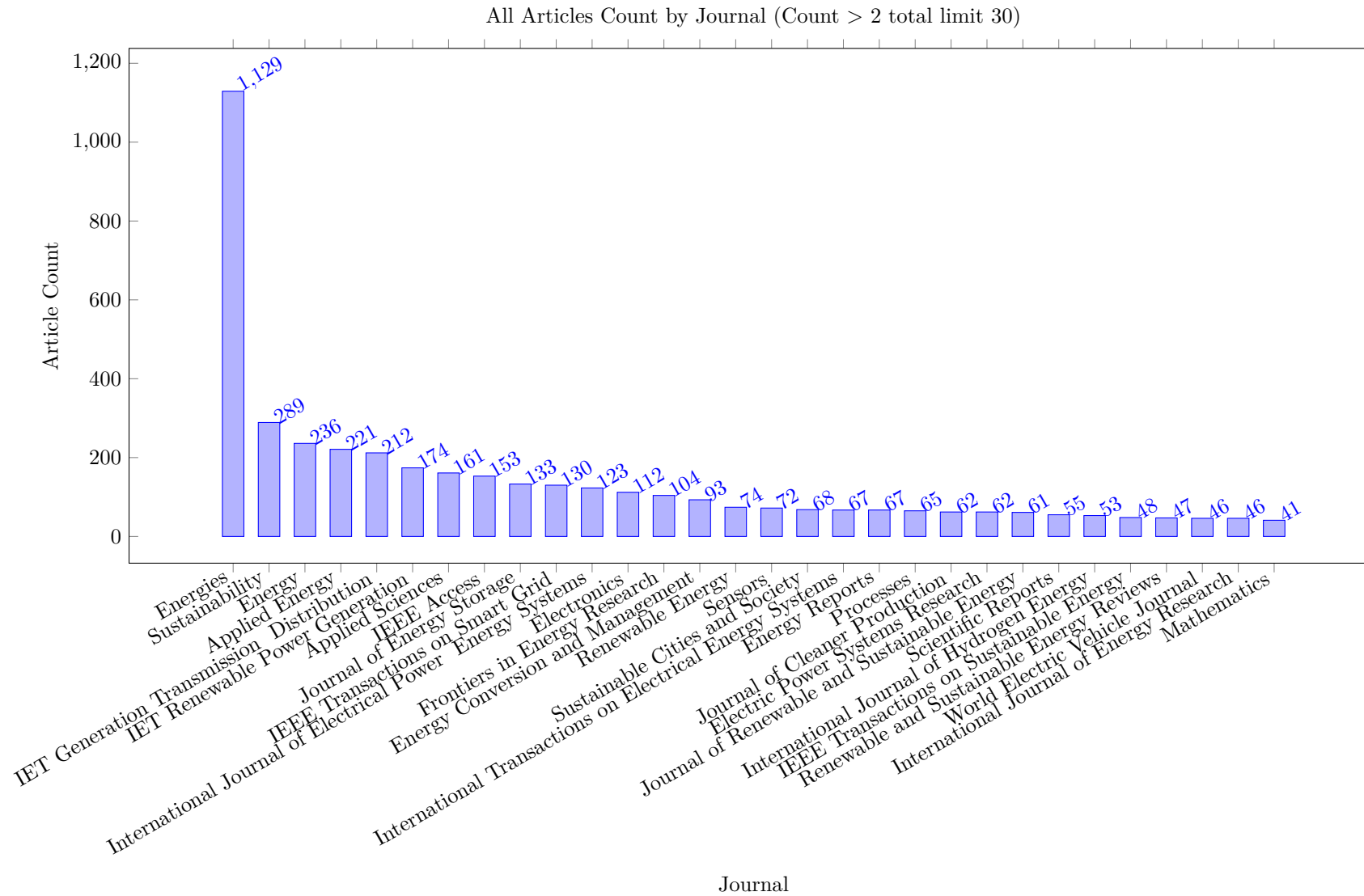


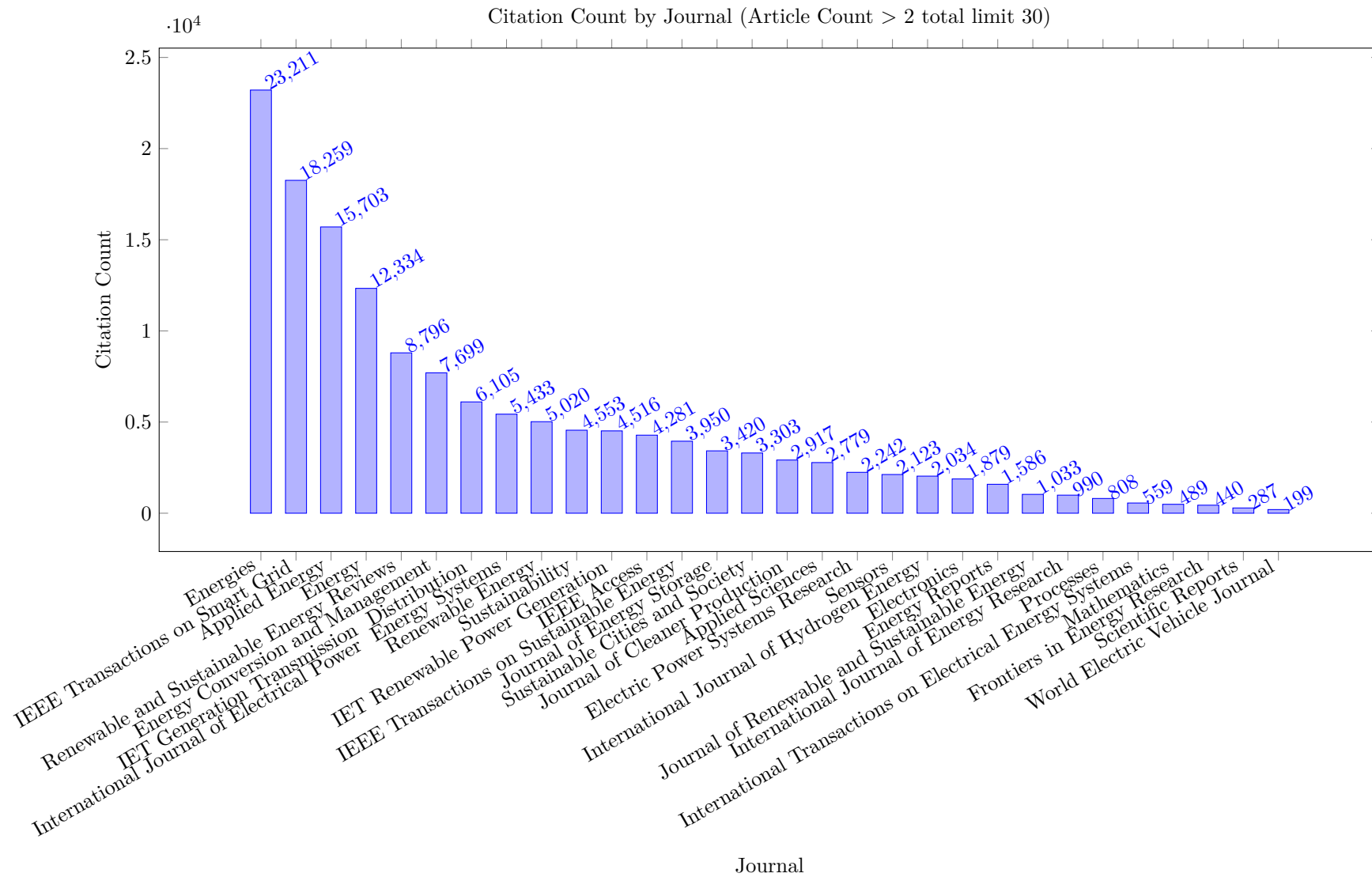


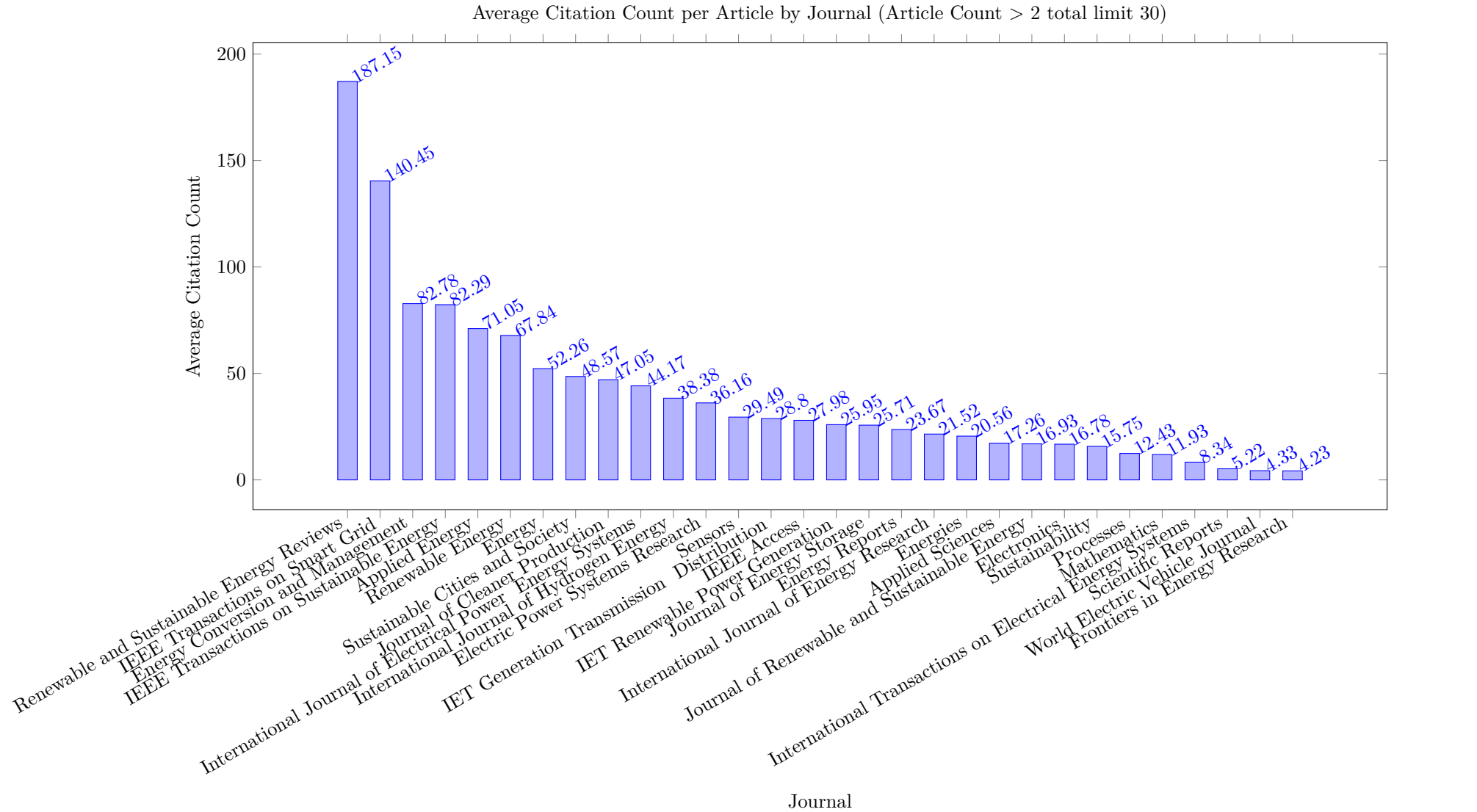


6.2 All Articles



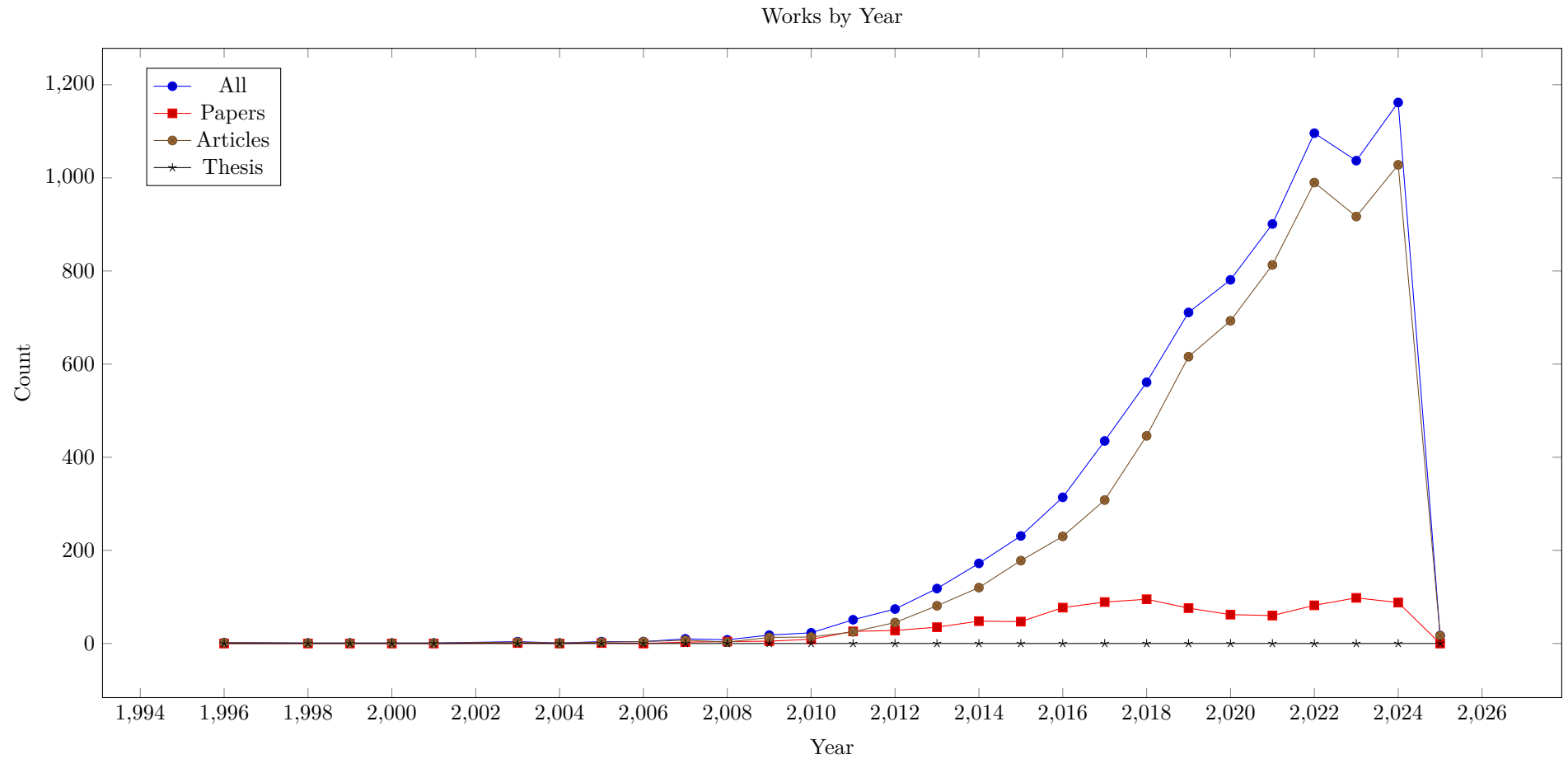


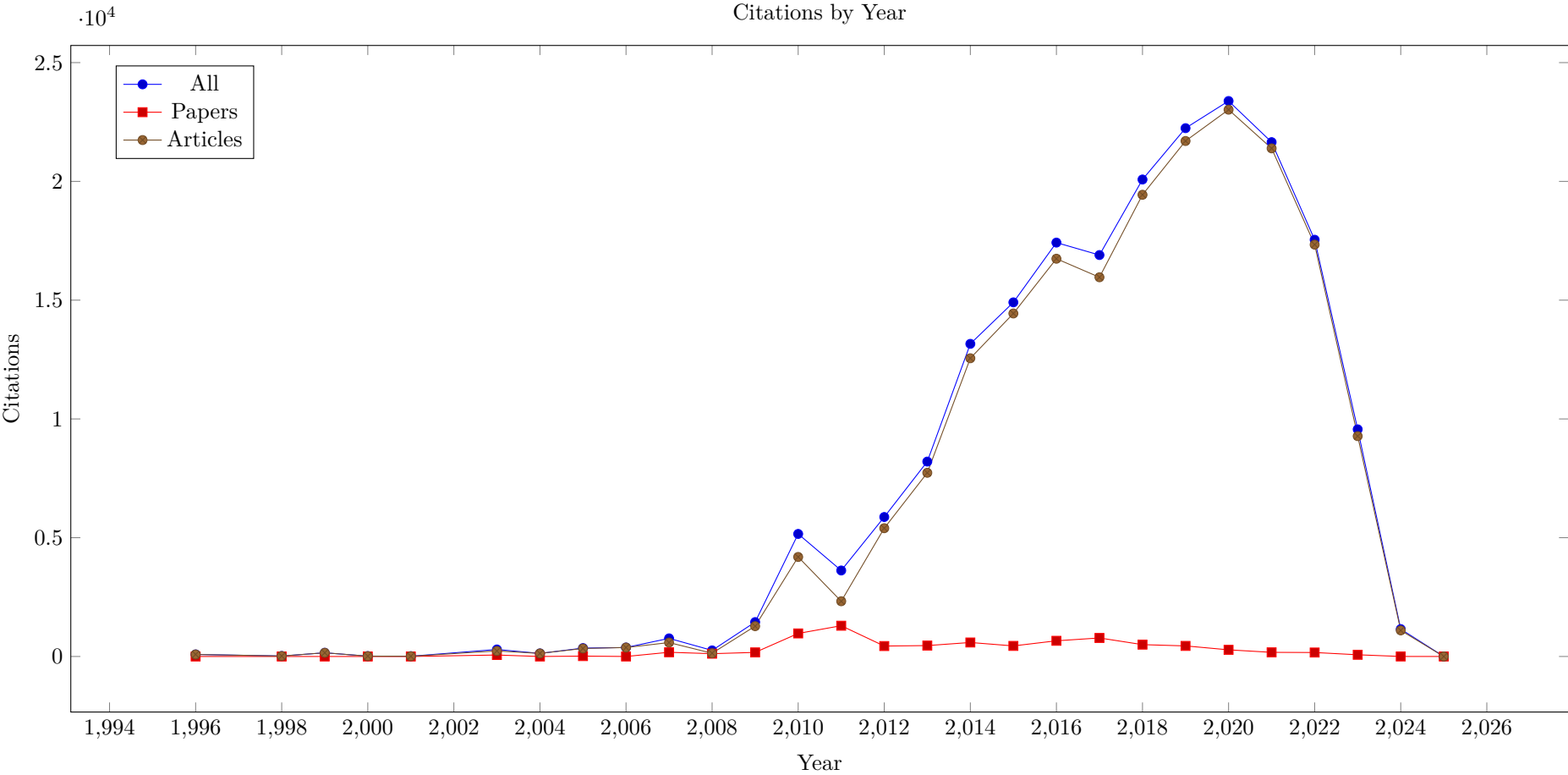


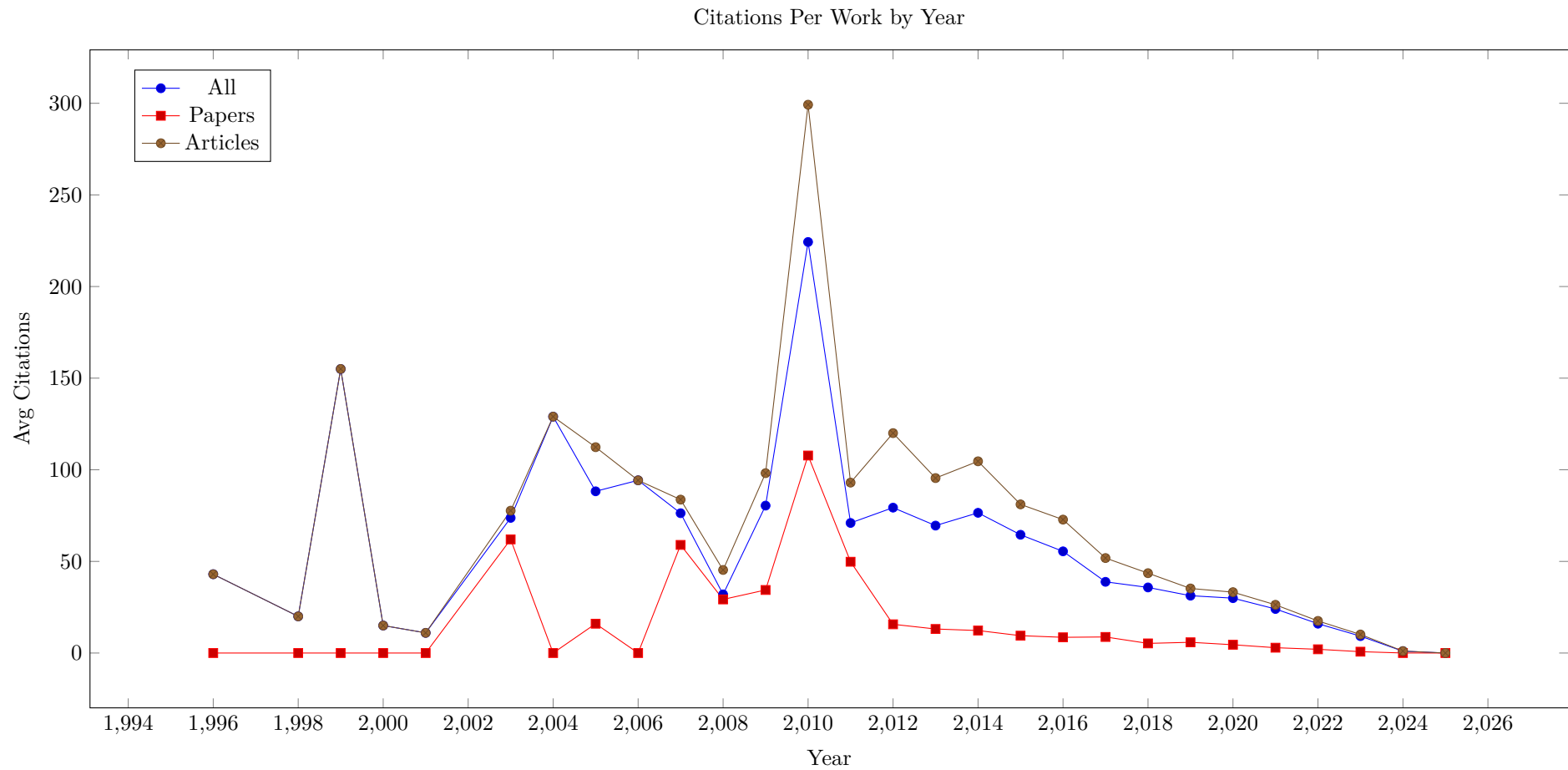


7 Works by Year

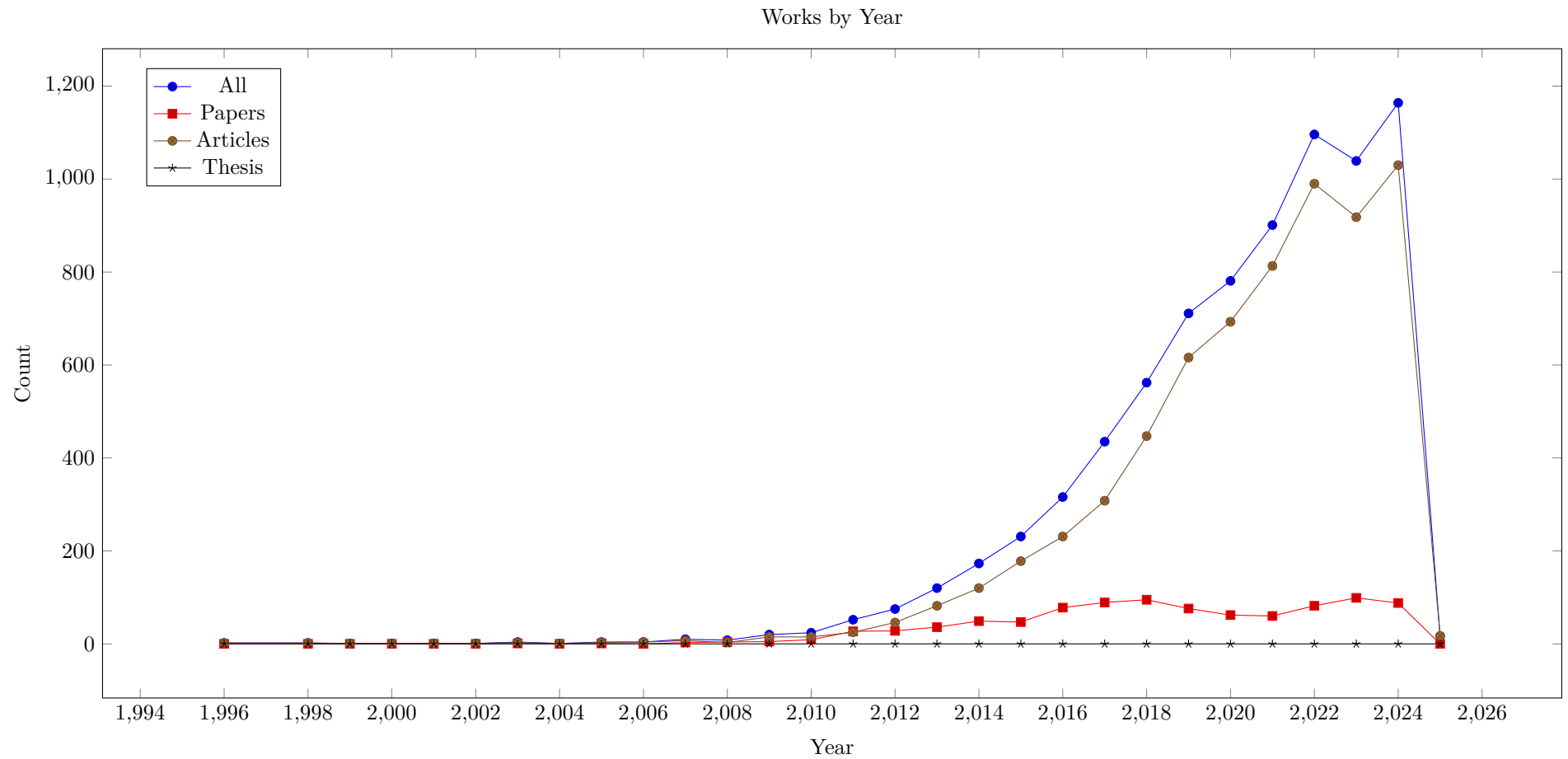
7.1 Relevant Works

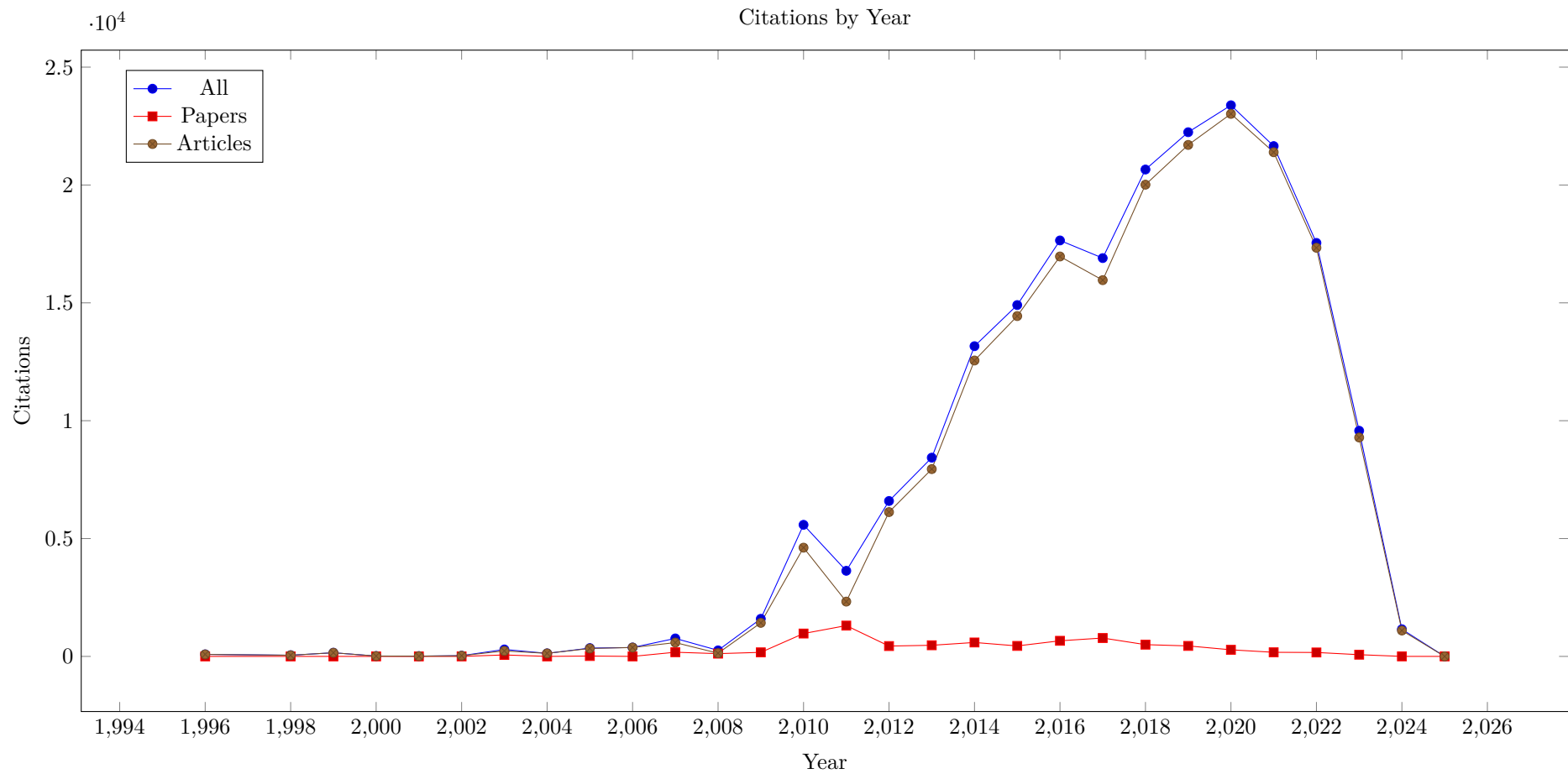


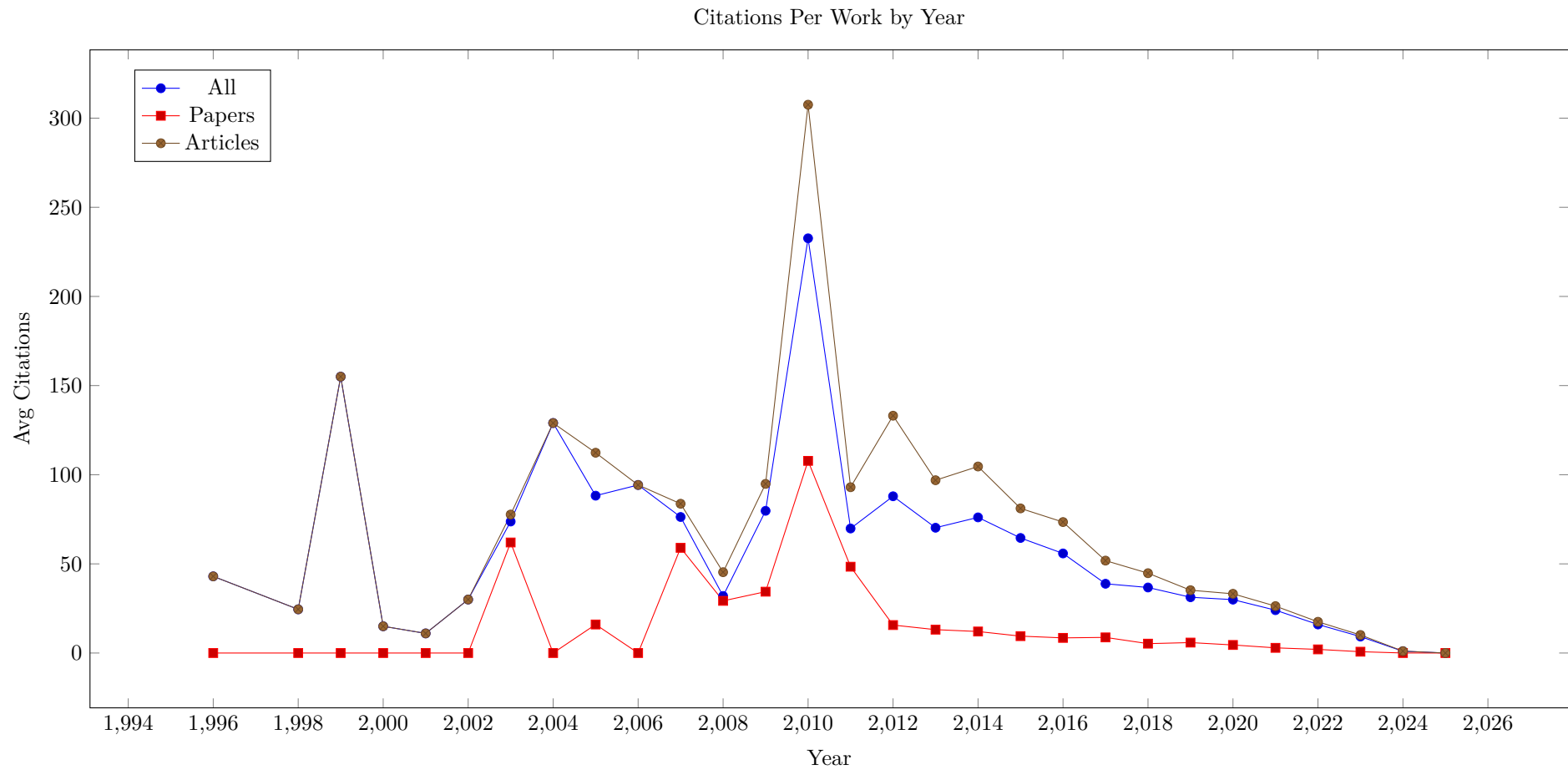




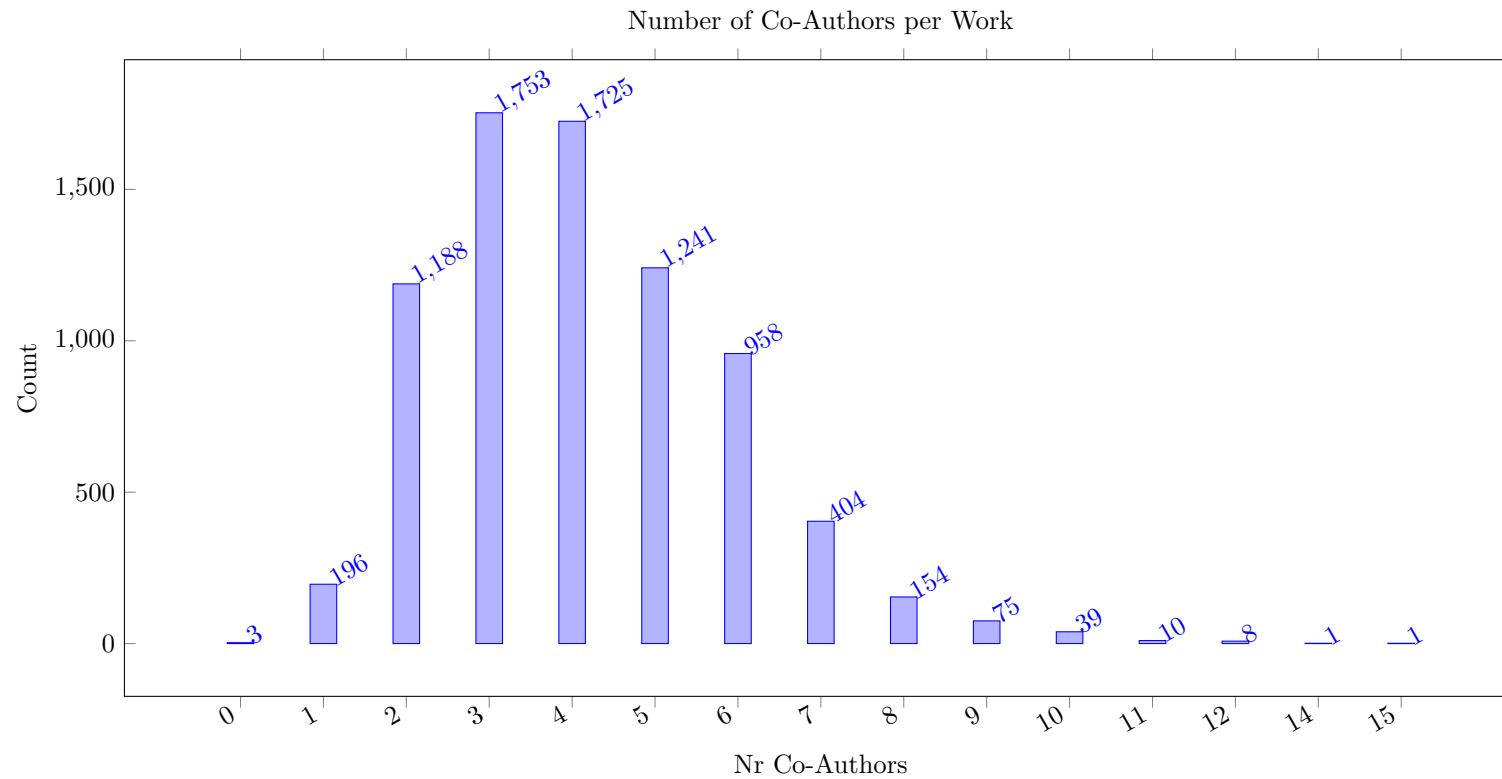
7.2 All Works



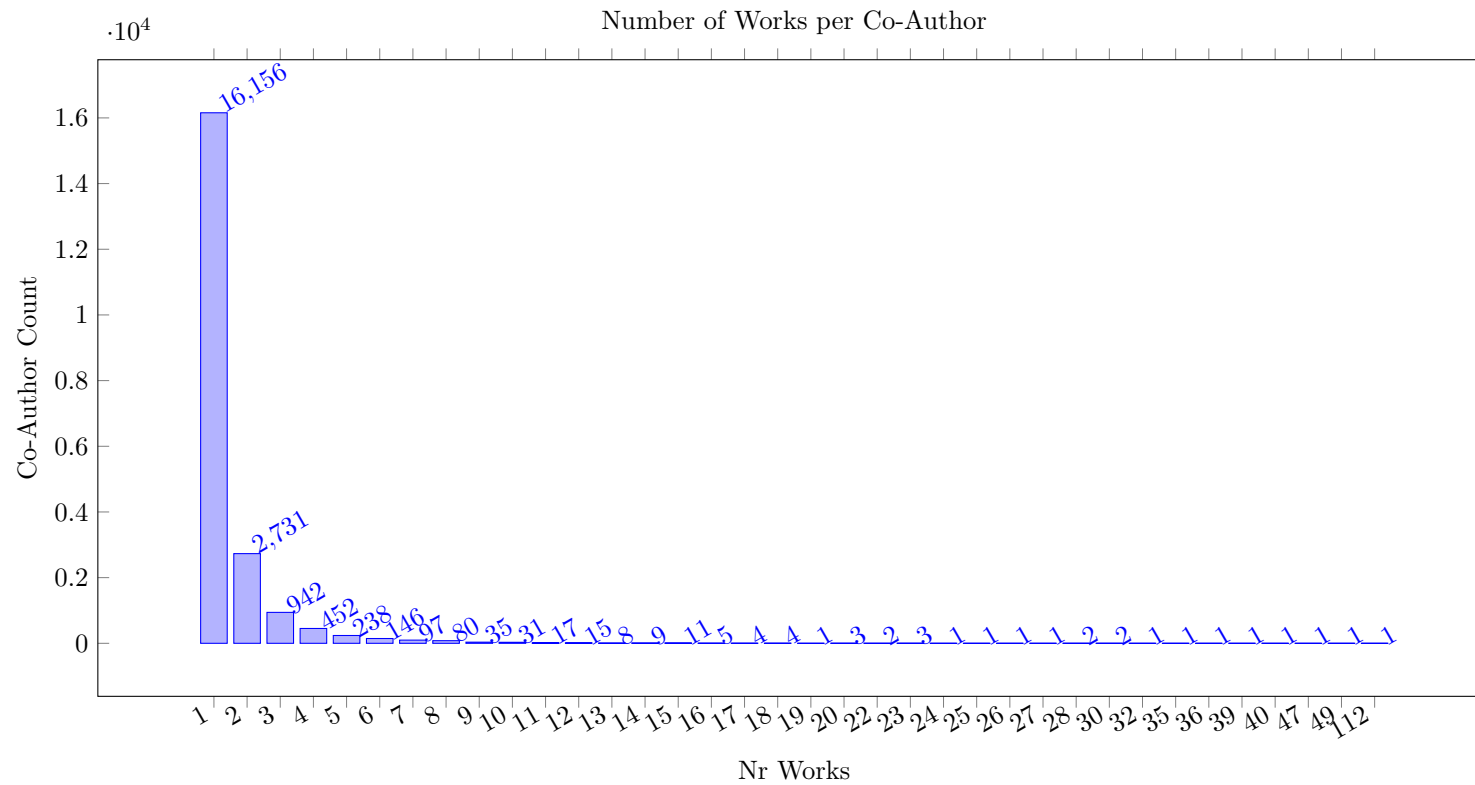




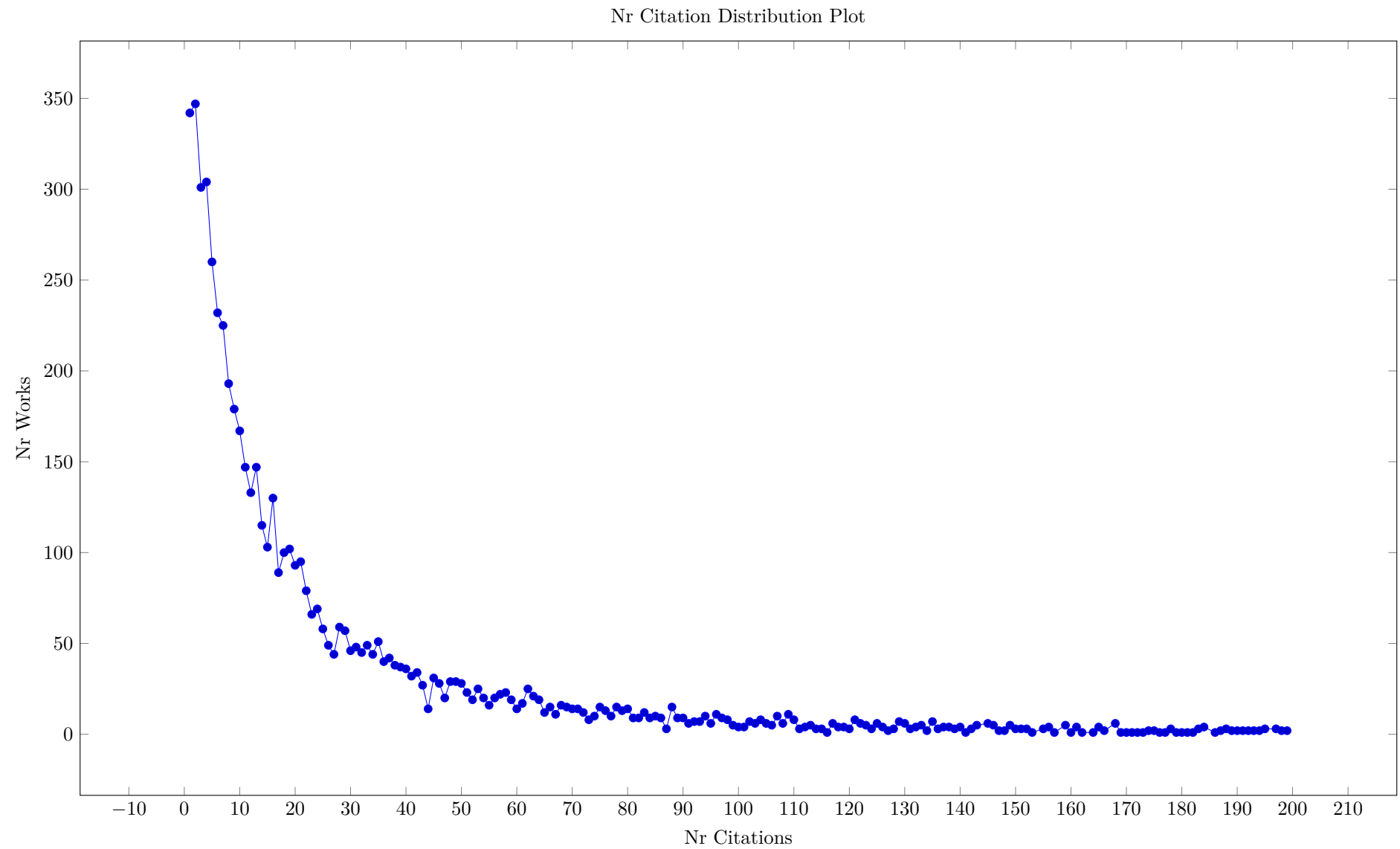
8 Number of Coauthors per Work



9 Number of Works per Author



10 Citation Distribution



11 Similarity Measures

Table 11: Heat Map based on rounded DotProduct Similarity of Concepts (high = similar)

From/To Total	Total	Other
Other		

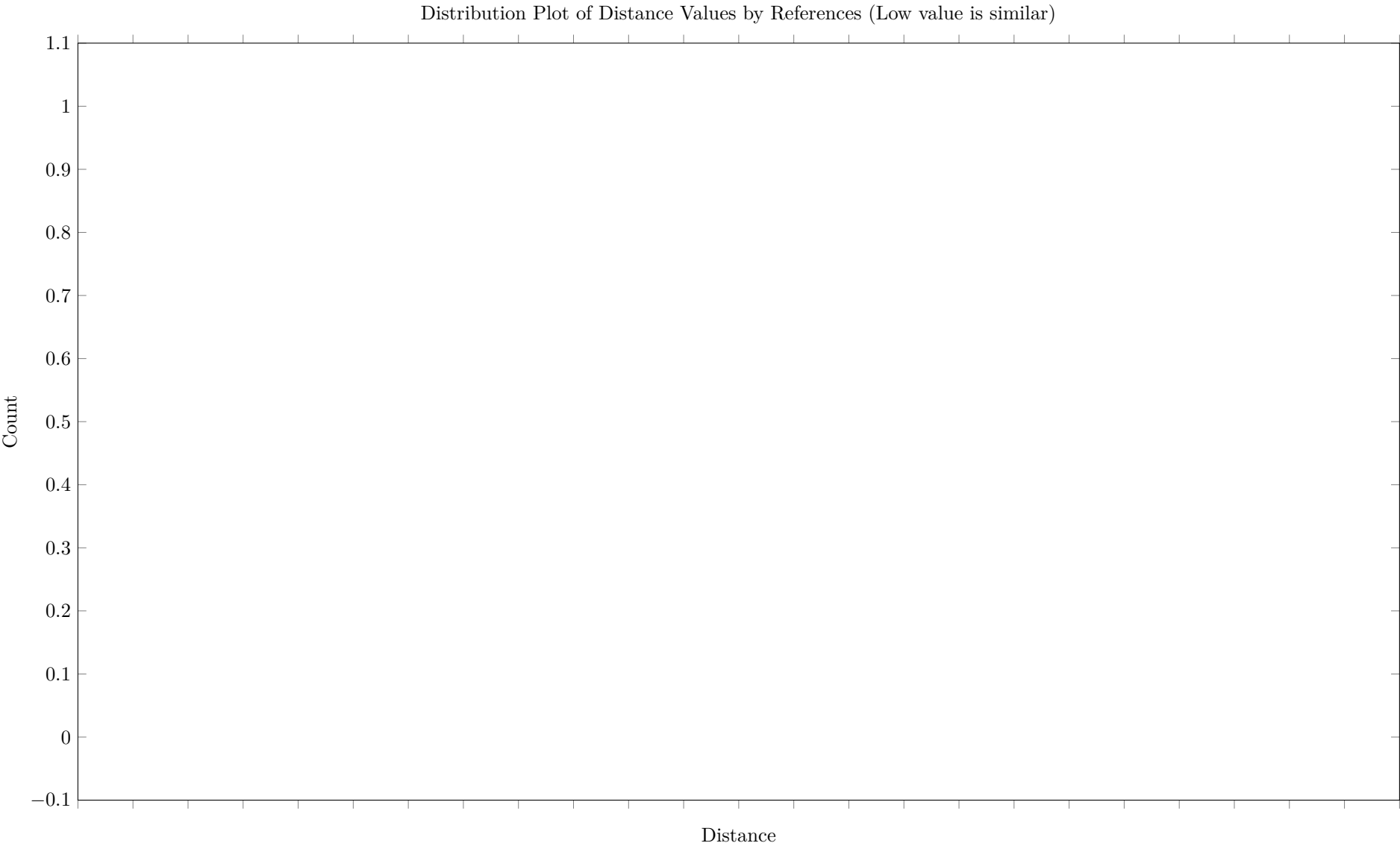
Table 12: Heat Map based on 100*Cosine Similarity of Concepts (high = similar)

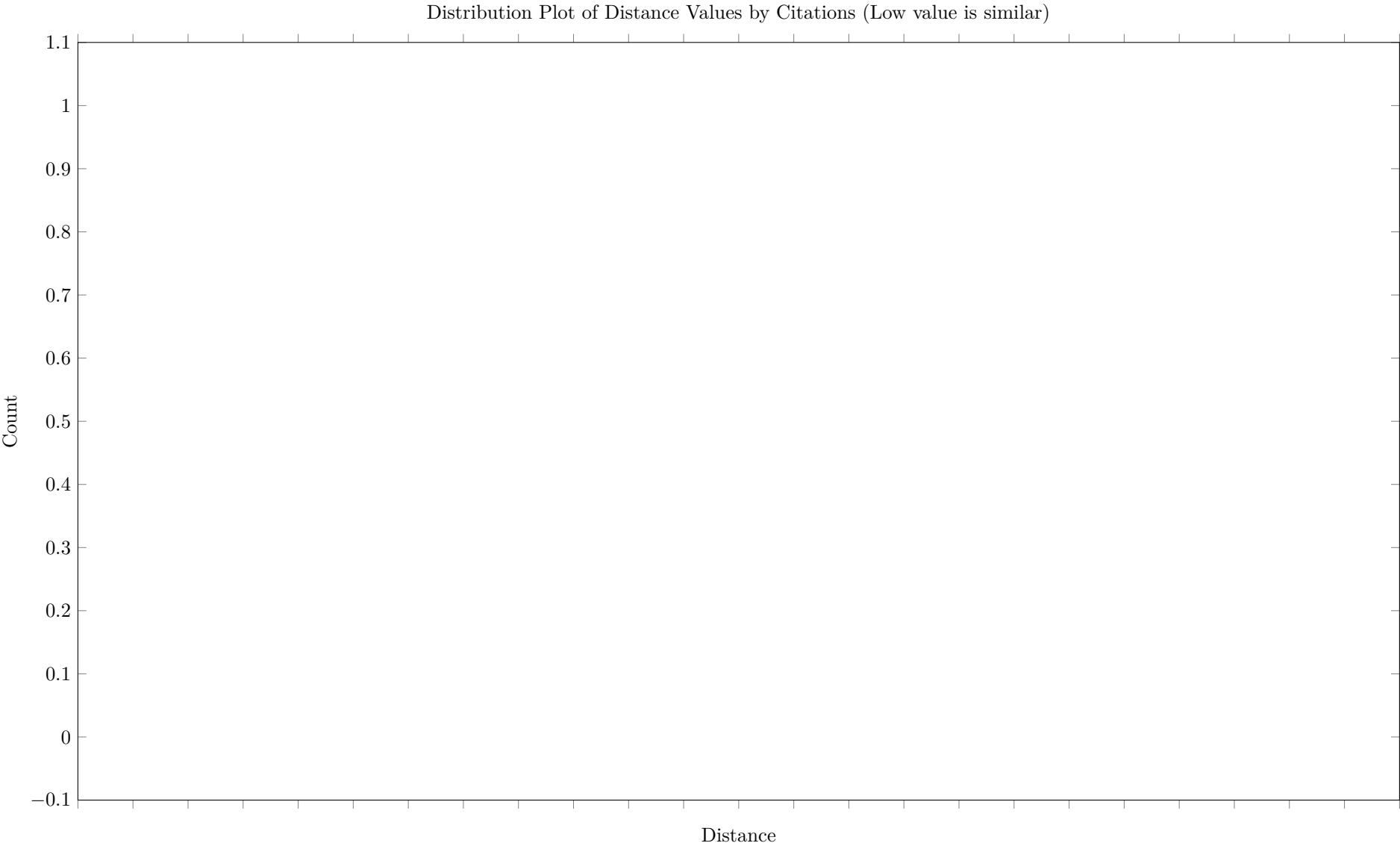
From/To Total	Total	Other
Other		

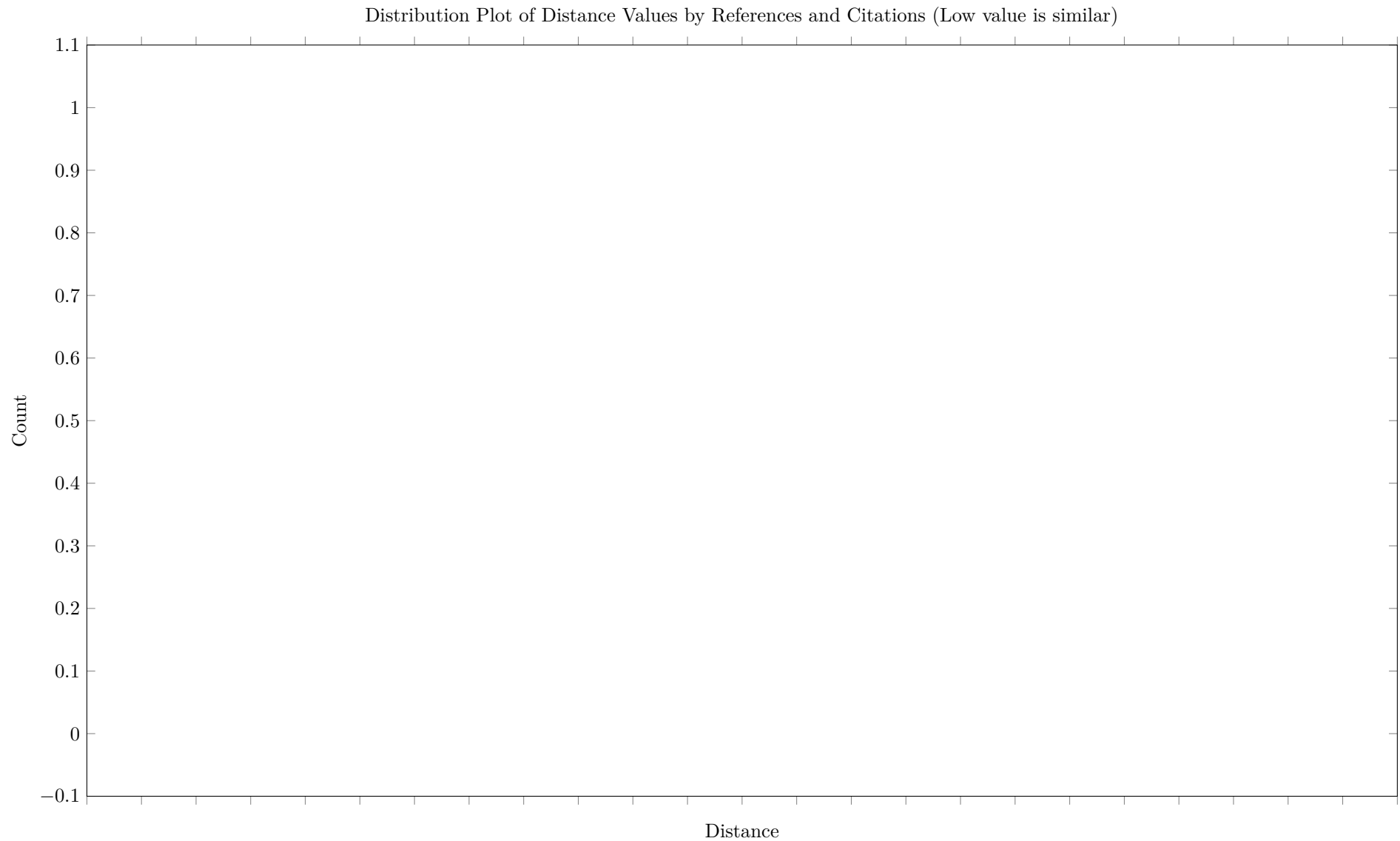
The following distribution plot shows the similarity values between two works based on citations and references counts. If either work does not have citation and reference values, then the similarity is set to NaN. The total similarity count is the average of the similarity for citations and for references. As value we compute the ratio of non-shared references (citations) to the sum of individual references (citations). So both the citation and reference similarity range between zero and one, and the average ranges between zero and one. Low values are very rare, as they require both works to be citing the same papers, and being cited by the same papers. A larger value indicates that items are less similar according to this measure. In the plot we group values into 0.1 wide value bins, so an entry for 0.2 includes values from 0.15 to 0.25.

We observe that low values of this similarity are often found for two works by the same authors that are close in time, where we assumes that the bibliographies in both papers is based on the same literature survey. If neither paper is widely cited, the similarity value is low.

The vast majority of paper pairs has a distance close to one, as their references and citations do not overlap much.

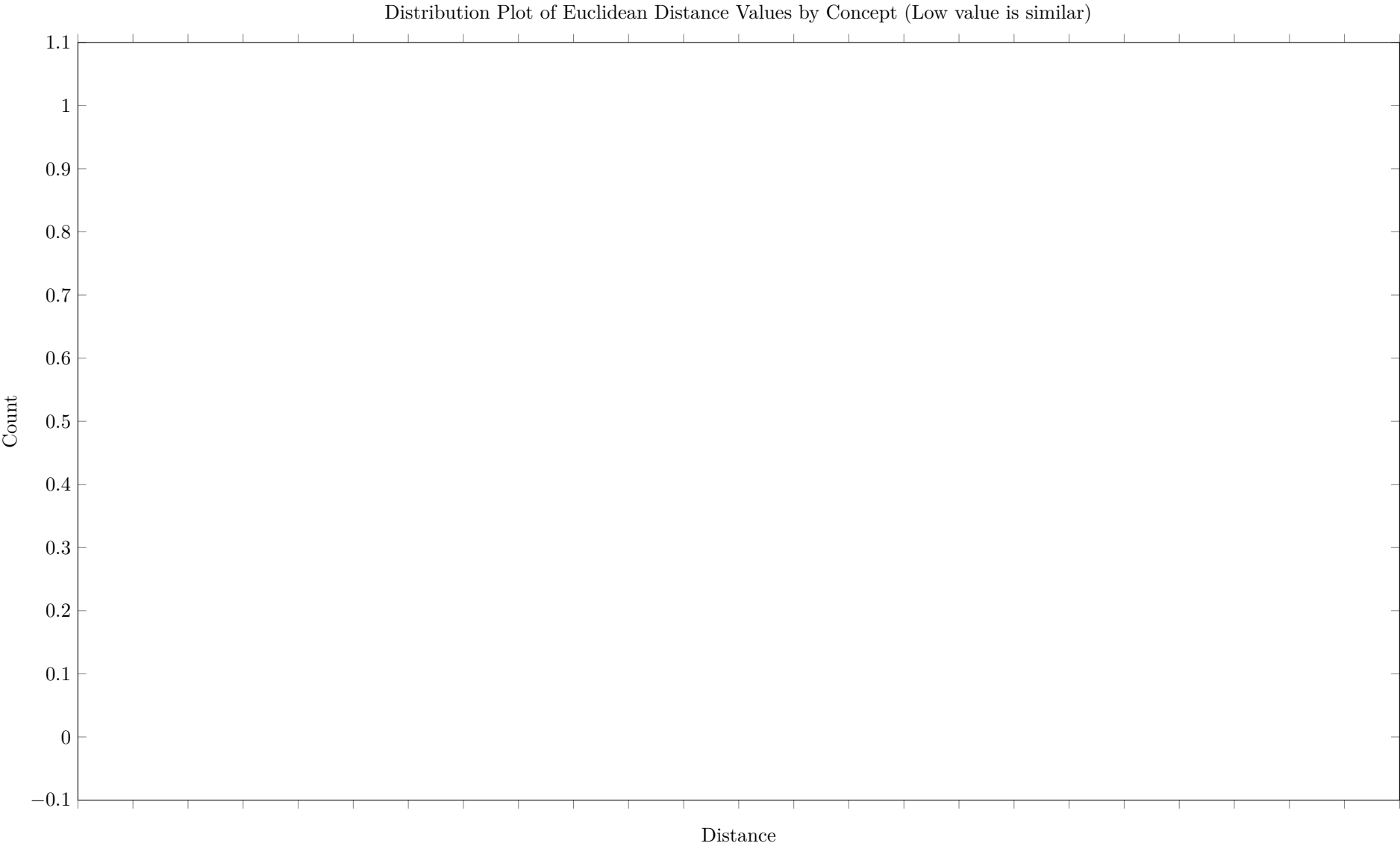


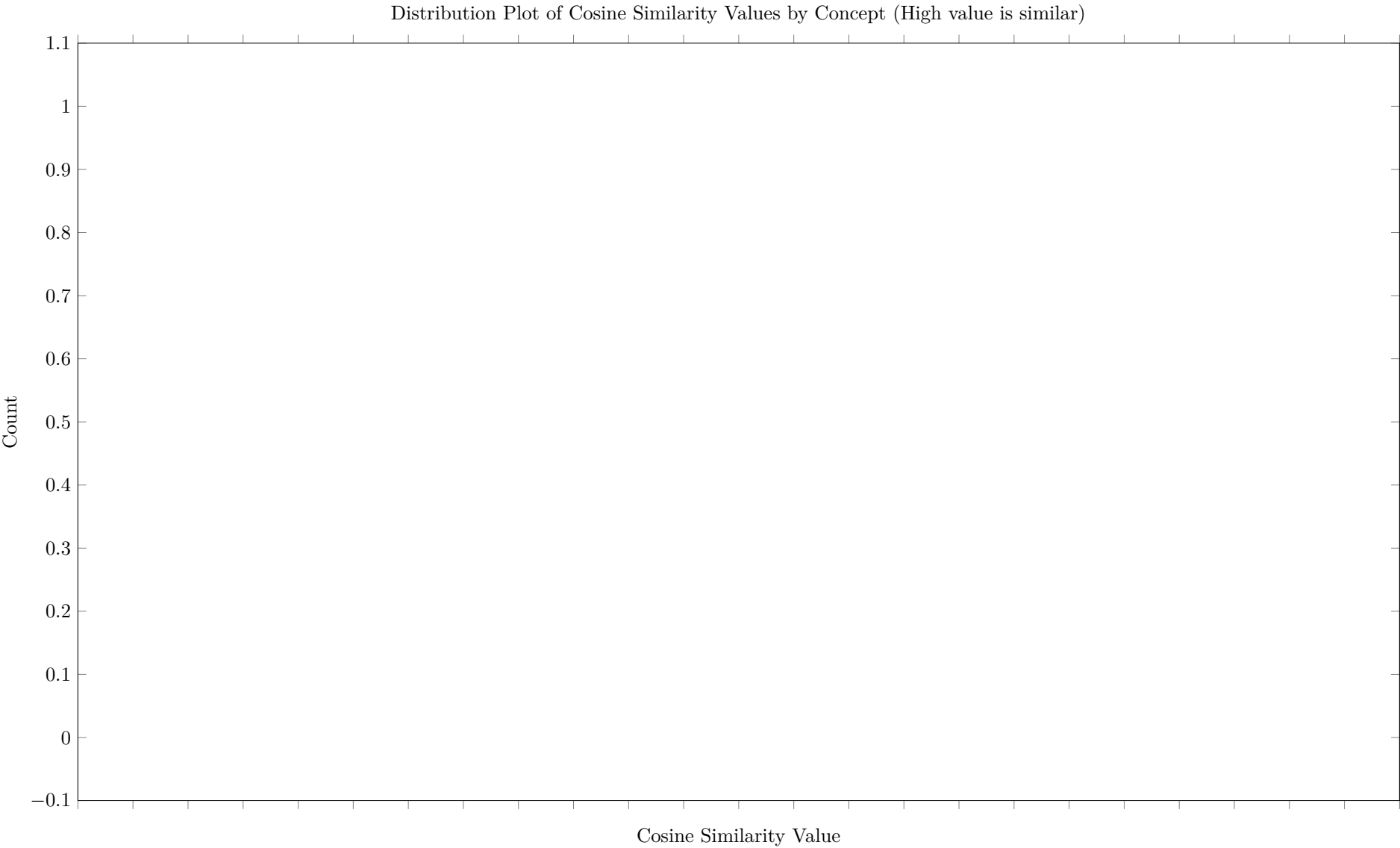


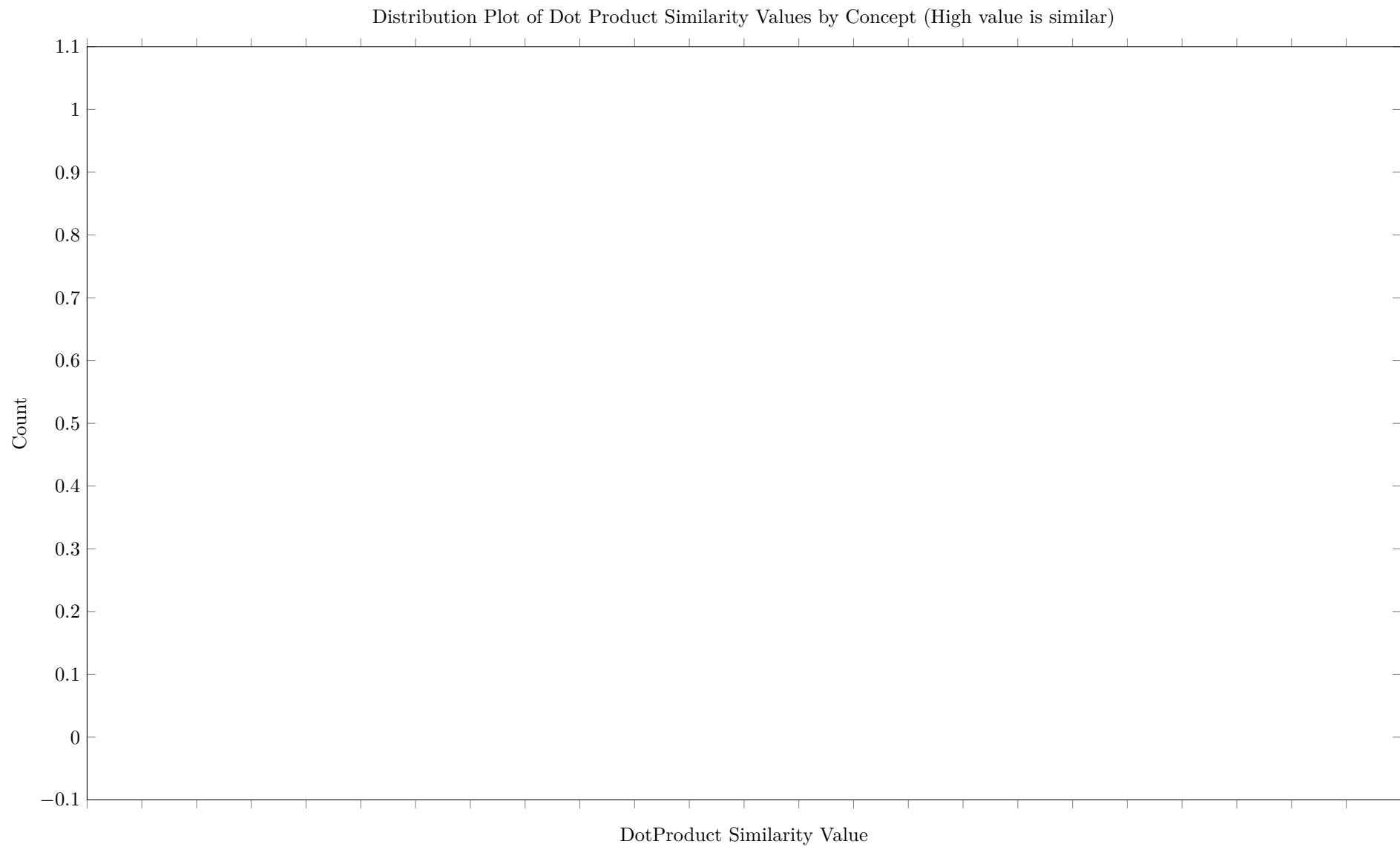


The similarity by concept uses the Euclidean distance between the feature vectors for two works. We translate the MatchLevel for each Concept into a linear

scale, and then calculate the distances as the square root of the sum of squared differences for each feature. The distribution plot below rounds the distances to integer values. Similarity values of this type are only calculated when both works have a local copy, from which we extract the features. If either work does not have a local copy, the similarity is set to be NaN.



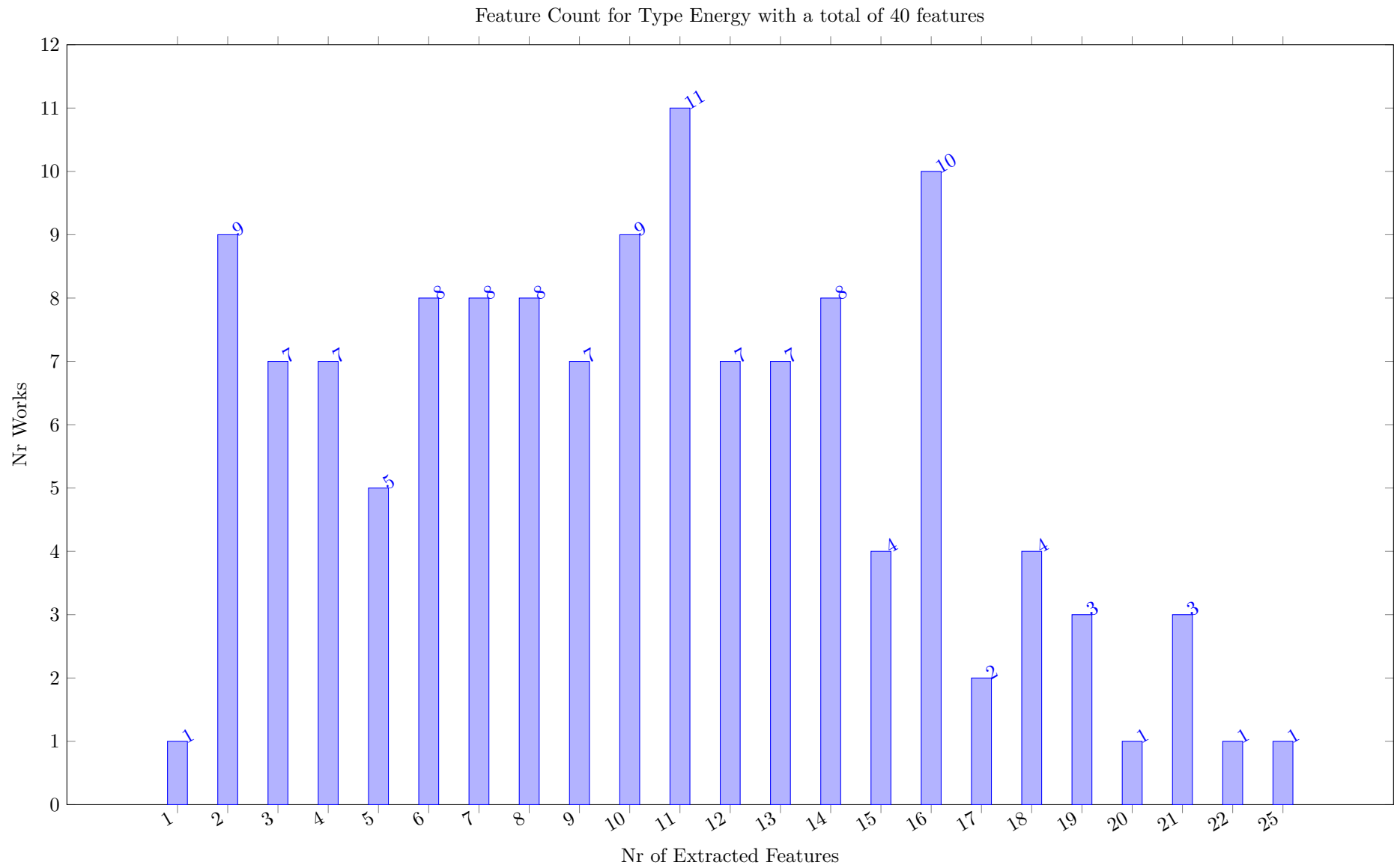


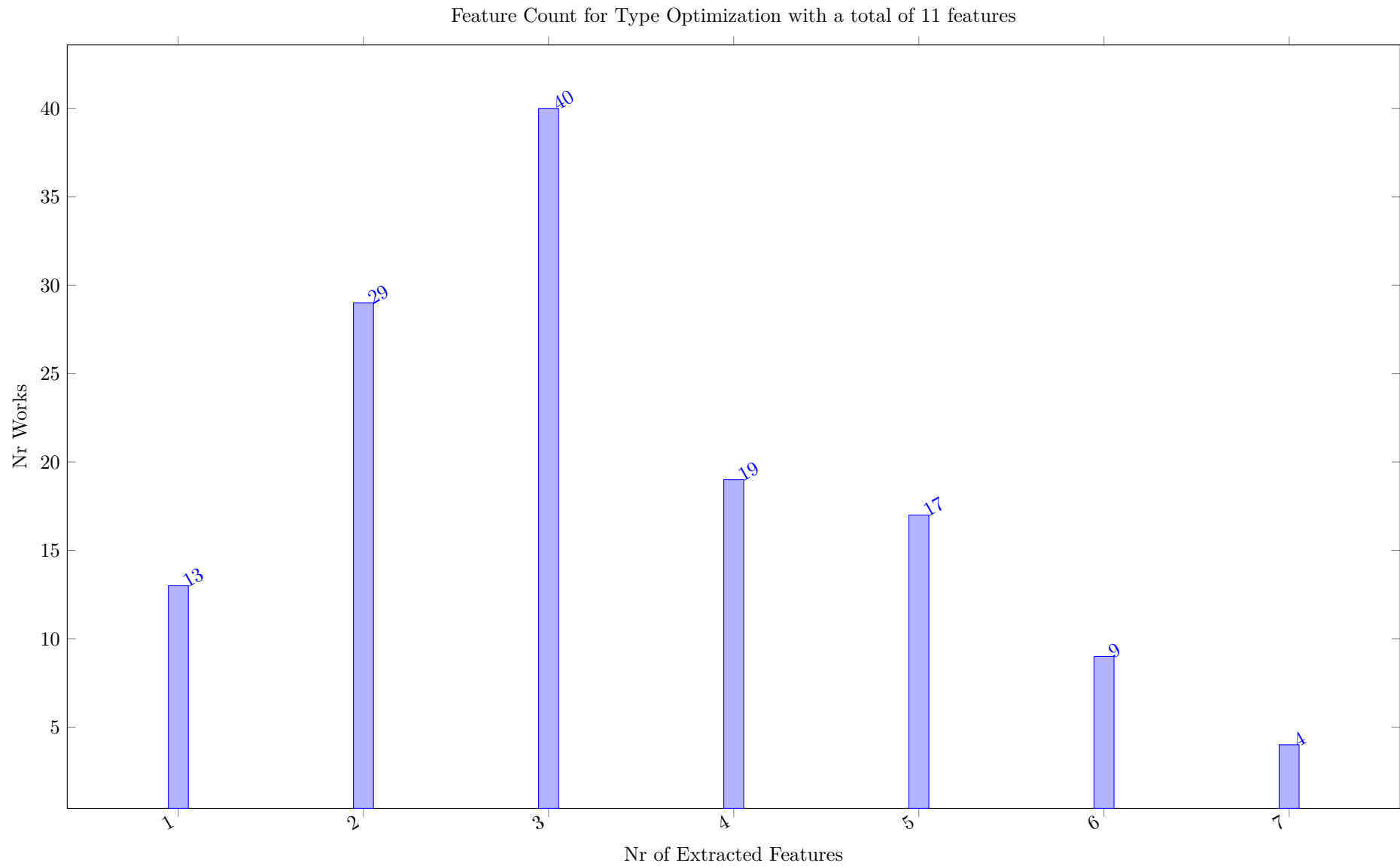


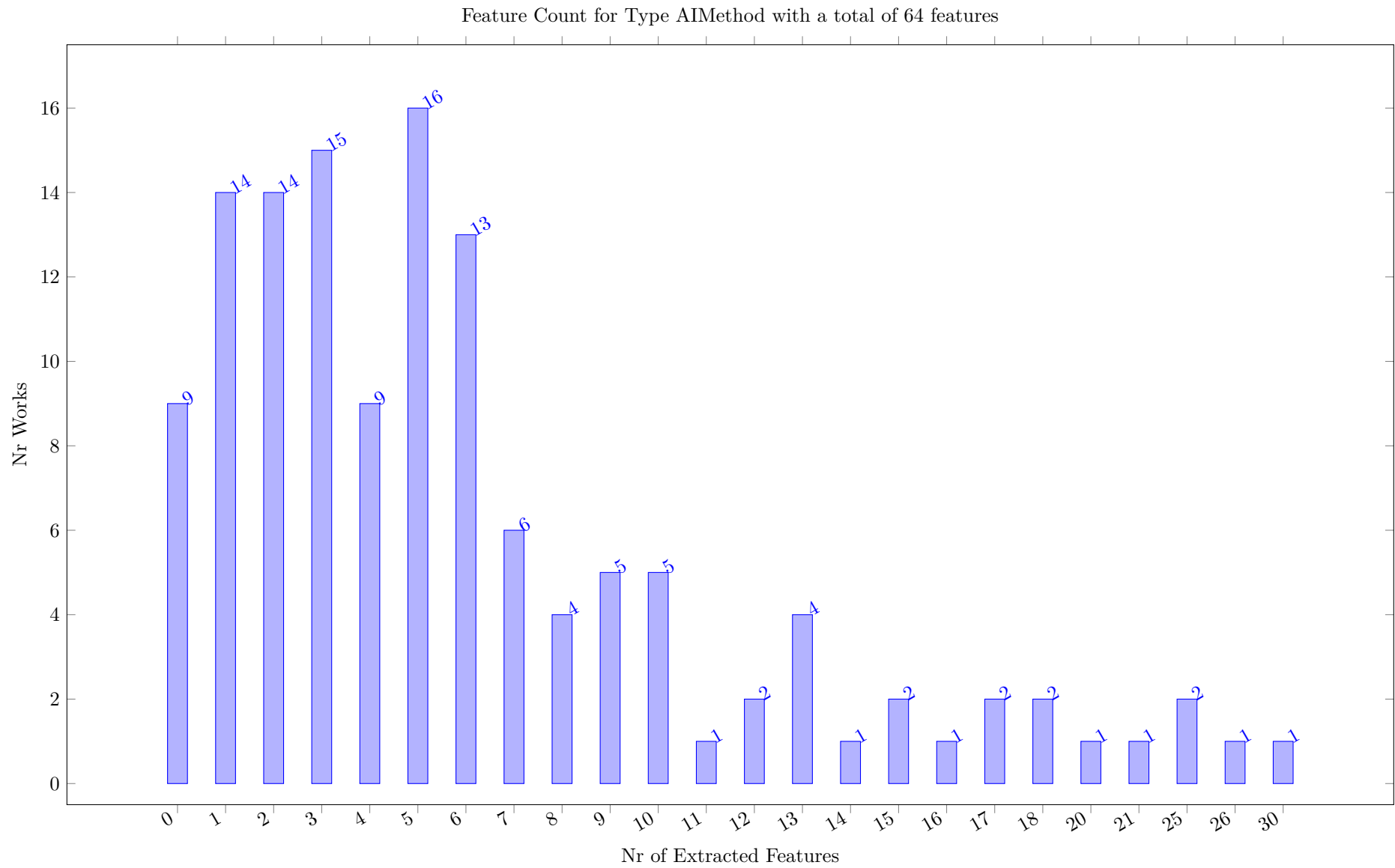
12 Concept Distribution

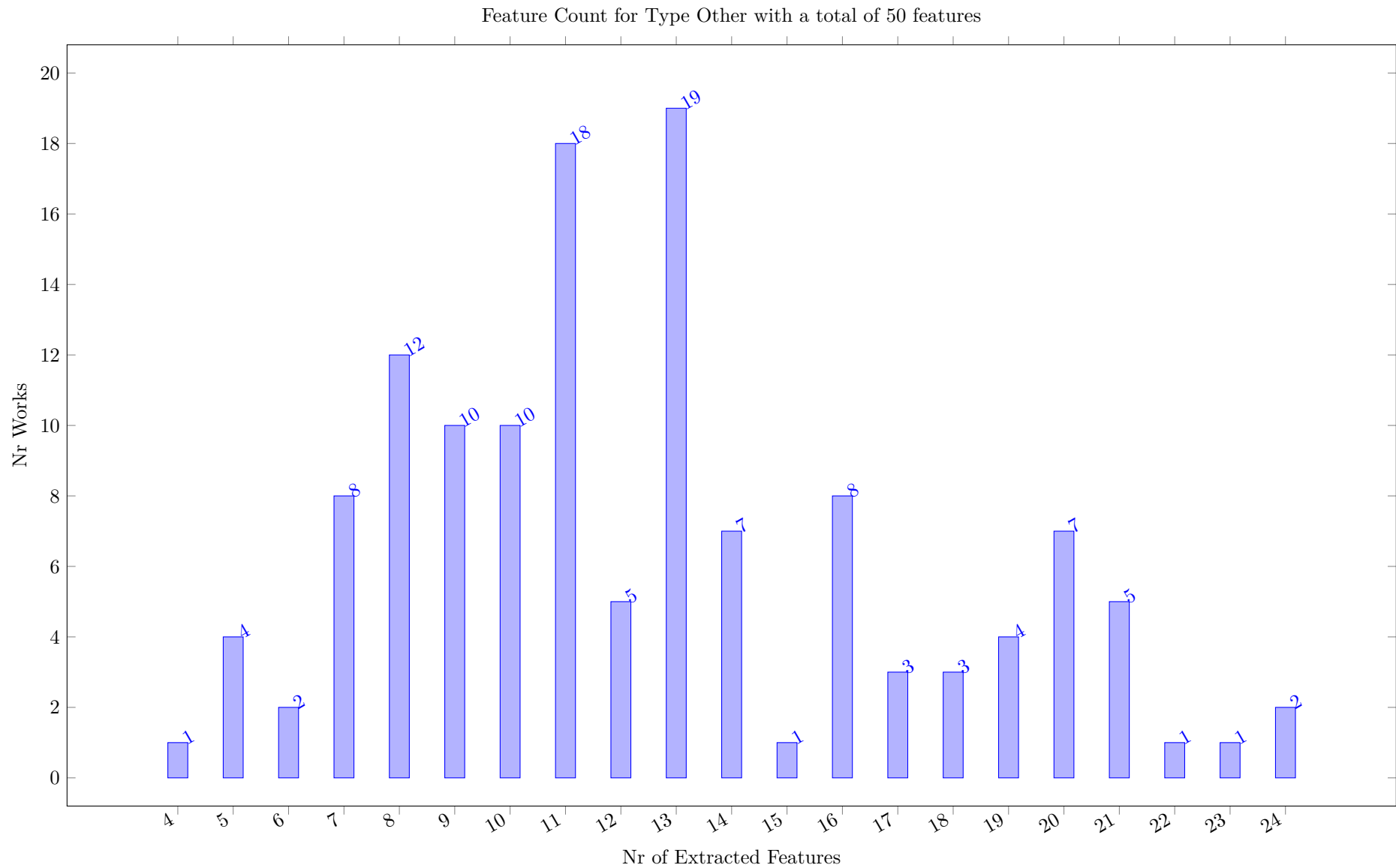
For each concept type, we count how many features are extracted by the individual works that do have a local copy, e.g. for which we can extract features. We can compare the number of features extracted to the number of concepts of a given type, which is stated in the title of the diagram.

A high count indicates that a work covers many of the concepts of the given type, a low count might mean that our ontology does not have relevant concepts for that work.







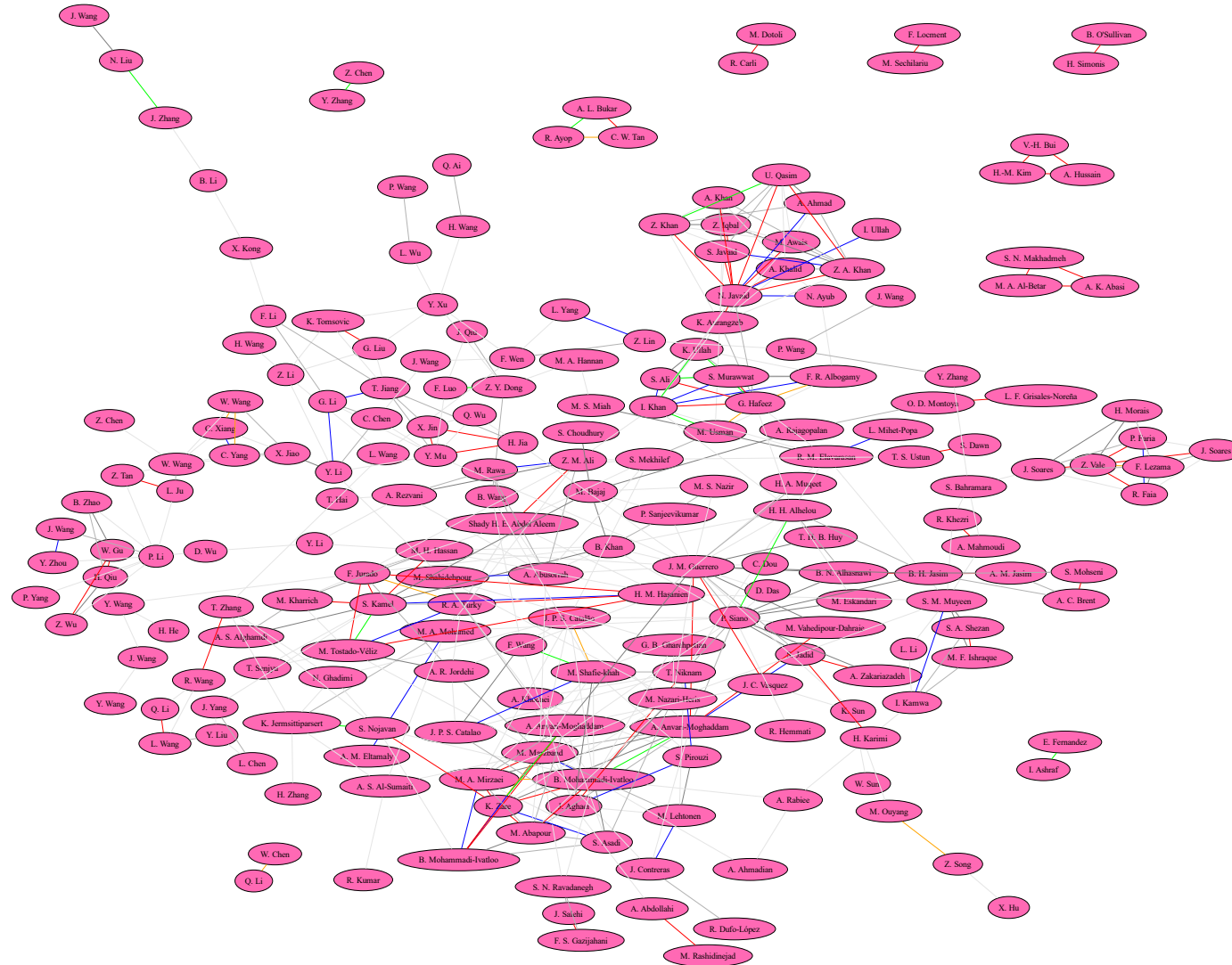


13 Coauthor graph

The coauthor plot is created by graphviz, and is based on the coauthor relations extracted from the author fields of the works. Authors with few works (less than `coauthorLimit`) are not shown, to avoid a cluttered view. Note that this analysis depends on the use of canonical forms of author names. If bib entries come from many different sources, we will need to check this manually. DBLP seems to be using ORCID values and typically identifies the authors of a work with a canonical representation of their name. Accents and umlauts are other sources of having multiple forms of the name of the same author. Note that the risk of two different authors using the same name should be low for very specific literature surveys, but cannot be checked automatically with the data sources currently used.

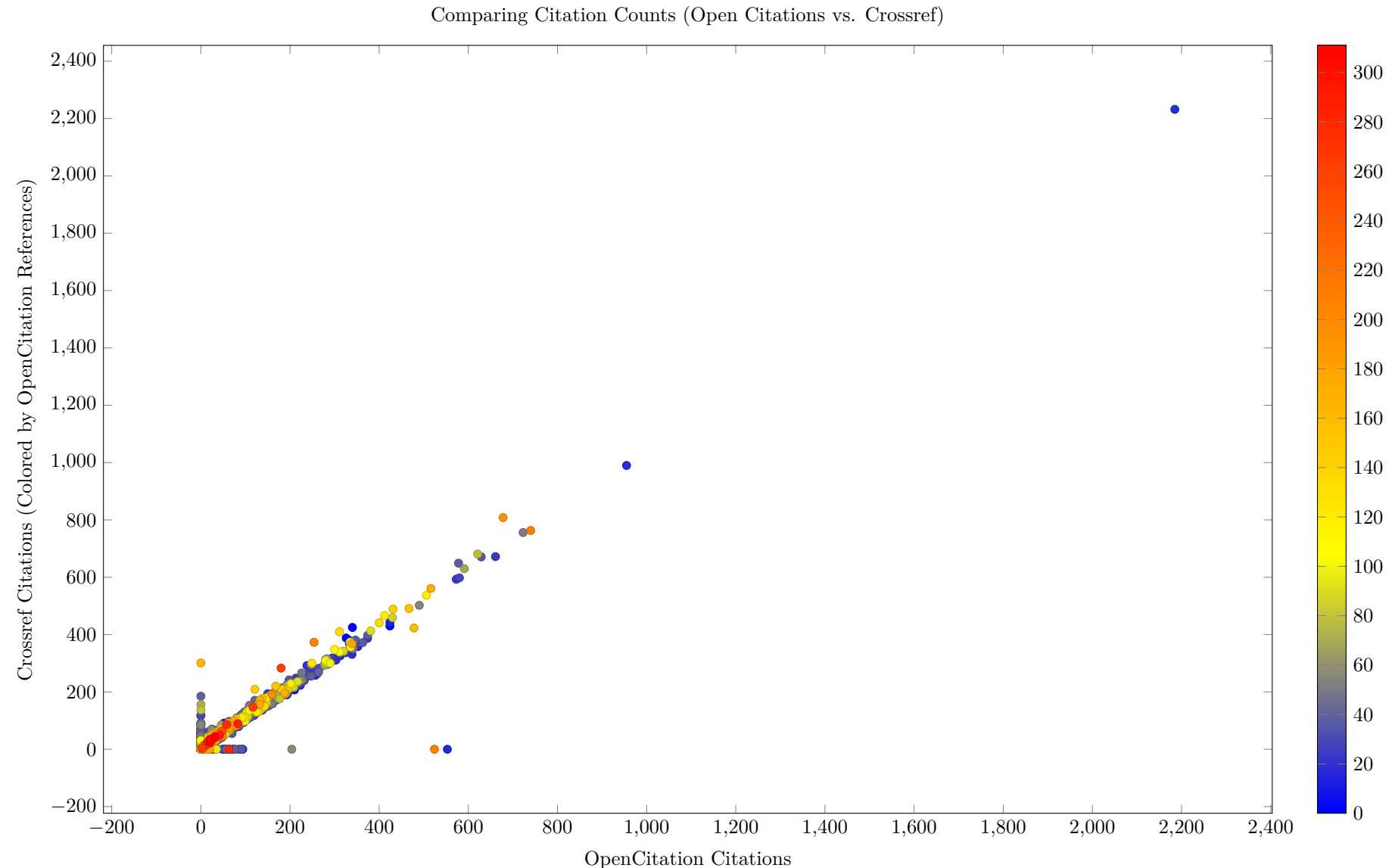
The plots can be made with different layout tools in graphviz, it seems that `fdp` produces the most consistent visually attractive plots for this type of display. This probably needs more work on parameter settings to be fully automated.

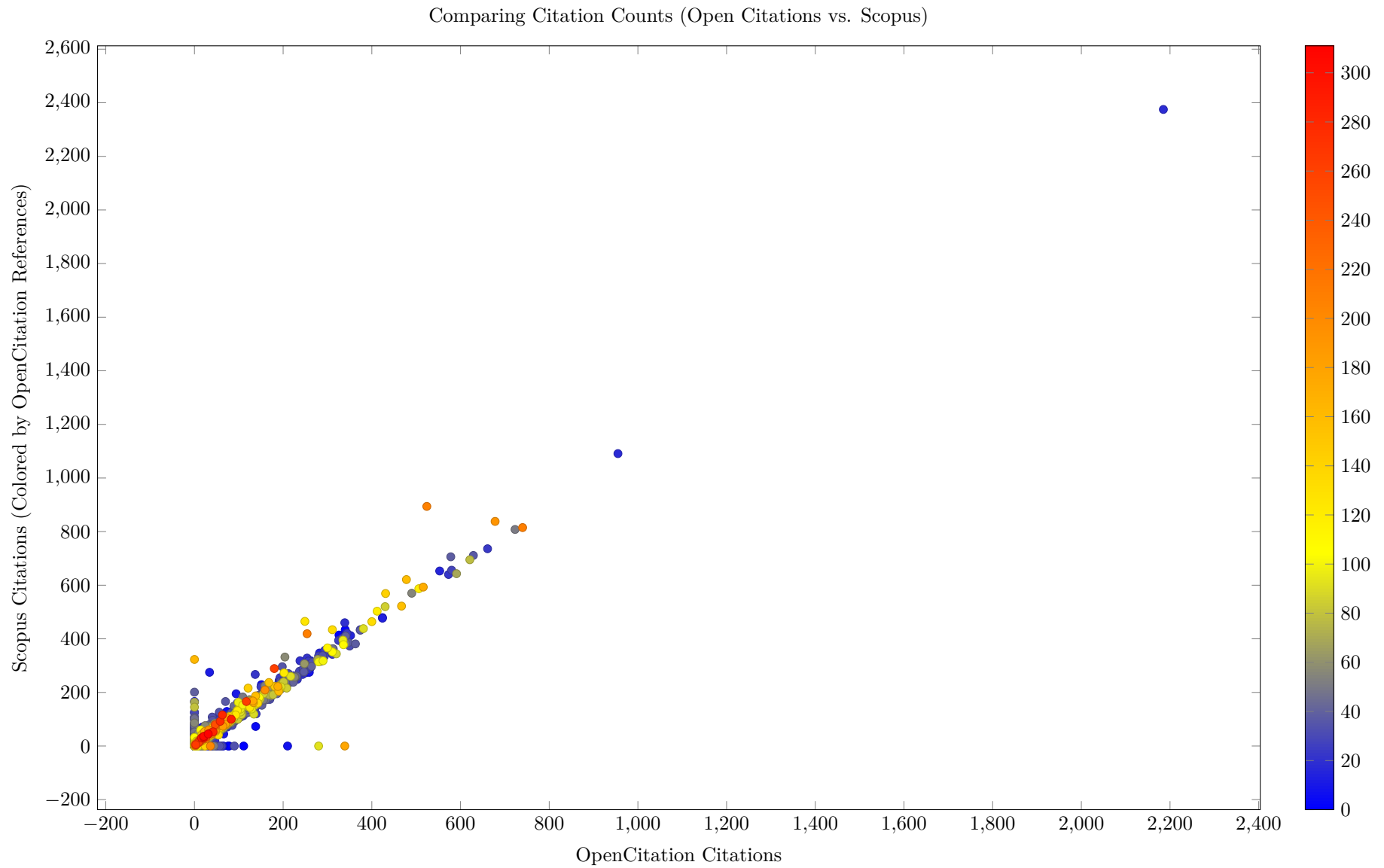
Figure 1: Coauthor Graph Drawn with fdp (Graphviz, CoauthorLimit = 8)

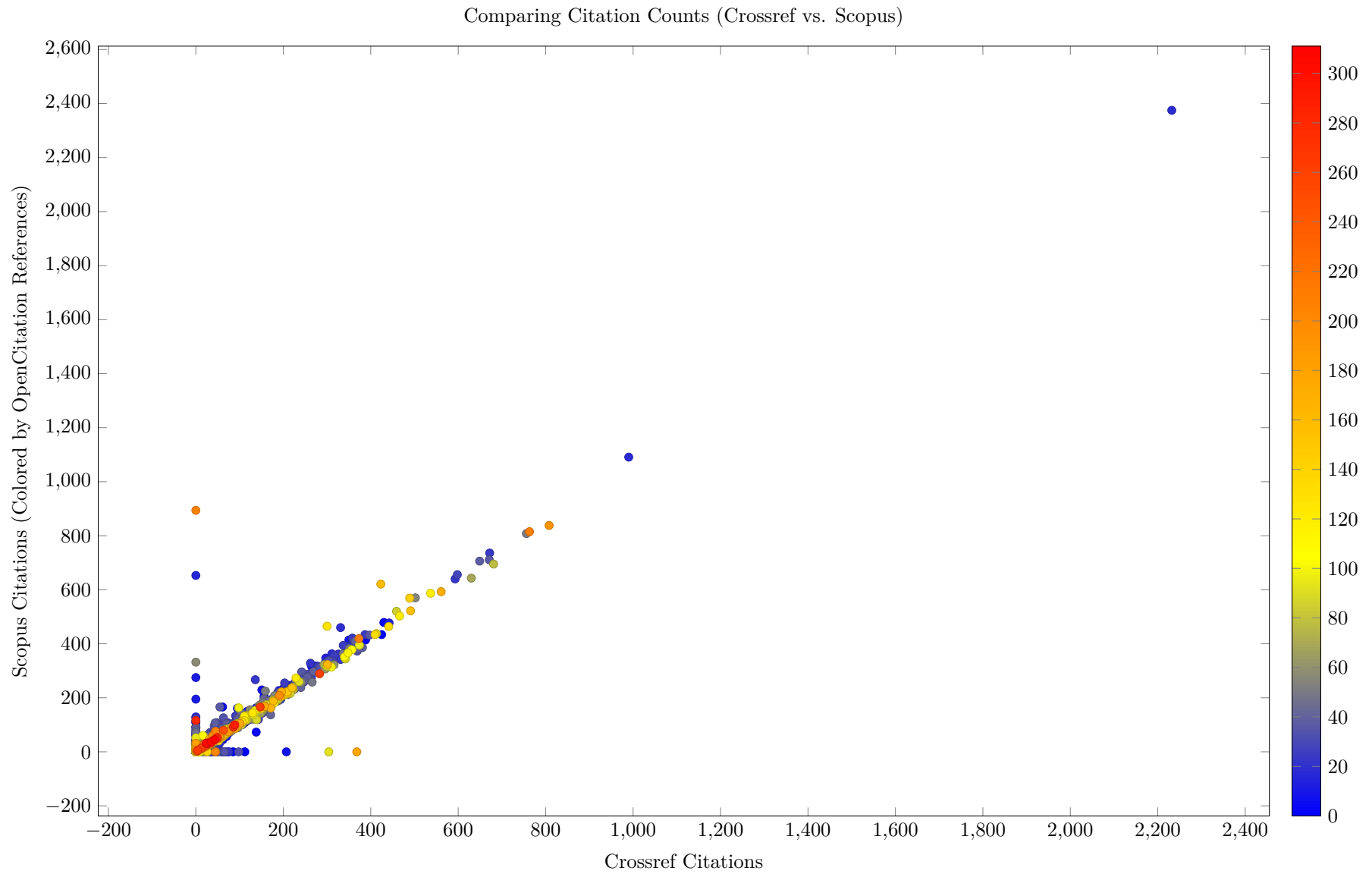


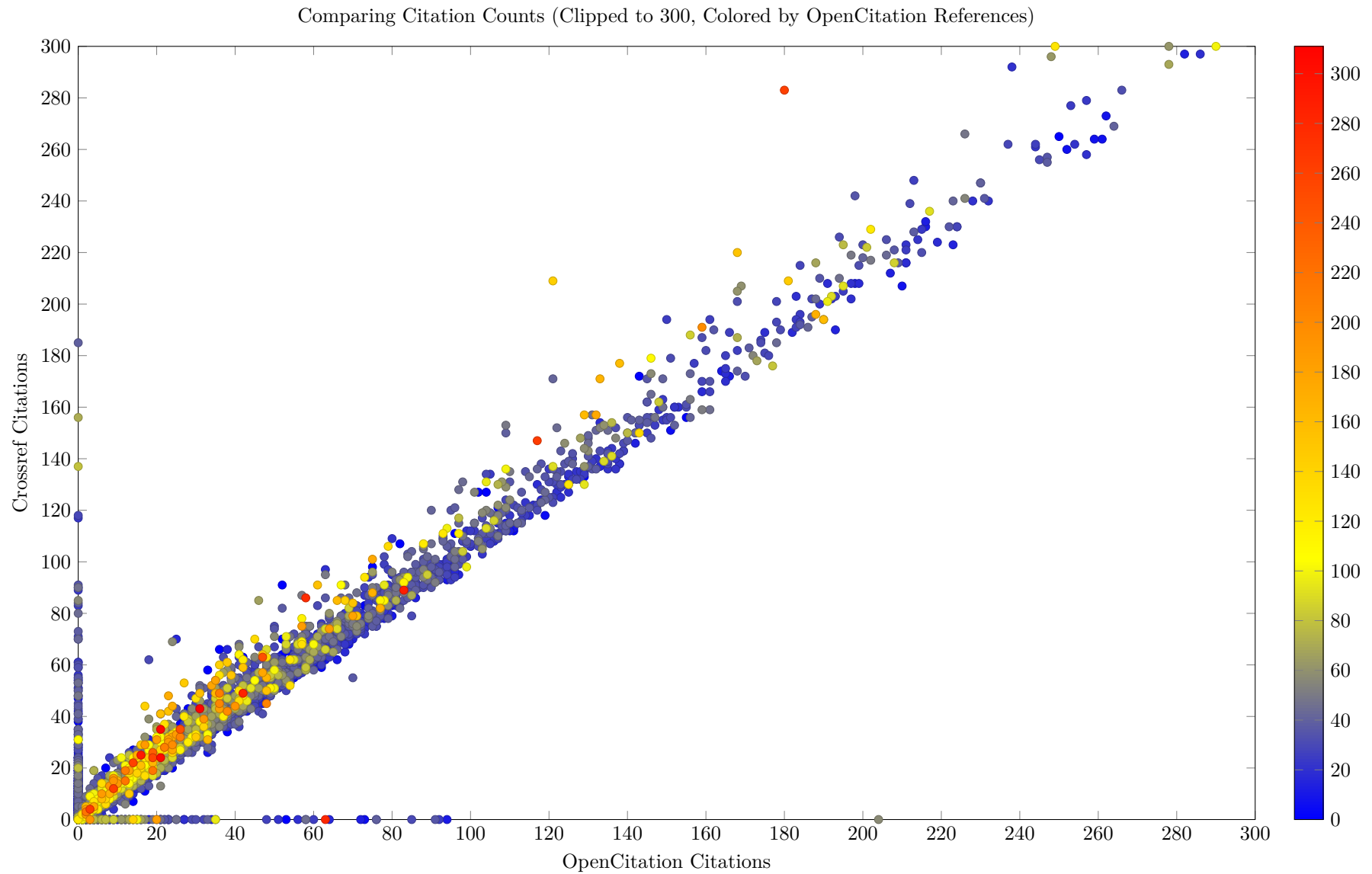
14 OpenCitations vs. Crossref Data vs. Scopus Data

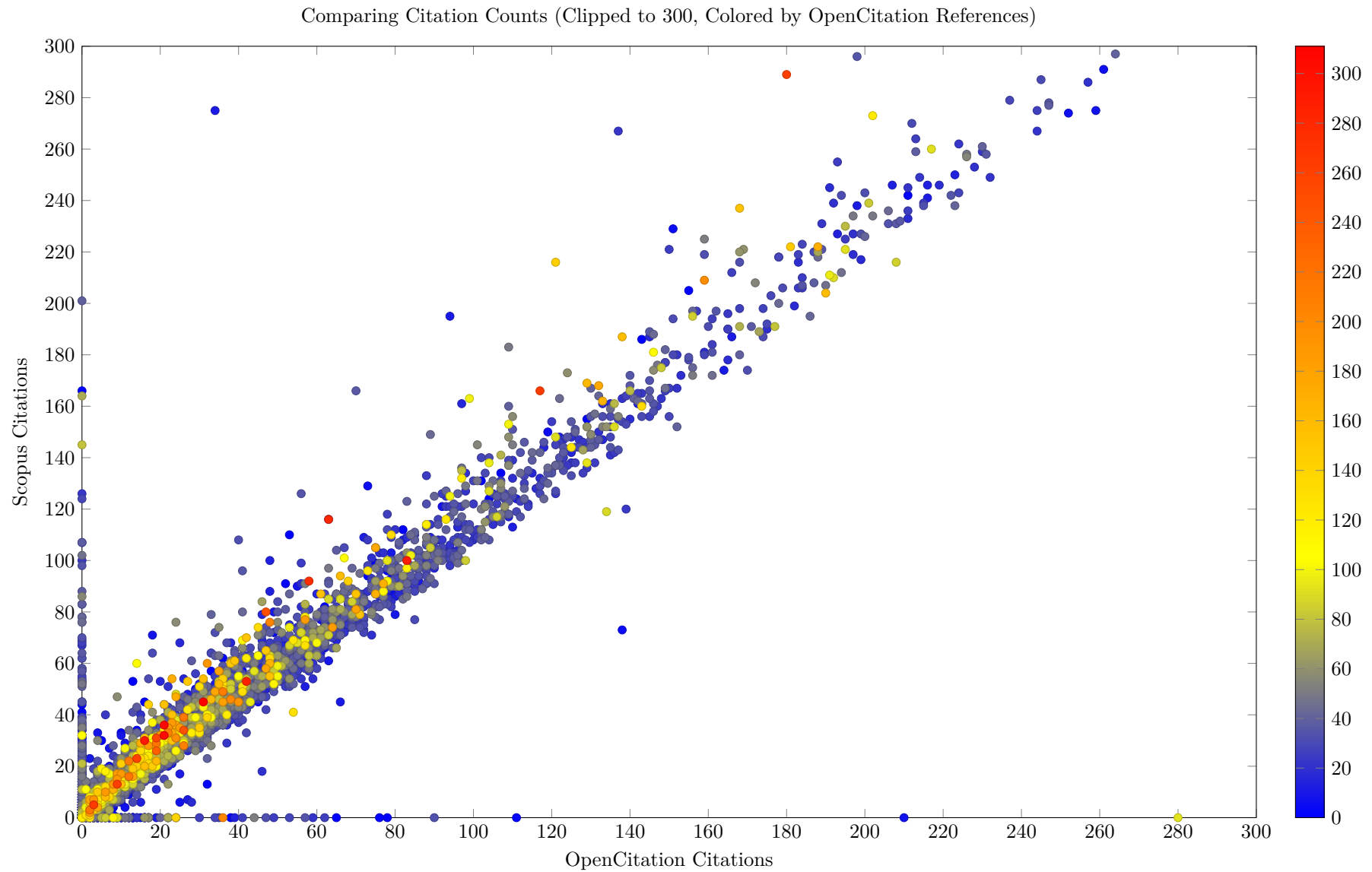
14.1 Citation Comparison

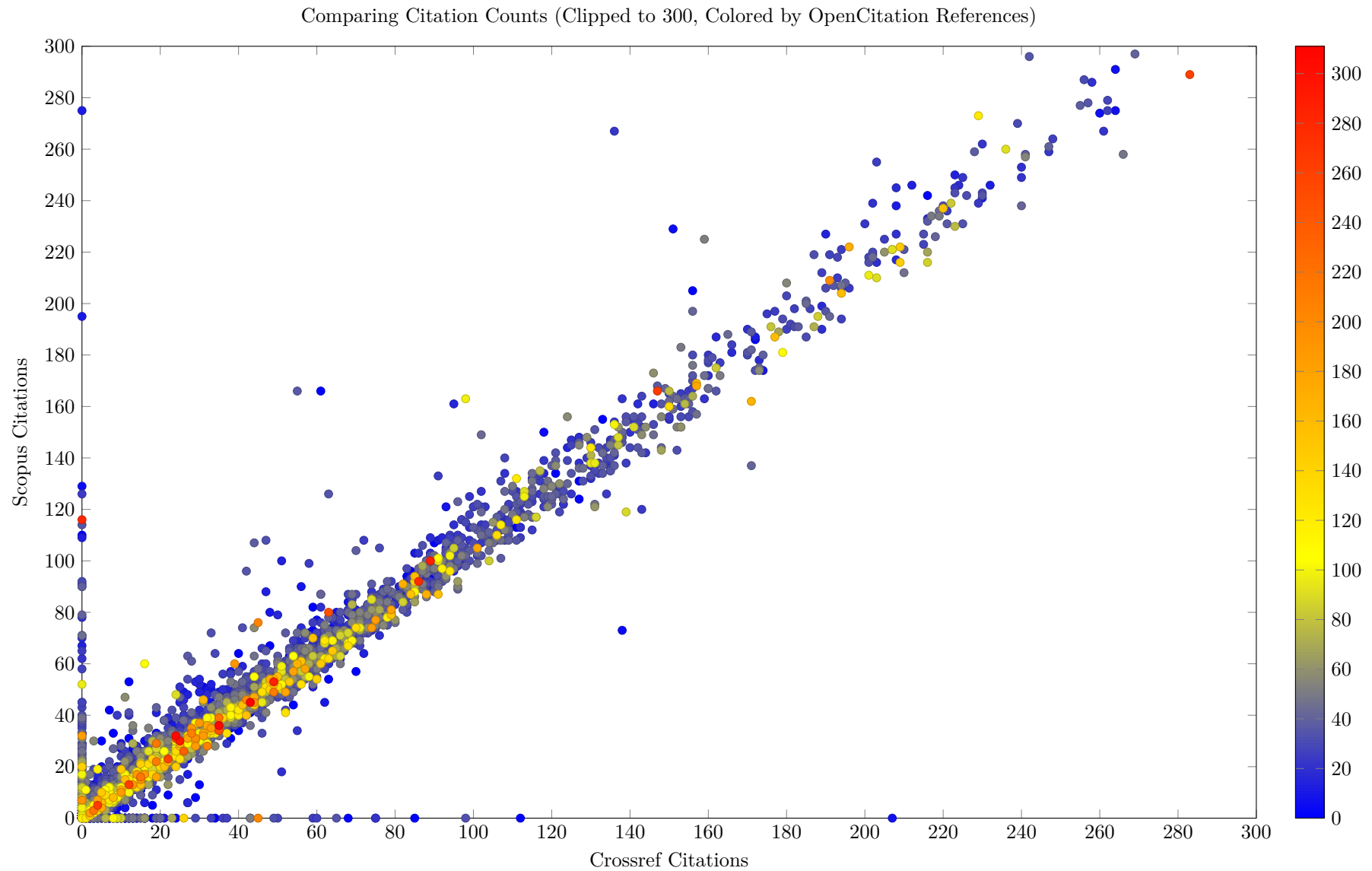




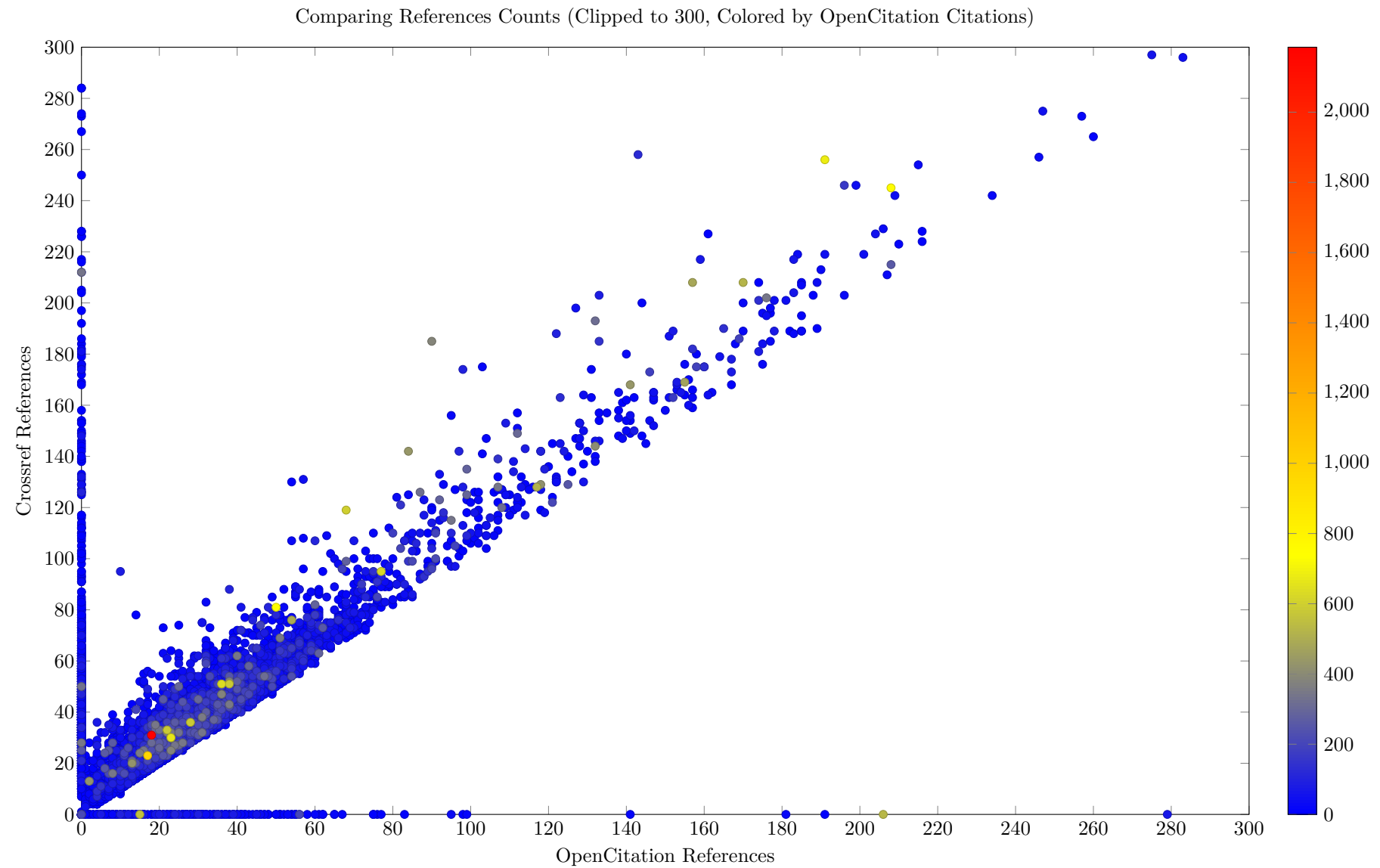




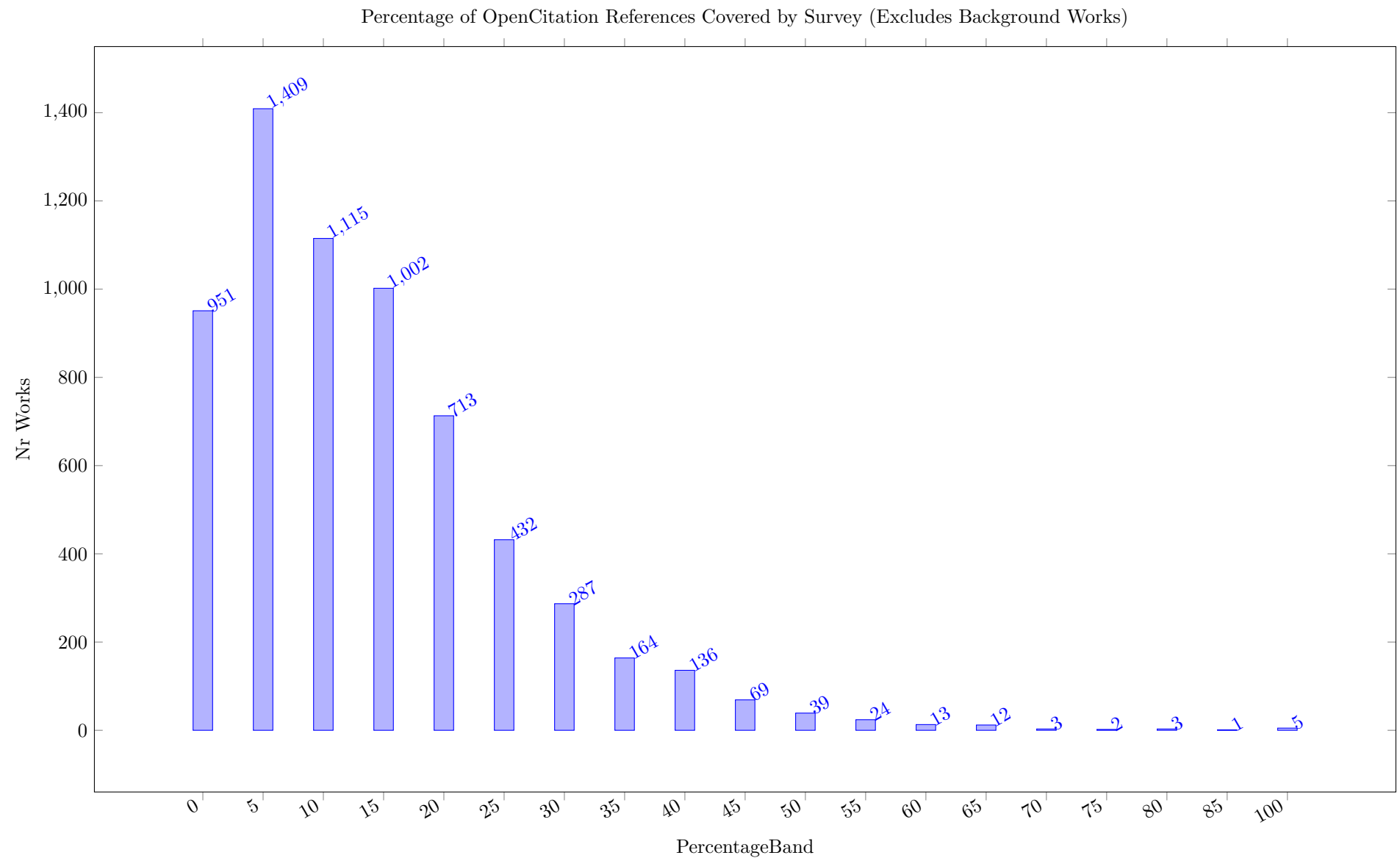


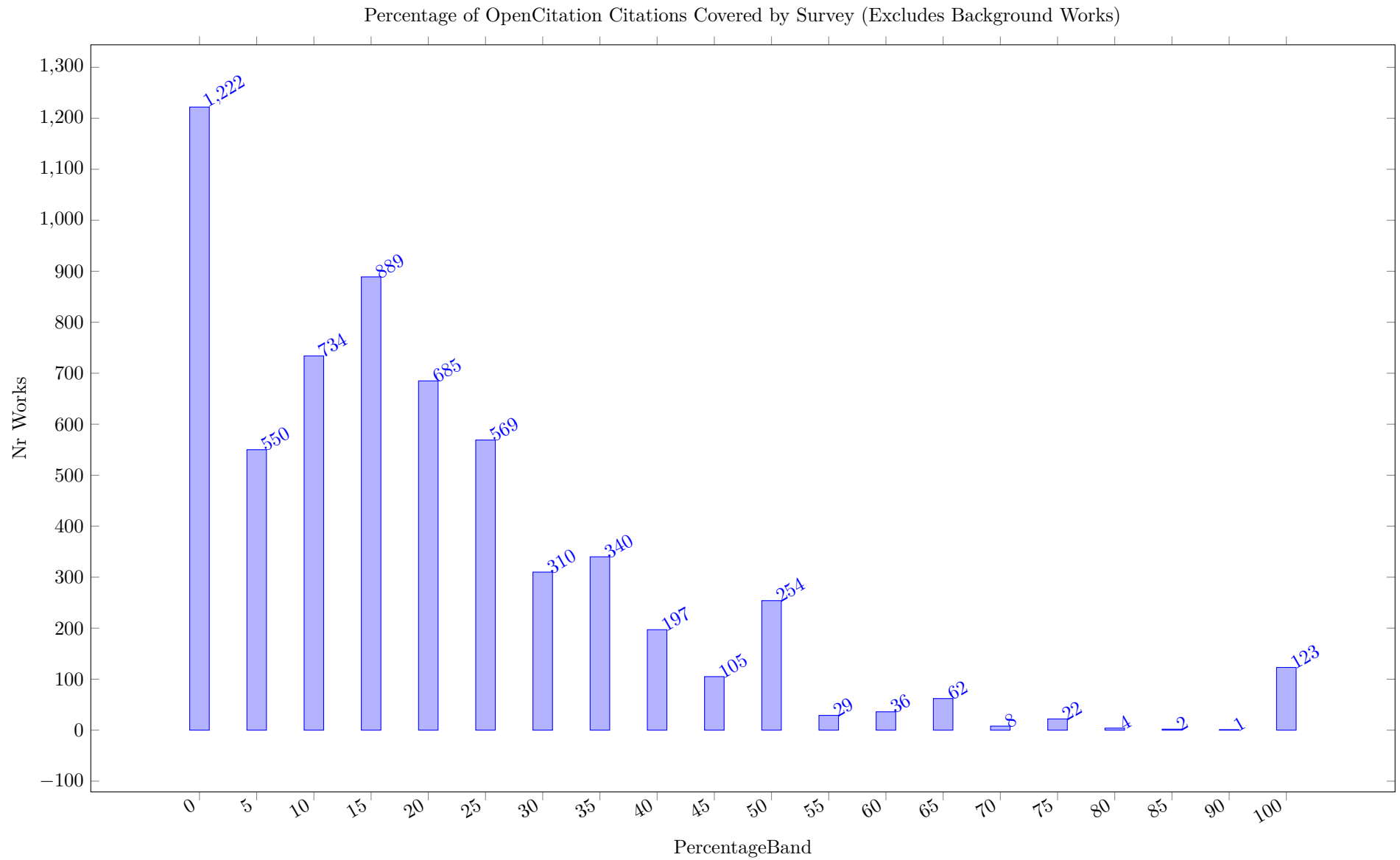


14.2 References Comparison



14.3 Percentage Cover







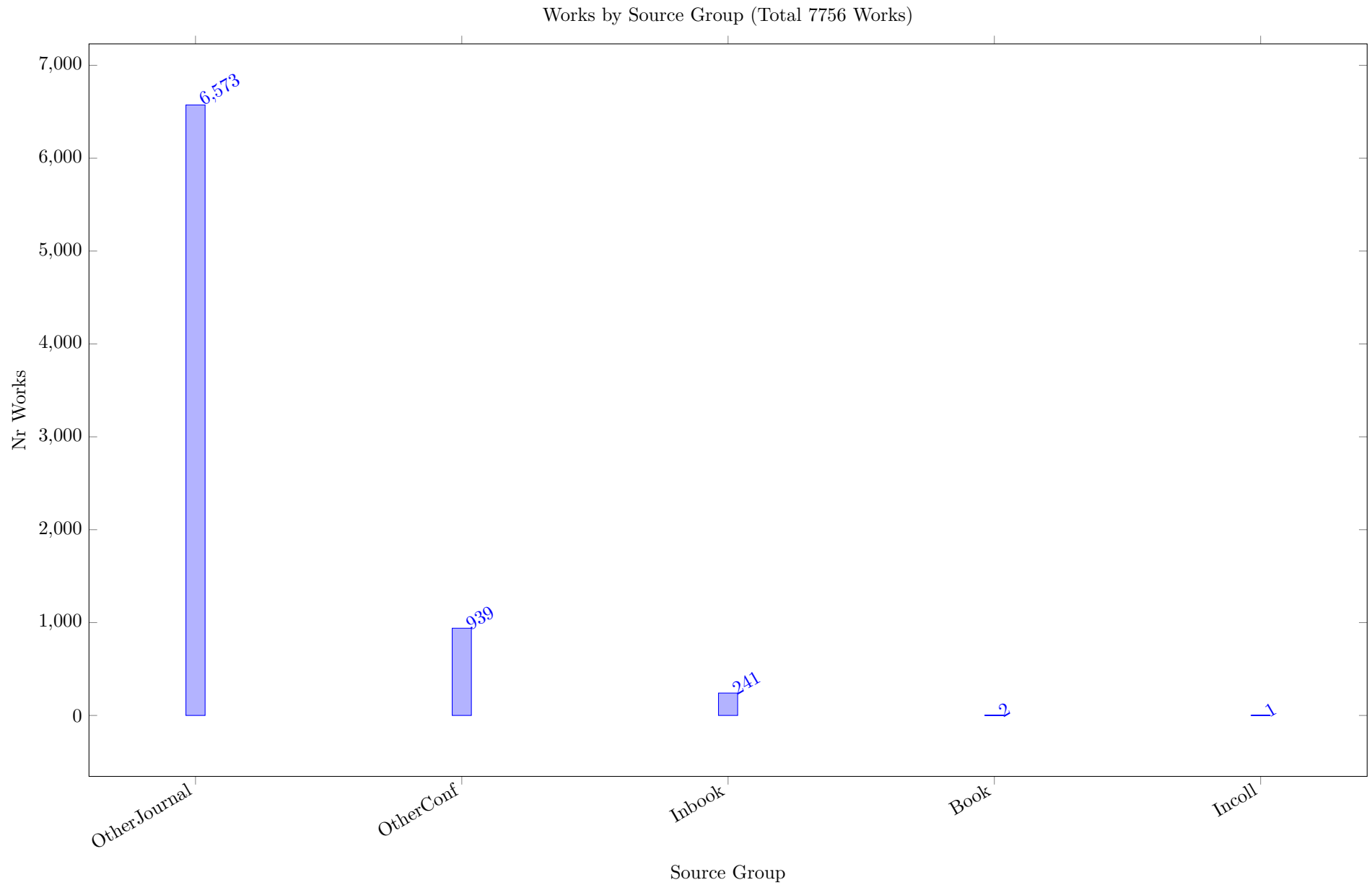
15 Citations by Year and Source Group

We have defined a number of source groups to group publications of a given type together, without using the full conference series and journal distinctions for grouping. The following table lists all defined source groups for this survey. Adding groups requires updates to the source code.

Table 13: Source Groups

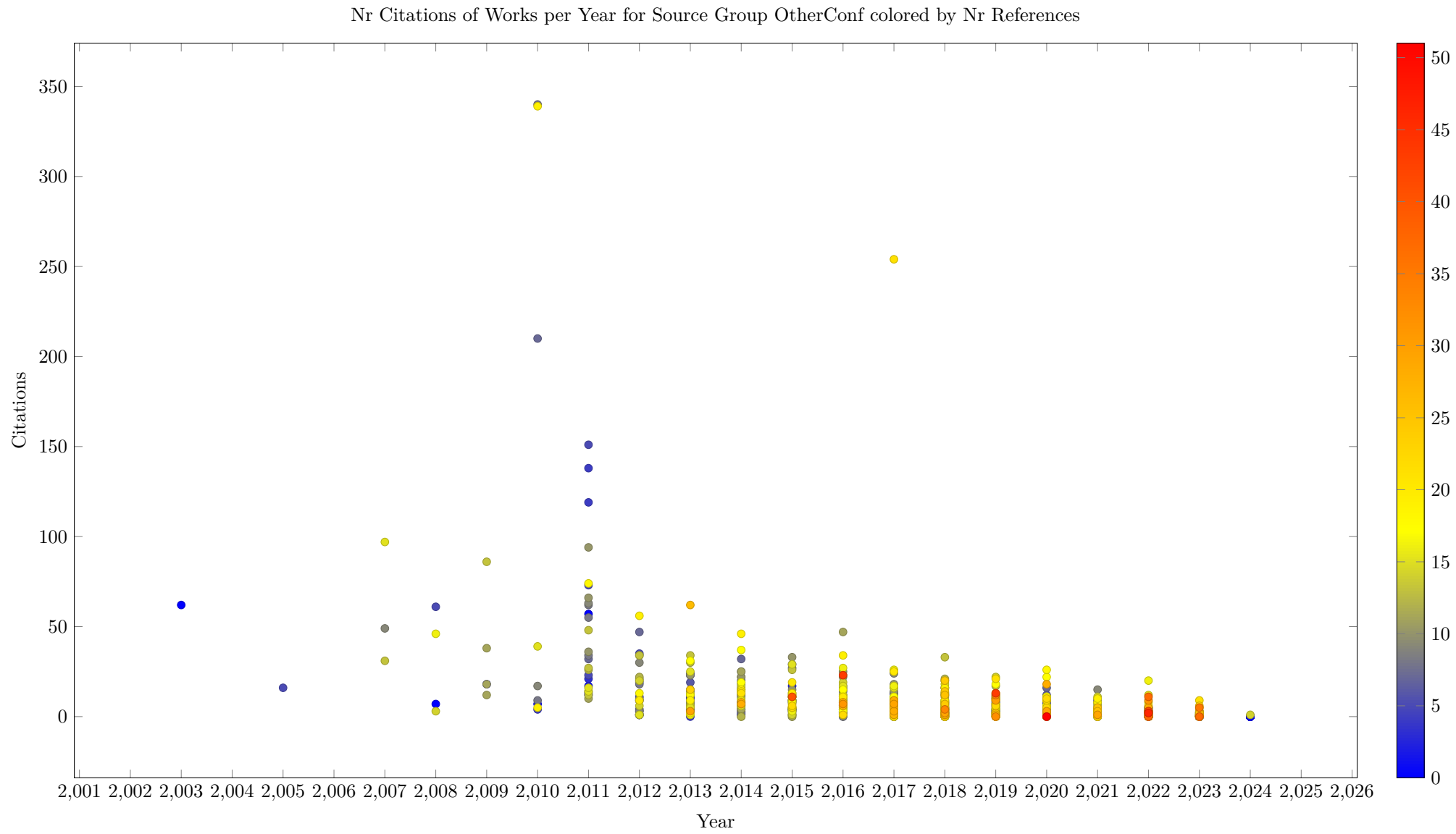
Name	Description
Background	Background Material
OtherConf	Any Conference
OtherJournal	Any Journal
Book	A book
Inbook	Chapter in a book
Incoll	Chapter in a collection
Thesis	A thesis
Other	Any other published work

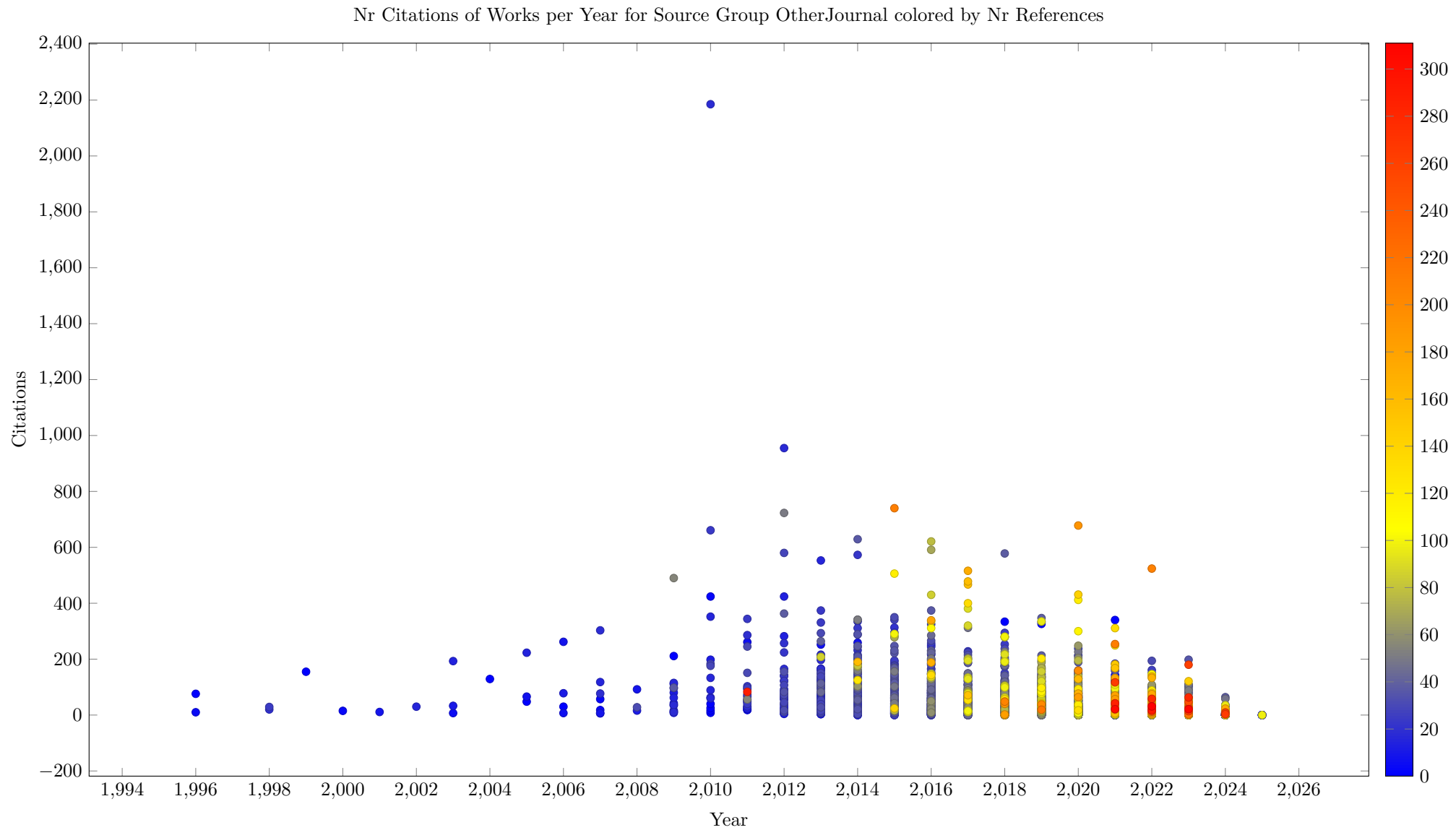
The first plot in this section shows how many works in each source group have been published. This considers the complete time period of the survey.

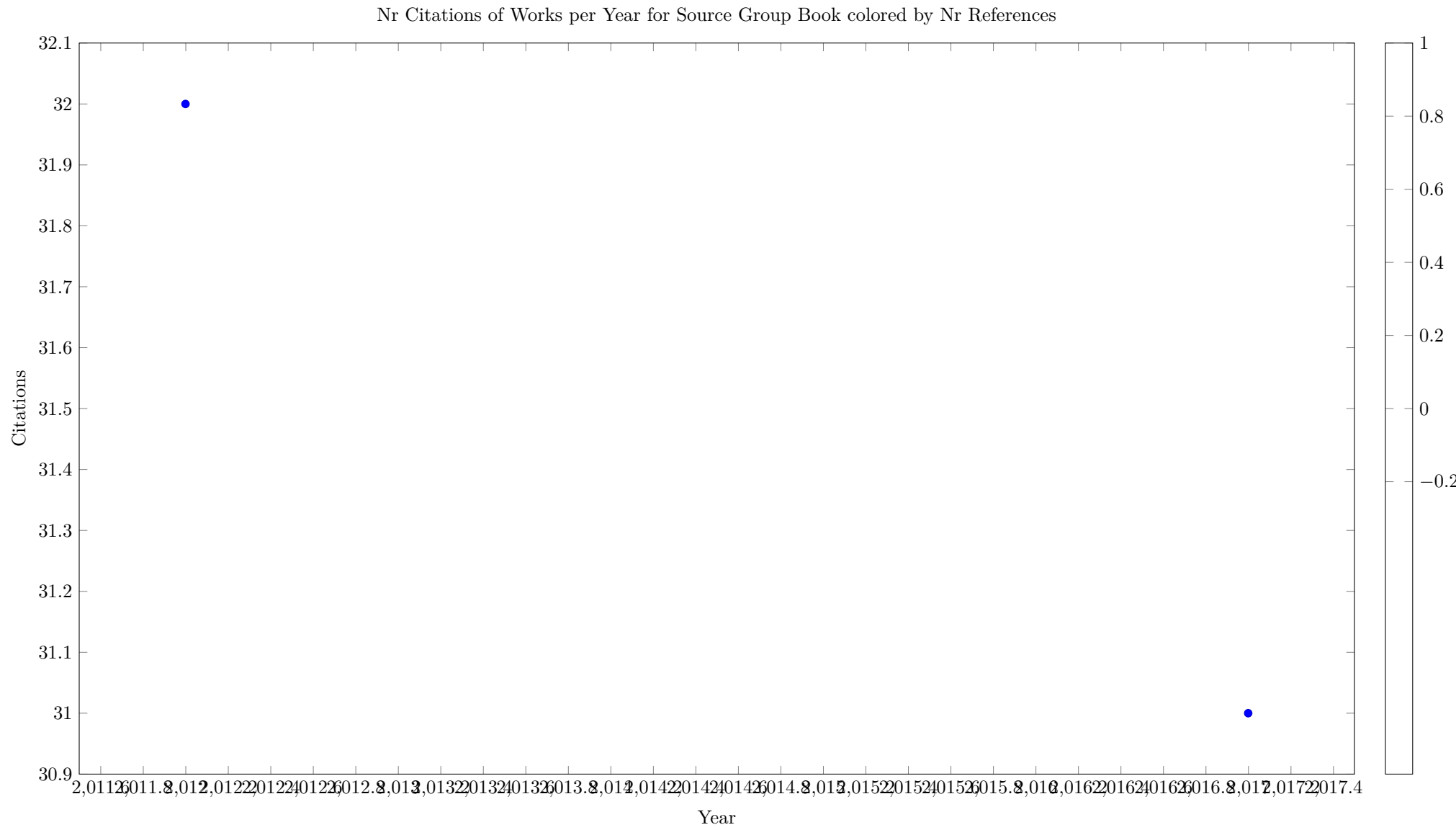


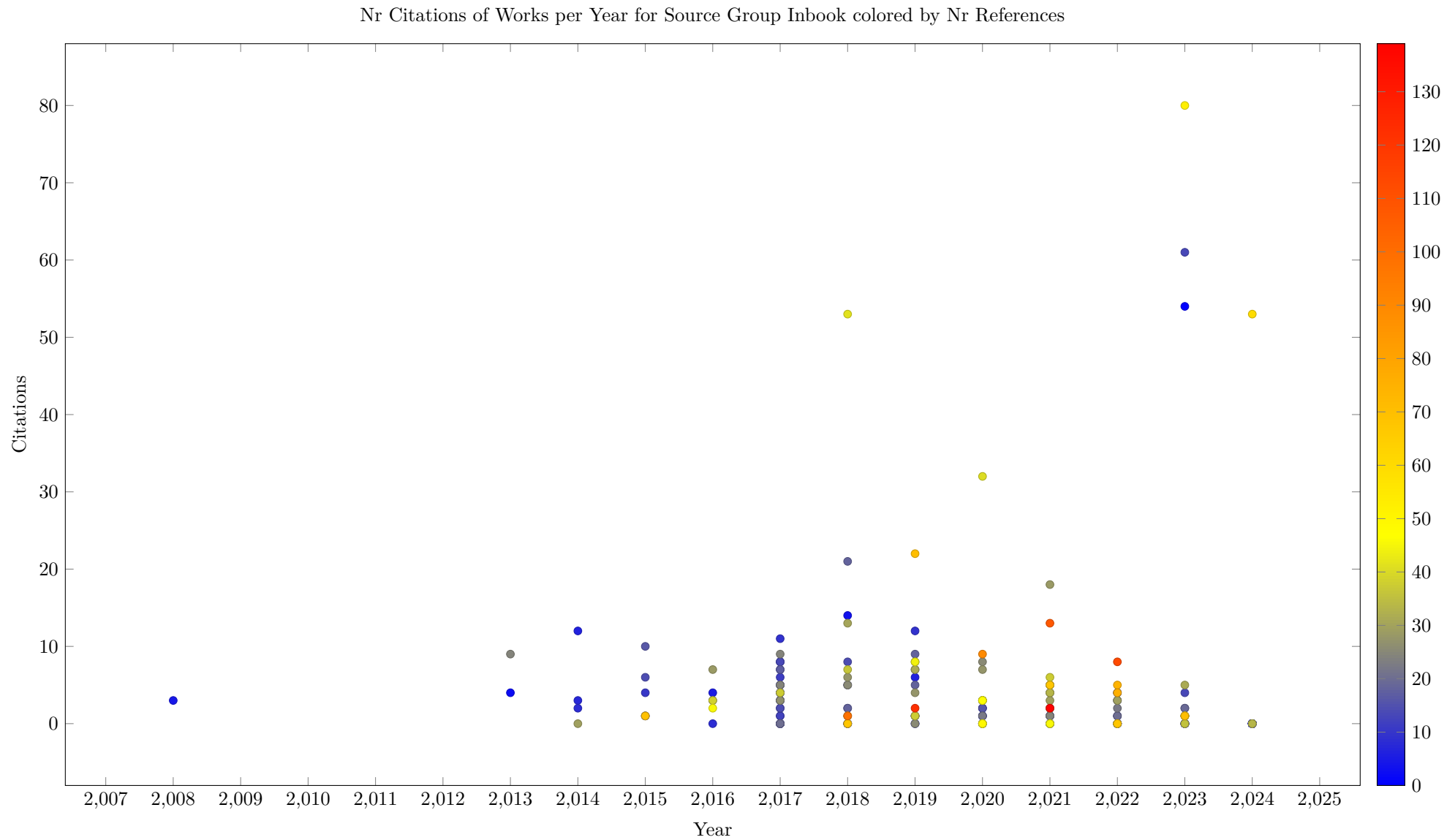
15.1 Source Group Citations by Year

We plot for each source group the number of citations obtained by papers published in a given year. This plot gives both an indication in which period the source group was active, and how significant the works in the source are. It is of course natural that more recent papers have fewer citations than papers published many years ago.









15.2 Reference Flows

The following table looks at references between source groups that are contained in the survey, i.e. where both the citing and the cited work is included in the survey. We show how many papers referred to in the group on the left belong to the group in the column.

Table 14: Reference Flows

	OtherConf	OtherJournal	Book	Inbook
OtherConf	300	1600		17
OtherJournal	1063	23688	4	142
Inbook	94	728		14
Incoll	3			

The entries in the previous table are not directly comparable, without knowing how many works are in group. The next table presents a normalized view, where we divide the flow count by the product of the group sizes. This produces a likelihood of a paper in the source group citing a paper in the target group, given as a percentage from 0 to 100. We can see that the likelihood does not depend on the prestige of the target, e.g. papers at AAAI are cited much less than papers in CP.

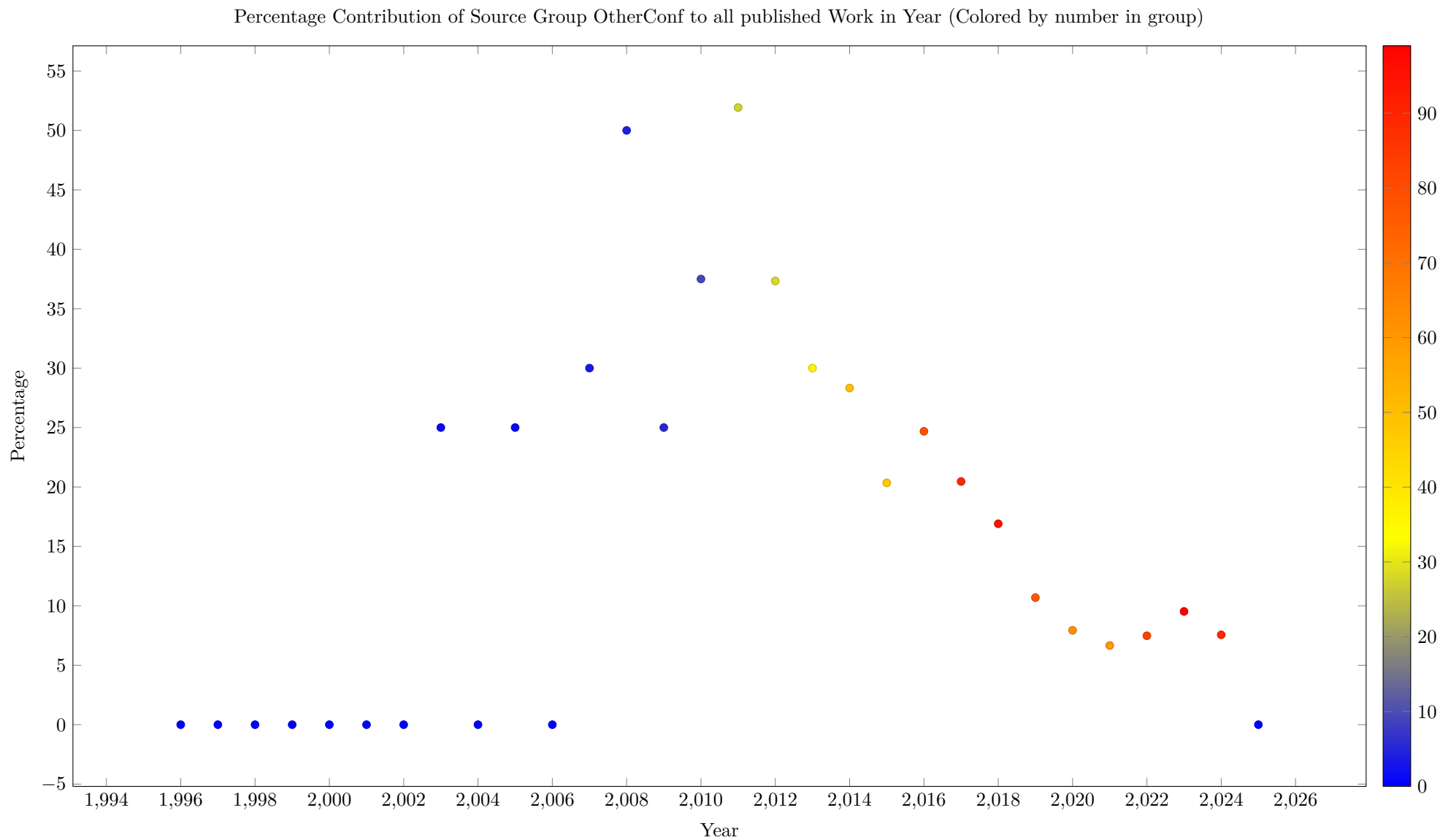
Note that the numbers are derived from the flows contained in the survey, which are based on the OpenCitation reference links. If such links are missing, or we are missing works in some group, then the results will be affected.

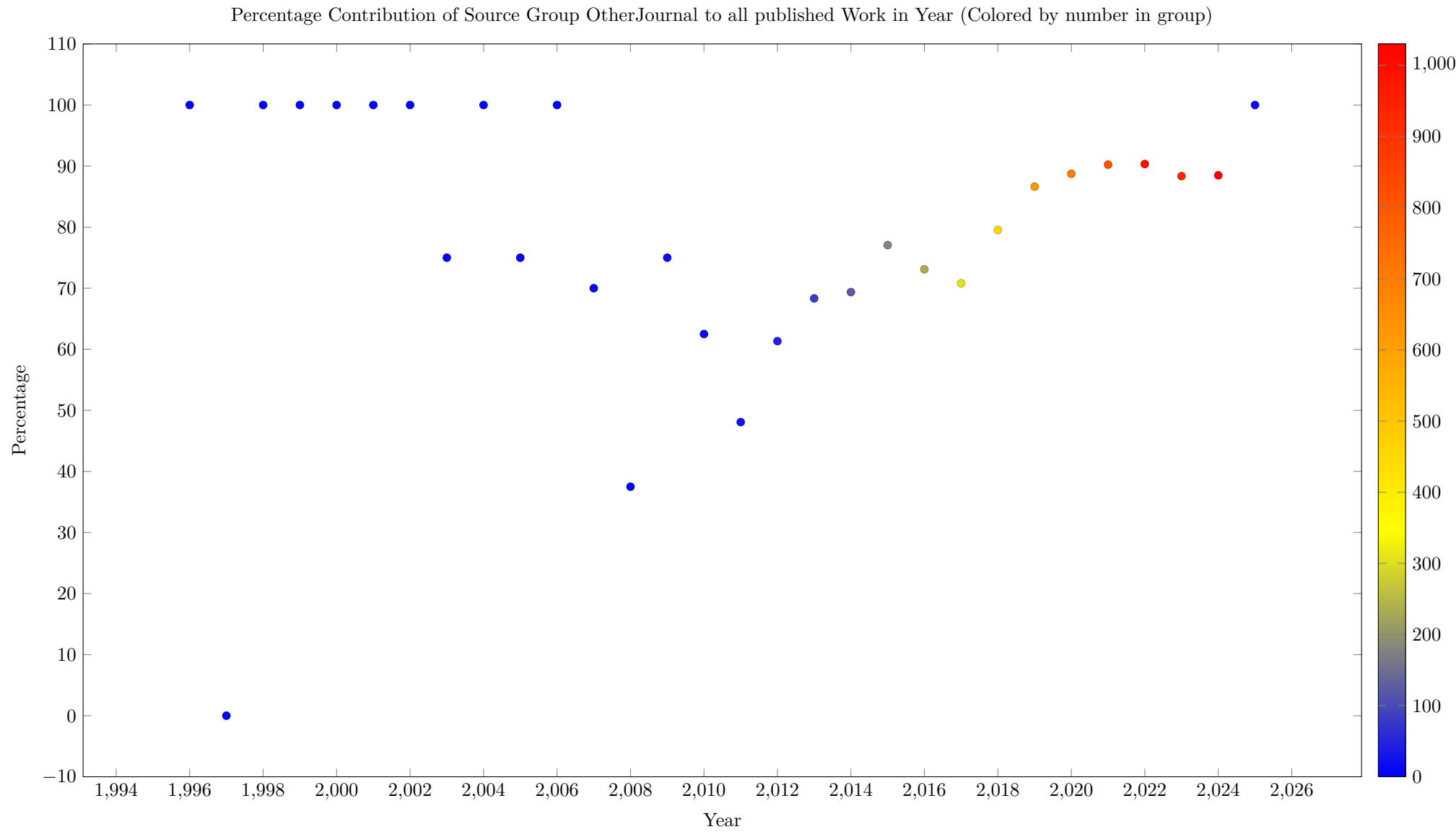
Table 15: Reference Flows Normalized

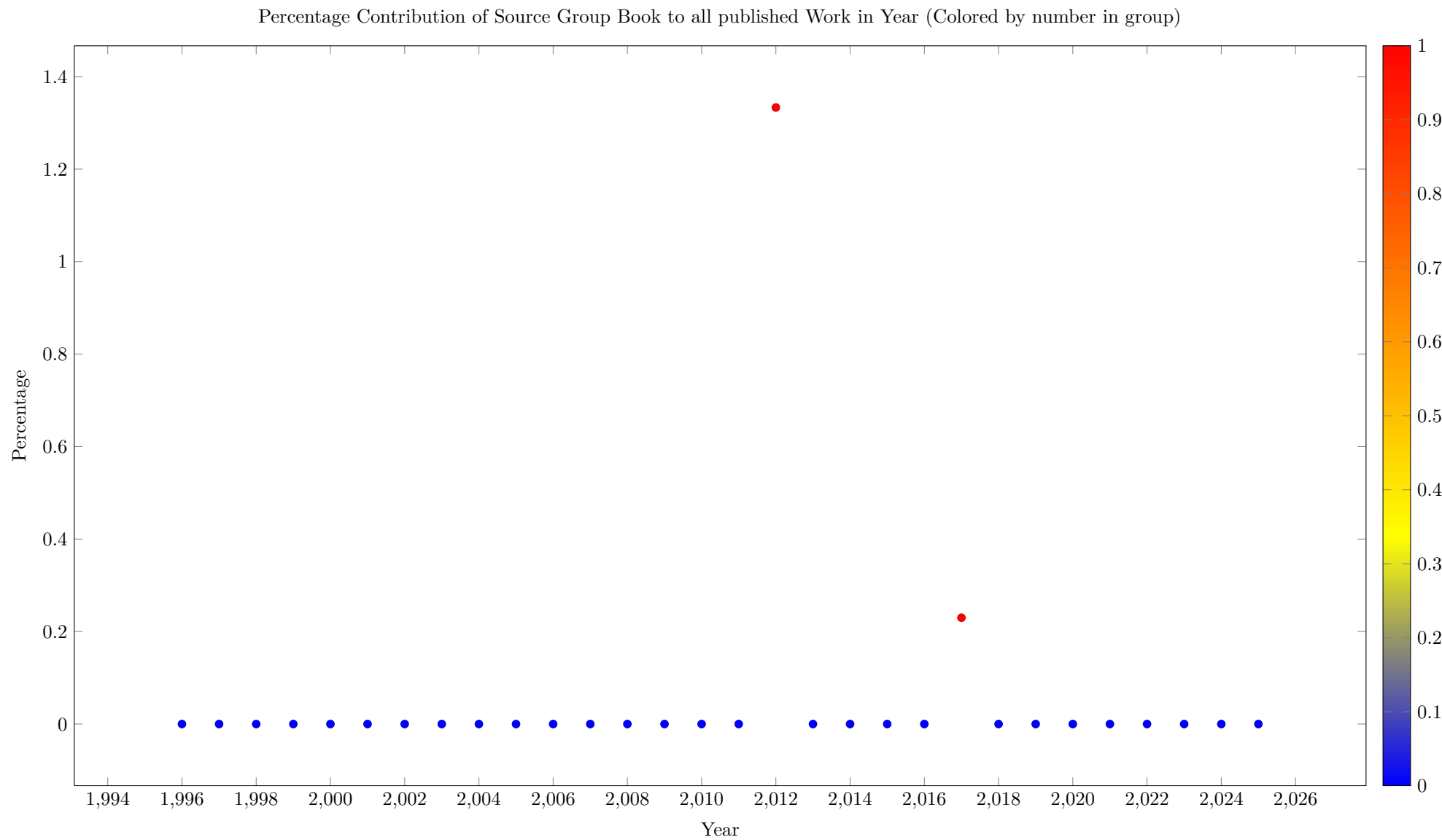
	OtherConf	OtherJournal	Book	Inbook
OtherConf	Infinity	Infinity		Infinity
OtherJournal	Infinity	Infinity	Infinity	Infinity
Inbook	Infinity	Infinity		Infinity
Incoll	Infinity			

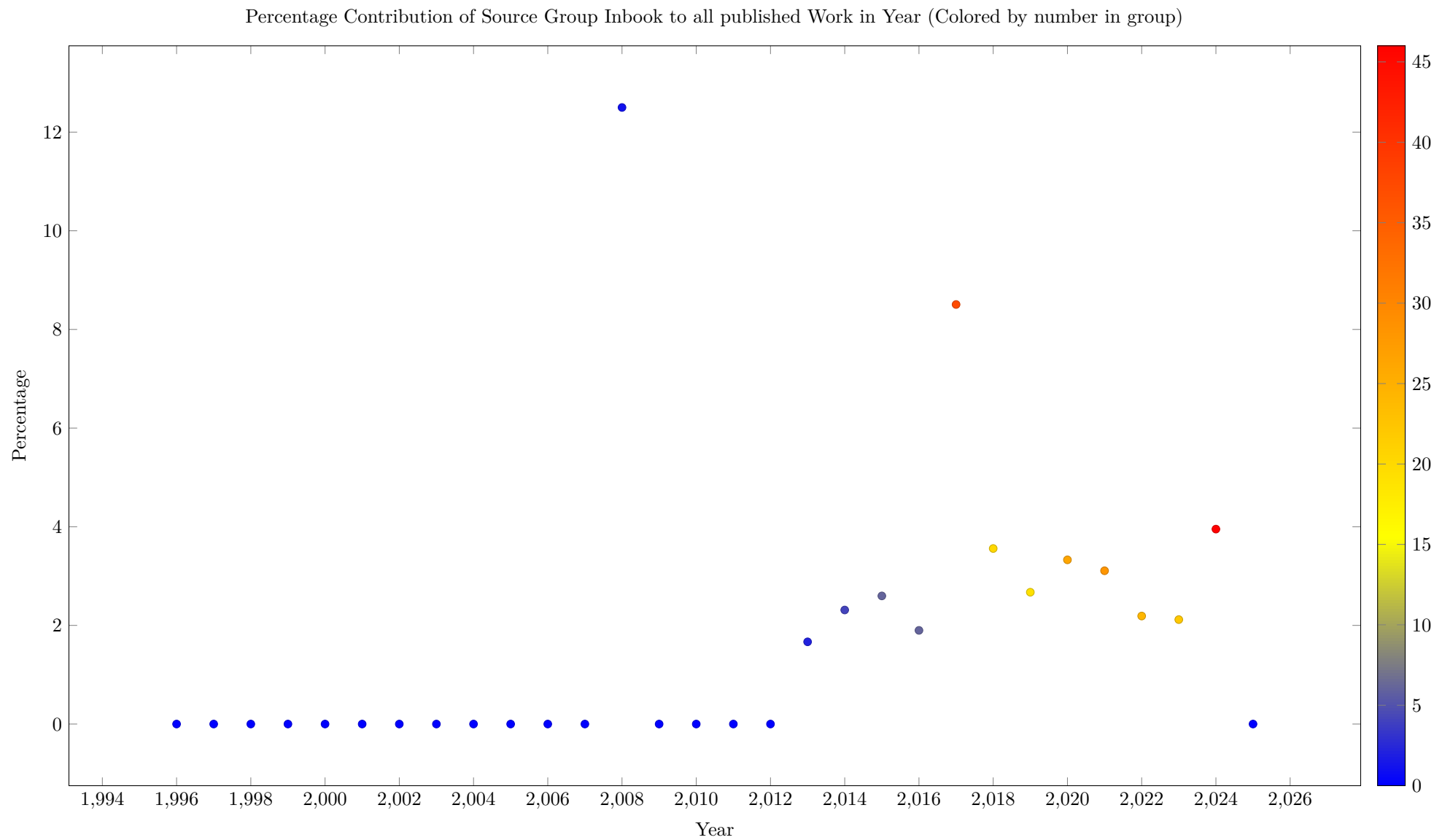
16 Contribution of Source Group to Total Works per Year

The following plots show the percentage of works published in a year belonging to a specific source group. This plot helps to understand how important that group is to the field over time

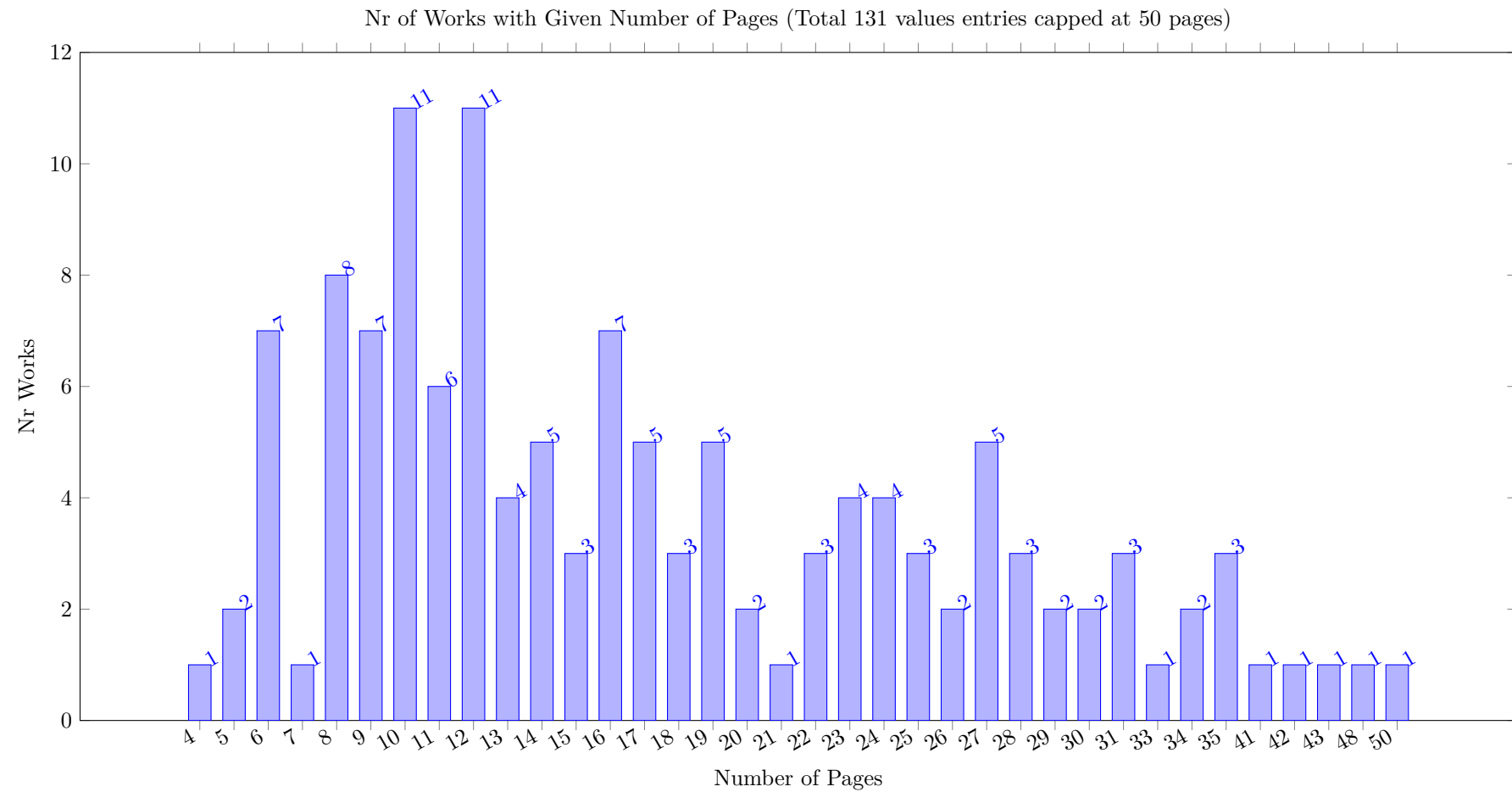




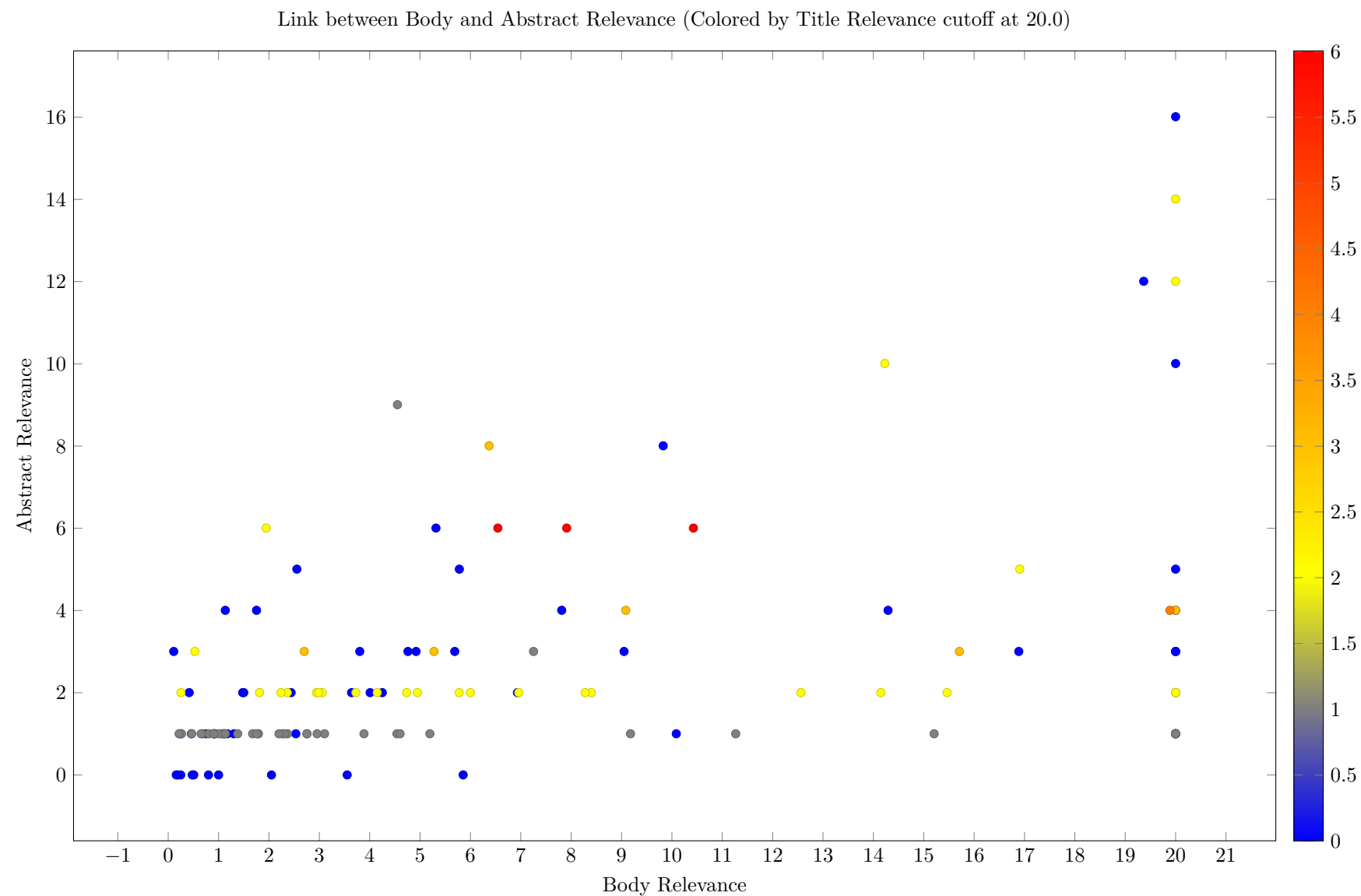


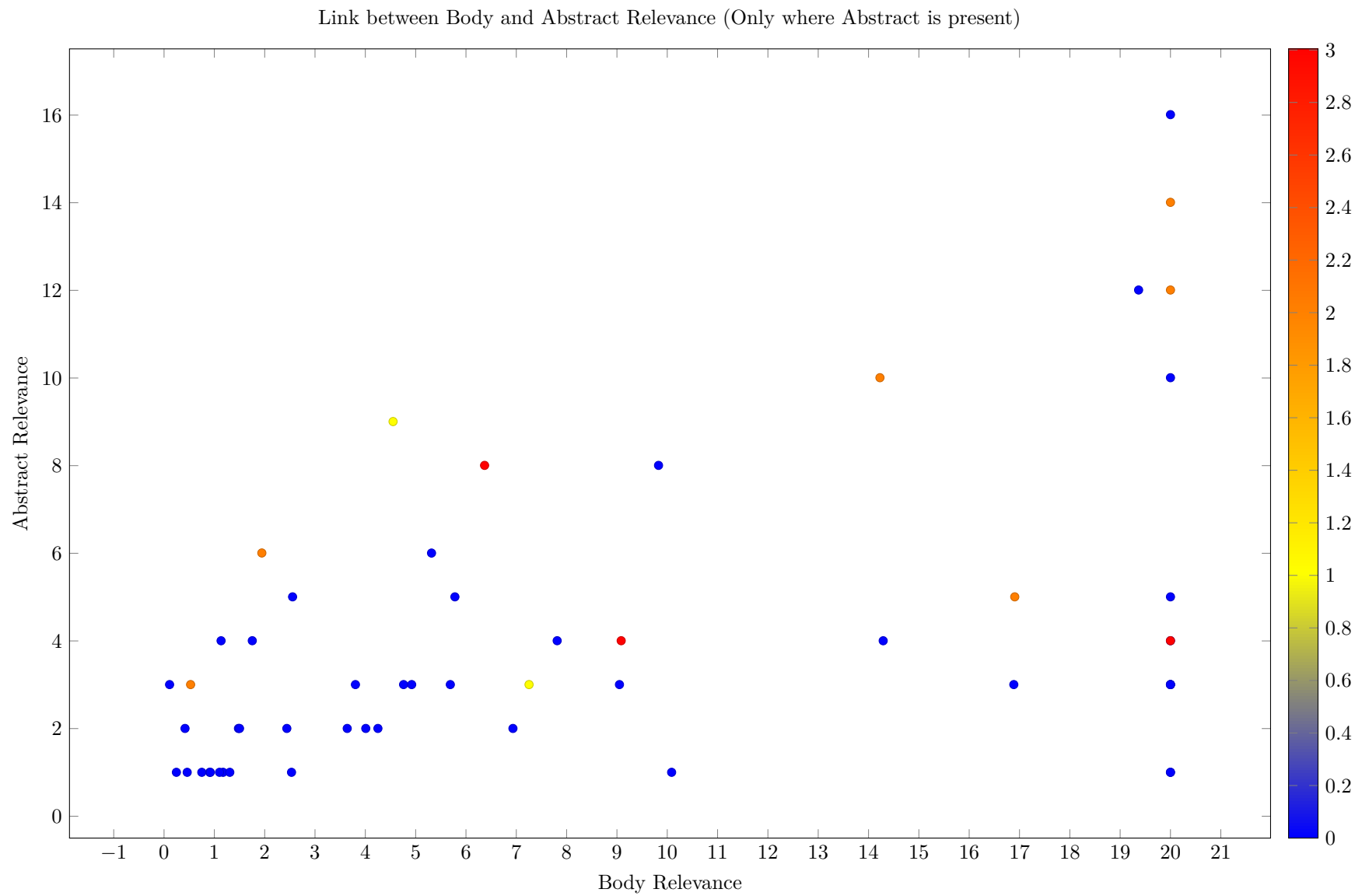


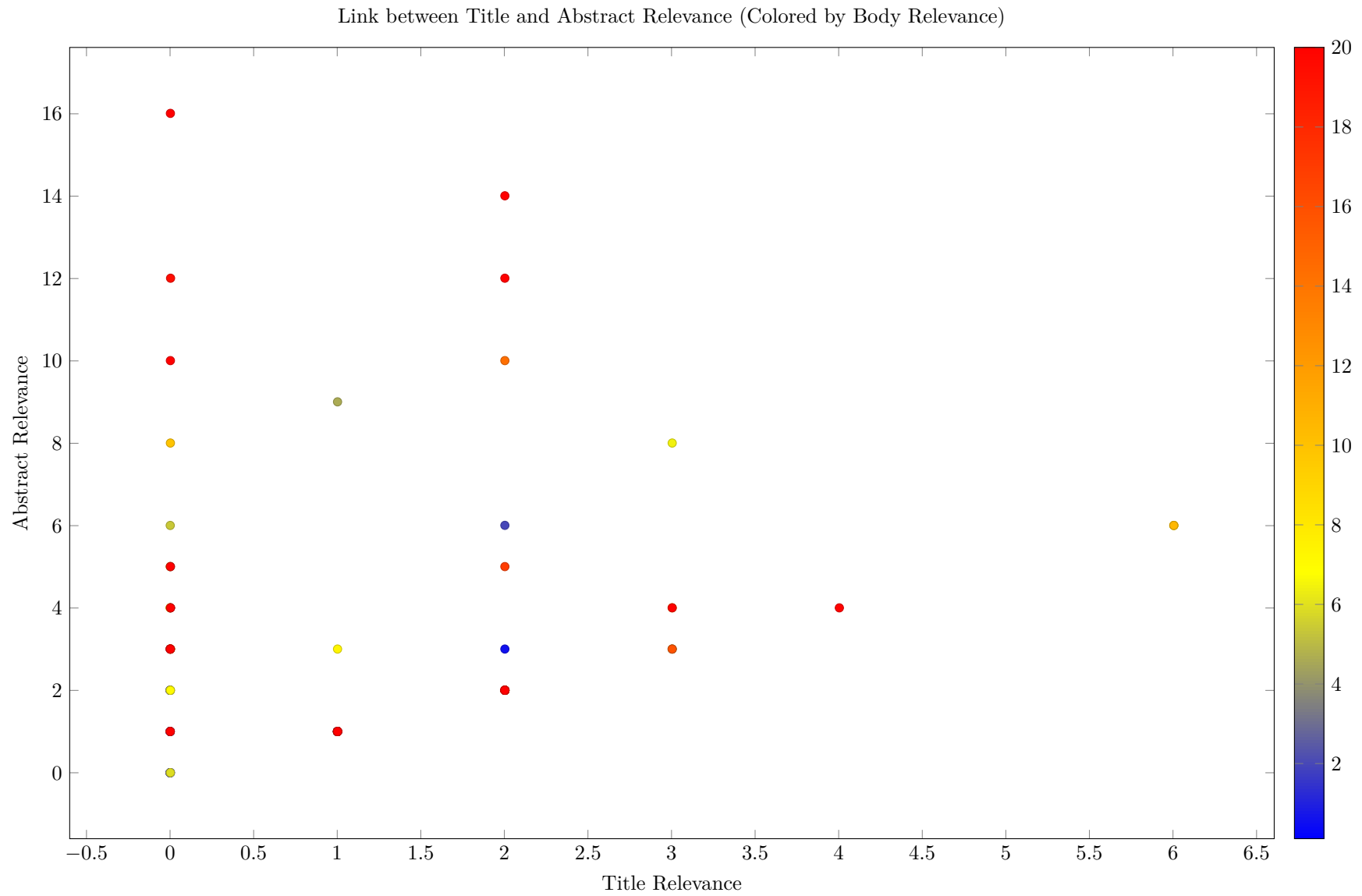
17 Page Length Distribution

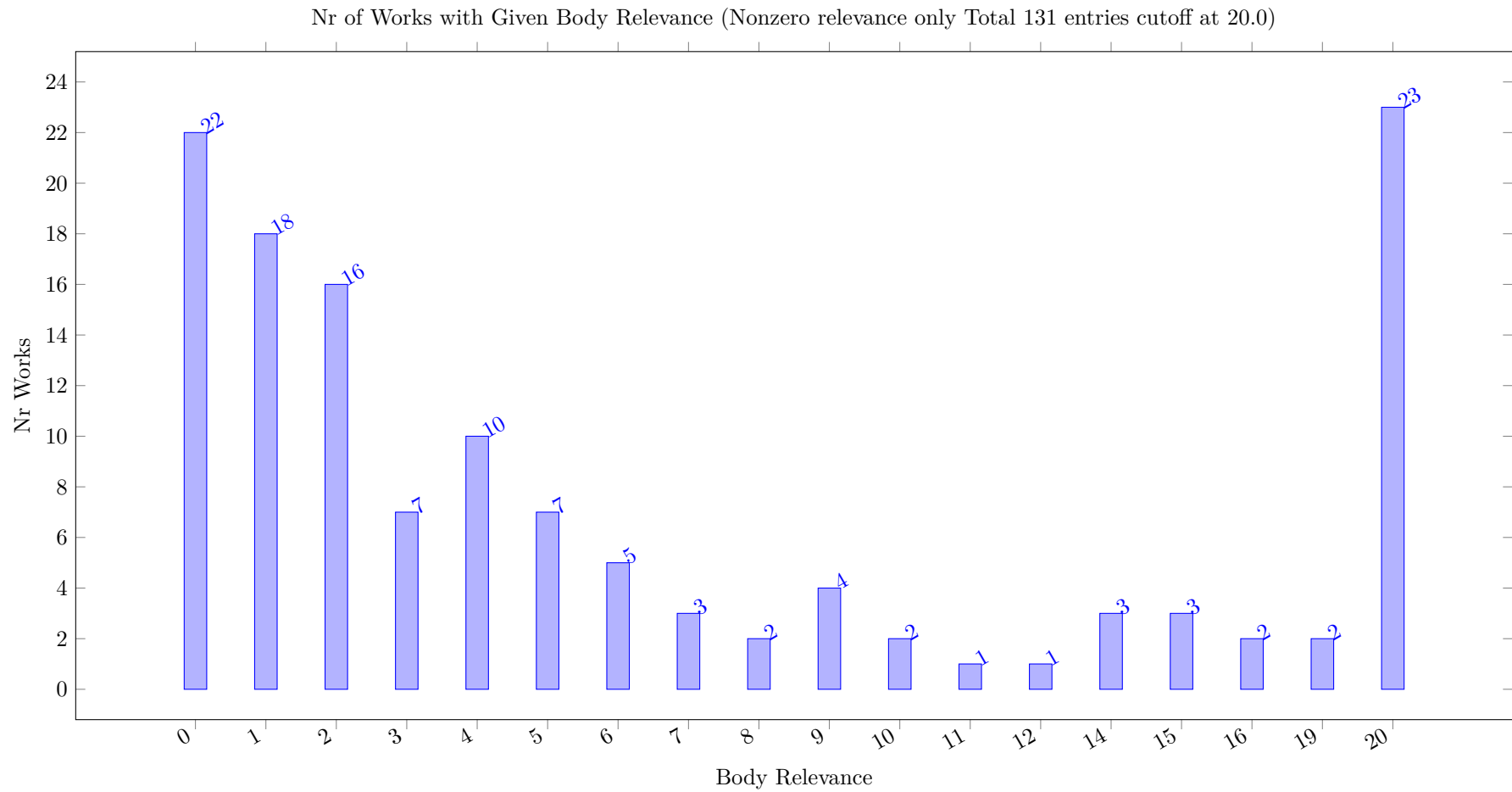


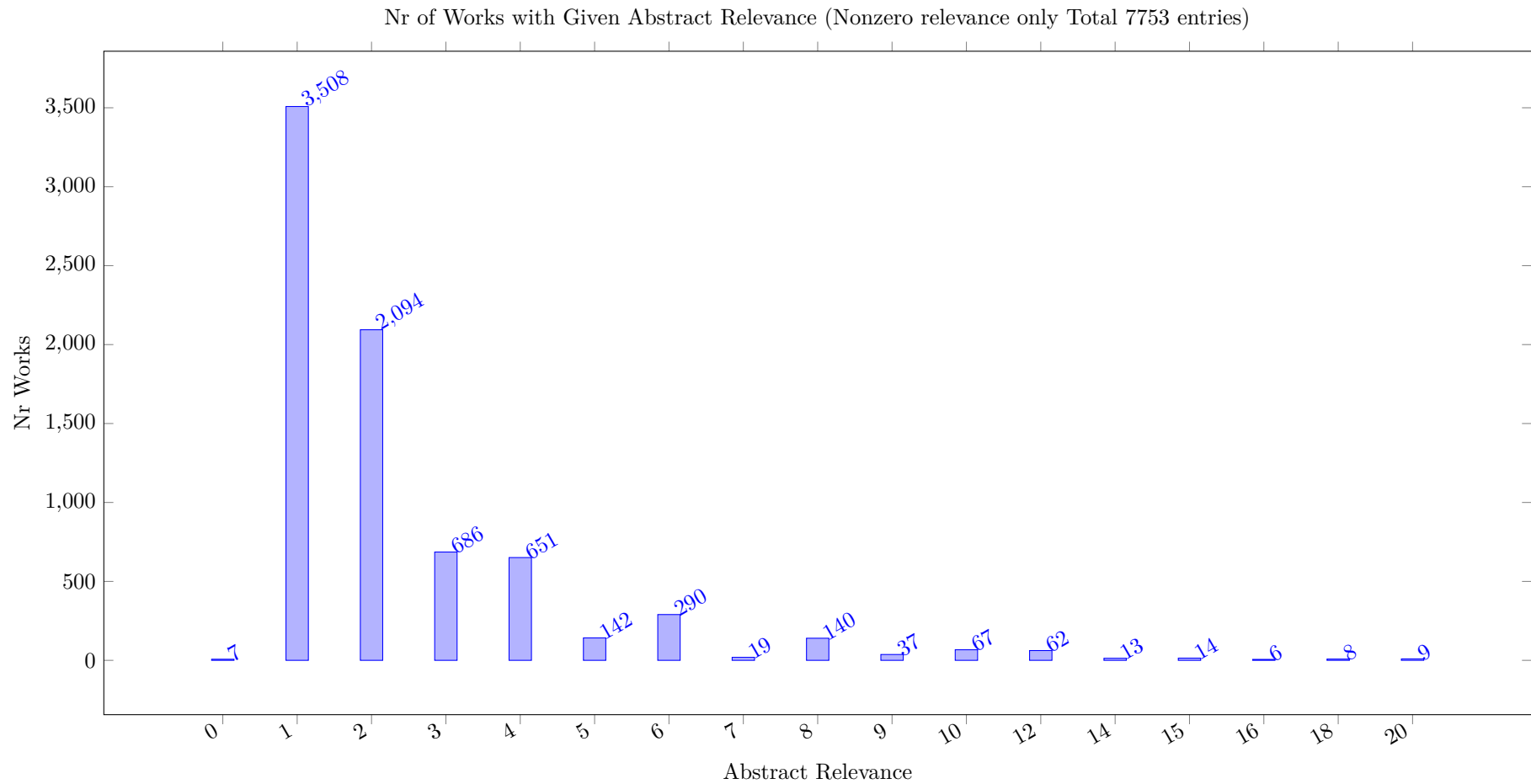
18 Relevance Distribution

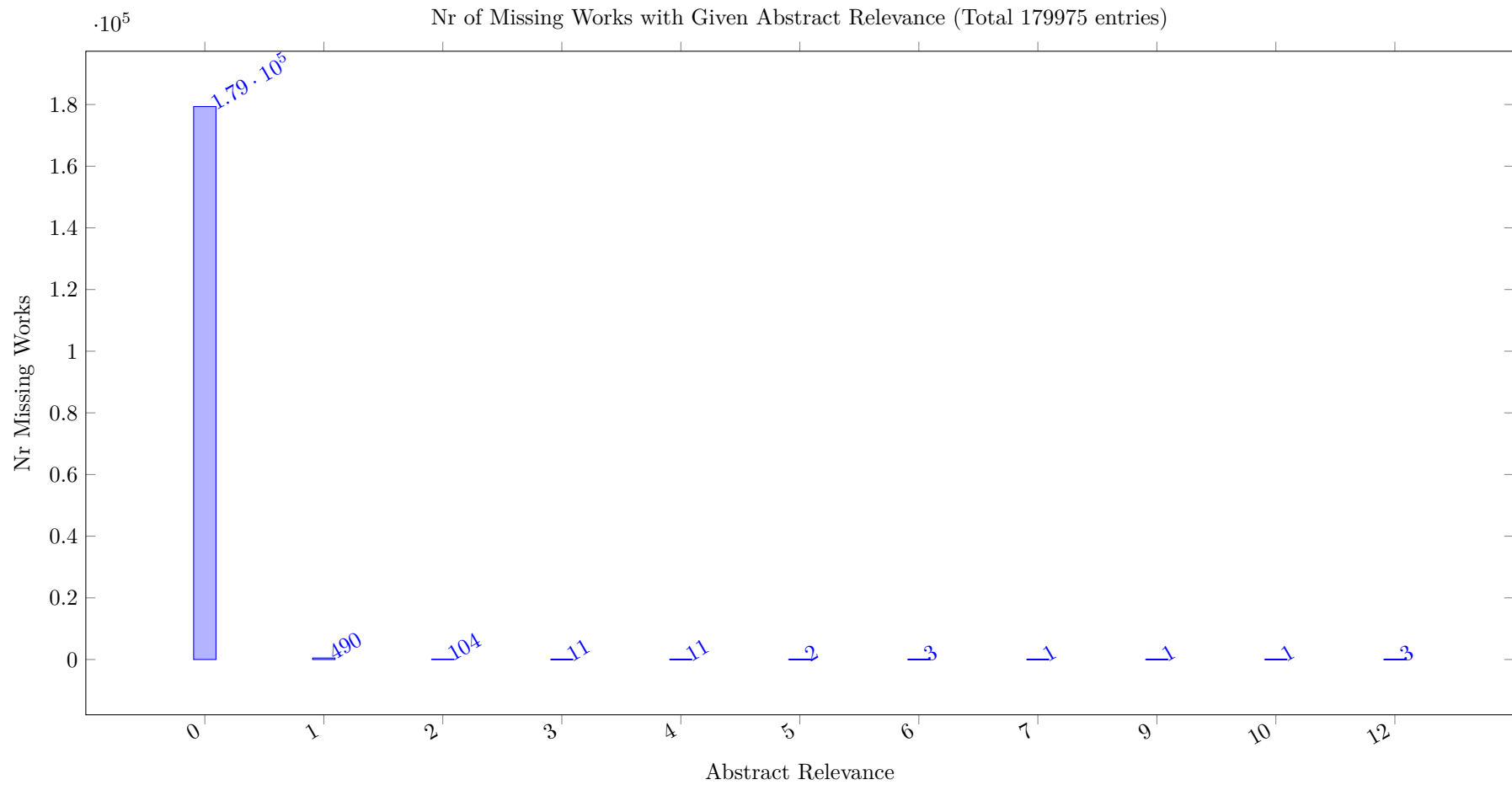


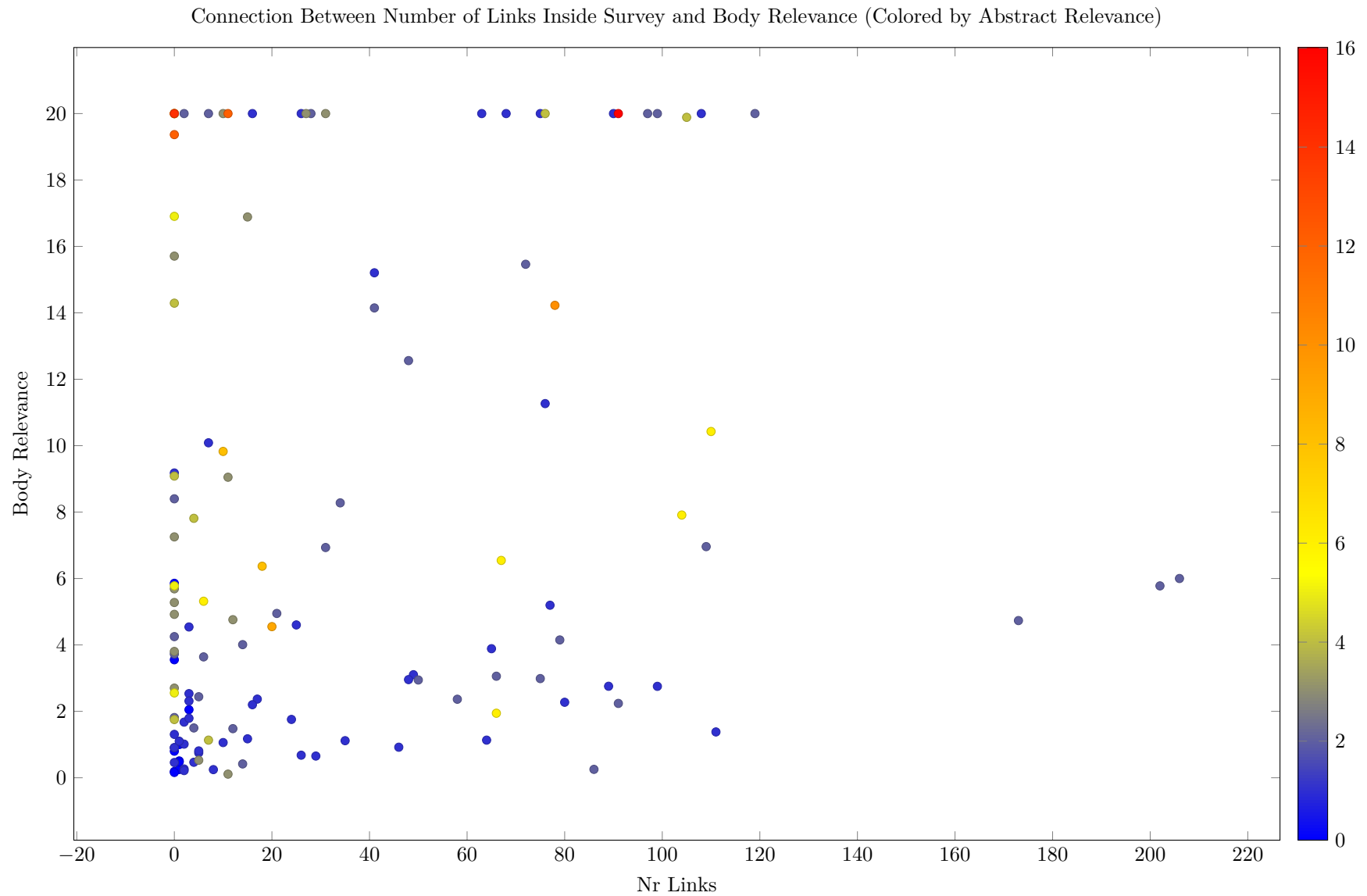


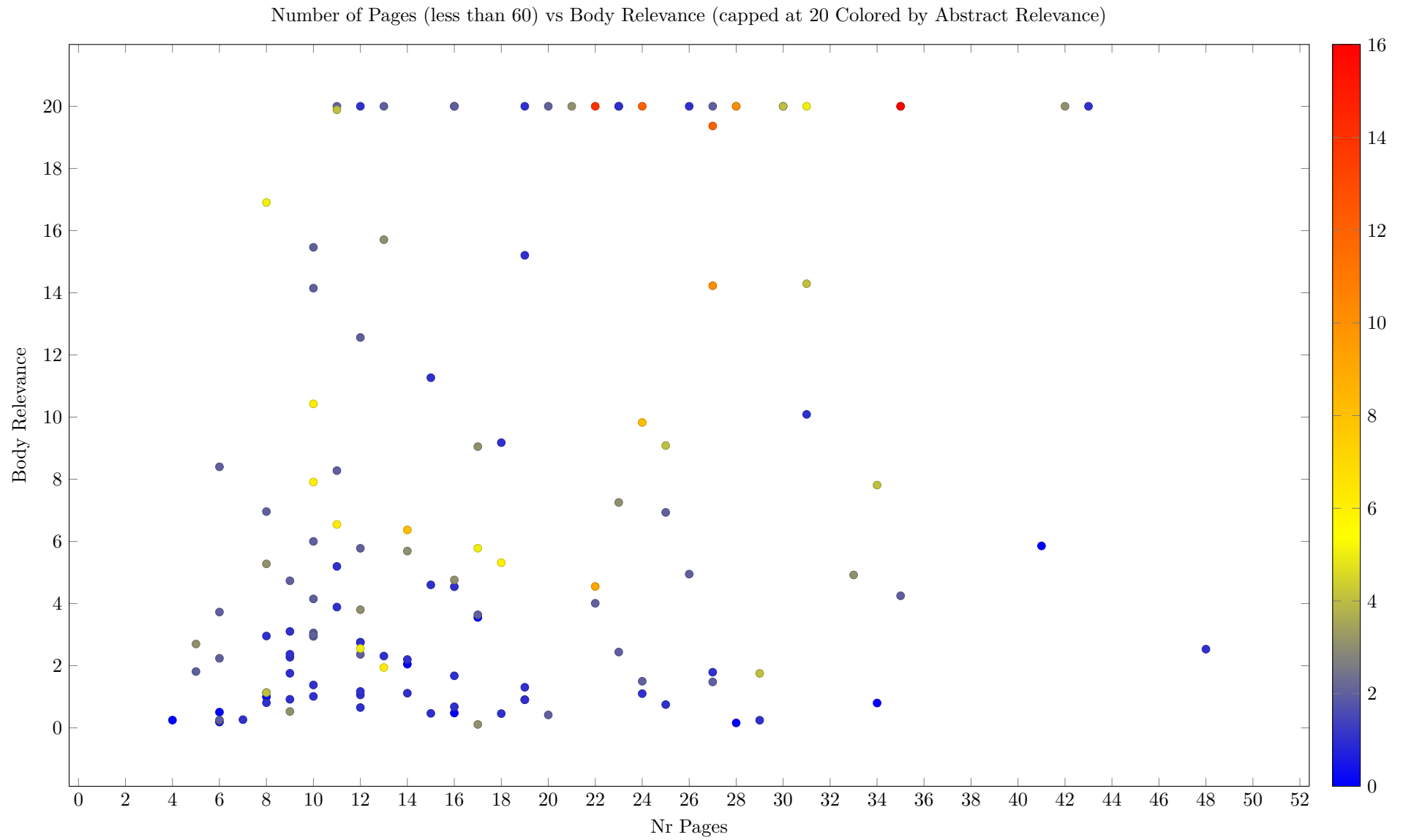












19 Most Important Publishers

